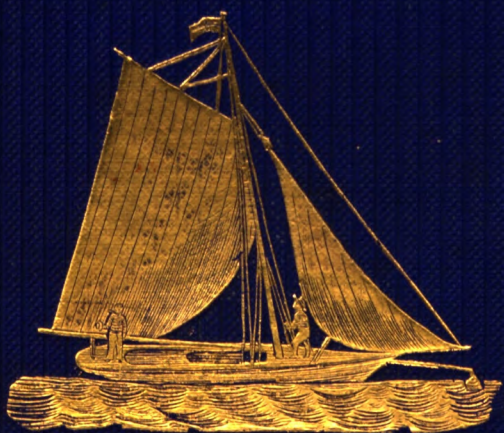
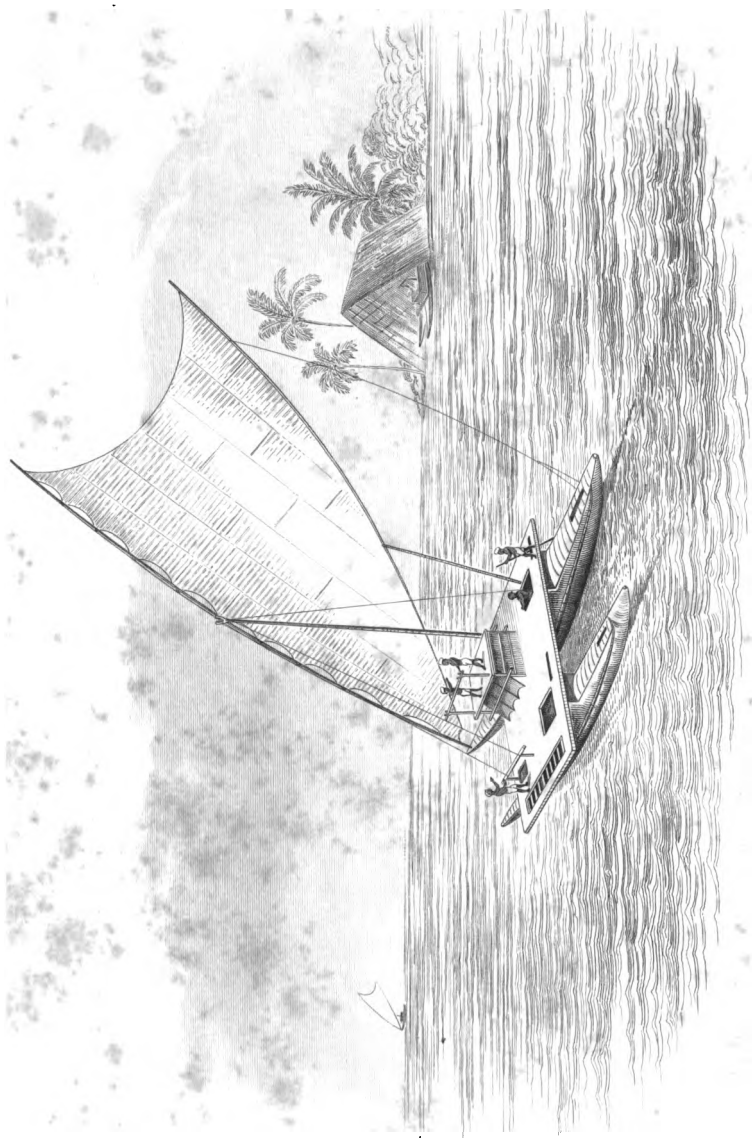




THE SAILING BOAT.

THIRD EDITION.



POLYNESIAN SAILING CANOE

See Page 246

THE SAILING BOAT:
A TREATISE ON
ENGLISH AND FOREIGN BOATS,
DESCRIPTIVE OF THE VARIOUS FORMS OF
BOATS AND SAILS OF EVERY NATION;
WITH PRACTICAL DIRECTIONS FOR
SAILING, MANAGEMENT, &c.

BY
H. C. FOLKARD, Esq.

THIRD EDITION.

LONDON:
LONGMAN, GREEN, LONGMAN, AND ROBERTS.

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(The Right of Translation is reserved by the Author.)

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PREFACE TO THIRD EDITION.

THIS little work has been a long time out of print, as the Author could not conveniently spare the time from professional occupations to devote that attention he has long wished to bestow upon it. The New Edition has been carefully revised, and it is hoped will be found an improvement upon the two previous ones. The extra matter contained in the New Edition being so very considerable, it was found necessary to print the work in a smaller type, so that two pages of the present edition contain, on an average, the amount of three of the previous ones; there are, besides, twenty-four additional pages of letter-press. The present work also contains double the number of engravings of either of the other editions. All the new illustrations, with the exception of the Land and Ice Boats, have been engraved from drawings by the Author. Many of the curious forms of Foreign Boats he sketched, with kind permission, from models in the Indian and United Service Museums; others from drawings, paintings, and photographs in his own possession; a few only having been copied from authentic works of voyages and travels.

Throughout the present work care has been taken to give accurate descriptions, as well as illustrations, of all the most curious and remarkable forms of boats

and sails to be met with in the various countries of the Universe. In the part relating to Ancient Boats, a few additional pages have been added: in the two portions—one descriptive of English Boats, and the other relating to the sailing and management of boats—only a few additions have been made, as those parts of the work contain purely the results of the Author's observation and long practical experience in the sailing and management of boats; and were written entirely without reference to any other work. In the part devoted to Foreign Boats, by far the most extensive additions have been made; indeed, more than two-thirds of it is entirely new matter, a good deal of which has been compiled from researches among authenticated works of voyages and travel.

The heavy expenses incurred about the numerous additional illustrations, render it impossible that the present edition can be offered to the public at the original price; therefore, instead of half-a-guinea, the future price will be twelve shillings and sixpence.

LONDON,
JANUARY, 1863.

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ERRATUM.

In page 2, last word of fourth foot-note, for "four," *read* two.

]

THE SAILING BOAT.

BOATS OF THE ANCIENTS.

" Illi robur et æs triplex,
Circa pectus erat qui fragilem truci
Commisit pelago ratem,
Primus :—————" HOR. 1 Od. iii. 9.

ANCIENT boat-building may be justly termed the stepping-stone to, and origin of ship-building; the vessels of the ancients being no larger than a ship's launch of the present day.

Though for centuries provided with boats and vessels of burthen, it was a long time before the boldest mariners dared trust themselves and their vessel far from land.

The earliest mode of navigation was by rafts, to which were afterwards added borders of wicker-work, covered with skins or leather. Of this kind were the Coracle of Ancient Britain, and the *Cymba sutilis* of Virgil.

It is mentioned by Homer, that the boat built by Ulysses was put together with wooden pegs instead of bolts; ¹ and that the gunwale was raised by hurdles of osiers to keep off the waves of the sea. ²

The Egyptians had boats of *terra cotta*, and leaves of the papyrus; the Indians of bamboo, or similar cane, from knot to knot, or of many pieces of rush. Skins were used for

¹ Od. v. 249.

² Ib. 256.

boats by the Romans and others, and the boatmen were called *Utricularii*.

To be represented in a boat was the Egyptian symbol of *apotheosis*; and many Emperors (as our Kings in a ship on their coins) are thus distinguished.³

The early Greeks are mentioned in history as the first who devoted attention to boat-building; the Trireme, Bireme, and other galleys were of their invention.⁴ The bows of some of the war galleys were ornamented with carved heads of boars and other ferocious animals, projecting four or five feet; beneath which, was a sharp iron spike or *rostrum*. It was in boats of this kind that the first naval action recorded in history was fought between the Greeks and their colonists, the inhabitants of Corfu.

The Romans afterwards improved upon the Trireme, and built a faster class of vessels, called the Liburni: these were more manageable than the others, and better adapted for sailing. The naval engagement at Actium, in which Augustus Cæsar was victorious over Antony, was fought and won in Liburni vessels.

An Etruscan boat has the prow turned up, stern flat and concave, with a hole in the side for the rudder. The latter is merely a long steering oar.⁵

In most ancient vessels, it is to be observed the prow was made in the form of a fish, dolphin, or some animal with the eyes very distinctly marked on both sides.

Vessels with oars preceded those with sails. Homer mentions masts, but not fixed, only put up as wanted.

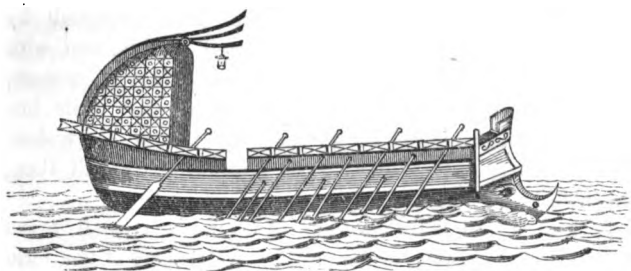
As to rudders, sometimes there were two to a vessel, at others four, two at the prow, and two at the stern.

The engraving at the top of page 3, represents a Trireme or boat of the ancients, with three benches of oars, after Bafius, Schæffer, and others.

³ Fosbroke's *Encyclo. Antiq.*, Charnock's *Marine Architecture*, &c.

⁴ The Trireme signifies a boat or galley with three benches of oars, one above another; the Bireme has four.

⁵ Kirke, pl. 18.



The ancient practice of rowing was as follows :—a director or person called *Celeustes*, gave the signal for the rowers to pull, and encouraged them by his song or cry. This song termed the *celeusma* was either sung by the rowers, played upon instruments, or effected by striking a gong, after the manner of the Chinese, Japanese, and others at the present day. Ossian and others mention the rowing song.—

“ And all the way to guide their chime,
With falling oars they kept the time.”

The commander of the rowers, called *Hortator*, *Remigum*, *Pausarius*, and *Portisculus*, was placed among them in the middle of the boat. He carried a staff with which he signalled by waving or otherwise when his voice could not be heard. The Anglo-Saxon, batswan, or boatswain, also used a staff, wherewith to direct the rowers.

The Greeks had boats called *ampheres*; these were long and narrow, and were rowed by one person with a pair of sculls. Rowing with the face to the prow is mentioned as customary with the ancients; but this may have been paddling, or pushing with a paddle or a setting pole.

The oar upon the Etruscan vases is in the form of a very narrow pyramid from top to bottom.

Baldarius was the inventor of oars, as applied to large fighting vessels.⁶

⁶ Aristoph. Ran. iv. 2. Rutil. Itin. i. 367. Pedian. ad Cicer. p. 37. Xenoph. l. 5. Polyb. i. 21. Encyc. Strutt's Sports, 70. Kirke's Hamil. Vases, pl. 18. Evelyn's Mem. i. 196.

Masts and sails are said to have been invented by Dædalus. Varro says they were invented by Isis, who, with an affection bolder than usually falls to the lot of women, sailed in quest of her son Harpocrates; so that while her maternal fondness urged her to the completion of her wishes, she appears to have displayed to the world arts, till then, unknown to it.⁷

In Stosch and the Florentine Museum, is a small vessel with oars, the prow of which ends in a *cheniscus* of the form of a swan's neck: precisely in the place of a mast and mizzen sail, are two large extended wings, proper to catch the wind, as if for flying. This clearly explains the fable of Dædalus.

The materials of which sails were anciently made were rushes, broom stuff, skins of animals, and the dried skins of the intestines of animals and fish; linen and hemp were afterwards used; indeed, from the time of Homer, linen was in use. The forms of ancient sails were various, square, circular, crescent shaped, and triangular, and the colours white, blue, purple, and sometimes curiously painted.

According to Pliny, they were at first set one above another on the same mast; and afterwards on two masts, at the stern and prow.

The sails set on the stern or mizzen mast were called *epidromus*; those on the fore mast at the prow *dolones*; at the top of the mast *thoracium*; stun sails, called *orthiax*, were also used occasionally in very light winds. Sometimes when two or more masts were used the sail of the main-mast was called *artemon*.

It is clear that both sails and oars were employed in many vessels of the ancients. Winckelman, however, observes that ships disposed for battle had neither sails nor yards.⁸

⁷ "Vela Isis rata primum suspendit, cum per mare Harpocratem filium audaci fœminea pietate perquireret: ita dum materna charitas suum desiderium festinat explere, mundi visa est ignota reserare."—Lib. v.

⁸ See also Stosch. Mus. Flor. t. ii. pl. lx. xi. 5. Enc. Mon. Ined. ii. 280. Pliny, L. ii. 48.

The Boats and vessels of the classical ancients were of many kinds; with ten, twenty, thirty, and up to 100 oars. Those distinct from war service were as under :⁹—

Actuariæ naves—Long and light vessels, propelled both by oars and sails: never manned by less than 20 rowers.

Annotinæ Frumentariæ—Provision vessels.

Calones—Boats for carrying wood.

Cercuri—Ships of burthen, both with sails and oars.

Celoces, or the Greek *Celetes*—Light vessels, used chiefly for piracy, with only two sets of oars, without deck or *rostra*.

Catascopia—Small despatch vessels, for carrying letters and reconnoitering.

Construtæ—Those which were entirely decked.

Cubiculatæ—Those with cabins and the conveniences of a house.

Fluviatiles—Boats of the river, as distinguished from those of the sea.

Gauli—Phœnician, and round for carriage.

Hippagines, or *Hippagogæ*—Transports for carrying horses and cavalry after the fleet.

Horiolæ—Small fishing boats.

Hornotinæ—Those built in a year.

Lenunculi—Small fishing boats.

Lentriæ. Pontones fluviatiles—Those employed exclusively upon rivers.

Lembus—Light and undecked, used chiefly on rivers, and on the sea by pirates.

Liburna, Liburnica—Light galliots, used both with sails and oars; from one to five ranks of rowers.

Lintres—Canoes made out of the trunk of a tree, and capable of carrying three persons.

Lintrarii—Boatmen.

Leves—Very light boats without decks.

⁹ Vide Pollux. Aulus Gellius, x. 25. Liv. xlv. 28. Plaut. Fulgent, Enc. Plaut. Rud. iv. 2. 5. Nonn. xiii. 8.

Lusoriæ—Pleasure boats and vessels, used by the guards of the boundaries of the empire in large rivers.

Myopara—A fly boat; a corsair's vessel.

Moneres Monocratæ—Modern galleys, and vessels with only one rank of oars.

Naves tabellarie—Advice boats.

Navigiolum ad animum oblectandum—A pleasure boat.

Onerariæ—Ships of burthen, both with sails and oars.

Orariæ, Littorariæ, Trabales—Coasting vessels.

Oriæ—Wherries, and very small fishing boats.

Parunculus—A small bark.

Phaselus—A small vessel, with sails and oars.

Prosumia—A small watch boat.

Piscatoriæ—Fisher boats.

Pontones—Ferry boats of a square form for carrying horses and carriages.

Plicatiles—Portable boats, built of wood and leather, in such a manner as to be capable of being taken to pieces and carried over land.

Præcursoriæ—Boats which preceded the fleets.

Piraticæ, Prædatoriæ, Prædaticæ—Long, swift, and light boats, used by pirates or picaroons.

Serilla—Boats or barges stuffed in the chinks with tow.

Sagitta, Saguntia—A kind of galley.

Scapha—A long boat.

Solutiles—Boats which fell to pieces of themselves, such as those in which Nero exposed Agrippina.

Stationariæ—Those which were moored or remained fixed at anchor.

Sutiles—Made of strong staves, and covered with leather.

Stlatæ—Broader than high; used by pirates.

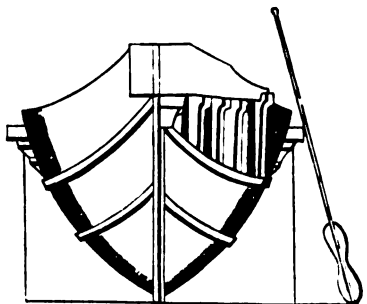
Trabariæ—Canoes (same as *Lintres*.)

Thalamagus—A yacht or vessel of parade and pleasure.

To the above list a few others might be added from Rosinus; but as the definitions are doubtful and various, the author has extended it no farther.

The engraved elevations of head and stern of a Roman

galley, with oar or paddle, are taken from a model presented many years ago to Greenwich Hospital by the late Lord Anson.



This model is one of the most reliable authorities we can refer to; it is made from one in marble, which was found in the Villa Mathei during the sixteenth century, and now stands before the Church of Santa Maria in Rome.

The model is not a war galley, but was probably used for commercial purposes, or for the transportation of warlike stores, provisions, and troops.

The Anglo-Saxons appear to have had *pleasure vessels*, if such be the proper meaning of *pleg-scip*, that is "play-ship." Some of these vessels are described as having ovens, fire-places, and other domestic conveniences; and boats covered with hides accompanied them.

The large sailing ships of the Anglo-Saxons were often called *Carikes*.¹⁰

There were also *gallyetis*,¹¹ which were probably a small sort of galley.

They had also *Crayers*, or small fishing boats; and *Balingers*, small sailing vessels.

Both sails and oars were sometimes used in the galleys of the Normans.

¹⁰ Grafton's Chron., p. 571.

¹¹ Caxton in Polychronicon, p. 409, cap. 14.

An ancient author, who wrote a history of King Richard the First, in rhyme, says of the King :—

“ Were the Maryners glad or wrothe,
He made them seyle and rowe bothe,
That the galley gedde so swyfte,
So doth the fowle by the lyfte.”¹²

Persons skilled in climbing the shrouds and rigging, were styled *funambuli*, as they were in the Classical *Æra*.

It is evident that canoes, made by hollowing out the trunk of a tree, were used by the Ancient Britons, several of which have been dug out of the earth in various parts of England within the last century.

It is stated in King's “*Munimenta Antiqua*,” that in a morass called Lockermoss, a very little distance from the Castle of Wardlaw, in Dumfries, an ancient canoe was dug up in the year 1736. This canoe was seven feet long, and dilated to a considerable breadth at one end : its paddle was found at the same time in the morass near to it.

Another canoe was found near Kiblain, eight feet eight inches in length, and two feet in breadth ; having a cavity of six feet seven inches in length, and of eleven inches in depth, the hollow of which had plainly been formed originally by means of fire.¹³

In the year 1720, several canoes similar to the above were dug up in the marshes of the River Medway, above Maidstone ; they were made of the trunks of trees hollowed, and one of these was so perfectly preserved, that it was afterwards actually used as a boat for some time.¹⁴

On the draining of Martine Mere, or Marton Lake, in Lancashire, a few years ago, there were found sunk at the bottom eight canoes ; each made of a single tree, which there is every reason to believe were used by the Ancient Britons

¹² MS. Harleian, 4690. ¹³ Pennant's Voyage to the Hebrides.

¹⁴ King's *Munimenta Antiqua*, vol. i. p. 20.

in fishing upon that lake, and which in size and shape were much like the American (Virginian) canoes.¹⁵

So also during the present century, on the draining of Whittlesea Mere, and in other parts of the Fen district, canoes in a very good state of preservation, have been excavated at a considerable depth from the surface.

BOATS OF THE ANCIENT EGYPTIANS.¹⁶

It appears that among the ancient Egyptians, there were two classes of boat-builders,—carpenters and wicker-workers. The boats made by the latter were used chiefly for fowling and fishing; they were made of osiers, and bound together with bands taken from the stalks of the papyrus, or cyperus.

Boats made from the papyrus are frequently mentioned by ancient writers. Isis is described by Plutarch, as going in search of the body of Osiris, “through the fenny country, in a bark made of the papyrus.”¹⁷ But they were so small and light, as to be easily carried from place to place on a man’s shoulders.¹⁸

They are also described by Strabo,¹⁹ at the cataracts of Syene, passing the falls in perfect safety, to the astonishment of the beholders; and Celsius affirms, that they were made of the papyrus.

Theophrastus mentions boats made of papyrus, and their sails and ropes of the rind of the same.²⁰

Pliny, also, speaks of boats woven of the papyrus,²¹ the rind being made into sails, curtains, matting, and ropes; and he

¹⁵ King’s *Munimenta Antiqua*, vol. i. p. 29.

¹⁶ Vide Champollion’s *Monuments l’Eyppte*. Also “*Manners and Customs of the Ancient Egyptians*,” by Sir Gardner Wilkinson, F.R.S., M.R.S.L., &c. Second Edition. Vol. iii. A.D 1842.

¹⁷ Plut. de Is. s. 18.

¹⁸ Achilles Tatius, lib. iv. Plin. v. 9.

¹⁹ Strabo, xvii. p. 562.

²⁰ Theophr. iv. 9.

²¹ “*Ex ipse quidem papyro navigia texunt.*” Plin. xiii. 11.

elsewhere observes, that the papyrus, the rush, and the reed, were all used for making boats, in Egypt.²²

Vessels made of bulrushes, are mentioned in the Old Testament.²³

The mode of binding and sewing boats with bands of the papyrus, are also alluded to by Lucan; "Conseritur bibula Memphitis cymba papyro."²⁴

With these evidences before us, and those of the sculptures of Thebes and Memphis, there can be no doubt but boats were anciently made of the material papyrus; and employed in various parts of Egypt, for fowling and fishing. Punts, and canoes, made of ozers and papyrus, are still used on the Nile, and the lakes of Egypt.

The Egyptian "boats of burden" were called *baris*; they are described by Herodotus, as made of a thorn wood, very similar to the lotus of Cyrene, from which a tear exudes, called gum. Of this tree, they cut planks measuring about two cubits, and having arranged them like bricks, they built the boat in the following manner:—They fastened the planks round firm, long pegs; and after this, they stretched over the surface a series of girths, but without any ribs, and the whole was bound within by bands of papyrus. A single rudder was then put through the keel; and a mast of thorn wood, and sails of the papyrus (rind) completed the rigging.

Pliny speaks of papyrus vessels crossing the sea, and visiting the island of Taprobane (Ceylon). "Quia papyraceis navibus armamentisque Nili peteretur (Taprobane)."²⁵ It may be, however, that he merely alludes to the sails of the vessels being made of the papyrus; for there is abundant evidence that the Egyptians had large boats of burthen, made of wooden planks.

Sir Gardner Wilkinson observes,—“We may be certain that the Egyptians had strong and well-built vessels for the purpose of trade, by sea; and for carrying merchandise, corn,

²² Plin. vi. 22, vii. 16.

²⁴ Lucan iv. 136.

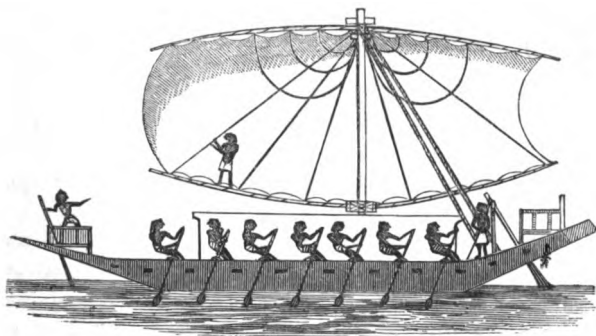
²³ Exod. ii. 8. Isaiah xvii. 2.

²⁵ Plin. vi. 22.

and other heavy commodities, on the Nile : and that even if they had been very bold and skilful navigators, they would not have ventured to India, nor have defeated the fleets of Phœnicia, in their paper vessels." ²⁶

MODE OF NAVIGATION OF THE VESSELS OF THE NILE.

Neither sails nor rudders were used with the canoes and punts of the ancient Egyptians ; they were propelled with paddles, in deep water, and pushed ahead with a pole, in shallow water. But the absence of a sail in the canoes, did not always depend on the size of the vessel ; for it appears, that many of their fishing canoes, some of which were very small, were frequently provided with a mast and sail.



The pleasure boats of the ancient Egyptians, were usually provided with two rudders, one on each side of the stern ; the rudder consisted of a long broad paddle, of great strength ; and the tiller, which formed part of the paddle, or rudder, was of enormous size, and supported by pillars or masts. The steersman moved the rudder by means of a rope fastened to the tiller head.

Other boats, though of large size, appear to have had but one rudder, and this was usually placed in a groove, or notch, in the centre of the stern.

²⁶ Manners and Customs of the Ancient Egyptians, vol. iii. p. 189.

The only kind of sail used by the ancient Egyptians, appears to have been a sort of square sail, with a yard both at the top and bottom.

The prow of the ancient Egyptian boats, was generally decorated with a painted eye on each side of the stem, a peculiarity which has been adopted by the Chinese, through centuries past, to the present day. The head and stern of the pleasure boats were generally ornamented with a painting, or a carving of a richly coloured flower.

Most of the ancient Egyptian boats are shown with a man standing at the prow with a pole in his hand, wherewith to sound the depth of the water and signal to the helmsman when near a shoal, or sand-bank. The precaution of the sounding-pole is still adopted in all the Nile boats, wherever the pilot is doubtful about the depth of water. The war galleys of the ancient Egyptians were also provided with a square sail, but it differed from the rig of the pleasure boats, inasmuch as it had no lower yard; the sail was therefore easier furled or brailed up, by means of four separate furling ropes, or brails, which, on being pulled, frapped the sail close to the upper yard, in four or five folds.

The square sails were always guided or trimmed by braces, or guy-ropes, called *pedes*; these were attached to the extreme ends of the upper yard of the sails. Some of the boats were rigged with a sort of sheer-mast, which consisted of two separate spars, placed wide apart below, but closing at the top, so as to leave sufficient space for the yard of the sail to be hoisted between them. The mast was secured by one fore stay and several back stays. None of the boats of the ancient Egyptians appear to have been fitted with more than one sail to each mast. When a single mast only was used, it was very firmly fixed and secured; the foot of it to a strong beam, or step, which extended across the whole breadth of the floor; it was also supported by strong knees and lashings, and finally secured by stout stays and shrouds, leading from the mast head to the sides and stern of the boat; and it appears, that all these were necessary to compensate for the

heavy yards and sail that were carried on the boat; and which, when hoisted and fairly set, were so considerably elevated, that the lower yard was full six feet above the gunwales.

When they wished to furl or lower this sail, the upper yard was let down, whilst the lower one continued stationary; and in this position the sail was furled, and remained until again required.

The yards consisted of two separate pieces, scarped and joined in the middle.

The sails of the grand pleasure boats were sometimes painted with gaudy colours, and embroidered with fanciful devices; but these sails were made of cloth, or linen, woven expressly for sails: the outer edges, or leeches of the sails, were strengthened with hems or borders, and sometimes with a small rope.

BOATS OF THE MODERN EGYPTIANS.

NILE BOATS.

THE Nile boats of the present day are not of very elegant proportions or graceful form of hull, although well suited to the purpose for which required: and the lofty pointed lateen sails make up for any deficiency in beauty. Some of the passenger Nile boats are from thirty to forty tons measurement, and are neatly fitted out, but often dirty; and, sometimes, swarming with fleas (very prevalent enemies in Egypt); these boats have a large cabin, standing high above the bulwarks, the roof of which forms two slopes similar to a cottage roof; the interior of the cabin is neat in arrangement; and there is, besides, a fore-castle for the crew, cooking-house, larder, and other accommodation. They usually carry two immense lateen sails, which drive them along at a rapid pace when the wind is fair, or aslant; when adverse, they use long heavy oars, or sweeps. Were these boats to put to sea, they would not be able to carry more than half the sail they set on the smooth waters of the Nile; but the Egyptian boatmen have their own peculiar method of management, which is

somewhat careless; for they console themselves, that if an accident occurs, the distance to the shore is not far. When a squall is observed approaching, the Egyptian pilot directs the attention of his crew to the halliards and brails, by the signal "*Arless ! arless !*" which means, "take care" or "be ready;" it is then surprising to witness the alacrity of the boatmen, who are otherwise a lazy, inactive race; but when aroused by warning of danger, just the reverse.

When the sails of the Nile boat require reefing, the lateen yard has to be lowered for that purpose (this is not the case with some of the improved lateens of the Mediterranean); and when required to come about, the yard has also to be lowered, and shifted to the other side of the mast. The Nile boatmen seldom go aloft, the sails are brailed up to the yard whilst standing, but lowered to be furled or frapped.

The various classes of boats of the Nile are:—

The *Djèrm* (Germ): the *Maádil*, *aggub* (Akkub): *Maash*, or *Rahleh*, *Dahabééh*, *Cangia* (Kangeh): *Kyás* (Kyáseh): *Sándal*, *seféence*, *garib* (Kárib): and *Maadééh*.²⁷

Of these, the largest are the *Germ*s, which are only employed on the Nile during the inundation, when the water is deep; or between Alexandria, Rosetta, and other ports on the Mediterranean.

During the summer they are laid up and covered with matting, to protect them from the sun.

These boats are enabled to carry a very large cargo of corn, in the transport of which they are chiefly engaged.

They are rigged with two masts, large lateen sails, and a foresail or jib.

The *Maádil*, or as it is sometimes called, *kyás*, is a similar kind of vessel, but much smaller.

The *Aggub* is used exclusively for carrying stone, and is remarkable among the boats of the Nile for its peculiarity of rig; for, whilst all the other sailing boats are rigged with lateen sails, the *Aggub* carries a square sail.

²⁷ Vide "Modern Egypt and Thebes," by Sir Gardner Wilkinson, F.R.S., M.R.S.L., F.R.G.S., &c. &c. Vol. i. A.D. 1843.

The remaining five are small open boats.

Sandal implies a small sort of Cangia or ship's boat.

The *Garib* is a fishing boat.

The *Maadééh* is a ferry boat.

The *Maash*, *Dahabééh*, and *Cangia*, are all passenger boats, and employed chiefly as such on the Nile; they are all furnished with cabins.

Of these, the *Maash* or *Rahleh* is the largest, and has the most lofty and commodious cabins.

The *Dahabééh* and the *Cangia* are so much alike, that it is sometimes difficult to distinguish between them; it differs only from the *Cangia* in being larger, and furnished with a gangway at the side of the cabin, extending to the steerage. They vary very much in size, the largest being generally fitted with two cabins. These boats are all rigged with two masts, and large lateen sails with very lofty peaks, in order to catch the wind when the water is low; the banks of the Nile being high; and when the wind fails, they are either rowed or tracked. The foremast is the smaller of the two, and is stepped in the bows of the boat; the mainmast is stepped amidships, and the mainsail is spread on an enormous lateen yard.

The *Sandal*, which is a small kind of *Cangia*, is rigged with one mast only, and a lateen sail.

The Nile boatmen are careless sailors, and much in the habit of making the mainsheet (*shoghóol*) fast; and to this, and the disproportionate size of the sails, may be attributed many of the accidents which occur to the smaller Nile boats. The passengers should insist on the boatmen holding the *shoghóol* in their hands, and keeping it *khálus*, i. e. free.

The keel of the Nile boats is of concave form, being deepest at the stem and stern, whilst there is scarcely any keel at all amidships; the advantage of which is, that when they get aground forward, by putting the helm to port or starboard, the hollow part clears the bank, and enables the boat to get off immediately.



THE CORACLE.

“ We dared to think, we dared to say, that he could frame a boat,
And many others said the same, but questioned—‘ would it float ? ’ ”

ELIZA COOK.

HISTORY tells us the coracle is one of the first forms of boat ever constructed. One of the most curious circumstances connected with the study of British boat-building is, that, at the present day, boats (*i.e.* coracles) are carefully made and constantly used, in parts of England and Wales, almost identical in size, form, construction, and materials, with those in use nearly nineteen centuries ago. And, however mean and insignificant the contrivance of the coracle may appear in the great yards of British boat-builders, and on the wide world of waters, it is a stubborn fact that the coracle is now, as it was then, fully adequate to the purposes of its possessor.

Coracles are much used in South Wales on the rivers Wye, Usk, Towey, and other favourite resorts of the fishers.

The English peasants term them cruckles. The original term is supposed to have been coriacle; and if so, was probably derived from the latin *corium*, a skin or hide, or *coriago*, hide bound. In shape, they are like the half of a walnut shell; but they vary in different counties, both in form and as to the material of which they are composed. They are generally now made of tarred canvas, strained over a frame of wicker or basket work, and are the lightest and most portable kind of boat that can be had: the general weight of one being about twelve pounds. The fishermen of the Wye usually make their own, but the cost of one of the best kind is only twenty or twenty-five shillings. In Cardiganshire the frame-work is covered with flannel, and dressed with tar. In the neighbourhood of Shrewsbury they are covered with canvas and painted. Some of the most fragile are merely covered with white linen, and dressed over with resin, varnish, and other compound. The most durable are coated with hides or skins; and such is the material of which the ancient ones were made; tarpaulin, and other waterproof material is now the more general article used; being well coated with pitch and tar within and without, so as to be perfectly water tight.

The common size is from three-and-a-half to four feet wide by two feet deep, and less than six feet long.

If the coracle gets pierced or injured by coming in contact with a stake or sharp stone, it may be quickly repaired with a bit of cloth or rag, and a daub of warm pitch.

The coracle is very useful to the salmon fisher, who has several different rivers or bits of water to fish in; one great advantage being, that it may be carried on the back as easily as a fishing basket, by means of a leather strap slung across the shoulders; placing the fisherman in a perfectly independent position as regards horses, ferrymen, and umbrella. When a large fish is hooked, the coracle is sometimes spun round as if on a pivot, and dragged a considerable distance, unless checked by the fisherman with his paddle.

It is a dangerous fragile bark in which to venture far from

land, and requires considerable practice to enable a man to balance himself safely: great caution and some little skill is also required in getting in and out, a slight lurch on either side will generally roll it over. The position in which a man sits, is facing the stem or fore part, when, by means of a double paddle, called a *sweep*, having a blade at each end, the coracle is rapidly propelled through the water. The sweep is held firmly with both hands in the middle, about one foot apart, and flourished something after the manner of a see-saw movement; dipping each blade in the water alternately, and thereby drawing the coracle ahead. When well practised, it is astonishing with what rapidity they may be made to skim the waters. This, however, is more generally the mode of propelling the larger ones. The smaller ones are propelled with a single-bladed paddle, the top of the handle of which the coracler places against his shoulder, and works it with his left hand, whilst holding his fishing rod with the right. Slow progress only can be made with the single bladed paddle; the sweep is the instrument to drive them ahead rapidly.

At the annual regatta at Monmouth, the coracle race is usually one of the most amusing and attractive matches of the day; and before it is concluded, at least half of the competitors are capsized in the scramble for pride of place.

Coracles are also used in various inland parts of Ireland, and for the same purposes as in Wales.

We are told that some of the coracles of the ancients, composed of two hides and a half, were large enough to contain three men with a week's provision; others were large enough only for an armed man and a rower.

As coracles of this construction are mentioned by Herodotus, the pattern was probably derived from the Phœnicians, particularly as Lidonian and Phœnician vessels were almost round in form.¹

In the time of Henry V., light boats, framed with wicker

¹ Gauli, p. 318.

or thin timber, and covered outside with leather, were carried by ship to the wars, to enable the soldiers to pass the waters which might be in their way on their marches. Such were also the *Naisselles* carried over to France in the wars by Edward III.² These were artfully made with prepared or boiled leather,³ and would each hold three men therein to fish or take their pleasure.

Hollingshed also alludes to Henry V. making great preparations for the war on his second expedition into France, by providing "boats covered with leather to passe over rivers."

Lucan, in describing the boats of the Ancient Britons, says they were made of osiers twisted and interwoven with each other, which were covered over with strong hides.⁴

"With twisted osiers the first boats were made,
O'er which the skins of slaughter'd beasts were laid;
With these the Britons on the oceans row,
And the Venetians on the swelling Po."

Pliny speaks of voyages of six days' sailing being performed in "vessels covered with leather round about, and well sowed."⁵

And again, in Book VII. he speaks of certain wicker boats being met with on the "British Ocean," made of twigs, covered with leather, and stitched round about.⁶

Strabo mentions a kind of boat, called *pécton*, apparently similar to the coracle; he describes it as "made of thongs, so as to resemble wicker-work."⁷

² Froissart's Chron. vol. ii.

³ "Faittees et ordonnees sy soubtillement de cuir bouilly."

⁴ Vide also Strutt's Horda, vol. i. Also Cæsar de Bello. Civ. lib. i. s. 54.

⁵ Ad eam Britannos vitilibus navigiis corio circumsutis navigare. Plin. Nat. Hist. Lib. iv. cap. 14.

⁶ Etiam nunc in Britannico oceano vitiles corio circumsutæ fiunt. Plin., Book vii. cap. 56, sec. 57.

⁷ Strabo, xvii. p. 562.

ENGLISH SAILING BOATS.

“ The breeze fills my sails, so adieu to the land !
My ensign’s unfurl’d, I’ve the helm at my hand.
What sport is more pure, what pleasure more sweet,
Than the sail and the breeze when kindly they meet.”

THE variety in English sailing boats is considerable, both in form and rig; and although some of the variety possess fewer advantages than others, it is often found that the nature of the coast, harbour, river, or other locality, requires a boat of a peculiar form or rig, in order to be useful and navigable. And a sailing boat, to be safe and serviceable, must be adapted to the waters in which it is to be employed; the most useful and reliable being those in general use by the resident boatmen, the form and rig of which will be found to possess some peculiar advantages, with reference to the nature of the waters and coast to which they belong: and generally, such advantages or peculiarities are the result of years of experience by those familiar with the navigation and locality of the waters.

A boat constructed and rigged for sailing on the smooth surface of inland waters, would be unsuited for the rough waters of a sea coast, and *vice versâ*. And, as some parts of the coast are much more dangerous than others, it is found that the ingenuity of the native boatmen has enabled them, from long experience, to design a form of boat admirably adapted to the locality and purposes required. For instance, look at the north country cobs, employed off the perilous north-east coast, by Flamborough Head, and such-like exposed parts. And again, the Yarmouth yawls; the feats of seamanship and daring constantly performed in those boats, are truly astonishing: whilst in boats of an ordinary form, it would be certain destruction to attempt such.

Boats for sea-going purposes, must have high bows; and the ballast should be trimmed rather farther aft than in river or smooth-water boats. It is also better for small sea-going boats to be so rigged as to carry a mizzen abaft the main-sail, which is of great assistance on coming about in a sea-way: life boats, and others exposed to rough weather, usually carry a mizzen-sail.

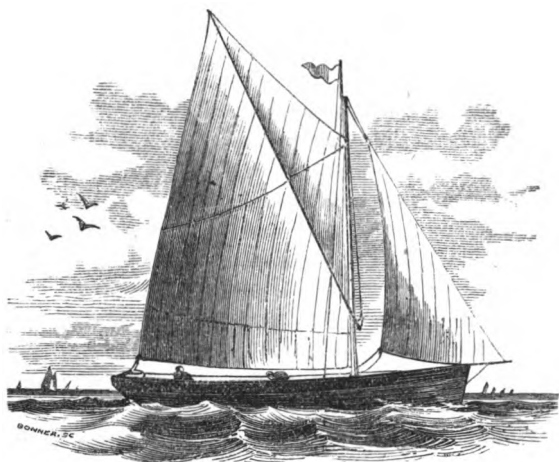
River boats should not be less than fifteen feet in length, by at least five feet in breadth: they should be sharp and fine at the bows, broad a-midships, and by no means narrow at the stern. Inexperienced persons are warned against the danger of setting sail in small boats of a narrow form of hull; such boats, though well suited to the oar, are unsafe with a sail.

Boats with narrow sterns are not desirable for sailing; those with tolerably broad sterns and good breadth of beam, are enabled to carry sail with far greater safety than those with wedge-like bottoms and narrow beam.

Short, wide boats are the safest and swiftest for beating to windward, in rivers and narrow channels.

Long boats are best adapted for wide waters, sailing on a bowline, and running free; and generally, in places where long tacks may be made.

Boats with a narrow deck, or water-way within the gun-wales, and all round the interior, are called "half decked boats," the object being to render them safer under sail, when listing over in a breeze. All open boats with large sails, should be so furnished. The half-deck or water-ridge is a great safe-guard, as well as convenience and comfort; it also makes the sailing boat more complete: but notwithstanding, if too much sail be set, and the sailor too venturesome, the boat is liable to be upset and sunk.



SAILING SKIFF.

THE SPRIT-SAIL AND FORE-SAIL RIG.

For small open sailing boats, there is no better, safer, nor more convenient rig than the sprit-sail and fore-sail; and to a tyro boat-sailor it is recommended as superior to many others, for boats under eighteen feet in length; when longer, some other rig is preferable, as the spreet required to set the sail will be found too heavy and too long to be convenient; and it is besides, in many respects, dangerous. Should it suddenly or accidentally slip out of the strop or snorter which holds the lower end to the mast, it would strike the floor or some other part so violently as to go through, and probably stave the boat and sink her on the spot.

The sprit-sail in shape resembles a cutter's main-sail, but has a sharper peak, which is raised by means of a small

spar called a *spreet*, the heel of which is set in a selvagee strop, or a grommet formed into a snorter; the top of the spreet is set in the peak eye; and so the sail is spread.

The usual sized sailing skiff is about fifteen or sixteen feet in length; such a boat should have an iron bumkin about a foot long, fitted to the stem: a short mast will suffice, as a good deal of the strain is taken from the top by the spreet, and the pressure thrown on the lower part of the mast: no boom need be used for the sprit-sail.

The fore-sail should have a stout rope attached to the fore-leech, which will answer the purpose of forestay; a single small rope on each side the mast, and made fast to the rim or gunwale will be sufficient for shrouds; but the boat will sail better and be safer without them.

The sprit-sail must be fitted with brails, wherewith to furl it at any moment and on any emergency; the brails should be so fitted as to catch the sail under the throat; and on the brail-rope being pulled, the sail is furled in the brails; they should at all times be kept clear and ready for use, so that the sail may be frapped up instantly. One great and dangerous error prevails among those who use this kind of sail: it is in setting the spreet; the invariable custom being to ship it *between* the sail and the brails, so that spreet and sail are both brailed up close to the mast when the rope is hauled taut; but this is as unsafe as it is erroneous; the spreet should always be set *outside* the brails, by which means the sail may be drawn closer to the mast, and much quicker and easier; the loose flapping at the top part of the sail is also avoided; whilst the spreet, instead of being drawn up, as it would be if within the brails, remains in its proper position.

The advantages of the plan recommended, need only to be tested to be appreciated. An instance came under my notice of a boat being upset solely from the spreet having been set within the brails; on the occasion alluded to, a heavy squall was observed approaching, and the brails were hauled taut to prevent an accident; but still the top of the sail was flapping about and filling with wind; besides which, the whole weight

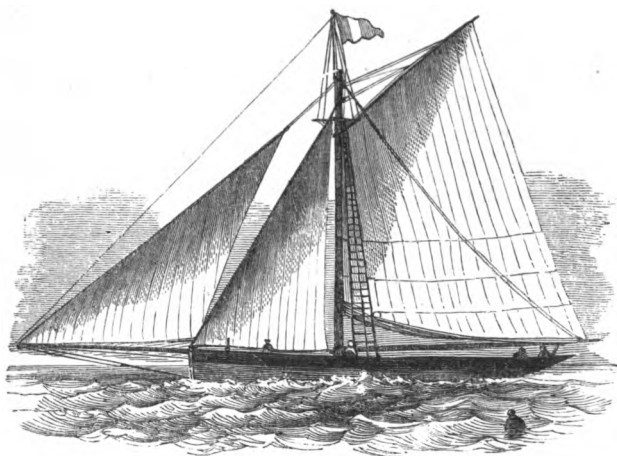
of the spreet was thus unavoidably thrown on the upper part of the mast, and even higher, by the lofty flapping peak; the consequence was, that when the squall struck the flapping sail, the upper pressure capsized the boat. Now, had the spreet been outside the brails, the flapping of the peak would have been avoided, and the spreet would have stood in the same position as if all sail was set; consequently, when the squall struck the boat, the whole pressure would have been thrown on the *lower* part of the mast, and not on the upper, as in the case of the accident.

On reefing the sprit-sail the selvagee must be slipped a little lower down the mast, which will bring the spreet lower, and generally admit of one reef; but when two or more reefs are necessary, and the heel of the spreet cannot be got down low enough, the top end must be taken out, and the eye of the peak lashed securely to the spreet, about a foot or so from the top; this mode of "reefing the spreet" will have rather an unsightly appearance, the top of the spreet projecting a foot or two beyond the peak; but it is better than the two other alternatives, the one of spoiling the spar by cutting it shorter, and the other incurring the danger of carrying the sail too high. It is advisable, however, to carry an extra spreet, of shorter length, when reefing is anticipated: but the sprit-sail should always be so made as to allow of the sail being close-reefed with one and the same spreet standing, and without unshipping, or taking it out of the selvagee. The boat should always lay *head to wind* when setting sail, and the same on shortening, or taking in sail. It will be found a difficult task (in fact almost impossible with a tolerably large sail) to set the spreet with a fresh breeze *abaft*.

The following are suitable dimensions for a suit of sails (sprit-sail and fore-sail) for a small open sailing boat, or skiff, fifteen feet in length, by five feet six inches in breadth.

Hoist of mainsail	7 feet	6 inches.
Head "	8 "	3 "
Aft leech "	12 "	0 "

Flap or foot of main-sail	9 feet	0 inches
Length of spreet	16 "	0 "
Fore-sail luff	10 "	6 "
" aft leech	7 "	10 "
" foot	5 "	9 "



THE CUTTER.

THE engraving represents a cutter yacht in a strong breeze, with her top-mast housed, and main-tack triced up. No rig is so much in favour among English yachtsmen, and sailors generally, as the cutter-rig, and none is better known and admired. Cutter yachts are more numerous, and have achieved more victories in sailing matches, than any other class of vessels in the world; they are also employed by the English Government in the revenue service, as the swiftest and most convenient form of vessel for

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chasing smugglers, and preventing the landing of contraband goods. The cutter may, therefore, be considered as possessing many, and great advantages for fast sailing: the rig comprises four sails—main-sail, fore-sail, jib, and gaff-top-sail: the main-sail is set by means of a gaff and boom; the outer end of the latter generally extends some little way over the stern; the fore-sail is set on the fore-stay: the bow-sprit is very long, and can be reefed or taken in when a small sized jib is used.

Considerable alterations and improvements have been made of late years in these vessels, so that the hull of a cutter of the present day bears little or no resemblance to one of ten years since.

The following may be relied on as suitable dimensions for a cutter-rigged sailing boat, half-decked, or provided with one foot water-ways all round the interior of the gunwale.

Dimensions of Boat.

Length over all	20 feet
Breadth, amidships	6 "
Depth	3 "
Length of keel	17 "

Dimensions of Spars.

Mast	19 feet	6 inches
Boom	14 "	6 "
Gaff	8 "	10 "
Bowsprit	14 "	(in all).

Dimensions of Sails.

Main-sail luff	12 feet	
" aft leech	17 "	
" head	8 "	8 inches.
" foot	14 "	0 "
Fore-sail luff	12 "	0 "
" foot	6 "	0 "
Jib luff	15 "	0 "
" foot	9 "	0 "

THE SLOOP.

THE sloop (anciently shallop) somewhat resembles the cutter-rig; the main-sail is precisely the same, but the fore-sail is set on a fore-stay, which is attached to the outer end of a standing bowsprit; if a jib is used, a jib-boom becomes necessary: the fore-sail is generally large, and, for fast sailing, should be laced to a boom. The sloop rig was, until recently, chiefly confined to small trading vessels, but since the Americans have improved upon it, and turned out several decided clippers, rigged as sloops, it has become rather a favourite rig for English pleasure boats, and small yachts; a boat with a long sharp bow is desirable for the sloop rig.

The sloop is as useful a rig for a small sailing boat as any, and as simple in management; no top-sail need be used unless the boat is sufficiently stiff to bear one; and more or less peak can be given to the main-sail, by simply pulling on, or easing the halliards.

THE SCHOONER SAILING BOAT.

THE schooner rig consists of two masts and three sails: viz. one main-sail, a spencer, or stay-sail, and fore-sail: it is a safe and handy rig for pleasure boats, particularly those of a long and narrow form of hull. Schooner-rigged boats have long been used by the native boatmen on various parts of the English coast; and since the arrival of the celebrated schooner yacht "America," various improvements have been made, and schooner-rigged pleasure boats of a superior class have become numerous.

The fore-sail of a schooner boat is usually set out beyond

the stem by means of a bumkin, or a small standing bowsprit; the spencer, or boom-fore-sail (which is the middle sail of the three) requires no boom; but the main-sail should be fitted with a boom and goose-neck. The two masts should be connected at the top by a jumper, or triatic stay; and the spencer peak should swing clear of the stay. In sailing a schooner boat, the fore-sheet is eased off on coming about, and the spencer kept a-weather to pay her off, in the same manner as a stay-fore-sail aboard a cutter. When running before the wind, the main-sail and spencer are boomed out on each side over the gunwale, in goose-winged form.

The schooner is the favourite rig for large yachts, particularly those used for long voyages and sea-going purposes.

There are many magnificent schooner yachts belonging to members of the various English yacht clubs, some of which are of beautiful form and construction; and the interior accommodation abounds with every comfort and luxury man can desire. Such vessels are truly an ornament to our seas, and the pride and boast of the pleasure squadron.

THE LUGGER.

ANCIENTLY the lug-sail was the only one used for sailing boats and small vessels. The *ancient* lug-sail was nothing more than a square piece of canvas, with a yard or spar at the top, and a sheet or guy at each of the lower corners. Modern sailors still find the lugger an exceedingly handy rig for boats, and have made sundry improvements in lug-sails, which are now cut in various shapes, with more or less peak, according to fancy, and the form of the boat for which intended.

Luggers have been from time immemorial, a favourite class of boats among beachmen, pilots, fishermen, and others; and as

such men have great experience in boat-sailing, and would naturally select the safest, most effective, and useful rig for general purposes, it may be fairly concluded that the lugger is the most convenient rig of all for an open coasting boat. The lug-sail requires less ropes and rigging than any other: in open boats, the single halliard rope serves as a stay to the mast, the only other rope required being the main-sheet.

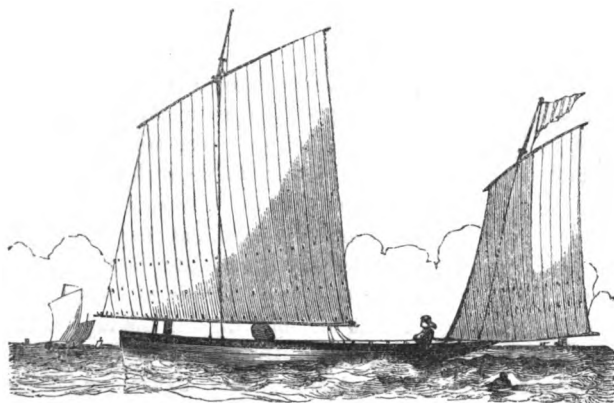
Pilots on approaching ships at sea, are enabled to lower a lug-sail in an instant, and strike the mast, to avoid collision with any part of the ship's rigging: a rope is then thrown them from the ship; with this they make the boat fast, and then quickly board the vessel. But in very rough weather and heavy seas, there would be great danger in running a boat alongside a vessel; so the pilot takes a turn round his waist, with the rope thrown him from the ship, and casts himself into the sea; he is then cautiously drawn aboard the vessel.

The common lug-sail needs no boom; the sail is suspended from a light stiff yard. A good and useful method for small boats, is, by setting lug-main-sail and fore-sail both as one; the tack of the sail, in such case, must be hooked to a tackle in the bows, or at the stem. When a separate fore-sail is used, the tack of the lug must be hooked at the lower part of the mast; or it may traverse a hawse just in front of the mast. There are various improved methods of hoisting, setting, and lowering this sail, and of "dipping the lug," as the crew of a man-of-war term it. Dipping the lug is a smart, active performance, although simply that of swinging the yard to the other side of the mast, at the instant of coming about. It requires at least two persons, one to slack the halliards a few inches, another to attend the tack, the main-sheet being also eased off; then, at the moment the boat is in stays, one of the crew jumps upon the thwart, and dexterously swings the fore-part of the yard to the other side of the mast; the halliards are then set taut, the tack secured, the main-sheet hauled in, and the boat is quickly on a fresh tack: when actively performed, this is a sailor-like manœuvre.

The old-fashioned luggers, when putting about on change of tack, lower the lug-sail, unhook the yard from the traveller, and shift the sail whilst down; which is a slow method, and a clumsy one.

The north country cobsles, which carry both lug-sail and fore-sail, are managed in the same manner as the sprit-sail and fore-sail; they sail on either tack without lowering or altering the sail, the yard remaining on the same side the mast as when first hoisted; there are, however, only a few inches of the yard before the mast: where this plan is adopted brails are generally fitted to the sail.

The yard for a common lug-sail, should be slung, so that one-third of its length only is before the mast.



IMPROVED RIG FOR A LUGGER.

A very ingenious and excellent mode of rigging a lugger as a pleasure-boat, or for general purposes, was suggested to me a few years ago, by a gentleman of great experience in boat sailing; and who, for a long time, had a boat in constant use rigged upon this plan. As I have had many opportunities of testing the merits of the rig, and also of examining the boat

alluded to, I am enabled to recommend it with confidence, as a safe and handy rig for a small boat. I also present my readers with an engraving of the design, from a drawing made by the inventor himself; showing a lugger so rigged, and under full sail. The method alluded to consists in the lug-sail being laced to a yard, which is slung in the usual way, and hoisted with a mast-traveller and halliards; the foot of the sail is also laced to a boom, and bowed down quite flat and taut, by means of a small tackle at the foot of the mast. By this method of rigging a lugger, there is no necessity for dipping the lug, or lowering it in any way on coming about; the man at the helm, can alone do all that is necessary; it is, in fact, almost equal as regards convenience for tacking and manœuvring, to a sheer-masted boat; the sail may be turned in any direction with ease and precision, and two or more sails can be used in the same boat, after the same manner.

THE SPLIT LUG.

This, though an old-fashioned mode of making the fore part of the lug-sail into a fore-sail, is an ingenious one. By this rig, the top of the lug is laced to the yard, and the tack of the main-part of the sail is made fast at the lower part of the mast; whilst the piece which forms the fore-sail is worked with fore-sheets, just as an ordinary fore-sail. The advantage is, that in tacking, the yard need not be dipped or lowered, but the boat may be worked to windward, with facility, without once lowering the yard or the sail.

SCOTCH LUGGERS.

These fishing boats are very numerous along the coast of Scotland, and the north of England. They are a strongly built craft, of great breadth of beam and considerable draught of water; the stem and stern posts are both nearly perpendicular, and the stern is not square, like English vessels, but sharp, and very much resembles the bows. They are rigged with two masts; the largest or main-mast is placed far for-

ward in the bows of the vessel, within a foot or two of the stem, and raking slightly aft : on this mast the main lug-sail is set. The mizzen-mast and sail are small in comparison with the wide spread of canvas forward ; but in very strong winds the main-lug is lowered, and the boat is sailed under mizzen-lug and small fore-sail. They are safe and fast sailing boats, and traverse the seas long distances from home, keeping company with the shoals of herrings, mackerel, and such fish as they go in quest of. Weeks and months are passed by the crew, at sea, in these boats ; as also by their companions the Dutch and French fishermen in their fishing vessels, round and about the English coast. And what is very remarkable and characteristic of ancient Scotland, the Scotch luggers are always manned by a race of people descended from the Scandinavian colonies established in many parts along the eastern coast of Scotland : the language spoken by these men quite confounds our south of England countrymen.

There is, also, another description of lugger belonging to the class of fishermen on the north-east coast of Scotland, differing from the others both as to construction and rig : this boat is very shallow, but of equal breadth of beam with the former ; a much longer hull, and raking stem and stern-post. Notwithstanding their being of less draught of water, they carry larger sails than the others, which are placed exactly the reverse of the other method of rig ; the small sail being placed forward, and the large or main-lug aft. The fishermen belonging to the latter boats are a distinct race from the others, being exclusively Celts : the others would deem it a derogation of their creed to have a Celt among the crew in one of the first described luggers, whilst it would be equally so on the part of the Celts to carry a Scandinavian in their boats.

It is also very remarkable that the superstitions prevailing among fishermen, pilots, and beachmen along the coast of Scotland, to which Sir Walter Scott, in his writings, has so frequently alluded, still exist. If an accident occurs to one of their boats, it is considered as foreboding some ill omen ;

and if a boat, new or old, is upset and any part of the crew drowned, that boat is no longer used ; but is either sent adrift or left ashore as an abandoned wreck. Only a few years ago, a small pilot-boat was capsized off the north-east coast, and part of her crew drowned ; the boat, although nearly new, was towed out to sea and cast adrift by the owners, who thus left themselves destitute of the means of subsistence. The superstition is so deeply rooted in the minds of those people, that they would rather remain ashore with starvation staring them in the face, than venture to sea in an "unlucky boat."

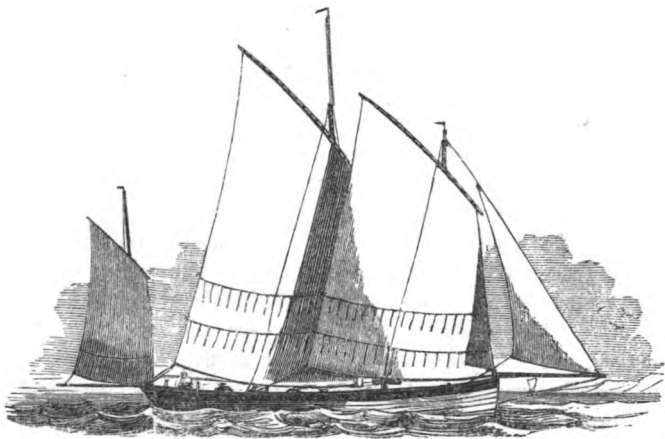
MOUNT'S BAY FISHING LUGGERS.

These boats have been much improved of late years, so that they now rank among the fastest and best fishing luggers on the English coast : they are very capacious and powerful sea boats, well adapted in rig and build, to the purposes for which they are chiefly employed, viz., mackerel, pilchard, and herring fishing. They have great breadth of beam in proportion to length, sharp bows, upright stems, and raking stern-posts : also, a forecastle deck, under which the crew occasionally sleep. The rig consists of two masts ; the main-mast is stepped "chock for'ard," in the very bows of the boat, and the mizzen-mast well aft ; the sails used are main-lug, mizzen-lug, fore-sail, and lug-top-sail, all of which are tanned for preservation. The hardy race of Cornish men who man these boats, make voyages of many hundreds of miles, to gain a precarious subsistence at their perilous calling.

If a storm overtakes them at sea, they disdain the idea of running for a harbour, and contrive to ride out the gale, by lashing spars and wooden gear together, in the form of a raft, and riding the vessel by it. The raft breaks the force of the waves, and prevents their running over and swamping the lugger. A few years since, a crew of five daring fellows undertook a voyage to Australia, in one of these luggers, and performed it safely ; taking the mail from the Cape of Good Hope, on their way.

The recent improvements in the Mount's Bay boats, may

be attributed to their desire to outvie the many rivals they have to contend with ; the chief of which are the French fishermen, from Cherbourg, Dieppe, and other parts, who come over in their fishing vessels, with the easterly winds of March, and harbour among the islands of Scilly. So numerous a fleet are these foreigners, that were they disposed to invade those small islands, their force would be quite strong enough to do so without powder and shot. Whilst there, they repair their nets, take in provisions, and large stores of salt for pickling herrings. The French boats are much larger than the Mount's Bay boats, being decked all over and fitted with cabins ; the largest of the French boats carry a crew of twenty-five men each, which is more than necessary, either for management of the boat or working the nets. But all French boats are compelled to carry a certain number of hands, according to tonnage. No ingenious inventions or improvements are to be learnt from a description of the rig of the French boats, which, to say the least of them, are certainly more picturesque and antiquated than suited to the taste of modern sailors.



HASTINGS LUGGER.

On several other exposed parts of the English coast, there are luggers to be met with of very admirable powers, both as regards swift sailing and sea-going qualities. But as there is so great a similarity in English coasting luggers, it would be superfluous and tedious to enter upon an individual description of those employed at every port round the English coast : suffice it to say, that besides those specially alluded to in other parts of these pages, those belonging to the most exposed parts of the coast, are invariably excellent sea-boats ; and the able manner in which they are managed in the hour of peril, has often been the subject of admiration by those who appreciate good seamanship.

DREDGING BOATS.

The fishing smacks and boats employed in dredging for cement stone, are now very numerous on the eastern coast ; they are rigged chiefly as cutters and sloops. The form and build of these vessels has undergone considerable improvement of late years, and there are among them many swift sailing and fine sea-boats. The seamen who manage them have constantly the very best experience ; being out at all seasons of the year, they are frequently exposed to gales and bad weather, and are sometimes compelled to heave part of their hard-earned cargo overboard to lighten the vessel, and then run for their lives to the nearest port. Some of the dredger-men manage their vessels in splendid style. They assemble in hundreds at favourite spots, and apparently all huddled together ; but seldom do they get foul of each other, although they appear to have many hair-breadth escapes. These boats are usually short beamy vessels, and can be turned about on the heel very quickly.

BOATS OF THE WESTERN COAST.

In Cardigan Bay, Morecombe Bay, and other parts of the western coast of England, many of the coasting boats are rigged as schooners. The liability to sudden squalls along that coast renders the fore and aft rig the safest and easiest to be handled; top sails are seldom used. Many of the fastest of these boats would be more than a match for some of the biggest of the eastern coast; they are excessively handy under sail, and particularly safe and easy in heavy seas.

SWANSEA PILOT BOATS.

The pilot boats at Swansea, and other adjacent localities, are in some instances, rigged with one gaff-sail only, no fore-sail being used, and seldom a mizzen; the mast is placed in the bows. These boats sail remarkably well, and possess excellent going qualities.

YAWLS.

The term *yawl* signifies a sharp, narrow boat, with stem and stern alike. There is a larger class of boats called yawls, which they have not yawl stems: they ought, more properly, to be designated yawl-rigged.

Small yawl boats carry a small mizzen-sail, in addition to the main-sail; the latter has no boom: it is a convenient rig for a fishing or pleasure yacht, as the annoyance of a boom is entirely obviated. The main-sail is also much smaller in

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Yawl-rigged boats carry a small mizzen-sail, in addition to the main-sail; the latter has no boom: it is a convenient rig for a fishing or pleasure yacht, as the annoyance of a boom swinging over-head from side to side when the vessel tacks, is entirely obviated. The main-sail is also much smaller in



YARMOUTH YAWL.

proportion to the tonnage of the boat, than a cutter's mainsail, and being all in-board is more readily reefed, and altogether more manageable than a large boom-sail; the mizzen is also of great assistance to the boat, as an auxiliary sail.

The yawl rig, is a neat and pretty one for pleasure boats.

YARMOUTH YAWLS.

These splendid boats are the admiration of all who visit Yarmouth, Lowestoff, Aldborough, and other watering places on the eastern coast.

They are the longest and swiftest yawls, if not the finest open boats in the world. Some of them are upwards of sixty feet in length,¹ and from ten to twelve feet in breadth. They have an exceedingly sharp and gracefully formed bow, which rises gradually from amidships; the stern is also somewhat elevated, so that they are much lower amidships than elsewhere. They are simply ballasted with bags of shingle, an inexhaustible supply of which may be had for the trouble of collecting it, all along the coast; the advantages of being so ballasted, are, that as the crew frequently have to throw all their ballast overboard, on beaching through a heavy surf, they have only to open the bags and shoot it out, which is merely the work of a few moments; the boat is then light and buoyant; and on touching the beach, the crew instantly jump out, and haul her up out of danger. The crew is generally a numerous one, and in a strong wind and heavy sea, all except those who have the management of the sails, squat down in the bottom, to preserve stability.

When the celebrated yacht, "America," was in the days of her glory, (having beaten the fastest clippers of the Royal Yacht Squadron,) the beachmen of Yarmouth sent the owner a challenge to race on the German Ocean for £200; but the owner of the yacht, probably not wishing to condescend to a race with an open sailing boat, couched his reply to the challenge in such terms as to put a final stop to the affair.

¹ The *Reindeer* is 69 feet.

The terms proposed on behalf of the "America," were, that she would not be permitted to race with the yawl for a less sum than 1000 guineas, an amount the poor beachmen could not stake, therefore no race took place. Such a sailing match would truly have been a singular one, on account of the unequal size of the competitors, the "America" being 200 tons, whilst the yawl was a mere open boat. Had the race taken place, it is believed the yawl would have come off victorious, unless it happened to blow a gale on the day of the match, in which case the yawl could not have competed with her; but in the "America's" favourite six knot breeze, or any wind in which the yawl could carry all her sails, the result would, it is believed, have terminated in favour of the yawl.

With a stiff breeze on their quarter, the Yarmouth yawls have been proved to sail *sixteen* knots an hour, a pace unequalled by any yacht or sailing vessel in the world: their beautiful proportions, large sails, lightness, and buoyancy, added to which, the exquisite seamanship of their crew, give them a superiority among the boating class, which has never yet been surpassed.

The yawls are rigged with three masts, on each of which they carry a lug-sail; the largest, or main-sail, being amidships: the next size forward, and called the fore-sail or fore-lug; and the smallest aft, called the mizzen. They are also provided with a bowsprit, and in fine weather they set a large jib. In a strong wind or a gale, the jib and bowsprit are dispensed with, and the lug-sails are all close reefed; sometimes the two smallest only are used, fore-sail and mizzen, in which case the main-mast is lowered, which enables the boat to ride easier in a heavy sea.

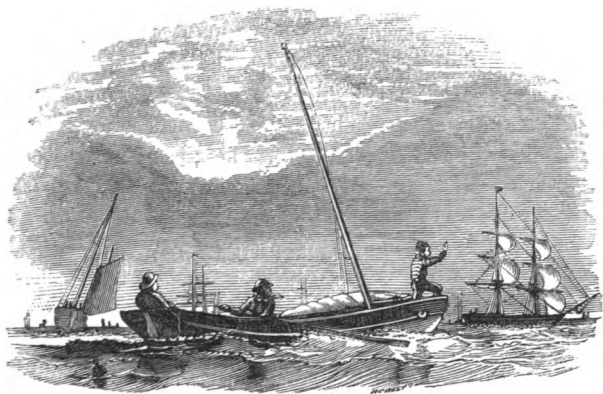
These yawls generally belong to companies or crews, each man being entitled to a share in the earnings of the boat. The brave fellows who man them are a hardy, fearless race, and venture to sea in the most perilous gales, to ships in distress; hundreds of lives have been saved by their daring exertions. It is no uncommon thing in a heavy sea, for two

or three of the crew to be constantly employed baling out the water, when the waves break over the bows of the boat. Such men are an honour to their country, and more credit is due to them than to the crews of life-boats, the latter being supposed to be safe, even if the boat should upset; but if the beachmens' yawl upsets, she sinks to the bottom. Nothing daunted by wind or weather, when they espy a ship at sea with a signal of distress flying, or a signal for a pilot to conduct them in safety to their destined harbour, they man their boat, and at all risks proceed towards her; sometimes a distance of ten or fifteen miles. It is almost incredible to witness the sea these boats go through. One grand caution they adopt in the management of these yawls, never if possible to allow a heavy sea to strike the boat a *broadside*, but always *stem on*. In turning, when the waves run high, they watch for a *smooth*; then, if rowing, all oars on one side pull one way, whilst the others back water; and the boat, although of such immense length, is quickly round.

Every facility is always at hand on the beach for launching these boats; such as cradles, fitted with rollers and other apparatus, provided by the boatmen for hauling them up out of the way of heavy seas.

The large coasting yawls are vessels of thirty and forty tons, decked and fitted with cabins; they cruise about off various parts of the English coast, sometimes in the roughest weather; and when many a large ship would gladly run for a harbour, these boats are beating about to render assistance to vessels in distress, and actually making good weather of it. These are unable to approach and land their crew on the beach, like the others, but can only do so in harbour.

The parts of the eastern coast off Norfolk and Suffolk abound with sands and shoals, some of which are many miles out at sea; the value of the services of such boats and crews as those described cannot be over-estimated.



THE COBLE.

THIS is undoubtedly the most perfect form of boat for putting to sea in rough weather of any yet invented (life-boats alone excepted). In all my travels, I have never seen more able-looking sea boats than some of the North Country Cobles. They are of Yorkshire origin; and from time immemorial have been renowned for superior qualities as safe sea boats, when under experienced management, but in unskilful hands they are as liable to accident as other boats.

Cobles are the favourite and prevailing form of boat all along the north-east coast of England. At Flamboro' Head on the Yorkshire coast, no other description of open boat is considered safe to put to sea in, and it is seldom that any other is used there in rough weather; but this is a very dangerous point; a north-east wind causing a tremendous sea.

The engraving at the top of the page represents a pilot coble with its sail struck, and about to approach a ship for the purpose of putting a pilot on board.

Cobles have a very high, flaring, but beautiful, bow, sharp and hollow; and a graceful wave-like form of rim, until it reaches the stern, which is the only part about them at all ugly, presenting a flat raking surface, not unlike the end of a large tree sawn asunder in a slanting direction. The rudder is deep and narrow, and reaches four feet or more below the bottom in a slanting direction, and, acting both as keel and rudder, is of infinite service in a heavy sea; the deep hold of the rudder enabling the steersman to keep the boat from broaching-to.

These boats have no keel aft; but the sharpness of the bow gives them a good fore-gripe. All along the other part of the bottom they have a very flat floor; indeed *quite* flat towards the stern, where the form of the boat is much narrower than amidships. The sides are bulged out, and incline inwards at the top and gunwale.

The coble is sailed under a large fore-and-aft lug-sail and small foresail; and, in fine weather, a jib is also used. The lug-sail is generally fitted upon the most approved plan, so that it need not be dipped or lowered on coming about.

There are three steps for the heel of the mast; in light winds it is stepped in an upright position; in a fair breeze it is stepped in the middle step, which inclines it aft, in a raking position; and when blowing a gale it is stepped in the third position, which is very raking: the aft rake of the mast tends to ease the bows of the boat immensely in a heavy sea, and to assist it in rising to the waves.

When a jib is used, the bowsprit is not placed in a horizontal position, but with the outer end tipped up like the jib-boom of a schooner; the advantage of which is obvious: if the bowsprit were run out horizontally, the boat, when pitching in a heavy sea, would soon carry it away; and the pressure of the sail acting on the bows would assist in burying the head of the boat under the waves; but, by inclining the bowsprit upwards, such an evil is avoided; and the sail acts as a lifting, as well as a powerful drawing sail.

The safety of the coble under sail, depends chiefly on the

adjustment of the main-sheet; in a strong wind it is never made fast, but has frequently to be eased and humoured to the waves, for safe management of the boat.

In small rivers and shallow waters, the coble would prove inferior as a sailing boat, on account of the deep rudder and high flaring bow: they are exclusively sea boats. Some persons have occasionally taken so great a fancy to these boats, as to insist on trying experiments with them in smooth water rivers, for which purpose all the lower part of the rudder extending below the boat, is cut off and added to the aft part; making it of the same shape as the rudder of a flat-bottomed sailing barge; but, independently of the want of a keel, the very high bow is an impediment to fast sailing in smooth water, from its catching the wind and causing the boat to blow to leeward.

When ballasted, the coble has a great hold of the water forward, but none whatever aft; and the deep diving rudder must be shipped before anything can be done with the boat under sail. The rudder is the only means at hand for preventing the boat broaching-to in a heavy sea; which a coble is very liable to do in the absence of a deep rudder, by reason of the fore-gripe of the bow.

In order to guard against such an evil, in some instances sliding keels, or centre boards, have been introduced among the North Country boatmen, and found to answer remarkably well; for when the keel is down, and the rudder shipped, the coble has all the properties of a keel-boat, both in working to windward and running in a heavy sea.

Cobles are used by all classes of boatmen on the north-east coast; there are fishing cobles, pilot cobles, salvage cobles, and pleasure cobles: the largest size are those employed in the herring fishery; some of these are from ten to fifteen tons burthen, and have a small cabin in the bows, or just abaft the mast. A few hours before dusk on a summer's eve during the herring season, a very interesting scene takes place, as several hundreds of these boats leave the different ports and harbours on the coast. Not a white sail is to be seen among the whole

fleet, all are tanned for preservation from wet, and to make them more durable. So highly esteemed is this tanners' liquid, that the fishing nets, and even the clothes worn by many of the coble-men, are dipped in the same odoriferous mixture. These fishermen often go many miles from the coast, and drift their nets in company with the Scotch luggers, Dutch, and other fishing boats; from all which the English and foreign markets are supplied with fish.

The North of England boatmen manage these cobbles with remarkable dexterity, and though often exposed to gales at sea on that perilous coast, they never come to grief.

As an open sea-going boat, the coble is unequalled by those of any other form I ever saw, with the exception only of the Yarmouth yawls described at page 37.

It is astonishing the heavy seas these cobbles will live in under skilful handling; and when they scud before a gale, they are so quick and lively that it is a very rare occurrence for a breaker to curl over the stern; as would be the case in a heavy sea with many other boats of a different form.

On approaching the beach, or going into shallow water, the rudder is unshipped, and the coble is backed in, stern first; the stern being of considerably less draught of water than the fore-part. When rowing these boats, whether before the wind or otherwise, they are generally propelled stern first, because they row easier and better in that position.

The coble oars are different to those of other boats, they consist of two separate parts; the loom of the oar being a square or flat-sided piece, and the blade, with a short portion of the loom, is usually joined to the other by two iron slides; which secure and hold the two parts firmly together: the flat part of the loom rests on the gunwale, and in that position will not admit of being *feathered* similar to an oar with a round loom. An iron ring is firmly attached to the oar at its proper equipoise, which fits over a single iron pin or thowl; so there is no danger of the oars going adrift, although left suspended over the side of the boat; and they may be turned either towards stem or stern without unshipping.

THE LATEEN RIG.

THIS is one of the prettiest, and most elegant modes of rigging a pleasure boat of any that is known. A lateen-rigged boat never fails to win admirers, if the sails are true lateens, well cut, and fairly set. But there is no rig in which the pretty effect so much depends on the shape and make of the sails.

The shape of a lateen sail is triangular, or precisely that of a large jib, or fore-sail; it requires a long tapering yard to spread it, one that will slightly bow or bend at the top when a stiff breeze fills the sail; a boom is sometimes used for lateen sails of a large size, but not always.

Boats sail very fast under this rig, particularly when close hauled; the one objection is, the very long yard that is required to set the sail fairly; often considerably longer than the boat, therefore somewhat cumbersome and inconvenient for stowing away. This inconvenience, however, only arises when the boat is rigged with a single lateen sail; when two are used (and such is a very general mode with lateen-rigged boats), the yards are not so long as to be inconvenient. The mast for the lateen rig rakes forward, or towards the bows of the boat; it is very short, and requires to be well secured by stays and shrouds. The lateen yards are not very often lowered in large boats, but the sails are brailed up, or frapped to the yard.

The lateen is considered a very safe and convenient rig for small open boats; for, although the peak of the sail stands very high, the principal breadth and body of the canvas is low—a great consideration in rigging all open boats. By the lateen rig, a large spread of canvas is made, with an exceedingly graceful and lofty peak, and yet the rig is a trusty and handy one. It is also worthy of remark, that

lateen-rigged boats sail remarkably well, and when running before the wind, if the boat is rigged with two lateens, they are "goose-winged," that is, boomed out over the gunwale, one on each side the boat.

A lateen-rigged boat, with the sails goose-winged, and running before a fresh breeze, is a very pretty object; when viewed from a distance it looks like an enormous bird skimming the waters, with its wings fully extended. When scudding before very light winds, the jib is occasionally slung as a topsail between the two peaks of the lateen yards.

The Mediterranean is said to be the birth-place of lateen sails; there they are met with in every variety, and among them are many fast sailing, and beautiful boats. Every one who has cruised in Mediterranean waters, and along the coast of Spain and Portugal, must have admired the lateen-rigged boats, and the lofty and graceful peaks to the sails; also the activity of the Spaniards who man them, the nimble manner in which they climb the slender yard, and perform other feats of seamanship, very amusing to English sailors. These boats are seldom seen without two or three of the crew clinging to some part of the yard, or at the top of the stumpy headed mast, where they sometimes remain for an hour or more at a time, but for what purpose it would be difficult to tell.

The lateen rig has been adopted by several other nations, although but little used in England; it is an excellent rig for a sailing boat where light airs and smooth waters prevail; but for strong winds and rough seas, there are other rigs which, probably, answer better. When a boom is attached to the lateen sail, the fore-end should be united to that of the yard by a small joint, or S; a hook should also be fitted to the lower part of the mast, so as to hold the boom close to the mast. By this method, the heel, or lower end of the yard, must not be secured at the stem of the boat; in fact, no other guys or fastenings will be required for the lower part of the sail, except the main-sheet, which will enable the sail to be easily worked and turned about in any direction by one person alone: a very flat and effective sail for working to

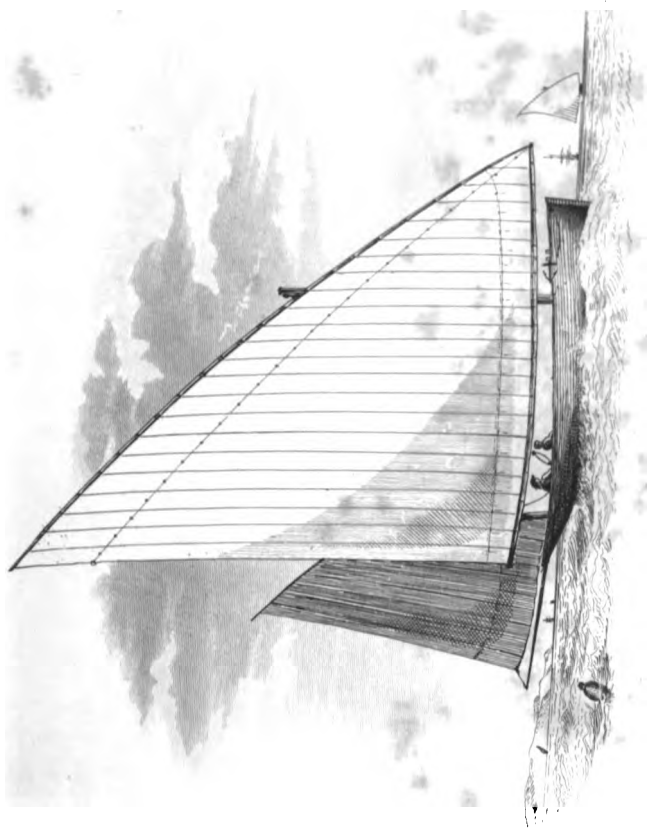
windward maybe set in this manner. Lateen sails are reefed, not at the foot as other sails, but along the fore-leech, close to the yard; the operation is performed by the Spaniards, in large boats, without lowering the yard.

THE LATEEN FORE-SAIL AND MIZZEN.

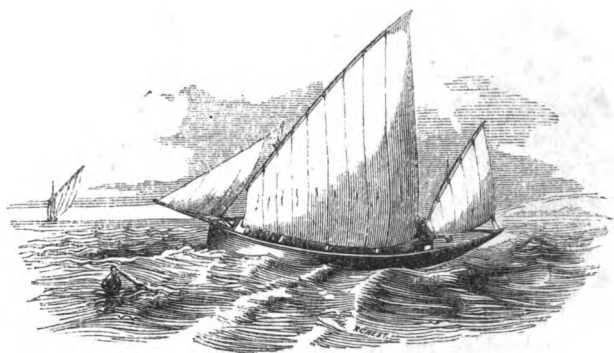
This rig is admirably adapted for narrow waters and rivers, where short tacking is unavoidable. The lateen fore-sail and lug-mizzen, as shewn in the engraving, is the best contrivance, and the most effective for such purposes of any I have ever seen.

Boats for turning to windward in narrow channels must be short and wide; a long rakish craft is unmanageable in such waters. The river Yare in Norfolk may boast of some of the fastest and prettiest boats, after this rig, of any in England: the largest are twenty-six feet on the ram, (or from stem to stern-post,) and ten feet beam; the smaller size are sixteen feet on deck, whilst the breadth of beam is eight feet six inches.

The lateen fore-sail, which is much the larger sail of the two, is peaked up by a yard of almost incredible length, in the larger sized boats; and the customary length of yard for the smaller size is from *forty-nine to fifty feet*! thus enabling them to set an immense sail with a very lofty peak; but notwithstanding this, the sail, if properly made, stands very flat, and the boat is extremely handy under such a rig, in staying, or any other manœuvre requiring quick movements. The safe management of a boat rigged in this manner may be done by one person alone. After the lateen fore-sail is once set, it is entirely under the control of the person attending the main-sheet, and may be turned about in any direction with ease and precision. An iron traveller is connected with the boom, at the lower part of the mast, and as this is the only fastening by which the sail is held down, it may be worked with almost equal facility to that of a revolving rigged boat. The boom is elevated a little above deck, so that nothing may incommode the crew; the mizzen works of itself, with little or no trouble.



This is altogether a most useful and ingenious rig for short tacking in smooth water, and there is none that can compete with it for beating up against wind in very narrow waters ; but for wide waters, and sea-going purposes, the Norfolk lateens were never intended. The main-sail is reefed in the same manner as other lateens, viz., along the fore-leech, close to the yard.



THE SETTEE RIG.

THE settee rig has now become old-fashioned in this country : it was formerly much in favour on the narrow rivers of the southern parts of England, and is still very much used in India and in some parts of China : it consists of a main-sail, the shape of which is something between a lug and a lateen sail ; having a very low hoist, but a lofty peak, therefore requiring a long yard. It answers well for boats of a narrow form, in which it is important to keep the pressure of canvas as low as possible : a short mast only is needed, consequently the fore-sail is short in the aft leech, and

largest at the foot. The settee main-sail is triangular in shape down to the reef points ; from which, to the foot, it is square ; therefore, when close-reefed, the settee is a perfect triangular sail, and very much like the lateen.

This rig is well adapted for a sailing punt, or any other light and narrow-shaped boat : no higher peak need be made to the main-sail than necessary ; but for a stiff and powerful boat the lofty peak is very effective in smooth water.

THE SHOULDER-OF-MUTTON SAIL.

So called from its resemblance to the flat side of a shoulder of mutton ; it is a rig best suited for smooth water, and seldom used on the coast, or in wide open rivers. The main-sail when properly shaped is neat and pretty ; whilst the tapering peak is found very effective under high cliffs, when the lower part of the sail is sheltered from the wind.

The shoulder-of-mutton sail requires neither spreet nor yard, it is set entirely by the mast, a light tapering spar, placed well forward in the bows of the boat. The rig very much resembles the Bermudian, and is in some respects exactly similar. The halliards need only be of small-sized rope, much smaller than for square-headed sails, and should be rove through a small block at the mast head, and securely hooked to the peak of the sail : no other halliards will be necessary for the main-sail of a small boat. The sail may be bent to the mast by small copper hoops, rope grommets, or otherwise ; but neither hoops nor grommets must extend higher up than the fore-halliard block ; which should be seized to the mast about three-fourths of its height from the stepping. A boom should be used for the main-sail : and the fore-sail, which is necessarily lofty, must be narrow, unless the boat is very stiff ; in which case a large and powerful fore-sail can be set.

The advantages derived from this rig are, that the heavy sway of spreet or yard is avoided; that on reefing the sail the widest and heaviest part is taken off the boat, whilst the light tapering peak is still maintained.

Boats under this rig sail well, are easily worked, and quickly reefed: brails may be fitted to the main-sail in the same manner as for other sails. The neatest and quickest method of stowing these sails is by furling them close to the mast, without lowering.

THE SAILING BARGE.

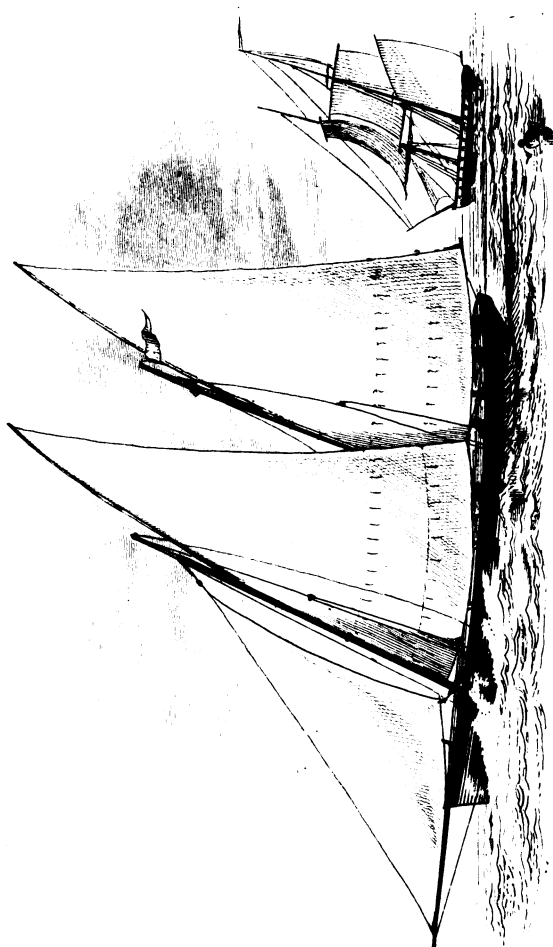
PLEASURE barges are chiefly employed in places where the water is shallow; and where the keel-bottomed boat would be found inconvenient. They are a comfortable form of vessel for pleasure parties, and well suited to the taste of the timid; being flat-bottomed, they do not careen to the wind so much as other vessels; and they have abundance of room for passengers or cargo; no ballast is necessary, the flat floor being sufficient to maintain stability under considerable pressure of canvas. The general form of rig is a gaff-main-sail without a boom, sprit-sail-mizzen, and a fore-sail; but sometimes they are rigged with a sprit-main-sail, fore-sail, and mizzen, similar to the London trading barges. The flat-bottomed sailing barge is, of necessity, fitted with lee boards on each side of the hull; the lee-ward-most of which is lowered when sailing on a side wind. These vessels require a very broad-shaped rudder, and are not generally so manageable, nor so fast under sail as boats with round bottoms and keels.

THE SLIDING-GUNTER.

(See Engraving.)

THE sliding-gunter is a rig often used for yachts' boats, and boats belonging to ships-of-war; it consists of a short mast, with a sail peaked up by a yard set parallel with the mast, so as to slide up and down, by means of two irons called gunter-irons; either one, two, or three masts and sails of this rig, may be fitted to a boat; two-masted sliding-gunters are more generally in use than single masted. The sliding-gunter is one of the safest rigs possible for open boats, and for general convenience it is almost unequalled; and this probably accounts for its being a favourite rig for first class ships' boats: and it is, besides, one by which the sails may be quickly set and as quickly furled: the masts being low, they are very convenient for unshipping: in case of using the boat for rowing; reefing this sail is also easily accomplished, the widest part, or that having the greatest pressure on the boat, being shortened to such an extent, that two reefs will generally take up half the sail. No better rig can be used for open boats in bad weather; and in fine weather, no rig looks prettier for a pleasure boat than a two-masted sliding-gunter: the sail should be well peaked up, and the luff laced close to the mast and yard.

The following dimensions and calculations for rigging and fitting a pleasure boat or ship's pinnace, upon this plan, may be relied on; they have all been carefully tested, and are therefore appended with confidence, as an accurate guide to any one who may be desirous of fitting out a two-masted boat of the size stated below, upon the most approved sliding-gunter plan.



SLIDING GUNTER.

Dimensions of Boat, Spars, &c.

Length of boat over all	-	28 feet 0 inches
Length on keel	-	26 " 6 "
Greatest breadth of beam	-	7 " $4\frac{1}{2}$ "

Fore-mast stepped	5 feet 4 inches from stem.
Main-mast	17 " 0 from do.
Length of fore-mast	20 " 2 inches, from head to heel.
Length of fore-yard	18 " 0
Distance between the gunter-irons	4 feet
Diameter of fore-mast (top)	$3\frac{1}{2}$ inches
" " (bottom)	$4\frac{1}{2}$ "

Length of main-mast (head to heel)	17 feet 3 inches.
Length of main-yard	- 19 " 8 "
Distance between gunter-irons	4 " 0 "
Diameter of main-mast, same as fore-mast.	

Length of main-boom	- 15 feet 0 inches.
Diameter do.	- 0 " $2\frac{1}{2}$ "

Diameter of fore and main yards, 3 inches at the stoutest or lower part; 2 inches at top.

Length of bowsprit, from tip to heel	11 feet 6 inches.
Do. part housed	- 4 " 6 "
Diameter of bowsprit, at gammon	0 " $3\frac{1}{2}$ "
Tapering at the tip to	- 0 " 2 "

The bowsprit may be run through the gammon, in the usual way, and the heel secured as in other boats.

The lower ends of the yards, from four feet downwards, should be at least three inches square, tapering gradually from the upper gunter-slide to the heel; the tapering to be on the aft side only, leaving it the same width, but reduced in thickness down to an inch-and-a-half; by this means the sail is brought closer to the mast, and stands better, less gap being left under the yard than when kept of the same sub-

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stance. The gunter-irons should be covered with leather, or at least, the circular part which slides up and down the mast; the square into which the yard is fitted, should be of flat iron, one inch deep, with one or two small holes for screws to secure it; the square for the upper iron should be three inches diameter, the circle five inches; the two being connected by a strong short neck, well welded into the other parts; the lower iron must be the same sized circle, but the square, three inches athwart-ships, and only one-and-a-half inch fore and aft. There is considerable strain upon the neck of the upper gunter-iron, it is, therefore, very important to have it strong and well made.

Dimensions of Sails.

Main-sail luff or fore leech	26 feet	0 inches.
“ aft leech -	25 “	5 “
“ foot -	12 “	10 “
Fore-sail luff or fore leech	27 “	0 “
“ aft leech -	28 “	3 “
“ foot -	11 “	0 “
Jib luff or fore leech -	15 “	0 “
“ aft leech -	10 “	6 “
“ foot -	8 “	3 “

There should be seven cloths in the foot of the main-sail, six in the fore-sail, and four and a half in the jib, each worked to twenty-two inches. The fore-sail and main-sail should be secured to the yards by small marline, which answers better than grommets; but from the lower iron of the yard downwards to the tack clew, a running lacing is best, and will be found most convenient for casting off when about to trice up the tack: small composition thimbles must be worked into the luff of these sails, to facilitate the working of the ratline, when fitted with a lacing. The brail-rope, if overhauled by a boat hook, will serve as a tack tricing line; and the fewer ropes the better: those necessary are,—a single rope for jib

halliards, which should reeve through a small block hooked to the mast head. ¶

Fore and main halliards should consist of a single whip purchase, one inch ; and a tye of one-and-a-half or one and three-quarter inch ; to one end of the tye a single block must be spliced to receive the whip purchase, the other end is rove from the fore side, through the sheave-hole at the mast head, passing through the upper ring of the gunter-iron, and round the neck of the lower gunter-iron, seizing it to its own part just above, between the yard and the mast ; a seizing should also be passed round the neck of the upper gunter-iron, to secure it to the tye. An eye must be spliced into one end of the whip purchase, and passed over a flat headed bolt in the fore part of the thwart, to which the mast is clamped,—the bolt answers the purpose of a cleet, and serves to belay the halliards when the sail is hoisted. There may be other methods of fitting the halliards, but this is the best I have seen.

A single rope on each side the mast will answer for shrouds ; which, by being brought a little forward of the mast will also act as a stay.

The brails, which catch the sail under the throat, in the same manner as for a sprit-sail, should lead through a small block or a bull's eye, strapped to the lower side of the lower gunter-iron ; the brails will also answer the purpose of down-haulers if required, and trice rope, as before mentioned. A single gun-tackle purchase is proper for fore and main sheets ; the upper or boom sheet-block must have an eye strap, large enough to admit the outer end of the main boom as far as the shoulder, which is left on the boom about one foot from the aft end, as a stop to the main sheet block strap.

The best plan of setting the main boom for this rig is, by pointing the aft end so as to fit into the clew of the main-sail, in the same way as a spreet fits into the peak eye of a sprit-sail ; the inner end of the boom is then secured by means of a lanyard rove through a half sheave in the boom end. By this method the goose-neck is dispensed with, and the foot of

the sail may be stretched taut, and will stand as flat as possible without lacing to the boom: it must be observed, however, that on setting sail, the same care must be taken to keep the foot of the sail taut, as is necessary in setting a sprit-sail; or the shaking of the sail, when the boat is in stays, may cause the clew to slip off the boom end: but a little practice soon renders this easy.

THE REVOLVING RIG.

THIS singular, but ingenious, mode of rigging boats and small yachts, is the invention of Molyneux Shuldham, Esq., Commander R.N., a gentleman who kindly furnished me with drawings, models, and other original information, developing the whole scheme and peculiarities of one of the most useful, but remarkable methods of rigging vessels and boats ever invented. As the invention has never been fully explained, or made known to the public in any other treatise, I am authorized by the inventor, to lay before my readers a brief history of the revolving rig, showing its originality and success under repeated trials at Woolwich and elsewhere, together with such information as I am enabled to give from personal observation of the boat itself, when under the able management of its inventor in Harwich Harbour, and on the rivers Stour, Deben, and Orwell. Many years ago the revolving rig was frequently submitted to public and practical tests, and was entirely successful under various trials, which called forth letters and reports from some of the most eminent officers in the navy, all acknowledging the extraordinary capabilities of boats rigged on the revolving plan. In August 1835, His Imperial Highness, the Grand Duke of Tuscany, presented the inventor with a valuable ring set with brilliants, as a mark of the high opinion his Highness entertained of the revolving rig.

The inventor repeatedly challenged the fastest boats on the Thames, and invariably beat them; particularly in working to windward: he would willingly have sailed with any boat in Europe with confident success; as the documents before me clearly testify.

Among those papers I also observe many interesting adventures relating to trials of speed; and some curious revolving manœuvres are recorded: many of these papers are official letters from the principal officers and commissioners of the navy. I have been fully authorized to insert some of them in these pages, with the names of those gentlemen appended; but, as I considered they would scarcely be of sufficient interest to my readers, have declined doing so: suffice it to say, they tend to show clearly the remarkable capabilities of the revolving method of rig, not only for simplicity of management, but also, because, by no other rig can the sails be made to stand so flat, or present such a drum-like surface to the wind.

The invention of a revolving mast originated entirely with Mr. Shuldham, and has not been adopted by any nation; the navy board were at fault in their endeavours to find out the non-originality of it; the Chinese¹ boats bore the nearest resemblance, some of which have a sheer-mast which revolves on part of a circle. Mr. Shuldham's first notion was to contrive the rig of a boat so as not only to sail well, but to be easiest handled. He conceived the idea that the most perfect sail would be that which would stand as flat as a board, and be a moveable plane with its axis on a line with it, and which could be placed in any position so as to receive the fullest or the least effect of the wind's power.

It was supposed when the yacht "America" entered our waters, that no Englishman had ever thought of lacing sails to booms, and making them stand as flat as boards, but it is a plan Mr. Shuldham seems to have adopted from his earliest experiences. On reference to the sketches entrusted to my

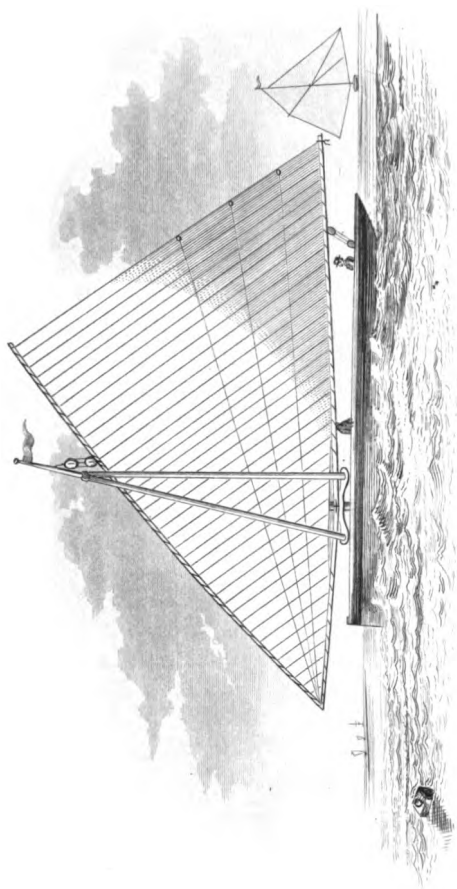
¹ The Soolo Pirate Boats are also rigged with a sheer mast.

care, I find that forty-five years back he rigged a vessel precisely like the late Mr. Steven's celebrated yacht the "Maria" of the New York Yacht Club, viz: a sloop with only two working sails, main-sail and fore-sail, the fore-sail or jib (for it may be called either) was laced to a boom, which was found very convenient, enabling the inventor to work to windward without any help, merely by leading two fore guys within reach at the helm. Mr. Shuldham discontinued the rig on account only of his having found it very troublesome in a heavy squall; this may not, however, be so in Mr. Steven's well manned yacht; but the other was short handed.

One gentleman, a captain in the navy, in his report upon trials of the revolving rig, says, that "for squalls, for convenience in reefing, and readiness in lowering the sail, I declare it to be the handiest rig I ever witnessed." Another navy captain in reporting a trial which took place at Brighton, adds, "I saw the plan tried here, and although the vessel it was tried upon was built for stowage, and not for speed, and she was leaky, and very objectionable as a trial vessel, yet, under these disadvantages, she went clear to windward of all the fast sailing crafts which were opposed to her; and fully established the superiority of the plan, as presenting a means of effecting a windward passage with greater speed and certainty, than any vessel of the rigs hitherto in use."

But I could fill my book with extracts of the kind, from reports of approbation from some of the first naval officers in the kingdom, after public trials of the revolving rig in their presence. I will, therefore, pass on to the more practical point of the revolving principle.

The principle of the invention, as shown by the engraving on the opposite page, consists of two or more spars affixed to a base, turning upon a strong iron pivot; the upper end of the spars being secured by a cap, so that the whole machinery of mast and sail depends for support on an iron spindle, stepped in an iron socket on the keelson. The original invention was confined to two spars only for the mast, which



REVOLVING CLIPPER.

is all that is necessary for small boats ; but for those of six tons and upwards, three or more spars, fixed in conical form, will answer better, and add stability and safety to the working of the boat ; as the mast then supports itself, and takes all the strain off the boom.

For larger, or decked boats, a top-mast may be fitted, and a jib-headed top-sail used to great advantage in light winds : also a ring-tailed boom and stern-sails, for running before light winds.

The boom and bowsprit must be contrived of two separate spars, and may extend as far over the stem and stern of the boat, as a cutter's boom and bowsprit, but with considerable spring at each end ; the main-sail, fore-sail, and jib, are maintained as one sail of triangular shape, laced all the way along the boom, and spread by a tough yard. The sail, when properly made, should stand just as flat as a board, and without a wrinkle ; and, as it moves *with* the mast, and not around it, is never deranged from its drum-like surface.

At right angles with the sheer, and on the revolving base, are fixed two iron rests, in which the boom lies, these serve to keep the outer ends of the boom up ; the aft rest should be much higher than the fore one, so as to keep the boom from dipping in the water when lying over in a breeze. The fore end of the yard is hinged to the fore end of the boom or bowsprit, by means of an iron joint or goose-neck ; the reefs are formed in radiating lines along the sail, from the fore end of the boom to the aft leech of the sail.

With regard to the due proportion of spars for the triangular sail, the boom, or that part of the spar abaft the mast, should be three-fifths of the whole, and the bowsprit, or fore part of the spar, two-fifths ; but for a lug-sail, the bowsprit should be a trifle more than one-third of the whole length : if there is less area of canvas forward, then the strain on the sheet, in sailing before the wind, would be greater, which it would be well to avoid.

The slings should be placed on the main-yard about two-thirds of its length from the fore end, leaving only one-third

to extend abaft the mast; by which means it is impossible for the peak of the sail to sway to leeward as the peak of a cutter's main-sail; it being kept in a direct angle with the boom, thereby defying any bend of the yard or belly to the sail: the length of the yard must be exactly three-fourths the length of the boom and bowsprit, so as to preserve a central pressure under every increase or reduction of sail.

The fore end of the boom (which forms the bowsprit) should be loaded at the outer end with lead, run into the spar, to counterpoise the weight of spars and sails to leeward: this may appear, at first sight, an injudicious plan, but, when carefully considered, it is not only practicable, but very ingenious.

The masts are supported in various ways according to the tonnage of the vessel. In open boats, a flat revolving mast without rigging will answer. For decked boats, or small vessels, a wooden or iron pivot supports the whole strain. In larger vessels, rollers are affixed to the circumference of the revolving base, which work between two circular sweeps or rings, firmly secured to the gunwales and deck, supporting the whole strain in every direction.

A very strong double or treble block is required for hoisting the revolver's sail; and this may be readily inferred, when it is remembered that the revolver's halliards are in lieu of the main, gaff, fore-sail, and jib-halliards of a cutter. The purchase should, therefore, be strong, and one that will overhaul easily. The sheers should be let into the mast-head with great nicety, rivetted to it, and well secured with hoops.

I have shown the revolving base above the bulwarks in the engraving at page 56; this has been done for the purpose of explanation; the revolving base is not of necessity fixed so high in the boat, on the contrary, in several of Captain Shulldham's boats, the revolving base was hidden below the bulwarks, and the sheers only were seen above the gunwale.

In rigging and fitting out a revolver, the novice must pay attention to the instructions here laid down; and he is

strongly advised not to deviate from them in any essential particular, as Captain Shuldham tried every conceivable plan, before perfecting the invention ; and all his best and most approved notions with regard to the rig, are here stated.

As a warning to any one who may venture on the experiment, it may be mentioned, that a boat-builder at Norwich wishing to rig a boat upon the revolving principle, with such deviations from Mr. Shuldham's plan as were considered improvements, actually made the boom *to droop aft*, in order to obtain the benefit of low canvas ; the consequence was, that the first squall the boat encountered laid her boom in the water, and she was upset.

It may be, that the rig would look much prettier if the boom might be rigged so as to hang in a horizontal line, parallel with the surface of the water ; but after many and repeated experiments, the inventor found such could not be accomplished in the revolver, consistent with the safety of the boat.

The boom requires to be of two parts, which meet at the pivot or base of the revolving mast, where they are each fitted with a goose-neck ; and thus, a bowsprit is entirely dispensed with, the fore part of the boom answering the same purpose.

It is one of the *main features in the revolving rig*, that *both parts of the boom should incline upwards, at the outer ends* ; as shown in the engraving. The author regrets that the engraving of the revolving clipper in the *two previous editions*, was inaccurate in that respect ; he has, therefore, substituted *an entirely new plate*, with a carefully executed engraving ; the sails and spars being shown in proportionate length, and upon a proper scale.

The inventor once tried the experiment of a revolving sail, made *to fit a yard of the same length as the boom* ; the consequence was, that with one reef down, the boat carried an unpleasant weather helm ; with two reefs, an increased one ; and with three, she could hardly be kept from luffing into the wind's eye : but by making the yard about four-fifths the length of the boom, the centre of the wind's pressure upon

the sail was the same in a longitudinal direction under all reduction of canvas.

This experiment at once convinced the inventor of the absolute necessity of maintaining a due proportion in the revolver's yard and boom.

AS TO THE MANAGEMENT OF REVOLVING RIGGED BOATS.

A revolving rigged boat, when understood, is much more easily handled than one of any other rig; but it requires another kind of seamanship, constant attendance at the halliards and sheet; for when the latter is let go, the sail points itself head to the wind. What sailor would suppose that the quickest way of putting a revolver about would be by easing off the main-sheet? And what sailor would dream of sailing or laying a vessel to—*stern foremost*? Again, what sailor would suppose a vessel could be 'hove-to' with all her flying kites set, with the wind-a-beam, on the quarter, or nearly right aft? And what sailor would imagine that it was a good plan to place some of the vessel's ballast at the end of the bowsprit? And again, what sailor would suppose that a vessel having *no way* upon her, could be put about? However all the above have been tried and proved, over and over again; the inventor, not having discovered them all at once, but by much experience. As the power is at will to carry as little sail as possible, and also much more than any other rig can possibly carry, a great deal of revolving seamanship must depend upon judging what amount of sail should be carried with prudence.

Many of the peculiarities of the revolving rig, at first sight, will strike the novice as curiously at variance with all other methods. One singular evolution inherent of the rig is, that the vessel may actually be hove-to, with all sail set, and the wind abeam or on the quarter, and kept in that position during pleasure; which is a great convenience when wishing to stop the vessel suddenly to avoid collision, or to allow of a boat's coming alongside to shift a passenger or goods from one

to the other : this may be done, and way given again to the revolver in a few seconds.

The vessel may be laid-to with her stern to windward, or may be sailed stern foremost : a two-masted revolver when close hauled, might in case of emergency, be guided or put about without the aid of a rudder, merely by a slight sway of the sail.

By the revolving method the dangerous practice of gybing the sail is optional, and may be entirely avoided ; the whole being worked by means of a bridle, both ends of which are made fast to the fore end of the boom ; thus the sail can be twisted on its axis, bringing it fore and aft or at right angles with the keel, with great velocity.

The revolving rig will answer for almost any description of sail : the shape is not limited to any one particular form more than another, the lug or any other sail can be worked on the revolving principle with equal facility.

For vessels navigating intricate channels, or crowded waters, the revolving rig is peculiarly adapted ; as the vessel may be almost instantaneously stopped, by all sail being reversed, so as to act in a contrary direction.

In the event of getting aground, the revolver's sails may all be thrown aback in an instant, to help her off ; it may, therefore, on such occasions, be found very convenient to sail a vessel stern foremost.

Another important feature is, that the sheer-mast may be easily and quickly lowered, without the necessity of unrigging or deranging a single rope, or unbending the sails ; a great consideration and convenience for passing under bridges.

A large spread of canvas can be set in running before the wind, and may be reduced with less labour, more simply and expeditiously than by any other mode.

Again, the nicety of its balance renders it impossible to be taken aback, and the worst apparently that can happen is, that the sail may fly fore and aft, or point itself to the wind. To the many advantages before stated, may be added the quickness and certainty of staying, either in a sea-way or in

smooth water ; the facility in wearing, and the small compass in which the evolution is performed.

There are one or two modes of reefing revolving sails, to which it may be well to call attention.

The first, and best, is by lacing the foot of the sail to a flat board, painted white, to correspond in colour with the sail ; this board is pivoted at both ends, and works in two iron stanchions at the extremities of boom and bowsprit, so as to be turned easily ; it is provided with self-acting catches, at each end, so that a reef can be taken by merely a half turn of the board, and by several turns the sail may be speedily furled :² by turning the board the reverse way the reefs may be shaken out ; the halliards must be slacked during the performance. If a sail of triangular shape is to be reefed in this manner, the reef board must also be triangular.

Another method of reefing quadrangular and triangular revolving sails, is without any reef-board or roller, but by lacing the reefs with an endless rope ; a pull of which will reef the sail on a small scale, and a few pulls on a large one.

The working of the revolving sails by means of the long sheet or bridle, requires some little experience ; but when once learnt, is simple and perfect. One end of the sheet should be made fast to the extreme end of the bowsprit, and rove through a single block at any convenient distance from the aft end of the boom ; the other end of the sheet should be passed to the end of the bowsprit, and made fast there, leaving as much slack in the sheet as to belay the bight, when the boom is squared or at right angles with the boat's hull. In wearing without gybing the sail, the sheet must be eased off or let go when the boat is nearly right before the wind ; whilst the boom is swinging over head, the man at the helm must lay hold on the other part of the sheet, and haul through quickly, which gives a bridle on the *other* side of the

² This mode of reefing sails appears to be very similar to that adopted by some of the islanders of the Eastern Archipelago. See *post* "Malay Jellores," &c.

boom, forming both a fore-guy and a sheet, which may be belayed or not.

In addition to the bridle, a short main-sheet might be used for working to windward, the standing part made fast amidships, and the fall leading through a single block affixed to the boom, can be instantly unrove in bearing up to run before the wind. The short sheet will be found convenient, although unnecessary, the bridle being sufficient to answer every purpose.

When sailing close-hauled, in a sheer-masted revolver, the top-sail only can be set, in addition to her large sail, because the others cannot be made to stand equally flat with them: it is in sailing with the wind on the quarter or abaft, that the advantage of additional sails will be felt.

The great objection to the rig, seems to be the room which the revolving base occupies in the boat.

There is one circumstance in regard to the revolving rig, that strikes me with surprise; which is, that it has not been more generally adopted: but even this may be partially accounted for by the invention being for many years patented, which rendered it an expensive undertaking to fit out a boat according to the original design. The patent, however, expired in the year 1839.

CENTRE BOARDS, SLIDING. AND REVOLVING KEELS.

CENTRE BOARDS and sliding keels are now common in half-decked sailing boats, both in England and America, particularly in those kept for match-sailing: the favourite boats being those of a very broad form, and with flat floors. Centre board boats have been in general use in this country since the year 1852, when a little yacht of $3\frac{1}{2}$ tons, named the *Truant*, was brought over from New York, and won almost every match she sailed in, beating all our little cutters and racing craft.

There are many objections to moveable keels in general,—

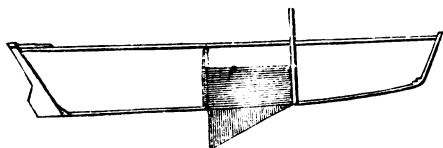
their liability to get out of order by being choked with sand or grit, the room they occupy in the boat, and the difficulty of keeping the keel-case water-tight.

Admiral (then Captain) Schank was the inventor of sliding keels, and there are several models of his invention in the United Service Museum.

Captain Schank's invention was publicly tested in the year 1791, on the *Trial* cutter, and highly laudatory reports thereon were made to the Admiralty. Captain Schank also, about that time, published a treatise on the subject.

The revolving keel, which is in some respects a modification or improvement upon the sliding keel, was invented by Mr. Shuldham, when prisoner of war at Verdun, in the year 1809; the model then made by him is deposited in the museum at Ipswich, Suffolk; a model was also exhibited in the Adelaide Gallery about the year 1829.

It has been asserted that revolving keels have been common in America for several years past: *centre boards* are in



Section showing Centre Board Keel.

fact nothing more than revolving keels; the probability is, that the idea of the revolving keel or centre board was taken from Mr. Shuldham's model in the Adelaide Gallery, and about that time found its way across the Atlantic. The revolving keel is necessarily quadrangular in shape, whereby it can be raised or lowered without injuring the boat by its leverage or twisting strain. Totally different from the sliding keel, it works or revolves upon a single pivot or bolt through the fore part of the keel; by which means the liability of getting jammed is avoided, particularly when required to be

worked in haste. The keel itself in Mr. Shuldham's invention is *lead*, which works in a water-tight wooden case, lined or ribbed with copper or zinc, for the purpose of adding strength to the case, keeping the keel clear of the wood-work, and decreasing the friction.

In the American models, the centre board keel is of a very hard and heavy kind of wood, shod with metal. The leaden revolving keel is adapted only to decked boats; for all open boats and small craft the revolving keel should be of wood: lead or iron would render a small boat unsafe in a sea-way.

The revolving keel is raised and lowered by means of a keel-rope passed round a sheave fixed to the upper part of the keel; which gives a double power: to the keel rope a tackle is affixed, consisting of a very small four-fold and a treble block, which will increase the power to twelve; by this means a keel weighing 5 cwt. and upwards may be raised and lowered with facility by the man at the helm: the utmost extent of raising, being up to the deck beams, which just brings its lower edge even with the keel of the boat. The centre of gravity is thus vertically moveable at pleasure.

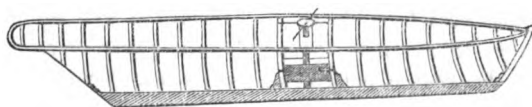
The advantages of centre boards, sliding, and revolving keels are many; in the first place, a great depth of keel may be put on the boat; which is of infinite service in turning to windward, particularly with flat sails: when running before the wind the keel may be drawn up, and the boat will travel much faster than with the keel down. In case of running sharply aground, the keel performs the part of a buffer, and saves the boat from damage. It may be well to mention here, that whatever be the mechanical power employed for raising the keel, it must be so contrived as to move upwards with the keel, when lifted by the boat running aground: on which occasion the vessel vibrates on a central point, so that without difficulty her head is placed off shore, when after raising the keel, the boat glides off into deep water.

The objection to centre boards, sliding, and revolving keels in regard to their general adoption in yachts and pleasure

boats is, that they absolutely divide the boat into two longitudinal halves, the keel case coming up quite to the deck, which renders the boat unfit for many purposes. But for a flat-floored sailing boat, or a small racing craft, or any other, where speed is the main object, the centre board or revolving keel is invaluable. So also for a shooting yacht; it being often necessary to go into very shallow water in order to obtain a shot at wild fowl. But the strength of the boat is materially weakened by cutting away the backbone and floor timbers. A boat, therefore, intended to be fitted with a centre board, sliding, or revolving keel, should be built expressly for the purpose, that proportionate strength may be given to allow room for the keel case, without weakening the keelson.

REVOLVING BALLAST.

The system of raising a portion of the ballast by means of a powerful screw apparatus, for drawing the ballast up to the deck beams, so as to alter the centre of gravity at pleasure, is another of Mr. Shulldham's inventions: the plan is very useful for steadying vessels in a sea-way, and preventing them from rolling so much as when all the ballast is under the floor. It should only be hove up when running before the wind; but as this method is better explained by illustration than profusion of words, the diagram below will more clearly elucidate the subject.



Section of Boat, showing Screw, with Ballast partly raised.

The lump of lead or iron ballast thus moveable, should be secured in a strong wooden or iron case, the screw passing through both. This method of raising the ballast enables

a vessel to run lighter and faster before the wind, particularly in a heavy sea ; when *on* a wind it must be lowered. There is a model of a revenue cutter in the Society of Arts Repository, for which Mr. Shuldhham was awarded a silver medal, for an ingenious method similar to this, but so contrived that when the vessel heeled over thirty degrees, the ballast lowered itself. The revolving lead-keel answers every purpose of raising and lowering ballast, and is in some respects superior to, and more simple in contrivance, than the revolving ballast.

TRIANGULAR YACHTS.

SOME few years since a remarkably curious boat, called the *Problem*, was exhibited by the inventor, Mr. Henry Dempster, H.E.I.C.S.: the hull of this vessel was of triangular shape ; the stern-post being made to rake at the same angle as the stem, so that both met and terminated in a triangular point under water ; and thus formed simply an angular keel. This model yacht was twenty feet in length, and six feet beam ; was iron built, and ballasted with lead. It was rigged with three masts, the mainmast being placed exactly in the centre, and in an upright position ; the fore-mast had considerable rake forward, and the mizzen-mast the same proportion of rake aft. Two square sails were set on the main-mast, one above the other, and a triangular sail on each of the other masts : these triangular sails were on a revolving principle, the booms being secured at the central gravity, one to a pivot on the stem, and the other to a similar pivot on the top of the stern-post ; by which means they would turn round and round, clear of the masts, and could be trimmed to any degree upon a circle. The sails thus possessed a double advantage, and with the help of the triangular hull could perform many rapid revolving evolutions, which no other boat could so quickly

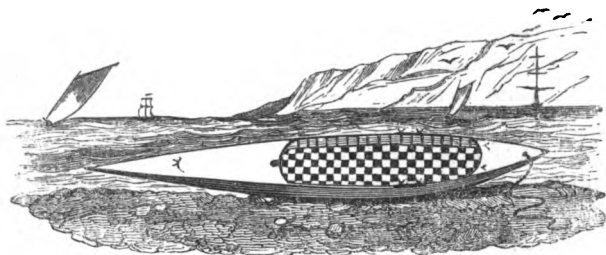
accomplish. Among other experiments by the inventor was a very remarkable one, tested at Newcastle, by driving two stakes into the ground at low water mark, to which a strong iron bar was lashed horizontally, in a similar manner to a leaping bar. A pole or gauge was then erected alongside the stakes, marked to feet and inches, to indicate the depth of water. When the tide rose sufficiently high to show that there was one foot and a half less water than the *Problem* required to sail clear of the obstruction (consequently that the vessel would strike it with her angular keel,) she was sailed *stem on* at the bar; a stiff breeze blowing at the time, when she went over it by *rise* and *fall*, similar to a horse jumping a gate—the performance was then repeated several times in presence of a large assemblage of spectators.

The advantages of a triangular hull (as stated by the inventor) are, that a vessel might be made particularly useful for narrow rivers, where much turning in a small compass is often necessary: but it will only answer in deep water; and the vessel must always be kept afloat; in fact, such a vessel could not lie aground. It has also been suggested that the invention is well adapted for trawling boats; and in fact any vessel where much turning is required.

The invention, however, of a triangular hull has never, I believe, been carried out in any other vessel than the inventor's *Problem*.

BOATS FOR WILD-FOWL SHOOTING.

THE GUNNING-PUNT.¹



THE engraving represents a round-bottomed gunning-punt, useful both for sailing or paddling; after the author's own invention.

A wild-fowling gunning-punt, for the purpose of carrying one individual sportsman, with a punt-gun, ammunition-box, shoulder-gun, and other requisites for punting, should be just large enough to be safe and serviceable, but nothing more. The smaller the boat is made to appear on the water, the greater will be the punter's chances of success: the more convenient it will be to manage, the quicker he can make up to birds, and the less will be the exertion necessary to propel it.

The size of a wild-fowling punt, however, must be in some proportion with the size of the gun-intended to be used, and also in proportion to the height and weight of the punter himself.

The most useful sized punt-gun for general purposes, is that carrying about $\frac{1}{2}$ -lb. of shot at a charge; and, as there

¹ See the "Wild Fowler, a Treatise on Ancient and Modern Wild-fowling, Historical and Practical," by H. C. Folkard, 1859.

are more punt-guns in use on the coast, of this size than any others, the recommendation would seem to be confirmed.

A punt to carry a gun of the size stated, with a man of ten or eleven stone weight, must be of the following dimensions:—

Length, over all	-	-	17 feet.
Breadth, amidships	-	-	2 feet 10 inches.
Ditto, ditto, at bottom	-	-	2 feet 8 inches.
Depth, at bows	-	-	4 inches.
Ditto, at stern	-	-	8 inches.

A boat of this description may be built either with a flat bottom or a flat floor: the distinction being, that one is perfectly flat as the bottom of a box—which is termed “flat-bottomed;”—the other, though said to be “flat-floored,” is built more like the bottom of a skiff, *i. e.*, clench-built; and the sides do not commence from an angle as in a flat-bottomed boat, but as round as a whale-boat; though it is a great desideratum to maintain the floor throughout as flat as it consistently can be, with due regard to shape; and, when ingeniously constructed, such is by far the best form of punt that can be used.

Colonel Hawker condemns all round-bottomed punts, such as are used at Southampton and Itchen ferry, as on a bad construction; and gives, as a reason, that they have unsteady bearings.

Now, without disputing the Colonel's assertion as to the Southampton and Itchen ferry gunning-punts, it is insisted that the reason of round-bottomed punts having unsteady bearings is, because they are *too* round at the bottom. If they be constructed with a long flat-floor, as flat as it is possible to make a clench-built punt, it will be safer, and even steadier, than a flat-bottomed punt.

The punt under consideration should be built with a spring in her bottom, fore-and-aft, of from $1\frac{1}{2}$ to 2 inches; or, in other words, the bottom of the punt, as taken from stem to stern, should form a section of a circle; and this, whether a flat

or round bottom: the object of which is, that in going into very shallow-water, when the bottom of the boat grazes the mud, it may nevertheless be pushed ahead many yards further; whilst a boat without any spring in her bottom would be set fast. This is sometimes a great consideration, when punting to birds on the ooze, as the tide flows towards them, when the success of a shot may depend on the chance of being able to push the punt a few yards farther ahead.

The fore-part of the punt, from stem to cross-piece, should be covered over with a very thin scantling of deal; adding no more to the weight, forward, than absolutely necessary, because of the heavy gun which has to be placed there.

The cross-piece alluded to, is placed just abaft the scantling, and marks the balance for tipping the gun.

The covered part of the punt may be finished with a semicircle, if the punter attaches any regard to appearances; and the edge should then be completed with a neat semicircular piece, rising an inch and a-half or two inches from the scantling-deck; which affords a sufficient screen to the punter, when making up to birds.

Another cross-piece of light wood should be placed athwart the gunwale, about $2\frac{1}{2}$ or 3 feet from the stem, before the scantling is put on; this is for the purpose of fixing the rest for the outer-end of the barrel. The rest should be a simple copper screw, with a small semicircular crutch in which to receive the barrel. It should be made in connection with a female screw, attached to the fore cross-piece; and the male screw should be of sufficient length to raise or lower the elevation of the gun from one to two inches.

As to the position of the chock, or strong-piece, to which is attached the necessary apparatus for checking the force of the recoil, it must depend entirely on the means intended to be employed. If the patent spiral recoil-spring is to be used, the interior of the punt must be fitted accordingly. But if the strain is to be thrown upon the stem-piece, the builder must take care to fix the same strong enough to receive it.

The fore-parts of gunning punts are sometimes left quite

open, and without any scantling or fore-deck ; but, when so constructed, they require to be rather deeper at the bows than the dimensions stated at page 70.

The gunning-punt may be built entirely of fir, or the upper strakes which are above water may be of fir, and the lower ones of elm ; if all fir, the boat will be so much the lighter, and more buoyant. A gunning-punt should never be built of oak, it is too heavy for the purpose. Many punters express astonishment that Colonel Hawker should have recommended such a material : during the latter days of the colonel's sporting career, however, he appears to have given preference to elm and fir.

The only objection to clench-built gunning-punts, with round bottoms, is, that in a breeze, the ripples on the water make a trickling noise against the planks of the punt, as it is propelled forward ; called in nautical language, " tell tales ;"² that is to say, the noise tells the helmsman of a breeze springing up.

I once heard an objection raised to clench-built punts ; the punter stating that but for the " tell tales," he could have heard the birds feeding, and so discovered their whereabouts ; whereas the noise of the water rippling against his clench-built punt, not only precluded him from so doing, but frightened the birds, and caused them to take wing. But I think the statement scarcely feasible, because, by resting a moment on the paddles, the " tell tales" are silent, and they certainly cannot be heard by birds beyond forty or fifty yards' range. I have never found the smallest inconvenience myself from the " tell tales ;" though a remedy may easily be found, if necessary, by substituting a carvel for a clench-built punt.

It is desirable that there should be no farther projection of the stem of the punt beyond the muzzle of the gun, than is absolutely necessary ; and this depends in a great measure on the form and construction of the punt, and may be materially

² This must not be confounded with " tell-tale," a portable mariner's compass.

provided against by continuing the floor as flat and far forward as possible; so that it may be more buoyant under the heavy weight of gun-metal, which depresses the head of an ordinary punt. The muzzle of the gun should never extend beyond the stem of the boat.³

The form and build of the gunning-punt differs slightly in almost every county around the coast.

The Hampshire punts have round sterns, and are short and narrow, the usual length being but fourteen feet, and the breadth at gunwales from two feet six inches to two feet eight inches, according to the size and strength of the punter. The recoil of the gun in the Hampshire punt, is received by a rope breeching, rove through a strong wooden knee, firmly fixed across the bottom planks of the punt, a little in advance of the gun's balance; and this method is also adopted in other counties, as the strongest place in the punt on which to throw the force of the recoil; though it may well be doubted whether the stem is not the best place through which to reeve the breechings, and receive the strain.

The eastern coast gunning-punts have attained considerable notoriety; particularly those in the neighbourhood of Maldon, Essex, the form of which is much in favour with those who resort to the pursuit of punting in winter as a means of subsistence.

The form of the Maldon gunning-punt is very similar to a coffin; simply flat-bottomed, with nearly upright sides, formed of two broad strakes. The usual length is sixteen and a half to seventeen feet, and the breadth two feet eight inches to two feet ten inches. They are very handy for setting to birds, because of the narrowness of their form, enabling the punter to use his arms freely, and, by means of the paddles, with considerable effect; but they are easily upset, and require very great care on the part of the occupant in all his movements. They have strong gunwales and cross-

³ For more minute information as to gunning-punts, the mode of using them, and wild-fowl shooting generally, see "The Wild-Fowler."

piece, and are the least costly of any gunning-punt that can be built. When intended for an extra-large gun, they are built in proportion—longer and wider—as may be required.

In loading the gun whilst afloat in one of these punts, it is usual to row ashore, or lash the punt alongside a larger vessel.

TWO-HANDED GUNNING-PUNTS.

Two-handed gunning-punts, or those constructed to carry two persons, are seldom used at the present day. Many years ago, when birds were more numerous and less wild than now, two-handed punts were more general. They have too formidable an appearance on the water to allow of much success; and although manned by two persons, they make slower progress through the water, and are much more sluggish to propel than single-handed punts. One man sits or kneels at the stern, in a most cramped and uncomfortable position, and sculls with an oar, whilst the other lies down to attend the gun; and if his arms are long enough to reach across the boat (which must of necessity be much wider than a single-handed punt), he assists with the paddles in propelling the boat towards the birds.

The advantages of two-handed punts are, that they carry a larger gun than others; sometimes a full-sized stanchion-gun, which throws from one and a-half to two pounds of shot at a charge, making fearful destruction among large numbers of wild fowl; and when loaded with mould-shot, they sweep the water from sixty to one hundred and twenty yards, spreading terrible slaughter among the feathered tribe.

The gun is generally fitted so as to be "tipped" with the facility of a smaller one; and flying shots are often made, just as the birds rise from the water, the man at the helm turning the boat skilfully with his oar, in the direction taken by the birds when rising in the air. There is but little use at the present day, for two-handed gunning-punts.⁴

⁴ *Vide* "The Wild-Fowler."

THE SAILING-PUNT.

The sailing punt enables the wild-fowl shooter to obviate a good deal of the hard work which he must necessarily encounter in a rowing-punt; and it is, besides, the most eligible boat for approaching wild-fowl in shallow water by daylight. It must be observed, however, that it is not every punt that can be sailed; a boat of peculiar and special construction is required for the purpose; for, so surely as an inexperienced hand attempts sailing an ordinary rowing-punt, such as is used for wild-fowling, so surely will he capsize himself. Of all the forms of gunning-boats, the punt is the least safe under sail, and the least manageable. A punt must be built specially for the purpose of carrying sail; and it may be so constructed as to be capable of being used either as a sailing or rowing-punt, and so that a strong man, with long arms, may propel it with paddles nearly as fast as an ordinary rowing-punt. And such is precisely the description of boat recommended; which, after trying several different forms, the author used many years, and found it not only a most comfortable, but serviceable boat for the purpose; and it looked no larger on the water than a rowing-punt.

The length of the sailing-punt, from stem to stern, should not exceed eighteen feet (if it be intended both for rowing and paddling), and the extreme breadth amidships three feet three inches, gradually tapering from the midship section towards the stern, and finishing with an upright stern-piece; the forepart of the punt should also taper gradually from the midship section to an upright stem-piece. The depth, forward, should be four-and-a-half inches, and aft, eight inches.

Neither water-decks nor wash-streaks are required for the sailing-punt, because no reasonable man would venture to set sail in so frail a bark in rough water.

The mast must be a shifting one, so that it may be readily shipped and unshipped at pleasure : a light piece of spar, about the size of a mop-handle, and from four to five feet in height. The mast-stepping hole may be made either through the scantling in front of the screen-piece, or just abaft it, and a little to the left of the punt, so as to be clear of the balance-rest on which the punt-gun lies. Neither shrouds nor stays should be fitted to the mast, either would make it dangerous ; because, should a heavy squall strike the sail, it is better for the tiny mast to go by the board, carrying all sail along with it, than to capsize the boat, which would be the result under such circumstances, if the mast were confined. It is better, therefore, to select a slender mast, that has not sufficient substance to overturn the punt. Never mind its bending in a strong breeze ; the carrying away of a mast now and then is far less to be regarded than the upsetting of a punt. The punt's mast must be fitted at the top with a small metal sheave for the halliards, which, after being led through the sheave-hole, are attached to an iron or copper traveller, sliding up and down the mast ; and a thumb-cleat being tacked on to the lower part of the mast, just above the punt's deck, all is ready for making fast the halliards whenever the sail is hoisted. The sail should be of lateen shape, and made of very light duck or white calico. It is unnecessary to give the exact dimensions of the sail, as that must be in suitable proportion to the stability of the punt ; but, as a general rule, the lateen-yard for spreading the sail may be fifteen feet for a punt of the size and length described.

The rudder-bands must also be sufficiently long to enable the sportsman to steer his craft, when lying at full length on the floor of the punt.

No other ballast will be required than that of the punter himself (who is virtually *the ballast*), the punt-gun, ammunition box, and other accoutrements ; all of which must be placed in exact position, so as to trim the boat to a nicety. If the sail is lightly made and fitted, and fairly proportioned, the punt will be quite safe in smooth water, in experienced hands.

But *one* sail only should be used, and that the lateen before mentioned. The rapidity with which a little boat of this kind skims along on a reach is astonishing; and the young sportsman will often be agreeably surprised at the easy and unsuspecting manner in which it may be run up to a flight of wild-fowl in a steady breeze; and if a shot cannot always be obtained before they take wing, by luffing the punt whilst they are rising, in the same manner as with a yacht or sailing-boat, an excellent flying shot may frequently be made.

It is not the strength of the wind which produces the danger; for if the water be smooth, and the punt built according to the directions here laid down (with keelson and flat long floor), sail may be carried fearlessly in an ordinary, and even a stiff breeze. But the inexperienced are warned against the peril of carrying sail on a punt in any but smooth water. The effect of venturing into rough water with a long low craft, whilst pressing her ahead, under sail, would be to drive her bows under water; and the weight of the gun on the head of the punt must tend to increase the danger. If the punter moves forward to lower the sail, his extra weight thrown suddenly forward would, in such a case, inevitably send the punt under water, head first; and, independently of such a glaring indiscretion, it is impossible to prevent the water from flying over the gunwales in a heavy sea. Therefore, the wild-fowler is cautioned *not to venture into rough water with the sailing-punt*, for a sportsman's life is supposed to be of more value than that of a duck.⁵

THE SHOOTING-BOAT.

This is a sailing boat used for wild-fowl shooting in large rivers and shallow bays, where the shooting-yacht is precluded from proceeding because of drawing too much water. Thus

⁵ See further and more fully in "The Wild-Fowler" as to sailing punts, and mode of using them.

the shooting-yacht and stanchion-gun are used for sea-going purposes, and the open sailing boat, with gun of equal proportions, for inland waters and shallows.

A boat for this purpose should be about twenty feet in length by seven feet beam; a shallow craft, with powerful bearings.

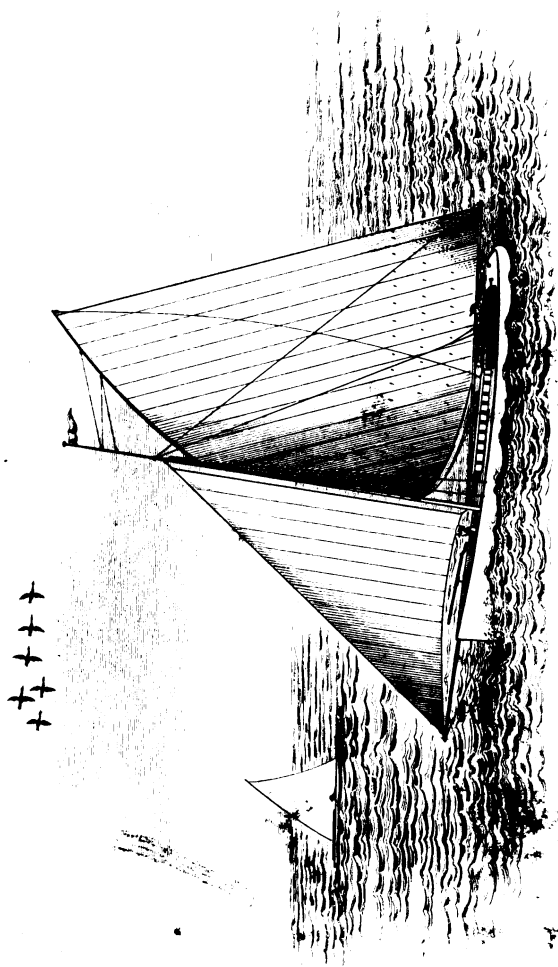
Stability is a great desideratum in a boat required for this diversion. It is not desirable that the boat should list on her side too much when under sail, as it interferes with the management of the stanchion-gun. A narrow deck-way of ten or twelve inches may be formed on each side of the boat, which should have no bulwarks; but the deck-way should be upon a level with the gunwales. The fore and aft part of the boat may also be partly covered in by a flush-deck, but in other respects it should be entirely open.

A boat of this description will require several hundred weight of iron ballast, which must be deposited with careful discretion beneath the platform.

The stanchion-gun should be fitted with chock and necessary recoiling apparatus upon the flush-deck, at the bows; and must be placed so as to swing clear of the forestay, that a shot may be fired from either bow.

The best form of rig for this boat is the "sloop-rig," (see engraving,) by which the forestay, instead of being made fast to the stem, stands farther out to the extreme end of the standing bowsprit; thus giving more room for the sportsman to work the stanchion-gun clear of ropes, and enabling him to place it in a more advantageous position.

Probably no better form of boat could be invented for traversing the shallow waters of inland bays and rivers, the resort of wild-fowl, than the American centre-board. Such a craft would carry the stanchion-gun well. The keel might be lowered at pleasure, or when beating up the channel of a river or in deep water; whilst the shallow form of the boat would enable the sportsman to go over almost any ooze or sand-bank with facility, where there might be a depth of one or two feet of water. There are several shooting-yachts con-



SHOOTING BOAT.

structed upon this principle ; those in which I have sailed appear to answer remarkably well.

THE WILD-FOWL CANOE.

This is a small boat, about twelve feet in length by three and a-half in breadth, and about fifteen inches deep in the fore part, by ten inches in the aft. It is clench-built in a similar manner to a skiff, and with a keelson ; but as flat in the floor as it is possible to make it, because of the occasional necessity of going into shallow water. The canoe is intended to carry two persons and a dog ; it is used for the purpose of going up creeks and under the banks of oozes at low water ; also when the tide is sufficiently high to bring the top rim of the bows of the canoe upon a level with the surface of the ooze ; when the sportsman may sometimes make a very profitable shot.

The manner of using the wild-fowl canoe is as follows :— When the sportsman has discovered wild-fowl feeding on the savannas, he places himself on his knees in the fore-part of the canoe, rests the barrel of the gun on the bow of the boat, and in that position remains as motionless as possible, whilst his companion cautiously sculls the canoe, with one oar, towards the birds : for this purpose, a sculling rowlock is fitted to the centre of the stern piece, through which the oar is thrust ; and the bows of the boat being higher than the stern, the movements of both men are concealed from view, and effective shots are sometimes made, particularly on moonlight nights.

When not actually approaching wild-fowl, one person sits facing the prow, to look out for sport and give directions to the other, who rows the boat with a pair of sculls, up creeks and rills, or wherever there may be a prospect of sport.

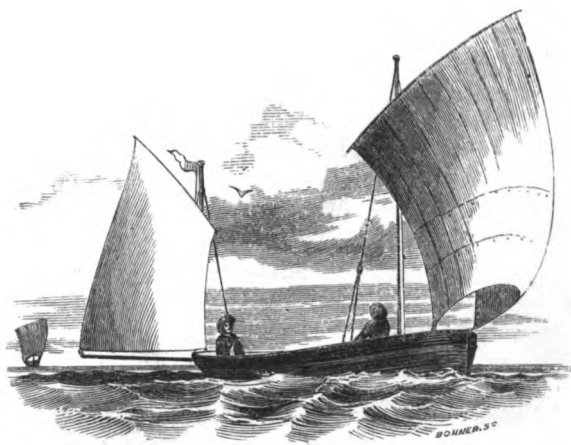
SHIPS' BOATS.

ALL large vessels are required by Act of Parliament, to carry a certain number of boats :—a first-rate man-of-war has six boats attached; but in merchantmen and other trading vessels, the number of boats is regulated according to tonnage. Ships' boats are classed under the heads—pinnaces, launches, gigs, barges, quarter-boats, jolly-boats, cutters, and long-boats.

The Pinnace, which is the largest of all, is generally from twenty to thirty feet in length, with proportionate breadth of beam : these are often fast-sailing, powerful boats, and are enabled to carry a great number of hands; there is no definite rig for the pinnace, two lug sails are those in most general use : the sliding-gunter is also a rig, frequently used for the pinnace (see the engraving at page 50). The masts of all ships' boats are made to *strike*, for convenience of stowage. The pinnace is rowed with ten or twelve oars, double-banked. Vessels going very long voyages carry decked pinnaces. When stowed away on deck, the pinnace is generally turned bottom upwards, and laid over the barge.

The Barge.—Ships-of-war carry a large boat of easy draught of water, called a barge ; it is generally very strongly built, and capable of accommodating a large company. The barge is occasionally sailed, but being broad and heavy is not usually so fast as other boats belonging to the ship.

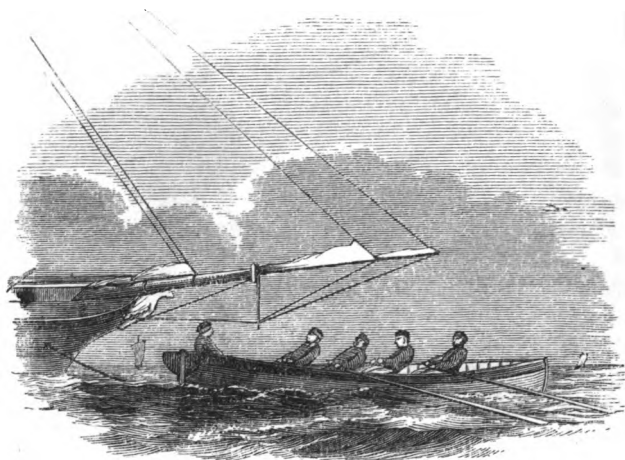
The Launch.—This is a strongly built boat, with very considerable breadth of beam, and very flat floor : it is used chiefly for carrying on board the ship's provisions, water, &c., for convenience of which, several of the thwarts amidships are fitted in such a manner as to be readily shifted, shipped, and unshipped : a lug-sail is used when required. The launch is occasionally half-decked, and equipped with a large gun at the bows, and other arms, when belonging to vessels-of-war.



Quarter Boat.—Large vessels generally carry two quarter boats, which are suspended one on each side, over the ship's quarters, by davits; they are used by the officers and crew, and are well-built and useful boats; they are sometimes called first and second cutters. A common rig for a quarter boat is, two lug-sails—main-lug, and mizzen-lug, as shewn in the engraving.

The Jolly Boat is strong, short, and clumsy-looking; used by the crew for running out the kedge, working the vessel in narrow waters, taking the sailors ashore, and other rough purposes: it is pulled by four oars.

The Long Boat is generally hauled up on deck, and laid bottom upwards, for use in cases of emergency, danger, or shipwreck; it is long and capacious, and belongs chiefly to trading vessels not requiring so large a boat as the pinnace.



GIGS AND OTHER SMALL BOATS.

Gig.—A boat of superior form, in shape resembling a well-built skiff, but longer and larger; an exceedingly smart-looking craft, varying in length from sixteen to twenty feet and upwards; and from three to six feet in breadth: it has a sharp bow and graceful proportions. Every large vessel carries (amongst others) a boat of this description, called the captain's gig (*see Engraving*), which is sometimes very tastefully decorated with brass mountings, gildings, mahogany back-board and fittings. The captain of a large vessel is generally very particular as to his gig, which is always the fastest and handsomest boat belonging to the ship; the sails are kept in good order and of snowy whiteness; the captain selects the best rowers from the ship's crew, to take him ashore in his gig.

The Gentleman's Pleasure Gig is a boat of the same form ;



but often much larger than those belonging to ships: it is rowed double-banked, that is, with two rowers on each thwart; sometimes eight or ten oars are used at a time: when sailed, the gig is rigged in various ways, according to the fancy of the owner. These boats may be classed among the handsomest of our rowing boats.

Beach Gigs.—These are employed at most of the favourite watering places on the English coast; they are strong, and well built, but of narrower form than those previously described. Beach gigs are sometimes thirty feet in length, and only three in breadth. No other than a small lug-sail is used in narrow-shaped gigs, and that but seldom; they are generally rowed.

The sea-going gigs used in the coast-guard service, are sometimes called galleys; they are superior boats, strongly built, and from twenty to thirty feet in length, and four or five in width; but some are much narrower. When sailed, the general rig is one or two lug-sails: in reaching and running with the wind free, few open boats can beat them, the coasting salvors or Yarmouth yawls excepted.

The Cutter (rowing boat).—The term cutter, in its general application, signifies a decked vessel of superior speed: there is, however, a small kind of boat used by the officers of large ships, called a cutter, in form similar to a gig, but with a sharper bow, whilst the beam or breadth is in proportion to that of a skiff. The cutter is a light little boat, and very fast, both for rowing and sailing. It is sometimes sailed with a sprit-sail and fore-sail, but more generally with a lug. Fast rowing gigs, with very sharp bows, are also sometimes called cutters.

The Skiff is a small light boat used on every lake and river in England: the skiff is so well known as to need but few remarks in explanation,—it varies in size from ten to sixteen feet; all boats of similar shape exceeding that length are termed gigs, wherries, ships' boats, &c.

A skiff, sixteen feet in length, would be about five feet breadth of beam: no boat is more generally in use than the

skiff, both for sailing and rowing, for pleasure and profit : a boat of general utility and unrivalled convenience, justly termed the "sailor's cradle."

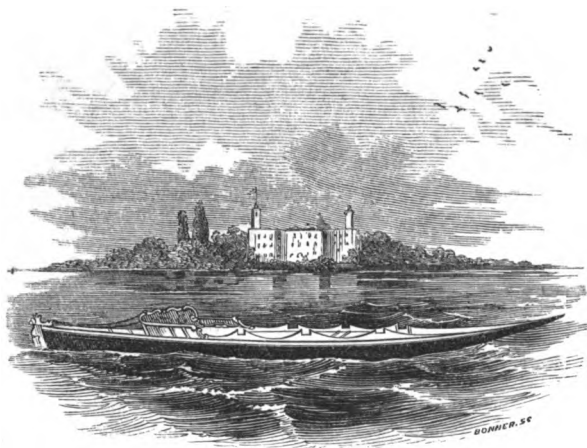
The "*sailing skiff*" is described at page 23, with the sprit-sail rig.

The Dingy.—A short, light, little skiff, very much the shape of a walnut-shell ; it seldom exceeds ten feet in length, is an accompaniment to small yachts and fishing vessels, to which it usually serves every purpose of long boat, jolly boat, and gig.

Galleys.—A galley signifies a boat of a long and narrow form, propelled by several oars. The wager-galley is exclusively a rowing boat, long and slender in shape, lightly built with a low hull and very sharp bow, rowed by four, six, or eight oars, according to length.

The Oxford and Cambridge eight-oared galleys, in which their distinguished annual rowing matches take place, are probably the lightest and most scientifically constructed rowing boats in the world. They are made from a solid tree of mahogany, or the best white pine, hollowed and formed with great skill, so that the sides are scarcely thicker than the edge of a half-crown piece, being afterwards strengthened internally with transverse ribs of tough wood. Light iron out-riggers are fitted to the sides, so as to give greater leverage for the oars. An eight-oared out-rigger of this kind is from sixty to seventy feet in length, and from two, to two feet four inches in breadth, at the widest part.

Thames Wherries.—Of all wherries in the world, none are so beautifully modelled or so light and graceful as the "trim-built wherries" of the Thames. They are constructed upon the most scientific and approved lines ; the numerous boat-builders daily endeavouring to outvie each other. These boats are truly a pattern for all nations, and are the especial admiration of foreigners, particularly the smaller craft, adapted for a party of five or six. They have a beautiful sharp bow, with graceful proportions, carrying the beam well aft, or gradually flaring from the bows, and finishing with a neat taper-



THAMES WHERRY,

(Presented to H.R.H. the PRINCE OF WALES.)

ing stern : a chair rail, for a support to the back, is prettily arranged round the stern-sheets, where the company sit.

Before the introduction of steam-boats on the Thames, wherries were the chief means of conveyance for passengers; but those were of larger size, and some were sailing vessels, and very inferior in shape to the smaller boats of modern days.

There are also numerous other rowing boats, of various light forms in daily use on the Thames, and at other places: among them may be mentioned Funnies, Randans, Gondolletes, &c., &c., many of which are of light and beautiful form and construction, whilst others are mere nutshells, and unworthy the name of boats.



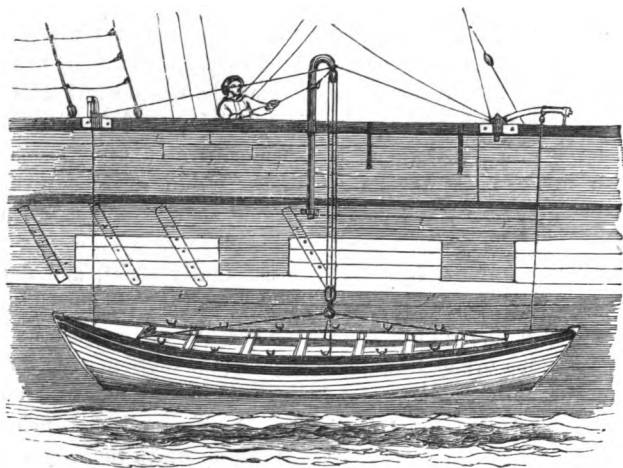
THE PETER-BOAT.

Hark! I hear a gentle splash, there's life upon the stream—
 Yes! yes! a whisper too methinks, or is't Dame Fancy's dream?
 Then softly drifting down the tide, a boat appears in sight;
 "Good luck t'ye wily fishermen, this is a 'catching' night!"
 The Peter-boat moves slowly on, the mullet net is spread;
 No coarse or bungling snare is that, but soft and fine as thread.
 The fisherman may crafty be, his scaly prize to get,
 But craftier still the mullet is, that gaily leaps the net.

THE Peter-boat is of very ancient origin; indeed, it is precisely the Roman *Amphiprora*. Peter-boats are much in favour with the fishermen of the English rivers. There are a few peculiarities in their construction, worthy of notice;—in the first place, they have neither gunwale nor top-rim, therefore present a very unfinished appearance; the top strake runs in a horizontal line from stem to stern, leaving no graceful fall amidships, nor any other improved lines to strike an admiring eye, but is simply what a fishing boat ought to be, *safe and serviceable*. Stem and stern are alike, after the form of a whale boat, but not so sharp, and not curved or elevated: the object of both ends being thus formed, is, that in working

the peter-net, the boat is as frequently propelled backwards as forwards, particularly whilst hauling. Having no keel, the boat can be turned quickly and in narrow compass. A well for live fish is built into the boat amidships, and divides it into two compartments, the fore-part being where the rower sits, the other for the fisherman and his nets: the well is generally about two feet wide at the base, and gradually tapering to one foot at the top, extending the whole width of the boat: the depth of the well is just sufficient to bring it upon a level with the thwarts; and, being covered over, forms a seat in the boat; the bottom of the well is pierced with a number of holes, to admit a constant ingress and egress of water: fish may be kept alive many days, or even weeks, in these wells, if the boat remains constantly afloat, and deep enough in the water. The stern sheets of the peter-boat are laid higher than in other boats, on account of the quantity of water which drains from the nets; but the division caused by the well, prevents the water from getting into the fore-part of the boat, where the rower sits. A small raised bench is placed in the very corner of the stern, on which the fisherman stands erect, when shooting the net: this elevated position enables him to lift the cork line high, that the net may run out freely.

The Doble (or Doval) is flat-bottomed, has stem and stern similar to a skiff; and when afloat, has more the appearance of the latter than a flat-bottomed boat. The easy draught of water renders this kind of boat serviceable for many purposes, for which other boats would be nearly useless; it is especially convenient for approaching shallow shores, and in tidal rivers where there is an extent of flat soil or ooze; and for river-fishing the doble is peculiarly adapted, and more used in such parts than any other boat; it is also very convenient for the navigation of narrow creeks, and for approaching the brink of rivers, where other boats of deeper draught of water could not go. When properly built they are pretty looking craft, and very safe whilst used for rowing purposes, but totally unfit for sailing.



THE WHALE BOAT

(With Clifford's Boat-lowering apparatus).

THE South Sea whale boat is strong, but of light and beautiful construction ; the head and stern are alike, both being of sharp wedge-like form, and slightly elevated by a graceful curve : it has considerable breadth of beam, and what is termed a very flat floor, with round sides, which render it safe and stiff on the water. These boats vary from twenty-six to twenty-eight feet in length, and are about five feet three inches breadth of beam : some are four-oared, others six-oared, and they are generally carvel built ; the stem and stern being, as before stated, a little elevated, the boat has the appearance of being low amidships ; but is so buoyantly built, that with the usual

crew of six, the draught of water is but a few inches. The fore part of the boat is covered with strong boarding; a deep circular cut is formed in the inner edge of the boarding, which serves as a rest and support to the headsman, when wielding the deadly harpoon: the stern end is also covered in a similar manner, but instead of an opening being cut in the boarding, two very strong wooden chocks or logger heads are firmly attached, which rise about six or eight inches above it; these are for checking the whale line when the fish tires.

No rudder is used for the whale boat, but a long oar with a wide blade, which works on a swivel row-lock, at the stern. The long oar gives greater leverage than the rudder, and the helmsman is thereby enabled to turn the boat much quicker, and with greater precision than by the aid of a rudder; an object of considerable importance in a whale boat. Generally, in the centre of the boat is a tub, containing the whale line, coiled in the most judicious and careful manner: this line, although small in size, is very strong, and of several hundred yards in length; the leading end is attached to the lance or harpoon, and all is laid clear, the line leading straight along up the middle of the boat, and through a hawse hole in the stem: the slightest hitch or kink in the line might be attended with the most disastrous consequences, and would probably upset the boat, when rapidly drawn out by the wounded whale. An idea may be formed of the velocity with which the line runs out, from the fact that, if it chafes against any wooden substance, the friction is so great as to envelope the harpooner in smoke; the wood snaps and cracks as if on a blazing fire, and unless preventive measures were used, would probably ignite; in such a case water is sprinkled upon the wood. The harpoon has always, until very recently, been thrown from the hand, but an instrument has since been brought forward, and is now in general use, by which it is fired from a harpoon gun; which is an useful invention, inasmuch as it may be pitched with great precision from one to two hundred yards, and the risk incurred on approaching too near the whale, is avoided.

Nicely arranged along the sides of the boat, under the gunwale, are various lances and other weapons necessary for the effectual capture of the whale; all of which being bright steel, and having sharp points or edges, are carefully encased in leather sheaths. The oars are muffled with some soft kind of matting or tow, to avoid noise; and if working in swivel rowlocks, are secured with a lanyard, being frequently required to be dropped alongside the boat, when they would be lost, unless secured. The instant the whale line slacks after being run out by a fish, the slack is hauled in and carefully coiled in a tub, as before.

LIFE-BOATS.

To give any thing like a history or description of the various forms of life-boats, that have been invented in this country, of late years, would far exceed the limits of our space; indeed, such a description would fill a volume of itself. It is no exaggeration to say the variety comprises *hundreds*; some of which are very valuable and useful inventions, whilst others are as erroneous in principle as if intended to promote disaster, rather than to relieve it. Still it is only due to those who have spent their time, their ingenuity, and their money about the invention of safe and serviceable life-boats, to say, that the art is one worthy of the highest encouragement, and the kindest and most humane consideration.

The National Life-Boat Society, is one which deservedly occupies a prominent position among the charitable institutions of this country; the services rendered by that society are highly meritorious, and worthy of the continued and increasing support which has been afforded it.

Through the medium of that institution (supported as it is entirely by donations and voluntary contributions), life-

boats are now stationed at most of the exposed and perilous parts of our coast; and splendid are the services, and brave and noble the deeds that have been performed in some of these boats, by the gallant fellows who man them.

Before the establishment of life-boats around the coast, an average of 700 persons perished annually on the shores of the United Kingdom, chiefly men whose lives were valuable to their country.

Although we have now many apparently safe, and entirely trustworthy inventions of life-boats, in which the crews who man them place implicit confidence, still it appears we have not yet reached the perfect principles of life-boat architecture.

Many years ago, when life-boats^{*} were first thought of, several attempts were made before one was constructed that could be relied on: the first invention worthy of notice, was by a Mr. Lukin, in the year 1785. But after the disastrous gales on the north coast of England in the autumn of the year 1789, when a great many lives were lost, because of there being no life-boats to put off to the rescue, a prize was offered for the best invention of a life-boat, capable of being launched from the shore in safety, to the aid of ships in distress: there were several candidates for the prize, which was gained by a Mr. Greathead; and in 1790 several boats were constructed upon his plan, which was much approved: numbers of lives were saved, and noble actions performed in perilous gales, by Mr. Greathead's boats.

In 1802, the Society of Arts wishing to make an acknowledgment of their approval of the invention, awarded to Mr. Greathead, fifty guineas and their gold medal; a grant of £1200 was awarded him by Government, and a similar amount was also presented to him by the Trinity-House; 300 lives having been saved off the coast of Tynemouth alone, solely by boats constructed on his plan.

Mr. Greathead's life-boat may therefore be termed the *original life-boat*; recent inventions are merely improvements upon it: it is made about thirty feet in length, ten in breadth,

and three feet three inches in depth amidships; both extremities being of precisely the same form, so that it goes through the water with either end foremost; and its shape, lengthways, is a curve, so formed that a line drawn from the top of one stem to that of the other, would be two feet and a half above the gunwale at midships. In this boat there are five thwarts, or seats for rowers, double banked, so that it must be manned with ten oars. It is cased and lined with cork, which gives it such buoyancy that it will float and be serviceable, though so damaged by hard knocks as to be almost in pieces; and this, the softness and elasticity of the cork is well calculated to prevent. The cork on the outside is four inches thick; and it reaches the whole length of the sheer or side of the boat: on the inside it is thicker, and the whole quantity is about seven cwt. It is firmly secured with slips or plates of copper, and fastened with copper nails. The advantages of this boat are stated to be, that its curvature gives great facility in turning; a single stroke of the steering oars (of which there is one at each end), moving it as though on a centre; that the covering of cork being immediately under the gunwale, gives great liveliness, or disposition to recover its balance after being suddenly canted aside by a heavy wave; and that its capability of going with either end foremost increases its manageability.

These life-boats continued in use, with a few slight improvements, until December 1849; when twenty of the most skilful pilots of the Tyne, were drowned off the coast of Northumberland, by the upsetting of a life-boat built from the invention; in this case, the boat having no buoyancy, either at the ends or gunwales, it became immersed, and sunk so deep in the water, when bottom upwards, as to leave no space for the crew to breathe; they were, therefore, drowned inside the boat, which must have been submerged until the water-tight deck at the bottom floated her.

In consequence of so lamentable an accident, and in the absence of efficient life-boats along the greater portion of the western coast of England, and the want of due reward and

encouragement to men going off to wrecks, who in many instances have received nothing; or, as in the case at Holy Island, in April 1850, when they received *one shilling and eightpence* each, for risking their own lives to save those of others; it was determined to make an effort to remedy some at least, of these evils; and the first and most obvious step was, the endeavour to introduce as improved and perfect a life-boat as possible. In furtherance of this object, His Grace the Duke of Northumberland originated a scheme, in order to draw the attention of boat builders to the subject, in all parts of the country; and, with a view of enlisting the best practical skill and talent, through the medium of public competition, his Grace offered a reward of one hundred guineas for the best model of a life-boat, which, in the judgment of a committee appointed for the purpose, should combine the most essential qualities for the preservation of life from shipwreck.

Accordingly, in October 1850, a notice to that effect was issued to the boat builders of this country, which pointed out the chief defects in existing boats, namely; that they had not the actual power of self-righting when capsized; that they did not free themselves quickly of water; and that they were too heavy for transporting along the shore: in other respects the form, construction, and fittings of the boat, were left entirely to the skill and judgment of the builder.

At the same time, a carefully selected committee was appointed to examine and decide on the models submitted.

The result was, that two hundred and eighty models were exhibited at Somerset-house; with plans, sections, and explanations of their peculiar merits. With these before them, the committee saw the difficulty that existed of ascertaining among so many, the relative qualities of each; and various examinations and experiments were made to bring out the peculiar characteristics of each boat; certain numerals were therefore given to each quality, in the order of importance in which it was regarded by the committee; and which are given in the subjoined standard:—

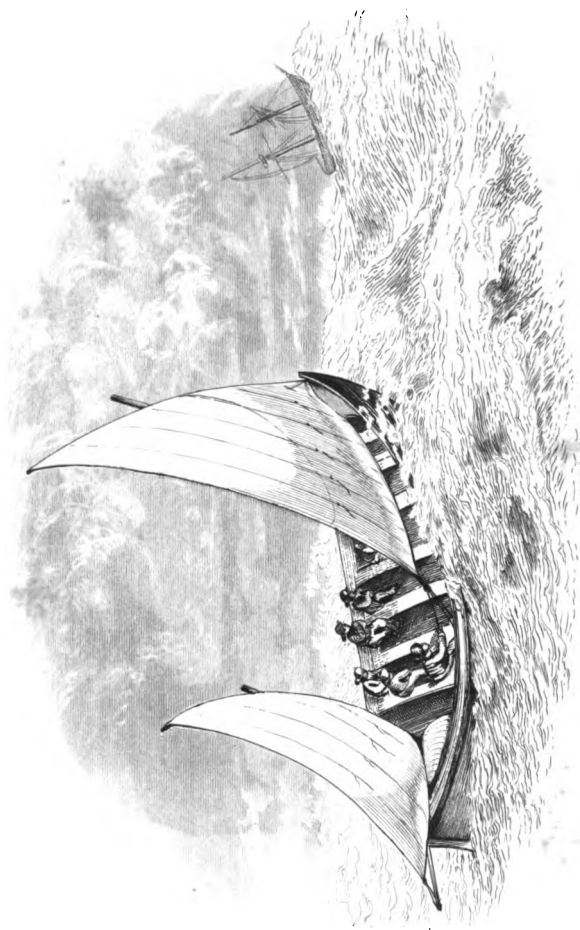
	Nos.
1 Qualities as a Rowing Boat in all weathers.....	20
2 " Sailing Boat.....	18
8 " As a Sea Boat: <i>i. e.</i> , stability, safety, buoyancy forward for launching through a surf, &c.....	10
4 Small internal capacity for water up to level of the thwarts	9
5 Means of freeing Boat from water readily	8
6 Extra buoyancy; its nature, amount, distribution, and mode of application	7
7 Power of self righting	9
8 Suitableness for beaching	4
9 Room for, and power of carrying passengers	3
10 Moderate weight for transport along shore	3
11 Protection from injury to the bottom	3
12 Ballast, as Iron 1, Water 2, Cork 3	3
13 Access to stem or stern	3
14 Timber heads for securing warps to	2
15 Fenders, Life lines, &c.	1

Total 103

It will be seen by the above formula, that the committee considered it an essential requisite in a life-boat, that it should be a good rowing boat; able to be put off the beach in any weather in which a boat can live at sea; as without the power of doing this, other good qualities are of no avail: to this then, was awarded the first number. As on the coasts of Norfolk and Suffolk, where wrecks generally occur on out-lying sands, all the life boats get off *under sail*: and it being evident some of the best models were prepared with this view, it was considered that these also were entitled to be placed on a par with boats built chiefly for rowing; but as rowing is a general practice around the coasts, and sailing an exception; a slight difference was made in favour of the former.

This formula arranged, all difficulty vanished. Each model was again examined; and the number, or proportion of the whole number, according to the merits of each boat, established conformably with the above standard.

It was then decided that the model by Mr. James Beeching, of Great Yarmouth, possessed the greatest proportion of the above requisites; the number being eighty-four of the hun-



NORTHUMBERLAND PRIZE LIFE BOAT.

dred. Six of the best models to which the highest numbers were affixed, were then for a third time examined, placed side by side, their points discussed, their qualities canvassed, and carefully compared one with another; the result was, a confirmation of the decision previously arrived at by the committee; who, therefore, made the award in favour of Mr. Beeching, and he was accordingly declared the successful competitor, and received the Duke's prize.

BEECHING'S LIFE-BOAT.

(See Engraving.)

The following is a description of Beeching's prize life-boat. The body of this boat is of the form usually given to a whale-boat; a slightly rounded floor, sides round in the fore and aft direction; a slight rake in stem and stern-post; clench built, of oak, and copper fastened.

Length 36 feet: breadth $9\frac{1}{2}$ feet; depth $3\frac{1}{2}$ feet; straight keel $3\frac{1}{2}$ inches deep; pulls twelve oars double-banked: has a cork fender round the outside, below the gunwale. Extra buoyancy is given by air-tight compartments under a deck, and a set of detached air-cases under the thwarts on each side, a raised air tank at each end, and cork fenders outside. Effective extra buoyancy 330 feet cubic, equal to $9\frac{1}{4}$ tons. For ballast, a water tank divided into compartments, built into the boat amidships, perfectly secure and water-tight. Means of freeing the boat from water in *twelve seconds*, tubes through the bottom. Rig, fore and mizzen lug-sails: to be steered by a rudder or oar, as required: would carry seventy persons.

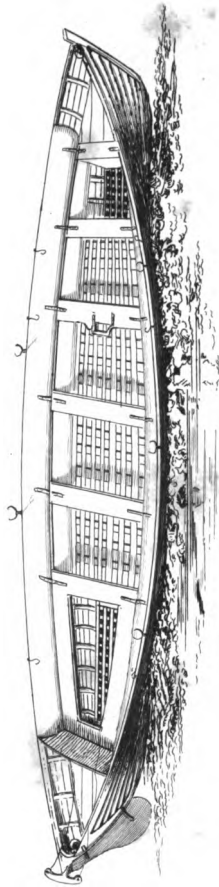
The very model that was exhibited for competition, has since been deposited in the United Service Museum.

Beeching's life-boats, then, came into general favour; seven were ordered to be built for the Royal Shipwrecked Fishermen and Mariners' Benevolent Society, besides others for various stations on the coast. Numerous trials were made with some of the boats, and highly satisfactory were the

opinions expressed as to their safe and efficient qualities : in many instances the beachmen declared their confidence to be so great in the powers and qualifications of Beeching's life-boats, that they would not hesitate to venture out in them during the hardest gale, or through the heaviest sea.

During the fearful gales of November 1852, these boats were put to several severe tests ; but strange to say, they did not then fully realize the expectations of the public. They had not long been in use, before two were upset during the same gale ; one at Lytham, and another at Carnarvon : the accident at Lytham proved fatal to all the crew, save one. Although it was stated that those accidents were mainly attributable to ill judgment, in carrying too great a pressure of sail, it appeared very remarkable after such extraordinary tests as the boats had gone through, and the flattering opinions that had been expressed by so many experienced and scientific men, that two of the first and best of these boats should be upset in such a manner as to be unable to right themselves. About two months after, a third was upset in a similar manner, at Rhyl, the boat evidently lacking sufficient power of self-righting ; by which disaster, six unfortunate but brave young fellows, met a watery grave. The subject was then taken into serious consideration, and an opinion demanded from Captain Washington, who was requested to inquire strictly into the cause of the accident, and report upon the safety of the boat ; the result of that inquiry tended to show the inefficiency of the boat, for some stations ; but it is much to be regretted that any deviation should have been made from the original model, which, in the opinion of the Northumberland Committee, possessed those essential powers of self-righting, when upset, and which had not been carefully carried out in the construction of the Rhyl boat.

These three successive accidents were alone sufficient to weaken the confidence of the beachmen in Beeching's life-boats, until some important alterations were made in their construction ; in order to give them those powers of self-



YACHT'S LIFE-BOAT.

righting which the committee had anticipated on awarding the Northumberland prize to Mr. Beeching.

The use of a life-boat is next to nothing, unless it can command the confidence of its crew; when once that confidence is won, a life-boat is doubly serviceable.

It should not be overlooked, that a life-boat which possesses the power of righting itself when upset, will be easily upset; that is to say, if it will turn over one way, it will the other. The Shields beachmen say, "We don't want one that rights herself when she capsizes; but we want one that will not capsize at all." And so it is, that a life-boat to command the confidence of its crew, should be so constructed that it be next to impossible to capsize it.

LAMB AND WHITE'S LIFE-BOATS.

The life-boats under the name of "Lamb and White's Patent," built by Messrs. T. and J. White, the eminent ship-builders at West Cowes, have now stood the test of many years service; they are well adapted for ships, and steam-boats, and are also admirably suited for yachts.

These boats are extensively used by the Peninsular and Oriental Steam Navigation Company; and also by the Royal Mail West India Steam Company, all their vessels being supplied with them. They have also been adopted in vessels sent on the Arctic expeditions; and by many public companies, merchants, and shipowners.

The capabilities of "Lamb and White's life-boats," have been frequently and severely tested, and in every instance proved deservedly meritorious; they are built of two thicknesses of plank, with prepared waterproof material of an adhesive nature interposed. The whole of the internal work, comprising the water-tight compartments, foremost and after bulk-heads and decks, are also in two thicknesses, with waterproof material interposed. The several compartments on each side, and forward and aft, are all built in the boat during its construction, forming double sides, and giving great

stiffness and strength ; which renders these boats particularly applicable for hoisting up and hanging in davits. They may be sailed in any weather, with or without ballast ; and will not capsize, owing to the centre of buoyancy when the lee side is immersed, being so near the gunwale ; and if water is shipped it acts as ballast, rendering them still more stable, from its being confined to the centre of the boat by the compartments on either side.

The primary object kept in view by this invention, has been to produce a boat which it is almost impossible to capsize ; because, should such an accident occur to *any* life-boat, the crew *must* (if in a heavy sea), be in imminent peril, whether the boat rights itself or not ; and whether or not the crew can swim.

If losing sight of this fact, the boat be so constructed as readily to *roll over*, and right itself again, she is liable to be rolled over and over again by the following seas, with greater facility when filled with water, because the buoyancy in the bottom then becomes a lifting power ; the danger of this need not be dwelt upon. On the other hand, these boats which possess such stability (without ballast), that it is almost impossible for them to be capsized, also possess this additional important qualification ; that supposing they are ever turned over, the crew can remain *under them in perfect safety*,⁶ as has been proved by experiment. This is explained by the buoyancy of the two ends ; which, in consequence of the sheer, will float the boat, so that the gunwale amidships will only be immersed three inches ; thus leaving the whole interior of the boat as free of water as a diving-bell ; so that the crew may remain in safty, fresh air being constantly admitted by each wave lifting the boat, as well as by valves in the bottom, which open inwards. Some of these boats are also fitted with a scuttle in the bottom, large enough to allow

⁶ This is no enviable situation to be placed in, but had the South Shields life-boat possessed such a qualification, the twenty pilots drowned in December, 1849, might probably have survived.

the crew to get through, if required. But the builders do not pretend to construct a boat which shall right itself at the sacrifice of the *main* qualification; the *secondary* one being only to be obtained by so altering the relative centres of gravity and buoyancy, as to make the boats dangerous, and liable to the very accident they are intended to prevent.

RICHARDSON'S TUBULAR LIFE-BOAT.

Of all the life-boats hitherto invented, probably none are safer or more serviceable than these: they have undergone, in the roughest weather, some of the severest tests that could possibly be applied to any life-boat; and in every instance were proved to possess superior and extraordinary capabilities. Some of the boats have been selected for service at the life-boat stations on the coast.

The tubular boat consists of two circular tubes two feet six inches in diameter, and forty feet long, gradually tapering at the ends, which are bent upwards and inwards, and so meeting each other, and giving the appearance of sheer. These tubes are composed of charcoal iron, one-sixteenth of an inch thick, strongly tinned and painted; and they each consist of twelve compartments occupied by air-proof bags, the divisions being of the same substance as the exterior. The plates are closely rivetted, other means being also taken to make them water and air tight; and throughout, each compartment is strengthened by an iron hoop and crossbars. The tubes are placed side by side, at a distance of three feet from each other, by which the curved ends are brought nearly in contact; and they are secured by strong, but light iron arches, one for each compartment, with stays and ties. Over the arched iron is placed a grating, extending nearly the whole length of the boat; and upon this grating rests a frame-work, forming seats for sixteen rowers (double banked), and supporting iron rowlocks, the boat being slightly outriggered with regard to the wooden frame, which rests above the centre of the tubes. It has been ascertained that the two tubes require a weight

of eight-and-three-quarters tons to submerge them, and bring the upper works to the water's edge: but even then the boat has proved perfectly manageable. There is sufficient space for stowing eighty persons in the boat, in addition to the crew; and this number would be less than the weight above mentioned. In the end compartments of each tube is fitted an air bag, of strong water-tight material, capable of supporting fifteen persons; which greatly adds to the buoyancy of the boat: any contrivance for enabling the boat to free itself from water is unnecessary, as none can by possibility be retained; and the danger of upsetting is totally out of the question; because, when the boat is lifted by the sea, so as to throw one tube out of the water, the wave would immediately have free course between the tubes; and, by its own action upon the second tube, compel a righting of the boat. As no flat surface is presented to the waves, the tubes can only be broken by contact with rocks, or some hard substance; and the filling of one or two compartments from an accident of this kind, would not materially impair the efficiency of the craft. Along the outside of each tube there is a substantial circular cork fender, which serves as a hold for any person in the water.

Finally, the tubular life-boat can neither be upset, swamped, nor water-logged, by any contingencies of winds or waves in the open seas; the whole attention of the crew can therefore be directed to, and fixed on the all important object—that of approaching the wreck speedily and carefully, and saving the lives of the helpless sufferers.

Among the tests to which the tubular boat has been submitted, were those of landing her passengers and crew, in a gale, on a lee shore, and going off again; towing, in stormy weather, behind a steamer, without any one on board, the commander of the steamer having *carte blanche* to upset, swamp, or tear her to pieces if he could; in all of which she proved perfectly successful.

Hitherto it has been found impossible to capsize the tubular boat, because the force of the sea passes off and through

the two tubes ; but if she were to be upset by any cause, her buoyancy would be greater than before, and she would become a safe and manageable raft, even bottom upwards.

In addition to the tests above stated, the inventor, and a small crew, performed a voyage in one of the tubular boats, from Liverpool to London (round the Land's End), during winter, encountering very severe weather on their passage.⁷

⁷ An amusing and spirited narrative of this voyage, and the singular adventures met with, are recorded in a small book published by W. Pickering, London, entitled "The Cruise of the Challenger Life-boat."

MANAGEMENT OF SAILING BOATS.

RUDIMENTS.

“D’ye mind me, a sailor should be every inch,
All as one as a piece of the ship,
And with her brave the world, without offering to flinch,
From the moment the anchor’s a-trip.”

DIBDIN.

No man can be called an able seaman, unless he knows how to haul down a reef-earring in a gale of wind, and lay out on a yard-arm in a tumbling sea; he should also thoroughly understand the use of the marlinspike, and be capable of making both long and short splices in a rope; should know how to turn in a dead eye, strop a block, pass seizings, and make all the ordinary knots in a neat, workmanlike manner: and every boat-sailor ought to be equally experienced, before he is entrusted with the control and management of a sailing-boat.

A rope is composed of three or more strands, each strand consists of a number of yarns twisted together; yarns in small ropes are sometimes called threads. Bolt rope is the strongest, and most tightly twisted, and is the best rope of all for working through blocks; it is also used for shrouds and stays: being heavier than other rope, it is the most expensive. Manilla rope is made of white flax, and is generally cheaper than hempen rope; it is clean looking, and much used in yachts and pleasure-boats, particularly for halliards; it requires to be well stretched when new, or it is very liable to kink.

Standing rigging are those ropes which are stationary, and seldom require alteration; as shrouds and stays.

Running rigging are ropes which reeve or work through blocks and sheave-holes, as halliards, braces, &c.

A very neat way of finishing off shrouds and stays, is by a simple process called "worming;" *i. e.* winding spun-yarn round the rope or stay, all along between the strands; by which means the divisions of the strands are hidden, and the rope has a smooth, neat looking appearance.

As a complete finish to a rope where a splice has been made, it should be done over with *service*; that is, winding spun-yarn or soft twine over the splice, with the aid of the *serving-board*. To perform such work neatly, two persons are required, one to put on the service with the serving-board, and another to pass the ball of spun-yarn or twine. Passing the ball is generally the office of the junior branches of the crew, aboard ships, who by this means learn the use of the serving-board. A larger tool than a serving-board is used aboard ship for large ropes, called the serving-mallet.

Splicing a rope is joining two ends together by weaving the strands one between the other, in a workmanlike manner; it cannot be performed neatly without considerable practice.

The main sheet.—This rope is so often mistaken for the main-sail, by inexperienced persons, that explanation cannot be too clearly given. The main sheet is *not a sail*, but a *rope*, by which the main-sail is hauled in, or eased off, at the aft part of the boat or vessel: it is that *rope* which is made fast at the outer end of the main-sail boom, and is the most important rope in the boat, when under way: it should always be kept clear from all entanglement, so as to be ready to be let go in an instant, and release the boat from the pressure of the main-sail, in case of a squall or other danger.

Fore-sheets are ropes of similar use as applied to the fore-sail; they are attached to the after clew of the fore-sail, and by the fore-sheets that sail is eased off or close hauled. When the fore-sail works on an iron hawse, and the fore-sheets are led aft to shift the sail from side to side, they are called bow-lines.

A sail extended by a yard and slung from the middle, is

called a square sail : a sail set upon a gaff or a stay, is called a fore-and-aft sail.

The upper part of every sail is the head ; the lower part, the foot and flap ; the fore-part, the luff and fore leech ; the aft part, the aft leech ; the top aft corner, the peak ; the lower fore corner, the tack ; the lower aft corner, the clew.

To luff is to bring the boat's head closer to the point from which the wind is blowing, which is done by putting the helm down or towards the lee side.

To gybe or wear (just the reverse of to luff), is to sway the sail over from one side of the boat to the other, when running free or before the wind ; it is performed by putting the helm up, or towards the windward side : and this is the most dangerous manœuvre of all, in boat-sailing, and requires caution in performance ; the sudden jerk of the whole weight of sail falling heavily from one side to the other, is likely to upset the boat, or carry away some part of the rigging, or the mast. The most judicious manner of gybing when under a heavy press of canvas in a strong wind, is by hauling in the main-sheet gradually ; then, when the sudden jerk comes, allow the boat to feel the main-sheet and draw out the slack. It is usual, when gybing in a very strong wind, to lower the peak, and sometimes to trice up the main-sail, which are good and wise precautions. If the sail is fitted with brails, as is generally the case in open boats, brail up the sail close to the mast, and then all may be gybed in safety.

By carrying a weather helm is implied, that in steering the boat on a straight course (close-hauled), the helm has to be kept a little towards the windward side, and on putting the helm down, the boat should instantly obey her helm, and shake the sail in the eye of the wind : if a boat refuses this in a breeze, she cannot be said to carry a weather helm ; and, as a moral certainty, it is because too much head-sail is set ; and until such is reduced, the boat will not be under safe command.

The *weather side* is that side of the sail against which the wind blows.

The *lee side* is the opposite side to the weather side.

The *port side* is the left hand side of the boat or vessel, looking forward from the stern.

The *starboard side* is the right hand side of the boat or vessel, looking forward from the stern.

The *port tack*.—A boat or vessel is on the port tack, when sailing with the wind blowing on the port side.

The *starboard tack*.—A boat or vessel is on the starboard tack, when sailing with the wind blowing on the starboard side.

To *port* the helm, to put it over to the port side.

To *starboard* the helm, to put it over to the starboard side.

To *tack*, to *put about*, or to *stay*, signifies turning the boat (head to wind), so that the wind blows on the other side of the sails. It is a term applied when beating to windward.

To sail *close hauled*, *on a wind*, or *by the wind*. These are terms which signify sailing as close to the wind as possible.⁴

Sailing large, *going free*, or *off the wind*.—These imply sailing with the wind on the beam or the quarter.

To *bear up*, or to *keep her away*, is to alter the boat's course by turning her head a little from the wind.

To *weather* any vessel, point, or object, is to sail to windward of it.

Wind a-beam.—The wind is a-beam when blowing full on one side of the boat, or at right angles with the keel.

Sailing, *beating*, or *working to windward*, is to proceed as close to the wind as possible; and, by tacking from side to side, to work your passage ahead, though the wind is against you.

Lying to.—Keeping the head of the vessel to the wind, with very little sail; and so arresting progress.

Hove to.—Keeping the vessel as nearly stationary as you can, by hauling one of the sails aweather, so that it acts against another.

To *box off*, is to haul a head sail aweather, so as to "pay the boat's head off," or turn her head; if through bad steering, she has come too near the wind.

The *buoy watches*, when it may be seen floating on the surface. It *does not watch*, when by the force of the current, or otherwise, it is held under water.

To *set up the rigging*, is to haul it taut by means of the lanyards and dead eyes.

To *weigh the anchor*.—To pull up, or lift it from the ground or bottom.

Under way.—A boat or vessel is under way, when moving ahead with the sails or any one of them set.

In reefing, tie the points without rolling the sail.

Never allow any person to stand on the thwarts or sit on the gunwale, when the boat is under way.

It is a general rule to sling the yard of a standing lug-sail, at 1-4th from the foremost yard-arm; and a dipping lug at 1-3rd.

In small boats, the sails should be managed without any person moving from his place.

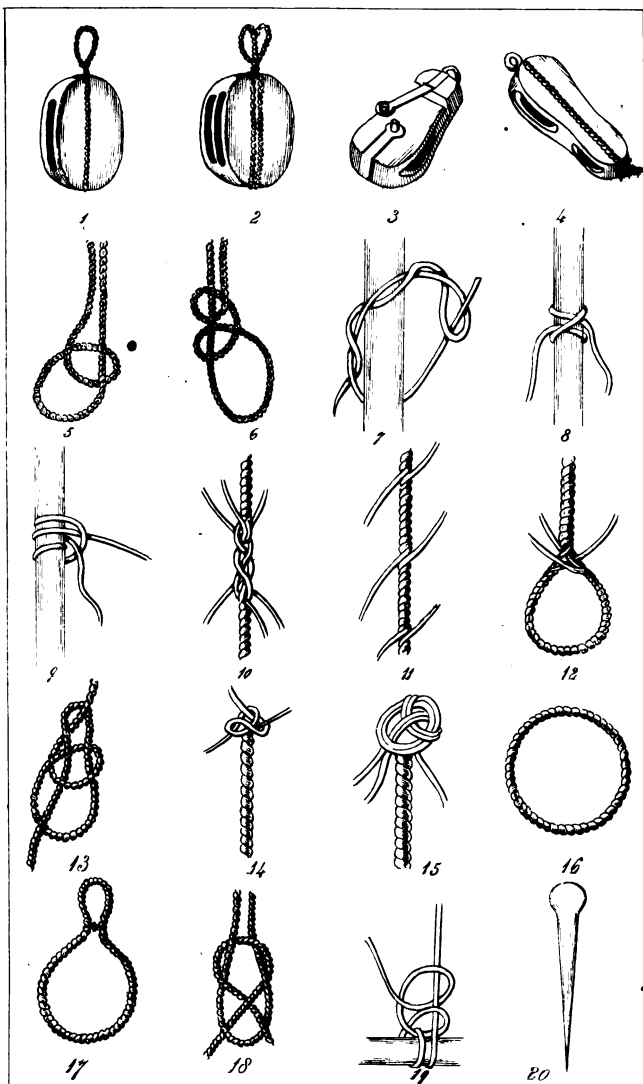
In lowering or taking in a lug or lateen sail, let go the halliards and haul down on the fore leech or luff of the sail.

One man should always stay by the helm until the sails are down. Accidents have frequently occurred through the helmsman leaving his place before the sails are lowered.

BOATS' BLOCKS, KNOTS, SPLICES, &c.

(See Engraving on opposite page.)

- | | |
|----------------------|-------------------------|
| 1.—A single block. | 11.—A long splice. |
| 2.—A double block. | 12.—An eye splice. |
| 3.—A snatch block. | 13.—A bowline knot. |
| 4.—A sister block. | 14.—A single wall knot. |
| 5.—A single hitch. | 15.—A double wall knot. |
| 6.—Two half hitches. | 16.—A grommet. |
| 7.—A timber hitch. | 17.—A snorter. |
| 8.—A clove hitch. | 18.—A sheet bend. |
| 9.—A rolling hitch. | 19.—A fisherman's bend. |
| 10.—A short splice. | 20.—A marlinspike. |



BLOCKS, TACKLES, PURCHASES, &c.

A boat's block is composed of three parts, the shell, the sheave, and the pin. The shell is the outer part or case containing the sheave; the sheave is the circular roller on which the rope runs; and the pin is the bolt which holds the sheave inside the shell. The best sheaves are made of brass; and such are the most durable; hollow brass sheaves being so much lighter than solid ones, are to be preferred for boats. Blocks are of various denominations, according to the purpose for which required; they are very important essentials in all sailing vessels.

For sailing-boats, blocks should be no larger than consistent with strength and convenience; when too large, they look heavy and clumsy, and incumber the boat unnecessarily.

Blocks are single, double, treble, or fourfold,—according to the number of sheaves.

A *running block* is one attached to the spar or other object to be raised or lowered.

A *standing block* is affixed to some permanent support.

A *snatch block* is a block with one sheave, having an iron hook at one end, and a clasp at the side to admit the bight of a rope, without the delay of reeving or unreeving the whole.

A *tail block* has a single sheave, is strapped with an eye splice and tail piece, for making fast temporarily to the mast or rigging.

A *long tackle block* has two sheaves, one above another; these are sometimes called sister-blocks.

A *fiddle block* consists of two single blocks one above the other, but both in one shell; the upper one being the largest: the object being for the upper rope to have play clear of the under one.

A *morticed block* is made by morticing out a block of wood or spar, and fixing a sheave in the aperture.

Double, treble, and fourfold blocks, are used where extra

power is required, and to ease the working of the rope : thus, a weight may be lifted by *one* man with the aid of a fourfold block, which *four* men could scarcely do with the aid of a single block. The main-sheet generally runs through one double block and one single ; thereby enabling a large sail to be trimmed by one person.

A *whip-purchase* is merely a rope rove through a block with one sheave : it is the smallest purchase of all.

A *tackle* is a purchase formed of two or more blocks, with a rope rove through for hoisting.

A *gun-tackle purchase* consists of two single blocks with a rope rove through both, one end of which is fast to the strap of the upper block.

A *luff-tackle purchase* is formed by a rope leading through a single and a double block, the end of the rope being fast to the top of the single block, and the fall leading from the double block.

All tackles have standing and running parts, the *standing part* is so much of the rope as remains between the sheave and part secured : the *running part* so much as works between the sheaves : the *fall* is the part laid hold of in hoisting or hauling.

Dead eye.—A circular piece of wood with three holes in it, and a groove cut round the outer edge for the shroud to lie in. It is used for turning in the ends of shrouds and backstays : the three holes are used for reeving the lanyard through, when setting up the shroud or backstay.

A *bull's eye* is a thick piece of wood of circular shape, with a hole through the middle, and a groove round the outer edge.

Improved spring thwart.—The spring thwart is very often fitted to ships' boats, because they are as frequently required for rowing as for sailing ; it may also in some instances be found convenient for pleasure-boats.

It is not a fixture as the other thwarts in the boat, but may be shipped and unshipped when necessary ; it rests upon the gunwale, and being much higher than the oars-men's thwarts,

and slightly bowing or springing upwards, is a greater support to the mast when stepped through or clamped to it. Ships' boats generally have a wash strake, which allows the spring thwart to be fitted with greater ease and convenience, by cutting away part of the strake sufficiently wide to drop the thwart down to the gunwale; then by bevelling away the under part of the wash strake to the thickness of the thwart, so as to admit of its sliding under the bevelling, it will remain firm: the cavity in the wash strake should be filled up by a woodlock or shifting piece. On the under part of the spring thwart a strong stop cleet should be nailed, so that the pressure caused by the mast, may be on the gunwale, and not on the wash-board; by this plan, bolts are dispensed with, which are always in the way and out of order for such purposes.

The trunk step.—This was invented for the purpose of avoiding the inconvenience that often exists in short open boats carrying a lofty mast. When the mast is lowered and the boat has to be rowed, it is sometimes found inconvenient for the mast to protrude over the bow or stern of the boat; and from its great length often incommodes the sitters. For boats of this description the *trunk step* is intended; and if the boat is fitted with a shifting spring thwart it will tend further to shorten the length of mast; as the trunk step may be carried several inches higher on account of the extra support derived from the elevated thwart. The trunk step usually stands about two feet above the gunwale of the boat; it consists of a solid piece of wood with a square groove or hollow, cut rather deeply into the top part, to receive the mast, and gradually diminishing in width downwards. Two strong flat iron clamps or trunk irons, are placed one quite at the top, and the other a foot lower on the trunk step, to hold the mast; the lower part of which should be made square to fit close and tight into the trunk; the mast will be better supported if the heel comes low enough to catch itself in the spring thwart. This is necessarily a heavy looking contrivance, but better than being put to inconvenience by too

long a mast, particularly when the boat is frequently in use for rowing purposes.

The mast clamp is considered a superior invention to the trunk step; it is one by which the mast may be raised and lowered in the boat by one person, and with great facility. The heel of the mast working on a pivot, and secured, when raised, by an iron pin; the whole performance being only the work of a few moments.

TRIMMING AND BALLASTING.

To trim a sailing boat, is to arrange the ballast so that the boat may sit upon the water in that position in which she may be sailed with the greatest safety, and at the fastest rate.

The trim of a boat depends on the right adjustment of ballast. It is a condition precedent to good sailing in every vessel, that the exact trim or proper water line should be correctly ascertained, and then, under no circumstances, should it be altered or deviated from in any degree whatsoever. A boat is supposed to be in trim, when with a moderate wind, she carries her helm amidships, or with a slight inclination to windward: that is to say, will sail a straight course without any or but little steering. In order to be in safe trim, the boat should carry "*a weather helm*," should come about quickly, and obey every turning of the helm, however slight. The heaviest and greatest quantity of ballast should be placed in the aft part of the boat, and amidships; whilst little or none should be laid before the mast; and none in the extreme end of the stern. In sea-going boats the rule is strictly adhered to: a vessel rises to the waves so much more buoyantly when her bows are not depressed with ballast. In smooth water, a small portion of ballast should be laid in front of the mast in some boats; and in long, narrow, sailing boats this is generally necessary.

The boat should not be laden too heavily with ballast, but merely weighted down to the true bearings or proper water line. It is a very necessary precaution, that, when the boat lists

over, the ballast may not slip or move. Should it fall suddenly into the lee bilge, a capsize must inevitably follow. Strictly speaking, the ballast should all be secured beneath the platform of sailing boats; but in very small boats, and those launched from the beach, this cannot be done, it should, however, in every boat, be lashed or secured, so that it cannot slip, when the boat lurches or pitches.

The best ballast for small boats is lead, but of all kinds the most expensive; it sometimes costs more than the hull of the boat. It is, however, preferable where expense is no object, as it may be stowed away in a very small compass, giving more room for accommodation, and rendering the boat stiffer under sail than by any other means; it creates neither rust nor dirt; and like silver and gold, will always realize its intrinsic worth, in whatever shape or quantity.

Every sailing boat must have more or less ballast according to its size and the form of its hull; a boat without ballast is unsafe and unmanageable. In ballasting a skiff, small flat shaped kegs of water are very convenient, and may be filled and emptied at pleasure; that is to say, when the crew consists of only one or two persons, the kegs should be filled; but when several persons are in the boat less ballast will be required, and some of the kegs may be emptied: again, a boat at starting may encounter fine weather and a light wind, and will require only part of her ballast; but a strong wind may come on before returning, when all the ballast will be needed: another advantage afforded by the kegs of water ballast is, that if a cording be put round them so as to form handles; in case of the boat being upset, an empty keg will answer every purpose of a life buoy; supposing each keg to be about the size of a two or three gallon cask, it would sustain two persons above water on their laying hold of the cords, one person on each side.

Bags made of tarred canvas, and filled with small shingle are often used for ballast, but they are not very durable. It is a good plan to put the shingle in wooden boxes, and place them along the floor of the boat; the boxes may be about one

foot in breadth, by three or four feet in length, and six inches deep. Iron ballast will be found to take up less room than shingle, it may be laid under the boat's floor; and if the iron be painted or tarred, and then covered with canvas, which may also be painted or oiled, the dirty rust and iron mould will be avoided. Sand ballast is also a weighty material, and may be stowed away very conveniently, particularly if put in bags or boxes.

In ballasting all kinds of small boats, that description of ballast which takes up least room in the boat is always to be preferred; the flat kegs before mentioned may be stowed away under the thwarts, and in the bilge of the boat.

HINTS AS TO RIGGING SAILING BOATS.

After the boat is launched and the ballast carefully adjusted, the next step is to ship and fit the mast, which should be no larger than necessary; and it had better be too small rather than too large; so as to snap off before upsetting the boat: or in other language, a mast that the canvas will carry away rather than capsize the boat. If this were constantly attended to in fitting out open sailing boats, accidents would be of less frequent occurrence. The sole cause of the lamentable accident, a few years ago, to the Lytham life-boat, may be attributed to the mast being too stout; so that even life-boats are not exempt from danger, when canvas is set on a disproportioned mast. When the boat is fitted with a mast of suitable size, it is a rule in setting sail, that so much canvas only as the mast can bear, is all the boat can carry. When too much sail is set, something must give way; an accident is inevitable unless the sheet is let go, or the sail immediately reefed. If the boat be upset, loss of life is generally the result; but when a mast "goes by the board," all sail standing, there the damage generally ends; except perhaps some slight breakage on the bulwarks or gunwale.

The mast for a sailing skiff or small pleasure boat, may be rigged with one or two small ropes for shrouds, which will be

a great protection to the mast; and when such are used, a small sized spar will answer the purpose better than a large one, and will be also better for the boat; and enable it to sail faster and more buoyantly than if incumbered with a heavy mast.

It must also be remembered that a sprit-sail needs but a very small mast, smaller than that required for other sails: the greatest strain upon a sprit-sail's mast is just at the heel of the spreet, or quite at the lower part of the mast. The strop or snorter, which holds the spreet to the mast, should never be more than one-third of the way up the fore-leech of the sail. The mast for a sprit-sail should be stoutest at the lower-part, tapering off gradually to the top; but for a lug-sail, lateen, and others, it should carry its size well up to the sheave-hole, or halliard-block. A rope called a *spreet heel-rope* should always be used for the sprit-sail; this rope must lead through a block at the mast head, one end being furnished with a loop, which receives the heel of the spreet, under the selvagee strop: so that on the spreet heel-rope being hauled taut, it takes part of the pressure of the spreet from the mast, and throws it forward. By this means too, the spreet will turn more readily on its heel when the boat is put about, or on gybing: the heel-rope will also be a means of preventing the constant chafe of the spreet against the mast, which cannot be avoided when the heel-rope is not used.

The length of the spreet must depend on the size of the sail; it should be no longer than necessary; the small end is the top, and that which is put into the peak-eye on setting sail; the largest end being the heel or lower part, which is set in a snorter or selvagee strop: the best spreets are small tough spars of white or red fir, that require little or no reducing; the same observations will apply to yards for lug and lateen sails, which should be neither larger nor heavier than absolutely requisite; for it is better to carry away a spar now and then, than to upset the boat by using those which are too large and heavy. Sailing skiffs are usually fitted with a small iron bumkin on which to hook the tack of the fore-sail: the bumkin is fitted to the stem, and secured by an iron pin which

passes through the stem and shoulder of the bumkin, and may be unshipped when not in use. Bumkins answer very well when not more than a foot or foot-and-a-half in length; when longer they are objectionable, on account of the weight of iron projecting over the stem. Iron bumkins are not so desirable as wooden bowsprits, because in the event of an accident occurring by running foul with the boat, a heavy blow would wrench the stem out rather than bend or break the bumkin; whilst a small bowsprit would give way, or snap off, and probably no further damage would occur to the boat beyond the breakage of a small spar.

A boom is seldom used with a sprit sail, except for short wide boats where the aft or main-sheet tack reaches beyond the stern; in which case a small boom is necessary; it should be fitted to the mast with a goose-neck, so that when the sail is brailed up, the boom may be drawn or turned up close to the mast.

The halliards, and all the ropes connected with the upper part of the mast, should be rove through small blocks; and in small boats where two or three blocks only are required, it is a very good plan to fit the top of the mast with a galvanized iron hoop, having several small eyes on which to hook the blocks; the eyes should stand out sufficiently far to prevent the chafing of the blocks against the mast; the *iron* hoop recommended for this purpose is preferable to *copper*: there being considerable strain on the outstanding eyes, they would be liable to break off if made of copper, unless of a larger size than neatness requires. It is a bad plan to reeve halliards through a sheave hole, especially a half sheave; because, if the rope be stiff or wet, it will be liable to get jammed: and unless the halliards run free and easy, the boat is exposed to danger; which is avoided by reeving the halliards through the blocks as before recommended; sheave holes and half sheaves are not recommended for sailing boats.

The running tackle or back stays are generally dispensed with in skiffs and small sailing boats, by fitting the shrouds a little abaft the mast.

The main-sheet should be rove through two blocks, one on the boom and another on the main-sheet hawse, which will ease the labour. In large boats double, treble, and fourfold blocks are sometimes used, which render the working of the main-sheet still more easy.

The lug-sail skiff requires no brails; the lug-sail should be lowered in a squall. Whenever a sail is not furnished with brails, extra attention should be paid to the halliards, to see that they are always clear and fairly rove.

It is necessary to caution young boat sailors against narrow boats, for sailing purposes: the narrow form of hull may be well adapted for rowing, but is at all times dangerous for sailing. Large fore-sails in small open boats are also attended with danger, and are more frequently the cause of accidents than the main-sail. It should always be borne in mind that the boat to be safe under sail, must carry a "weather-helm;" and such will not be the case if the fore-sail be too large. Short wide boats rigged as sloops or cutters, or with sprit-sail and fore-sail, are well suited for turning to windward in narrow channels: long boats are fastest for long reaches, sailing on a bow-line, and running before the wind.

For small or open boats, it is not considered a safe plan for the foresail to work on an iron hawse: the fore-hawse should only be used in decked boats, and those where a jib is set, as well as a fore-sail.

SETTING SAIL.

On setting sail, first see that the bobstay is secure, and the bowsprit bowed down at the outer end; then cast off the main-sail lashing, and look to the main-sheet and halliards; unfurl the fore-sail, and lay it out ready to hoist; haul out the jib on the bowsprit, and run the jib-sheets to their berths: having got all three sails ready, hoist the main-sail before you set the head-sails; make the main halliards fast on one side the mast, and the peak halliards on the other; the peak should not be entirely set up, until the main is well set; then hoist the fore-sail, and slip from the moorings. Having got

away clear, set the fore-sail fairly, and hoist the peak of the main-sail; whilst under way, run the jib up, and trim the sheets; then coil the halliards neatly and separately, so that they are each and all ready at an instant's notice. If the course is to windward, or on a reach, bowse down the main-tack; if before the wind, the main-tack may be cast off, and the sheet given freely.

If on slipping from the moorings among crowded shipping, it is necessary to turn the boat quickly round, and there is but very small space in which to do it; having set the head sails, haul them both aweather, drop the peak of the main-sail, ease off the main-sheet, and put up the helm; the boat will then turn as if on a pivot, if she is in proper trim.

REEFING.

"One night, as we drove with two reefs in the main-sail,
And the scud came on low'ring upon a lee shore,
Jack went up aloft for to hand the top-ga'nt sail,
A spray washed him off, and we ne'er saw him more."

DIBDIN.

This is a performance with which every person should be thoroughly familiar, before venturing to leave the shore and take the command of a sailing boat, for he knows not how soon he may be called on to reef the sails, in order to insure a safe return.

Reefing is generally required in a strong wind, or heavy sea, and should always be done carefully and with as little delay as possible. In a sailing match, the boat is kept on without reefing, as long as she can stagger under her sails, but on such occasions there are always plenty of hands aboard to assist in the management of the sails. When about to reef the sails, luff the boat up close to the wind, but not so as to allow her to come about; ease off the jib-sheet, or if you intend to set a smaller jib take in the other the first thing, then haul the fore-sail aweather, and make it fast; haul in the main-sheet as close as possible, and the boat will be "laid

to:" now lower the peak, then the main, sufficiently for the intended reef; cast off the main tack, and begin at the earings, or outer end, by hauling down the reef and securing it with the reef-earings, to the boom; then tie up the points with reef-knots, all along the sail, and make the main-tack fast; set up the peak, cast off the main-sheet, trim the fore-sheets, and the boat will then be under a single reefed main-sail: another reef may be hauled down in a similar manner, and the boat will then be under a double or two-reefed main-sail; and so of a third reef. Never tie up the points of a second or third reef, until the first or lower ones have been secured; you may then shake out the reefs one at a time, as the wind decreases. Always look to your reef-tackle before setting sail, and see that the earings are sound, and strong; for it is seldom necessary to reef except in heavy winds, which try the strength of ropes and tackle.

FURLING THE SAILS.

This is also frequently termed "stowing the sails;" it merely implies rolling them up neatly, and in a sailor-like manner.

The *main-sail* is furled as follows.—The sail being lowered down into the boat, the gaff and boom should be placed close together, one on the top of the other; then lift the flap of the sail over the boom, and lay the aft leech over the flap, hauling it taut from the gaff end; keep all taut whilst another hand rolls up the loose sail neatly, and close to the gaff. The sail should not be rolled *round* the boom, nor *round* the gaff, but *close up to* the latter. Then pass three or four small lashings round the sail and over the gaff, and, having secured them, the main-sail will be furled.

The *fore-sail* may be furled in the following manner.—This sail is generally fixed to, and traverses up and down the forestay by means of brass thimbles; it should be let down to the stem of the boat, rolled up, and an oil-skin coat spread over it. In small boats the fore-sail is not attached to the

forestay in this manner, but the fore leech of the sail forms the forestay; in which case the fore-tack may be unhooked and the sail rolled up in the main-sail.

Jibs and gaff-top-sails are generally kept in the fore-castle, or in the cabin, or whichever is dryest and most convenient. In small sailing-boats, having neither cabin nor fore-castle, they are sometimes rolled up in the main-sail.

After the main-sail is furled, a waterproof coating should be put over it, extending from the mast to the boom end: but it is not a good plan to leave the sails coated and furled for any length of time, a few days will sometimes incur the risk of mildew; therefore, they should be exposed to the air as often as possible.

The *sprit-sail* is generally furled by rolling it up close to the mast—not round it—after taking out the spreet, and without lowering the main halliards; the fore-sail should be rolled up inside the sprit-sail: it is a neat method, but no coat can be fitted over a sail furled in this manner.

When the sails are wet, they should be loosely furled; unless they can be spread out to dry. New sails should be well and frequently wetted with salt water, to preserve them from mildew. In damp or wet weather, sails require much attention, and must be frequently spread and exposed to the air during the driest part of the day; if neglected for any length of time (although under an oil-skin coat), they are very liable to be disfigured and otherwise injured with mildew.

SAILING TO WINDWARD.

“ Thus tars at sea, like swabs at home,
By tack and tack are biass'd,
The furthest way about we roam,
To bring us home the nighest.”

DIBDIN.

This is the most interesting performance connected with boat-sailing; the tyro should spare no pains to make himself

master of it: and with that view he must thoroughly accustom himself to the use of the tiller, and practise sailing in all the varieties of light airs, gentle breezes and strong winds. The art of sailing the boat against wind by sundry zig-zag performances, requires considerable skill, a watchful eye, and frequent practice; for although the rudiments of the art may be learnt in a few lessons, the art itself, which is the perfection of boat-sailing, can only be acquired from long experience. The principal thing for the helmsman to attend to in sailing to windward with effect, is to watch the fore-leech of the main-sail; or that part nearest the mast. The boat should be sailed as close to the wind as possible without shaking this part of the sail; which is always the first to quiver to the breeze, and warn the keen-eyed sailor of too close a luff. Many sailors steer entirely by a vane at the mast head; and some cannot sail a vessel without one; but such is an uncertain guide, and at all times an improper one; some steer by the ripple on the water, which may be all very well when the wind is light, but at best a very uncertain guide. The young boatman who wishes to become skilful in the art of sailing to windward, will practise his eye upon the fore-leech of the main-sail, which is always the most faithful and unerring indicator; the instant the slightest waver is perceptible in this part of the sail, he should bear up a little; the least motion of the tiller will suffice, if the boat be in proper trim; it often happens that the wind is unsteady, blowing sometimes in a continued strong breeze for two or three minutes or more, and immediately afterwards a light wind follows; but this occurs chiefly with easterly winds, which are generally irregular; on such occasions, great advantages may be gained by watching the effect of the wind upon the luff of the main-sail; the boat can be sailed two points nearer the wind in some of the gusts than in others; then is the time for the skilled helmsman to wedge his way to windward in the race, and leave his opponents in the lurch.

The greater the force of the wind, the closer the boat may be sailed to it when in smooth water; and in racing, or

match-sailing, an experienced helmsman will so narrowly watch the wind, that should the slightest variation occur, he will take advantage, by sailing his boat up to it with all possible precision; but never so as to allow any part of the sail to quiver or hinder the boat's progress. It is a well known maxim in sailing to windward, to "keep her full," that is, keep the sails full of wind, and do not allow any part of them to quiver; for a shaking or quivering sail is of no assistance to the boat.

In turning to windward in a very narrow channel, it is best to take in the jib, and work the boat under main-sail and fore-sail; the jib is the sail that sticks the boat ashore in channels which are so narrow that the boat is no sooner round and on her way, than it becomes necessary to put about again. On putting the helm down to come about, it should not be put over too suddenly, but gradually, that the boat may obey it quicker: some people are so impatient in getting the boat round, that the helm is jammed over all at once, and oftentimes the boat misses stays in consequence; which not only causes delay and vexation, but danger as well. Another practice equally erroneous is, putting the helm up before the boat is fairly round, by which means she loses way, or lays head to-wind, without going ahead; and it then becomes necessary to haul the fore-sail aweather, to pay her off, or the main-sheet must be eased to allow her to veer off into the wind.

In sailing to windward, a good deal of discretion is required as to the proper sized jib to set: as a general rule, a boat will go to windward better with a small jib, than with a large one. Many a race has been lost through carrying too large a jib when working to windward; the effect of a large jib is to sag the boat to leeward, particularly when she has but little way on her; this may be proved whenever the boat is put about; and it is a good rule, that the jib-sheet should never be trimmed until the boat is fairly going ahead.

Gaff-top-sails are seldom of any assistance to a boat when

working to windward, except in very light airs, and under cliffs; but they are of great service in reaching and running with a free wind.

But withal, it is impossible to bring out the utmost speed of which a boat is capable, unless a number of points are first carefully attended to; such as ballasting, rigging, setting and shifting sails, trimming the sheets, &c.; each of which is almost an art of itself, and cannot possibly be learnt without considerable attention and experience; but with these combined, a good sailor will frequently have the satisfaction of being foremost in the race: and experience, with skill, may make many an old-fashioned vessel beat a new one.

REACHING AND SAILING ON A BOWLINE.

By reaching is meant sailing with a side wind: the boat is said to be on a reach when the main sheet is not quite close-hauled; and on a bowline, when the sheets are eased off in such a manner that all the sails draw powerfully.

The principal points to attend to when sailing on a bowline, are to trim the main and jib sheets to a nicety; not so that they stand too slack, or cause any part of the sail to flap or quiver, but in such a manner that every inch of sail is felt by the boat. The greatest speed at which a boat can sail, is brought out when sailing on a bowline.

Long boats are fastest for this branch of sailing; a boat with a long keel and narrow form of hull, if judiciously ballasted, will pass many a sea-going boat in smooth water, with a reaching or bowline wind.

The fastest description of boats for reaching, are large sailing galleys or yawls, with one, two, or three lug-sails; and such, in smooth water, would be formidable opponents to many a racing yacht; but such narrow boats are not recommended to inexperienced hands; they are much in favour with pilots, who are obliged to hasten out to vessels, and are generally well manned and beautifully handled;

besides, they seldom go out with a smaller crew than five or six.

Should a heavy squall strike the boat when reaching in a strong wind, she must be instantly luffed up; but if the squall is very sudden, and there is not sufficient sea room for luffing, the main sheet should be slacked at the instant, by which means the sail will be eased, and the boat will quickly recover itself. In bearing up, or wearing round, when on a reach, if the wind is strong or squally, the peak should be lowered, and the main sheet slacked, or there is danger of carrying away the mast. If the boat has running tackle or backstays, they will be a great support to the mast on such occasions.

When a boat is capsized in wearing, the water first comes in over the lee bow, which is sometimes nearly driven under water in hazardous sailing.

SCUDDING, OR RUNNING FROM THE WIND.

"A wet sheet and a flowing sea,
A wind that follows fast
And fills the white and rustling sails,
And bends the gallant mast."

CUNNINGHAM.

This branch of the art requires to be performed with caution, particularly during a strong wind, or in squally weather; although to the inexperienced it may appear the most simple and easiest mode of sailing, and the least fraught with danger, as the boat travels more upon an even keel, and without lying over on her side, as she sometimes does in turning to windward; but experience tells us scudding is, of all sailing, the most perilous; because of the risk of the main-sail suddenly gybing. A back stay is necessary when running before a strong wind, in order to protect the mast from being carried away. The shrouds at the sides are no protection when sailing with the wind abaft.

The main-sheet should be run out as far as the boom requires, the running tackle cast off to leeward, and set up to windward; the sail should be allowed to blow out freely, but not so as to chafe against the shrouds, the main tack should be cast off, in order that the main-sail may hold the better wind; a watchful eye must be kept upon the sail, and attention paid to the wind; for should the sail be suddenly gybed or blown over to the other side of the boat when the main-sheet is all run out, the boat is almost certain to be capsized, or the mast carried away if there is much wind. But in order to avoid this, when running directly before the wind, should the sail exhibit the slightest symptoms of being taken aback, by wavering to leeward, instantly put the helm down a little, and avoid a gybe if possible. The boom swinging suddenly over is apt to knock anybody overboard who does not happen to be on the look out; the person steering should, therefore, always give warning when the boom is coming over.

Should a squall strike the main-sail fairly, when running before the wind, and appear too heavy for the boat, ease her by putting the helm down immediately, and bringing her up into the wind; and so you will probably save the mast from being carried away. The boat should be steered as straight a course as possible when scudding. The fore-sail will be of little or no use when going directly before the wind, unless boomed out; the jib-sheets should be slacked, and the sail allowed to draw.

It is customary when the wind is fresh and squally, to lower the peak of the main-sail, or to trice up the main-tack; either of which proceedings will be the means of taking some of the strain off the mast; but the safest precaution is to reef the main-sail, and take in the jib. The danger is increased by scudding in a heavy sea; and a boat is generally hard to steer when she pitches. The safety of the boat and crew, when scudding in a heavy sea, or strong wind, depends entirely on the watchfulness

and skill of the helmsman ; who, on observing the least inclination of the sail to gybe, should instantly ease down the helm.

If the main-sail has no boom, it will be the more liable to gybe ; therefore, extra caution must be used when running before the wind.

The young sailor must always bear in mind, that if a squall strikes the sail, and the boat is in danger, he must put the *helm down*, that is, push it towards the same side as the sail is ; to put the helm *up*, is to push it from the sail ; and this precaution must be distinctly impressed upon his mind, unless he has a particular wish

TO CAPSIZE THE BOAT.

To perform which, he has nothing to do but put the boat before the wind (which is the scudding position), slack out the main-sheet, and make it fast to leeward, put the helm up, and allow the sail to gybe, and then, over goes the boat ! This is the neatest mode known to the author of capsizing a boat, and giving yourself and crew a complete turn out, and good ducking.

BOAT IN STAYS.

A boat or vessel is "in stays" when the helm is put down to bring her about, and the sails are all shaking in the eye of the wind ; but directly the fore-sail has "payed off" the head, and brought the boat round, she is on a fresh tack or reach ; the port or starboard tack, or port or starboard reach. When a boat is in stays in squally weather, it is a critical time ; should the wind take the sails aback, or a squall strike them, there is danger of upsetting the boat, which has then no way on, *i.e.* is not going through the water, therefore will not, when thrown suddenly on her side, answer to the helm. The sheets should be clear and free, lying in coils, so as to be ready to let go instantly,

in case of peril; the jib-sheet especially, should be kept slack and in hand, until the boat is fairly round and has recovered way. Long boats are always more sluggish in stays than short ones; the short, beamy, little craft is quickly round, generally before danger can touch her; whilst the long rakish craft takes a tremendous sweep in twisting, shooting ahead considerably in the performance, and soon fore-reaches upon her shorter antagonist.

Should the boat miss stays in a squall, the main-sheet and jib-sheet must be let go or slacked, and the fore-sail drawn to windward; after which, if the squall throws the boat so flat on her side as to be in momentary danger, let go the halliards, if possible, in time to save the boat from capsizing.

MISSING STAYS.

This term implies a failure on the part of the boat to answer her helm, or "come about," when the tiller is put down.

When a boat misses stays in a light wind, there is seldom any danger attending it; but when it occurs in a strong wind or squall, or in a heavy sea, there is always more or less risk; for the boat, when in stays, has no way on her; consequently, she is very liable to be capsized on the wind suddenly striking the sails. It is necessary, therefore, to attend to the sheets, and see that they are all clear, and not made fast until the boat is fairly on her way again.

It is a matter of prudence on the part of the helmsman of a sailing boat, to avoid, if possible, putting about in a squall, as it is also in a heavy sea, for fear of the risk incurred, and the danger attendant on a boat which misses stays. If a boat is judiciously ballasted, and in good trim, she will seldom miss stays in smooth water.

TO BRING UP AT MOORINGS.

"I've heard, cried out one, that you tars tack and tack,
And at sea what strange dangers befel you ;
But I don't know what's moorings—what don't you, cries Jack ?
Man your ear-tackle then, and I'll tell you."

DIEDIN.

A vessel riding by two or more anchors in different directions is said to be moored. A boat's moorings consist of a strong mooring-chain, the two ends of which are anchored in different directions, and a buoy is attached to a smaller chain, or bridle, that is secured to the centre of the mooring-chain.

Nothing looks more like mis-management than getting into a muddle with the sailing boat when it is to be stopped, brought up at a particular spot, or dropped alongside a yacht: bungling hands when endeavouring to do this, sometimes run foul of neighbouring boats, and in their ineffectual attempts to bring up at moorings, keep hoisting and lowering the fore-sail and peak, hauling the boom over, dropping anchor, jamming their fingers, working themselves up into a state of perspiration and excitement, getting their legs entangled in the ropes, and toppling head foremost in the boat, and sometimes overboard; to the amusement and ridicule of bystanders, whose hearty laughs at such lubberly seamanship increase the difficulties of the unskilful crew.

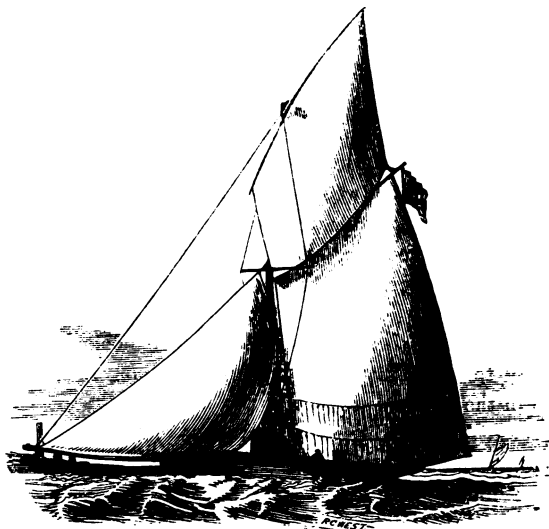
An experienced sailing master, however, is enabled to bring up at moorings, or drop his boat alongside another, with as much ease as a practised coachman drives round a lawn, and stops in front of the door-step.

Bringing up at moorings in a crowded harbour, is one of the prettiest manœuvres in boat sailing; a skilled and confident sailor comes boldly up to his berth with all sail standing, and performs the task without making a scratch upon his boat.

It should be borne in mind, that the greater the length of the boat, the more sweep it requires in coming about : a light sailing skiff requires but little room, and may easily be laid alongside after practice ; but a boat heavily ballasted, and of a larger size than an ordinary skiff, cannot always be stopped in a moment when a fresh breeze is blowing. On coming up to moorings, the distance required for the sweep must be measured with the eye, and, if in a tide-way, allowance made for strength of current ; luff the boat boldly alongside in a good sweep, and, after a little experience, it will be an easy matter to lay her alongside with all sail standing, but, of course, shaking in the eye of the wind ; it is usual, however, first to take in the jib, that the boat may have less way on her ; the fore-sail should be standing until the moment before the moorings are hooked, so as to be ready to haul a-weather, and pay her off in case of a failure in the manœuvre.

Never attempt to lay a boat on, or too near a lee shore in a heavy swell, or a strong wind ; and on going up to a berth at moorings, never do so before the wind, but always to windward ; if it happens that there is but short turning room, and the moorings cannot be approached in any other way than by running before the wind, lower all sails except the fore-sail, and run up as slowly as possible, hook the buoy with a boat-hook, and make fast quickly.

To steer the boat when going through the water stern first, the helm must be put in the same direction as that in which the head is required to be turned. When drifting with the current in a tideway, the aft part of the boat being deeper than the fore part, the stern will have a tendency to drive faster than the head ; in which case, and in order to retain a proper control over the boat, less aft sail must be set, and more head sail.



ENGLISH RACING CUTTER.

MATCH SAILING.

“ With the wind right on end, and waves running high,
The spray o'er his vessel flies free ;
But the sailor proclaims as he welcomes the blast,
'Tis a bonnie boys' race on the sea ! ”

Those only who have taken part in a spirited sailing-match, and shared the struggles and the victory, can truly appreciate and enjoy the pleasures and excitement of a contest between sailing-vessels ; when every inch of the rippling surface is as closely contested as if life and death were pending the result.

The daring efforts that are made in strong winds and heavy seas, by an undaunted crew, and their earnest struggles for victory, have often won public admiration and applause.

There is no truer test of skill and daring in a sailor, and skilful seamanship in a skipper, than the fact of his sailing and winning a hard-fought sailing match, in a gale of wind :

“The mast may be bending, and threat’ning the gale ;
The gunnel borne down deep a lee :
But the stoutest of hearts, and most daring of men,
Win the perilous race on the sea.”

In a sailing match, there must be activity and readiness among the crew, and skill and daring at the helm.

Before the starting gun is fired, every man should be at his post, rope in hand, ready to perform the duty allotted to him; for, after the gun is fired, every moment’s delay is a lost moment: and often the chances in a hard fought *time race*, are much in favour of the yacht that succeeds in first getting the lead.

The boat that can be sailed fastest to windward, is considered the best boat, in all sailing matches; and is generally the winner of the race-cup. Although very much of the success is often attributed to the form of the hull, much more depends on the skill of the helmsman: and when we consider how slight a touch of the helm will put a well-trimmed boat out of its course, and turn it a point or more off the wind, it is easy to understand how sailing matches may be won and lost through the superior skill of the one, and the least inattention or want of skill on the part of the other. We can also, by the same test, account for the fact, of more than one of our most famous racing cutters having been signally defeated, time after time, when in the possession of one owner; but on changing hands, turning out the very clipper of clippers.

And besides, too, on the cut or fit of the sail, the success or failure of the sailing master’s exertions in a race, often depends. The sails must stand as flat as a wall, and without ruck or quivering, if great and distinguished performances are expected of the boat.

The importance attached to trimming and ballasting racing-

boats, cannot be overrated. These, however, as well as other less important considerations, are preliminary steps to which due attention must be paid before coming to the starting point with any chance of success in a sailing match.

A few years ago, a highly disreputable system prevailed in match-sailing, that of *ballast-trimming* : but which, I rejoice to say, is now entirely discountenanced by all English yacht clubs : the sanction of such a pernicious and dangerous practice, led to the introduction of a class of vessels unworthy the name of English yachts, and which were very appropriately termed "*sailing machines*." These vessels, to all appearances, when under sail, were enabled to carry a great pressure of canvas, although of narrow form of hull ; and they were rigged with spars wholly out of proportion to their tonnage : a peep into the cabin, however, would disclose the whole secret, and thus an astonished beholder might discover why, and how it was, that such long narrow boats could hold themselves up under such an amazing pressure of canvas ; whilst the more sturdy and stiff-looking coasting vessel of the same tonnage, found about half as much sail a sufficient quantum.

In that cabin of deception were three or four men trimming heavy bags of shot—real shot, such as sportsmen use to wing the feathered tribe ;—and those they shifted from side to side, or from bilge to bilge, to "hold her up ;" or to "keep her stiff," as they termed it. In the absence of bags of shot, were long bars of lead, or iron, which they lugged from one side of the boat to the other, and secured from slipping, by slides or shifting boards : the four men then cringed over to windward, or laid down upon the shot bags, until they heard the order, "bout !" when the shifting ballast was instantly trimmed to the other side ; and again they coiled themselves up on the shot bags, as before. When there was much turning to windward, ballast trimming was hard work ; in fact, the hardest work aboard the yacht : as much as a ton of shifting-ballast was but an ordinary quantum for a ten ton racing-cutter.

Such proceedings are now strictly prohibited by the rules and sailing regulations of all the yacht clubs in the kingdom.

Boats used for racing, require to be very strongly built ; or the immense strain caused by extra pressure of sail, soon tells with effect upon some part of the hull.

When racing in small open boats, the crew should sit as low down in the boat as possible ; if on the floor, so much the better.

When putting the boat about, the helm should not be put down too sharply, but slowly and steadily ; by which means the boat will shoot farther ahead in turning, come about quicker, and be less likely to miss stays.

Every man should be somewhat master of the art of sailing before he aspires to the helm in a sailing match ; therefore, all preliminaries as to setting sail smartly, canting round for windward berth, &c., should be first well learnt and practised. Many a race has been lost by carrying too large a jib *on a* wind, and as many more by large gaff-top-sails ; when, with a jib-headed top-sail and a smaller jib, the boat would have sailed faster and made less lee-way. But in reaching, running, or sailing with the wind free ; the boat should be allowed all the sail she can carry ; set your balloon jibs and balloon gaff-top-sails as large as the weather will allow ; always acting with a judicious regard to the safety of the spars and vessel. I have often observed a leading boat drop astern of the others, in turning to windward under too large a gaff-top-sail and balloon jib ; be the wind ever so light, those butterfly sails will not answer *on the wind* ; on the contrary, they retard the progress of the boat, and deny her a fair chance.

In tacking to windward, in a race, great advantages may sometimes be gained by manœuvring ; but nautical manœuvres can only be well learnt by practice and experience, and are not a subject for book-teaching.

In match-sailing, when there is abundance of sea-room, less skill is required, consequently less manœuvring ; but in rivers and narrow channels, every move of the opponents must be watched, and every chance embraced. If two boats be standing on the same tack, and the hindmost (although perhaps the

faster,) cannot pass her opponent to windward, by skilfully tacking an instant before her opponent, the next tack, may, if the boat be quick in stays, reverse the order of things, and place the hindermost first; and he must be an unskilful sailing-master who, with the faster boat, cannot accomplish this manœuvre in the course of one, two, or three tacks.

When beating up against tide, the boat should be kept as much out of the current as possible; particularly when putting about: but if the tide be in favour, keep in it, and tack where the current runs strongest.

As soon as the sails are set, the sheets trimmed, and the halliards coiled and laid in their berths, every man aboard the racing boat should sit or squat; and so remain, without moving, until ordered to do so by the sailing-master. The crew should not sit down altogether in one part of the boat, but judiciously distribute themselves about the vessel, so as not to disturb her trim, or depress her bow or stern. In small yachts and open boats, this is an important consideration; the weight of one man in the wrong place, may make a material difference in the trim, and consequently in the sailing of the boat.

It is a customary rule in all sailing matches, that the sails of competing vessels should not be skeeted (*i. e.* wetted), except when the vessel is on a wind; the sailing-master should, therefore, take care to do it at the proper time. In dry and sultry weather the sails become soft and expansive, and a sprinkling of salt-water not only fills up the pores of the canvas and enables it to hold a better wind, but also makes the sails stand flatter, and more drum-like, for going to windward.

When on a wind, the sailing-master who watches the wind, and the luff of the main-sail closest, obtains the greatest advantage; for it often happens that there are slight variations in the wind, in the course of an hour or less; and at some moments, he is enabled to lay a much better course than at others; he should take advantage of all these, and wedge his way to windward; constantly creeping as close to the wind

as possible, so long as he can do so without shaking the wind out of the sails. The fault, in those unaccustomed to match-sailing, generally lies in their too great fear of not keeping the sails full; and so, having once placed the vessel close to the wind, they are careful to steer a straight course; and regardless of any variation in the wind, they steer as straight as an arrow for some object ahead, or on shore. Now, this is undoubtedly an excellent plan when running free; but the man who so steers a racing vessel when *on a wind*, will very seldom succeed in bringing her in first at the winning goal. He should watch constantly for variations in the wind, by keeping his eye on the luff of the main-sail; and by the most delicate touch of the helm, he will be enabled occasionally to squeeze his clipper into the very eye of the wind: it is only by strict attention to such opportunities, and instantly embracing them, that they can be turned to so good advantage: but, be it remembered, they are assuredly golden opportunities, when attentively and actively seized.

In trimming the sheets of a boat or vessel engaged in a sailing-match, a good deal of discretion is required: a pull of an inch too much on the jib-sheet, when on a wind, may render the sail far less effective; and, instead of acting as a powerful drawing-sail, half its power may be lost; and it then becomes a "lee driver."

When sailing on a bow-line, care should be taken that the sheets are trimmed so that every inch of the sails draw, and assist the boat in the most effective manner.

When running before the wind, pay out the main-sheet freely, haul the jib aweather, ease off the sheet, and allow all the sails to draw powerfully, and assist the boat to their utmost. It is a mistaken notion to set up the shrouds of the mast too taut, in a racing vessel; the mast should have a little play. If too confined, the vessel will appear as if benumbed, when close hauled; whereas, by easing the weather-shrouds and giving the mast more play, the boat will be released, as it were, from its bonds, and appear as lively under sail as a bird on wing.

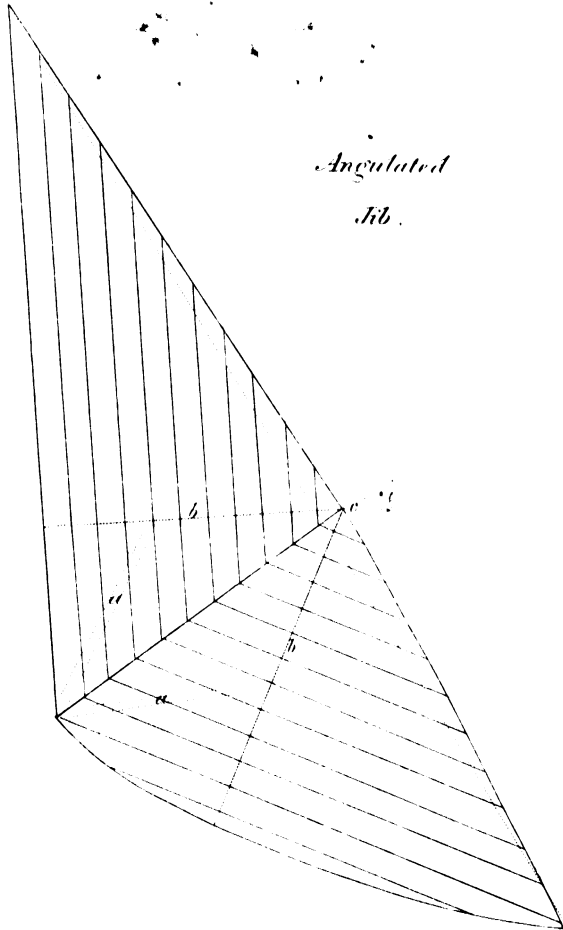
THE ANGULATED JIB.

It must be obvious to those who have given the slightest attention to the making and standing of sails, that jibs, when made according to the common method, do not retain, when set, so *flat a surface* as fore-sails, boom-main-sails, and gaff-top-sails; and it has been generally admitted, that if jibs and other trilateral sails could be made on a principle that would ensure their standing flat, many advantages would be gained.

The late Mr. Matthew Orr (of the firm "Orr, Hunter, and Co.," of Greenock,) was the inventor of the angulated jib, which, from its construction, avoids the defect of the knuckle, and makes a stronger and flatter sail than any previously produced.

Every seaman is aware that the sailing qualities of a vessel materially depend on the cut or flatness of the sails, and particularly on jibs that will trim to the same angle as the other sails. In regard to boom-main-sails and fore-sails, they are made to stand flat without difficulty, and are frequently laced at the foot to booms; but a difficulty long existed in getting large jibs to stand equally flat with the other sails, particularly after much usage, by which the bag or bellying is increased to such an extent, as to render them almost useless when close hauled. From a consideration of these facts, Mr. Orr was led to a different method of making all sails, the foot of which formed an obtuse angle with the aft leech, and termed by him the "angulated method," the principle of which consists in a new arrangement or combination of the materials used, in a manner calculated to produce a more favourable effect of the power acting on them, and consequently their more advantageous impulse on the vessel; and although the plan deviates from long established custom, its utility and superiority have been confirmed by frequent tests and long experience. The principle of the angulated method

*Angulated
Fib.*



is to place the cloths in such a manner as to do away with the knuckle, by binding the warp threads so that they are all acted upon by the strain from the sheet. (*See Engraving*). The strain bands, "a a," are carried from the clew to meet the straight threads "b b," running from the point "c." Angulated sails are also made with less cloth, as they do not require so much roach on the fore leech; and the angulated jib can be made to trim to the same angle as the other sails, as it stretches equally, presenting a flat and comparatively even surface to the wind; it also lasts much longer, and requires less trimming of the sheets than the common jib, the strain of which is from the clew to the stay, whereby it forms a bag in the upper and lower part of the sail; and when half worn is, in consequence, particularly difficult and troublesome to keep properly trimmed when close hauled.

The angulated jib has a further advantage over the common one in being less liable to shake; and when the vessel is in stays, with the sheet to windward, the angulated jib takes effect sooner than the other.

AS TO THE MANAGEMENT OF SAILING BOATS IN SQUALLS.

"But again we pressed on her, the gale still increasing;
Not a squall now and then, but a squall without ceasing."

On a wide expanse of water, signs of a squall may generally be seen on the surface some few moments before it reaches and strikes the sails of the boat; and at sea, a squall may sometimes be seen at a distance of many miles; in which case, there is plenty of time for shortening sail before its effects are felt: but in narrow rivers, and when sailing close under the land, squalls often come down upon the boat with all the suddenness imaginable, and such are the most dangerous of all squalls: and it is, besides, difficult to

suggest a means of avoiding their dangerous effects on an open boat, except, that double caution should be taken that the main-sheet is always ready to be let go in an instant, whenever you are sailing along the coast; for squalls sometimes come sweeping down the valleys, with great force, and often catch the boat in a calm, as it emerges from the shelter of some high cliff or mound; and many and sad are the accidents that have arisen from such squalls.

When a squall is observed approaching, the peak should be lowered, or the sail brailed up; if only a light or little squall, allow it to just strike the sails, then luff the boat up to it, but not so as to lose all way; keep the boat going, or she will not answer the helm: if a very heavy squall, the spreet should be taken out, and the fore sheet slacked.

It is easy to distinguish a little squall from a heavy one; the light one flits over the water like a dark cloud, but a heavy one or "white squall," brings with it a fierce-looking, white crest upon the tops of the waves:

"'Luff! luff!' was the shout; 'a white squall to wind'ard!'
Then we eas'd her a moment, tho' her progress was hinder'd."

When threatened with a white squall, it is safest to let the fore-sail run down, and to drop the peak of the main-sail; and also take in the jib, unless it is a small one.

In an ordinary squall, if the sails are reefed, and the boat be a safe and powerful one, there is nothing to fear, under judicious management. The boat may be conducted through it with safety, by "sailing her narrow," *i.e.* so close to the wind that the fore-leeches of the sails just begin to shake; but great caution is necessary, lest the sails be taken aback.

In all cases, it is of the highest importance to keep good way on the boat; for if the boat loses way, or is stationary, the squall will tell upon it with double force and treble danger, to that which would be incurred if it were moving rapidly ahead.

If a squall strikes the sails whilst the boat is running free, always *put the helm down*; and this is a very important prin-

ciple in boat sailing : for if you make a mistake and *put it up*, the squall will most likely capsize the boat.

CAUSES OF BOATS CAPSIZING.

Notwithstanding the numerous and melancholy accidents that occur, year after year, through the mismanagement and upsetting of sailing-boats ; there are persons who will not take warning therefrom, but persist in rushing headlong into dangers, which, with ordinary prudence, they might certainly avoid.

The casualties that occur, are not always occasioned by stress of weather, but are mainly attributable to causes within control.

Boats are not so frequently capsized on account of large sails, strong winds, and heavy seas, as they are from mismanagement or carelessness : by far the greater portion of accidents occur in fine or moderate weather. Among the principal *causes of boats capsizing*, are inattention to the main-sheet and jib-sheet ; wrong adjustment of sail, particularly head sails or those before the mast ; disproportionate spars ; improper trim of hull, whereby the boat carries a lee-helm instead of a weather-helm ; insufficient ballast ; the ballast shifting and rolling over to leeward ; the jamming of a rope, whereby it is checked in running ; the sail not coming down freely ; ill-fitting blocks ; reckless pressure of sail ; overcrowding the boat with people ; intoxication ; standing up in the boat ; leaning over the gunwale ; and various other causes hereinafter explained.

Every person who ventures on the water in an open sailing boat, ought to know, that the most important rope, and that on which the safety of the boat often depends, is the *main-sheet* ; next in importance to which are, the *fore-sheets* ; or if two head sails, the *jib-sheets*. All the halliards, and indeed every rope belonging to the sails, should be laid in a separate coil, so as to be ready to be run out without obstruction, in case of sudden emergency ; but more particularly the *main-*

sheet, which should never be made fast, except in the slightest and most simple manner. Care should be taken that it be not entangled, or in any way hidden from view; and this, whether the boat is sailing before the wind, on the wind, reaching, or otherwise. In many cases, when boats are capsized, the reason is, simply, that the person attending the sheets, in his confusion at the moment of danger, from fright, inattention, or some other unpardonable cause, fails to slack the rope until too late; and at a period when his own weight, and probably that of the other inmates of the boat, suddenly jerked over to the leeward side, actually accelerate the upsetting, and deny the boat a chance of righting.

It sometimes occurs that the coil or fall of the sheet becomes entangled or twisted about something in the boat, so as to render it impossible to be let go suddenly. Many persons may consider this as very unlikely to occur, but there are others who know it has unfortunately happened too many times to need any comment to prove its probability; *e.g.*, suppose the fall or end of the sheet to be lying in a neat, round coil at the bottom of the boat; the part leading from the clew of the sail being uppermost, and apparently, all clear and ready for running out in an instant: now the chances are, that after sailing a short time, this rope becomes slightly deranged; particularly if there is much of a rolling motion, or many persons in the boat, or any circumstance occurring to call the attention of the person attending the sheet, to some other object; the neatly coiled main-sheet is then forgotten, becomes entangled or foul of something; and if required to be slacked, meets with some impediment to check its course through the main-sheet block, and thereby the boat is capsized. A rope when drawn rapidly through a block, assumes a meandering or corkscrew form, and is very liable to catch round something or other in the boat, such as an oar, a boat-hook, a cleat, or person's foot; any slight check from either of which, may cause an obstruction, and consequently a capsize. It is not unfrequently the case, that a rope, although neatly coiled, becomes kinked on getting

wet, particularly if new ; a wet rope is also liable to swell ; in either of these cases, the sheave of the block may be choked, or the rope jammed between the parts of the shell.

New rope, if not well stretched and the turns taken out before reeving through the blocks, will be liable to twist, in such a manner as to stop its running freely ; therefore, simple as the precaution may appear, it is a very important one to be observed in fitting new ropes to a sailing-boat, particularly those used for *sheets* ; where a temporary obstruction may occasion the most disastrous consequences : for if the sheet be not instantaneously cleared, when the boat is struck by a squall, it must inevitably be capsized. Perhaps the most effectual manner of clearing away the obstruction, in such a case, would be, to cut with a knife and cut away the sheet ; an experiment which has, ere now, saved boat and crew from destruction, even after having been thrown flat on beam ends, and the water pouring over the gunwale.

Boats passing under the lee of large vessels, in squally weather, are very liable to be upset, on the instant after passing the vessel. The boat having lost the wind out of her sails, has no way on ; and if a squall then strikes her, there is great danger of a capsize, unless the sheets are slack and clear.

In moderate weather, or during a steady breeze, with a clear sky, and when not likely to be squally, seamen and boatmen are frequently inclined to take what is termed a "slippery hitch" in the sheet : this is done by twisting the bight of the rope once round its own part. A careful sailor, however, will never, under any circumstances, allow the main-sheet to be belayed ; he either holds or orders the slack to be held in the hand.

It sometimes becomes necessary, in light winds, to row and sail at the same time, either on account of a foul tide, or from lack of wind ; but such a proceeding is highly incautious, if the sheet has to be made fast, and no one left in charge at the helm. The more prudent course would be either to lower the sails, and depend entirely on the oars, or to dis-

pense with the use of oars, and trust to the sails. If oars are used whilst sailing, they should be employed on the windward side of the boat: there is considerable risk of upsetting the boat, through catching the oar under water, with the flat side of the blade uppermost, if the boat lays over or suddenly catches a breeze when the oars are employed on the lee side.

Among other causes of boats capsizing, and one as likely to occur as any, is, when the ballast is merely placed upon the floor without any platform over it, or other means of keeping it secure from shifting; then, if the boat lists over in a sea-way, a slight puff may cause a more than ordinary lurch, when the ballast slips from the windward bilge to the leeward, and no effort can then prevent the inevitable result. When running with the wind fair abaft, the sail is more liable to gybe without the boom than with it; a watchful eye should therefore be kept upon the sail, and the main-sheet must not be made fast; for should the sail gybe and the sheet be belayed to the leeward side, it will assuredly upset the boat, even if only a moderate wind. There is not a more effectual way of capsizing a boat under sail.

Boats may also be upset by having too large and heavy a mast, which gives too much leeward pressure, and materially weakens the stability of the boat; a mast should not be stiff and unyielding, but rather flexible, and not a shade stouter than necessary to sustain the pressure of the sails in a stiff breeze.

It may be considered a singular assertion, but it is nevertheless a correct one, that many a small racing-yacht of the present day, if stripped of sails, stores, and *ballast*, leaving nothing but the empty hull and spars, may be completely capsized, solely by the weight of the mast swaying over on either side, although in perfectly smooth water. This experiment, however, does not prove that the spars are too large, nor does it prove any bad quality in a yacht; on the contrary, the more "cranky" the hull, when empty, the stiffer it will be when ballasted.

The action of the sea upon a boat running into a heavy

surf, may be thus described:—when on the top of a heavy wave or roller, the bows are lifted high out of the water; then, as the sea recedes, the boat is hurled forward, and the bows are buried under water; when, the sea acting powerfully on her head and fore-gripe, twists her round, broadside to the waves, —called “broaching to;” and the sea then runs over the gunwale into the boat; the next motion that inevitably follows, is a heavy lurch on the other side, and another sea breaks completely over, and fills or capsizes the boat. This may happen either under sail or oars. There is considerable difficulty in preventing a boat from broaching-to, when stem and stern are alternately lifted out of the water by the waves; and should the boat broach-to and meet a very heavy roller, broadside on, the chances are fifty to one that she will be swamped.

Drogues^a are now a good deal used on the Eastern Coast, in both sailing and rowing boats; they serve to check the boat's way, and keep her *end on* to the waves; and are, therefore, of great assistance to the crew, in preventing the boat from broaching-to. Experience teaches, that when a heavy breaker follows the boat up astern, it is useless to attempt running away from it: then a question naturally arises, what must be done on the impulse of the moment? “For your lives, men! back her astern; hard at it every one of you! and let the man in the stern-sheets creep forward a moment, to lighten the boat's stern!” By this effort the wave strikes the boat kindly, and passes on; but if allowed to follow her up astern, so surely as such an experiment is tried, the sea will either curl over the stern, or the boat will broach-to and take it over the gunwale.

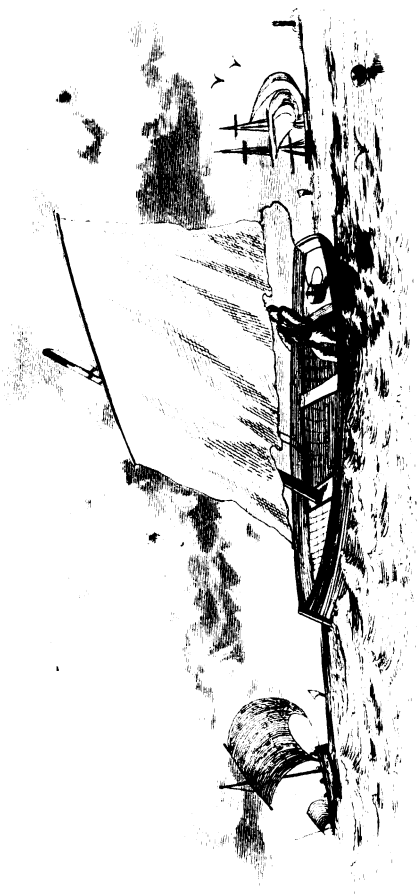
^a A drogue is a conical-shaped collapsible bag, about two feet in diameter at the mouth, and four feet six inches in length. When towed by the mouth, the drogue fills with water, and draws heavily; thereby checking the progress of the boat. A tripping-line is made fast to the apex or pointed end, and by slacking the towing-rope and hauling on the tripping-line, the drogue collapses, and may be drawn on board very easily.

It is much to be regretted, that the crews of wrecked vessels who take to their boats in moments bordering on despair, should recklessly endeavour to gain the shore amidst the fury of the gale, driving their boat through heavy surfs, ignorant of the risk they incur; and, as a dead certainty in such a case, the boat must be swamped. Now, if they could only command sufficient presence of mind to back their boat when heavy seas threaten them astern, and keep her bows on when pulling in the teeth of the gale, they might often land in safety. It is, however, more advisable to keep out at sea, during a gale, provided the boat be kept stem on, than to incur the risk of forcing her through breakers. A boat will not rise so buoyantly over surf, as over an unbroken wave.

Short boats, with high sides, are not equal to long ones with low sides, in a heavy sea: the short boat would be thrown end over end, whilst a long one, under skilful management, may be taken through the surges in safety.

Many practical and experienced sailors are ignorant of the principles or necessary acquirements for managing an open boat in a heavy sea; and when the hour of danger arrives, no wonder at their courage forsaking them, as they abandon the wreck and hasten to their certain doom in an open boat.

No class of men, either at sea or ashore in our Maritime Isle, or in nations far and distant, understand the management of boats in a sea way, so well as those fishermen and boatmen who pursue their daily avocations on the most exposed parts of our coast: these men learn from daily experience, the safest mode of conducting a boat through the difficulties and dangers before mentioned: and they have found, and well proved, that the safest plan, when a wave threatens them astern, is, to face the danger boldly, and drive the stern of the boat at the very crest of the wave, with all the impetus the oars can give. And they are always careful in heavy seas, to keep the bows and stern as buoyant as possible; suffering no one to sit there, and no ballast to be stowed in either of the ends.



A SQUALL.

BEADON'S IMPROVED PATENT⁹ SELF-ACTING BOAT-SAFETY-REEL.

The self-acting boat-safety-reel is the invention of Captain George Beadon, R.N.; it is intended as a means of saving boats from capsizing, when suddenly struck by squalls or heavy gusts of wind. The inventor has spared neither time nor expense, to make the reel as perfect as possible.

The engraving represents a boat struck by a squall; the main-sheet being released by the self-acting safety-reel, thereby saving the boat from capsizing. The action of the reel appears to be simple, but infallible; there are no cogs or catches of any description; it consists of some secret contrivance contained in a small box, rather ornamental than otherwise, fixed to the thwart of the boat; and so contrived, that when the boat careens over to a certain extent, the sheet instantly releases itself, and flies off the reel; it requires no care, and cannot, without design, get injured. The reel can be instantly unshipped from the thwart, and thrown aside in the boat; it is very durable, and will last a great number of years,—quite as long as a well built boat. Its dimensions are so inconsiderable, that it occupies but little space, varying according to the length and size of the sheet.

The boat-safety-reel is so constructed that it can be regulated at pleasure, so as not to permit the boat to heel over further than to any desired bearings, without letting go the sheet; which instantaneously relieves the boat from the pressure of the wind acting upon the sail; so that safety does not depend on individual attention, and no movement need be made on the lee side of the boat. The apprehension of the most timid may thus be consulted.

The coil or spare end of the sheet is carried on a reel, from whence it runs off freely when required; and being protected from any obstruction, there is no possibility of its becoming foul or entangled.

⁹ The patent has recently expired.

The sheet, when on the reel, cannot twist or kink, and not being so much exposed to the wet, will not be liable to choke the block by swelling.

The boat-safety-reel being self-acting, is independent of all attendance, so that all hands may be plying the oars, when it is necessary to pull and sail together; instead of having one or two hands solely to attend main-sheet, halliards, &c.

There being an extra power combined with the safety-reel, for hauling aft the sheet, the sails of the largest boats can be trimmed with ease.

AS TO THE MANAGEMENT OF SAILING-BOATS IN A GALE.

"It blew great guns, when gallant Tom,
Was taking in a sail,
And squalls came on in sight of home,
That strengthened to a gale."

DIBDIN.

In order to manage an open sailing-boat, or a small yacht, with safety, in a gale, a man must have had considerable experience in boat-sailing: there is always, more or less danger to be apprehended; and those to whom the control of the boat is intrusted, should possess skill, nerve, and discretion. But with all these, and years of experience combined, it is often beyond the power of man to prevent accidents under such trying circumstances of wind and weather.

No experienced sailor would, under ordinary circumstances, venture to set sail in a gale: but there are occasions, when unexpected gales overtake sailing-boats as well as ships, and at such times it would be well to know how to manage the boat with the greatest chance of safety. It is always of great importance that the crew should have confidence in their boat, and also in the man at the helm: firmness of nerve, decision, and good judgment, are highly essential qualifications, at such a time: many boat accidents occur through fear, indiscretion or mismanagement. The man at the helm should

be bold, but wary : he will require all his skill and all his daring ; for, as his courage forsakes him the danger increases : the moment he gets unnerved he becomes bewildered ; and on the approach of danger, before he can do what is really necessary, perhaps the boat is capsized or the mast carried away.

On signs of an approaching gale being detected, attention should be immediately turned to the sails, which should be reefed as close and snugly as possible ; and all head-sails that can be dispensed with, should be lowered and taken in : all the canvas must be inboard ; and if the boat will obey her helm without a jib, it will be advisable to draw the bowsprit inboard, to save it from being carried away when the boat pitches. If the little craft will not "wend" without a jib, a spit-fire should be set, or the very stoutest and smallest little head-sail that can be found. If the boat is a two or three-masted lugger or schooner, or if it has a mizzen, the main-sail should be lowered and entirely dispensed with ; for the boat may be sailed under a fore-sail and mizzen ; or, if preferred, these may both be lowered ; and the boat sailed under a close-reefed main-sail, or a try-sail, if furnished with one.

It is necessary to attend to the trim of the boat, under the trying circumstances of a gale, and to see that there is no ballast before the mast, in the bows of the boat ; or in the extreme end of the stern. The head and stern must be kept as light and buoyant as possible. The ballast should be all judiciously spread over the floor of the boat, and a platform firmly secured upon it, so that no part of it can shift, though the boat should lurch or pitch ever so heavily.

Another very important caution is, to attend to the mainsheet, and see that it is never, for a moment, made fast ; but always in a clear coil, free and ready to run out on any sudden emergency.

In a heavy sea, with a gale or strong wind, the boat should not be sailed so near the wind, nor the sails set quite so flat, as if in smooth water with a moderate wind.

Take care to keep the sails full, for directly they shake, the

H

boat loses way ; and it is one of the most important principles of boat-sailing (particularly in time of danger or difficulty), to keep good way on the boat ; in which case, she will the more readily obey her helm. Whenever the boat has no way on her, she is in danger, if the sea is heavy and the wind strong.

Another precaution is necessary to be observed, particularly when scudding, or sailing with the wind abeam, in a heavy sea ;—*take care that the sail does not gybe* : if it should, the boat will assuredly be capsized, or the mast carried away, unless the main-sheet be perfectly free, and let go at the critical juncture.

Whenever signs are observed of an approaching gale, it is always best to shorten sail in time, and prepare for the worst.

Let us now suppose a crew of three persons in a boat, overtaken by a gale ; the boat having three sails set,—main-sail, fore-sail, and jib ; and no harbour is nearer than ten miles, and that dead to windward ; there is every prospect of an increasing gale, so let the fore-sail run down, take one reef in the main-sail, and set a smaller jib, take a second reef in the main-sail, and try her to windward : the mast bends like a twig, and the little bowsprit threatens every instant to snap off ; luff the boat up, and set the very smallest jib you have ; but in all these movements let the man at the helm keep the main-sheet clear, and ready to ease off at any moment ; he must not leave the helm an instant, let the two other hands attend the sails ; haul the jib-sheet aweather, whilst a third or fourth (if there are so many) and last reef is hauled down in the main-sail : the waves are now running high, and the boat is pitching heavily ; try her cautiously to windward :—she flies through it madly, and must be eased or luffed a little, as the approaching waves meet her ; let one hand stand by¹⁰ the main-sheet, and one at the jib-sheet, whilst the other remains at the helm cautiously watching the

¹⁰ To “stand by” does not strictly imply that the man should be on his legs ; it is far better that he should be sitting or kneeling, in a small boat : a man may “stand by” a rope, in any position : that is, —be ready to haul in or let go.

threatening waves. In luffing to the heaviest, the least motion of the tiller will generally suffice: be careful not to allow the boat to lose all way, or she will not answer to her helm: take advantage of every smooth sea, which usually follows three bouncing waves, to get good way on; keep her full and keep her at it, and only ease the helm on the approach of a heavy wave that is likely to drive the bows of the boat under; then luff, as it were, into the very crest of the wave, which will impede the boat's progress for a second or more; and such must be regained by bearing up instantly, but slightly, to get the sails full again, that the boat may not roll over into the trough of the seas. Be not frightened at the boat's rising and falling with the waves, so long as she answers to her helm; if she can be kept from broaching-to there is nothing to fear; on that hangs all the danger. No one should give orders but the man at the helm; and he should bawl out so that his voice may not be lost in the wind; his orders should be instantly obeyed; as in his position he can see exactly what the boat can bear, and when she can bear it no longer: the crew should cringe down as low as possible in the bottom of the boat: every rope should lie in coils, clear and free from kinks and hangings; and every movement of the crew should, if possible, be performed without standing up.

The boat should not be sailed so near the wind as if in smooth water; but the sails must be kept full; and it will be found, that the faster the boat goes, the quicker she will obey her helm: many boats are upset by large fore-sails, therefore do without them in a gale, if possible, and set a storm-jib instead; but if no jib, set a very small fore-sail. If the sea increases so much that it is impossible to prevent the waves from breaking over into the boat, it will no longer be judicious to attempt turning to windward.

Avoid putting the boat about, unless it can be done in smooth water; the most perilous time for open boats in a heavy sea is, when in stays. It will, therefore, be more prudent to bear up, and if a port can be reached with the

wind abeam, it will be a safer mode of sailing than running directly before the wind. If, in scudding, it should be found that the boat has too much sail, lower the peak of the sail a little, according to your judgment. If, in running before the wind, the sea travels faster than the boat, and threatens to run over the stern, shake a reef or two out of the main-sail, and run her as fast as possible ; but not so as to bury the bows under water : get into smooth water quickly to get out of danger.

Boats and vessels are often wrecked through the rudder being unshipped, or carried away by a heavy sea. All sea-going boats should, therefore, have their rudders hung and secured in the safest manner possible.

TO RIDE OUT A GALE AT SEA.

This may appear like a bold undertaking in an open boat, or small sailing-vessel : it is, however, sometimes not only practicable, but the only means of saving the boat from being swamped, and consequently the crew from being drowned. When any attempt to reach the land, or force the boat ahead, would be certain destruction, then is the time to consider the best mode of keeping her afloat, and averting the surrounding dangers. This may be done by lashing a few spars together, and casting them overboard ; allowing the boat to ride by them from a rope, at the bows, made fast to the middle part of the spars. In this manner the boat's head may be kept to the wind, and she may be prevented from broaching-to. With good sea room, an open boat may so ride out a heavy gale. It is astonishing how a raft of the kind breaks the force of the sea, and so fights the battle of the waves for the boat ; and this contrivance may be used whether the boat be laid-to under a small sail, or without one. In the absence of spars, or anything wherewith to form a raft, a loosened sail attached to a yard, or an oar, will answer the same purpose ; and if a sail be used, a weight may be suspended from the clew, which will impede the

drift of the boat. If the water is not very deep, and you have a small anchor with sufficient rope, the raft may be anchored, and the boat will still ride in safety.¹¹

SAILING CHARIOTS.

“ But who is this, what thing of sea or land?
Female of sex it seems,
That so bedeckt, ornate, and gay,
Comes this way sailing
Like a stately ship
Of Tarsus, bound for th’ Isles
Of Javan or Gadier,
With all her bravery on, and tackle trim,
Sails filled and streamers waving,
Courtied by all the winds that hold them play.”

MILTON.

It appears, from many authorities, that sailing chariots are a very early invention: though there is but little trace of their having been much used in this country; and for reasons which it is not difficult to imagine; for nothing can be more certain to frighten horses, and cause accidents to passengers on the road, than a sailing chariot.

¹¹ When the “Flora Temple,” a first-class ship, was wrecked in the China seas on her passage from Macao to Havannah, in October, 1859, and 850 Coolies were drowned; throughout the seven days whilst the gale lasted, the long boat, with thirty-one hands aboard, including the Captain, was *hove-to under a close-reefed main-sail, with a bucket veered out to twenty fathoms*. The boat was an open one, but they rode out the gale at sea in the manner stated, and their lives were saved, though they suffered severely from exposure to the wind and spray.

Sailing chariots have long been known and used in Holland. In "A Description of Holland," published in 1743, are contained a few remarks on one belonging to Prince Maurice, which he kept at Scheveling, a village in the neighbourhood of the Hague. This chariot is said to have been made by Stephinus, a great mathematician. The form of it is stated to be plain and simple, but resembling a boat moved upon four wheels of an equal size, and steered by a rudder placed between the two hindmost wheels: it had two sails. The mode of stopping the chariot was, either by luffing, or lowering the sails.

It is also stated that, in the space of two hours, it would pass from Scheveling to Putten, a distance of forty-two miles.

Walchius¹² also speaks of the Scheveling sailing chariot; and affirms it to be of such prodigious swiftness in its motion, and yet of so great a capacity as to its burthen, that it far exceeded, in swiftness, any ship under sail, with ever so fair a wind. That, in the space of a few hours, it would convey six or ten persons twenty or thirty German miles; and that with little labour to him who sits at the helm, who may easily guide the course of it as he pleases.

Milton, in his "Paradise Lost," thus speaks of sailing chariots being employed on the barren plains of China:—

" But in his way lights on the barren plains
Of Sericana, where Chineses drive
With sails and wind their cannie waggons light."

Bishop Wilkins¹³ gives a chapter on sailing chariots; in which he says, they are commonly used on the Champion plains of China. He also speaks of the Scheveling chariot.

Sailing chariots are also mentioned by Grotius in several of his epigrams.¹⁴ And Hondius, in one of his large maps of

¹² *Fabularum Decas.* Fab. 9.

¹³ "Mathematical Magic," by Bishop Wilkins, 1680. Book ii. cap. ii.

¹⁴ *Gro. Eps.* v. xix. xx. and xxi.

Asia, gives a conjectural description of such as are used in China.

Bishop Wilkins gives two illustrations of sailing chariots; one of which, containing seven persons, is a cumbersome boat-like contrivance, placed upon four wheels, all of the same size, and fitted with two masts and square sails; the larger one being placed aft, and the smaller one in front. This chariot appears to be provided with a rudder, which terminates at the lower end in a point or spike, and which, it is presumed, can be raised or lowered at pleasure.

Wilkins's second illustration is a suggestive one, which he thought might be more conveniently framed with moveable or revolving sails, shaped after the manner of a volute propeller, so as to impel the chariot with a force proportionally equivalent to that of a windmill. The fore-wheels in this are much smaller than the hind ones. But the revolving sails are so contrived as to catch the wind from any quarter, and set the volute in motion, and, consequently, the wheels of the carriage. Now, it appears that, if these volute sails could be used in a similar manner, and with similar effect to the wheel of a screw steamer, a good deal of power might be got out of them.

And, with sails so contrived, the chariot might be made to go against wind as well as with a side or back wind; and the occupant or driver, by a very simple machinery, could, at any moment, stop the chariot by stopping or reversing the volute.

SAILING BARROWS.

It is stated in the Journal of Van Braam Houckgeest, in his "Embassy to China," that sailing barrows are used in China; but the sailing apparatus is merely an additional contrivance to relieve the toils of the barrow-men when the wind is fair. These sailing barrows are described as having a little mast, very neatly inserted in a hole or step, cut in the

fore-part of the barrow : to this mast is attached a sail made of matting, or more commonly of canvas, five or six feet high, and three or four wide ; with reef-tackle, yards, and braces, like those of the Chinese boats. The braces lead to the shafts of the barrow, and by means of them the conductor trims his sail.

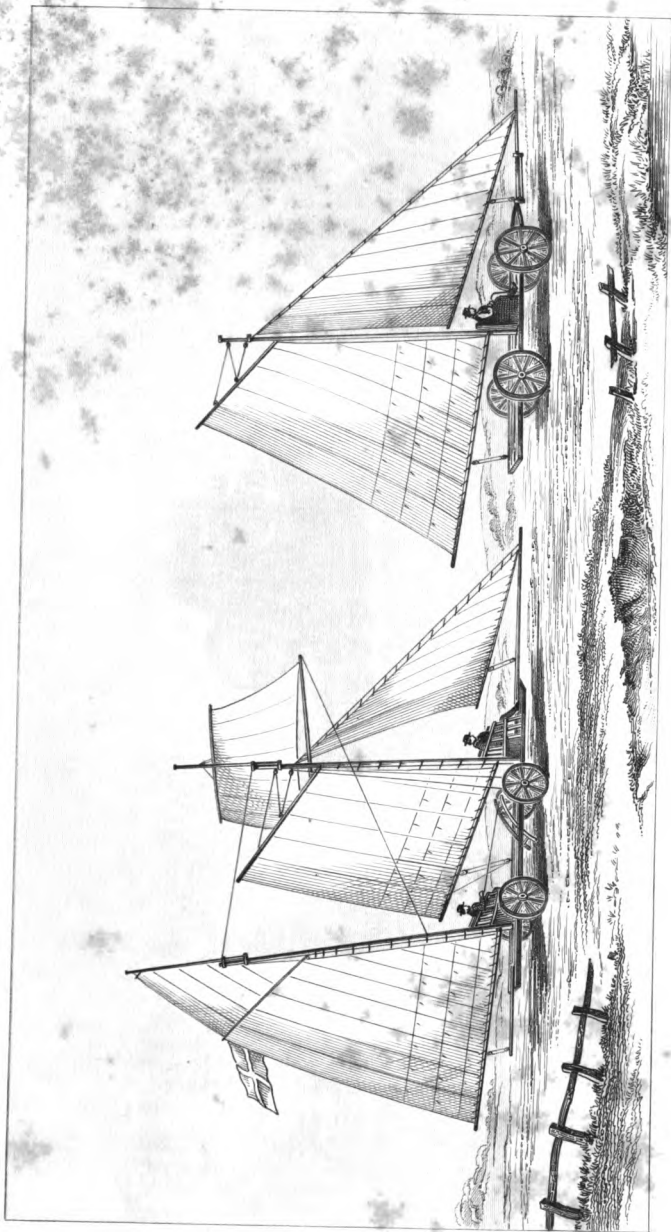
Van Braam says, he could not help admiring the contrivance, and felt real pleasure in seeing a score of them rolling along one after another.¹⁵

THE SHULDHAM LAND-SAILING-BOAT.

This is a modern invention by Commander Molyneux Shuldham, the author of many useful and valuable nautical contrivances. It is one which afforded considerable amusement to himself and friends when prisoners of war at Verdun : they having been desired by General Wirion (the Commandant of the English prisoners) to discontinue the use of their sailing and rowing boats on the river Meuse, because of a complaint and petition of the fishermen, that "the boats frightened all the fish away!" But Mr. Shuldham, determined not to be deprived of the pleasures of a sail, invented a land-sailing-boat. The first was with one mast, the other a schooner ; both being represented in the engraving on the opposite page, which is made from drawings kindly furnished by Mr. Shuldham for the express purpose of this work. The description, which he also supplied for the same purpose, is given as nearly as possible in his own words.

There is no doubt that wind power can be used with advantage in various ways, and on many occasions ; and land-sailing-boats may afford a good deal of sport and

¹⁵ "Je ne pouvais m'empêcher d'admirer cette combinaison, et je goûtais un plaisir réel en voyant une vingtaine de ces brouettes voilières cinglant l'une à la suite de l'autre." P. 150. tom. i.



LAND-SAILING BOATS.

amusement, on plains, commons, and hard sands by the sea side, and elsewhere.

DIMENSIONS OF THE SCHOONER-RIGGED LAND-BOAT.

Length end ways, between the axle-trees	5 feet	1½ inches.
Height of the wheels	2 "	4 "
Extreme distance from the lower part of one wheel to that of the other, across	5 "	1½ "
To the upper part of ditto ¹⁶	4 "	3 "
Extreme breadth of the frame	2 "	1½ "
Length of the axle-trees	3 "	9½ "
Length of the spreaders for the rigging.	5 "	4 "
Length of each mast to the hounds ...	13 "	0 "
Mast head	1 "	8½ "
Distance between the masts	7 "	4 "
Length of the fore-top-mast	9 "	0½ "
Top-gallant-mast	2 "	9 "
Top-sail yard	6 "	0 "

Weight of the boat, including masts, sails, &c., 500 lbs.

The general speed of this boat, on a rather rough high road, was seven or eight miles per hour; but with a fresh breeze, it would keep pace, and sometimes beat a horse at full gallop. A boat of double the size would sail much faster, but the inventor adapted the breadth of beam to the ruts of the high roads, so as to be steered and handled easily by one person alone at the helm. The boat was stopped, when required, by an anchor formed of two handspikes framed together, so as to revolve on a bolt fixed amidships in the boat; a weight being affixed to the upper ends sufficient to balance and keep their lower ends clear of the ground. To the upper ends was affixed a small tackle, which was hooked on to the right arm of the steersman's seat, so that when he wished to anchor, he had only to pull upon the tackle-fall with his right

¹⁶ The wheels were dished inwards.

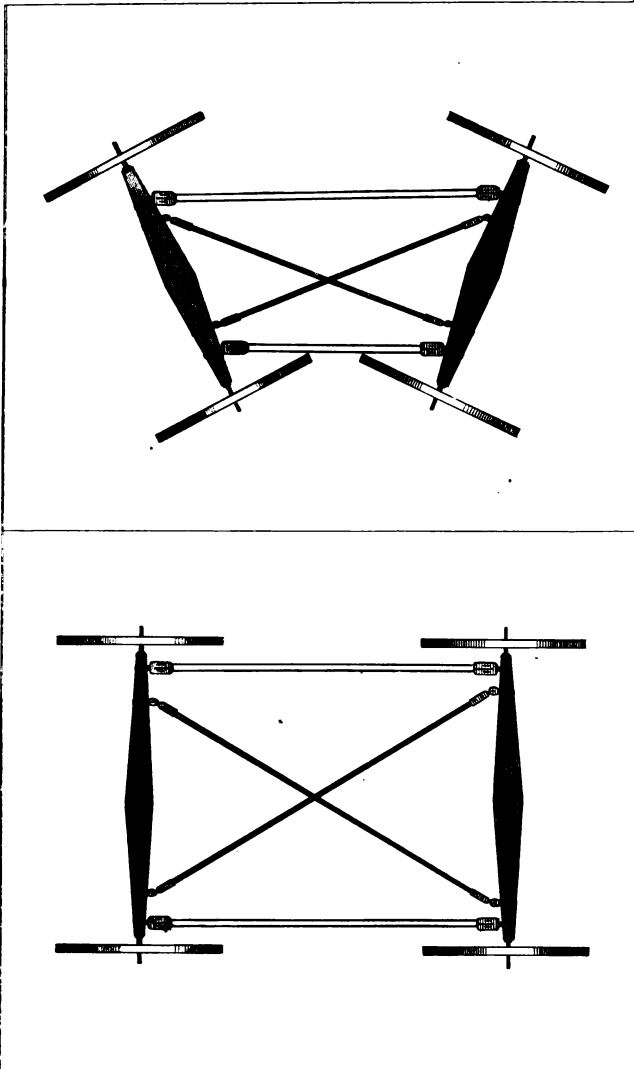
hand, whilst steering with the left; the double handspikes would then scrape on the ground, and eventually stop the boat without the necessity of shortening sail.

The greatest peculiarity, and indeed one of the most important features in Captain Shuldham's land-sailing-boat, is the steering apparatus, which is partially explained in the opposite illustration. Two double pulley blocks were fixed to each of the axles, forming two tackles. Suppose the starboard tackle to be the first rove, and the fall led through a single pulley fixed to the frame of the boat, and led upwards through another pulley fixed across the end of the fore-arm of the steersman's seat, then led athwart and downwards on the port side, and then reeve the other tackle; hook the blocks to their respective axletrees, haul the fall moderately tight, and make it fast. As the fall passes conveniently in front of the helmsman, he can, by clasping it with both hands, and moving them *to and fro* sideways, steer with the greatest ease. The mechanical purchase of the tackle is eight-fold; whilst the overhauling friction of the blocks is so great, that the concussions of the wheels, on passing over stones or rough roads, are scarcely felt in steering.

In turning quickly, as in tacking or veering, the fall of the tackles must be worked hand over hand.

The diagonal braces of the steering apparatus were absolutely needed, for, without them, the axletrees could not be brought parallel to each other when the boat was required to be steered straight. The diagonals were of ash, and sufficiently elastic to accommodate themselves to the shorter distance which the axletrees were from each other when at right angles, to what they were when parallel.

The inventor's plan of the steering apparatus seems to have fully answered his expectations; all the wheels being rudders, the boat was very lively, and could be turned so quick that the inventor used to tack her so as to fill the sails, and a slight push ahead would then set her off again.



LAND-BOAT STEERING APPARATUS

When the boat was sailed on a fair road, she made good way about five points from the wind; and, with sufficient space, was worked to windward with ease.

The inventor specially remarks upon the advantages of the wheels being *dished inwards*, as the stability of the boat is not only much increased thereby, but also the strength of the wheels; in consequence of much of the strain being lateral upon the lee wheels when the boat is sailing upon a wind, or with the wind abeam. To please critics, the inventor once tried the experiment of placing them upright, to the detriment of the boat's sailing, and was soon glad to replace them in their original *dishing* position.

With reference to the application of the steering apparatus to other vehicles, it may be as well to remark, that the late Admiral Sir Sidney Smith took out a French patent for it, which, however, he applied to what he termed a "Balance Carriage," mounted upon six wheels; the diagonals being applied to the fore and the after axletrees; but such an arrangement could not answer, because the carriage would turn just twice as quick as the shaft horse, or the pole horses. It appears that Captain Shuldham was so fully aware of this, that whenever he had to be towed in his land-sailing-boat he only required a tow rope, independent of the boat's being directed, excepting by steerage.

The great objection to using a land-sailing-boat on the high road is, the undeniable fact of its frightening every horse that comes near it. It appears, the inventor, in spite of all his precautions, was the cause of several accidents; on one occasion he upset a farmer's wife in a ditch, and this provoked the neighbouring farmers to declare war against the boat, which they did by pelting it with stones, and riddling all the sails. On another occasion, when forcing the boat up one of the steepest hills on the road to Paris, the horse of a French cavalry officer was so much frightened that it reared bolt upright, and threw its rider.

Once, when sailing the boat on the top of Mount St. Michael, the inventor was caught in a heavy squall, which

upset the boat ; but it seems to have turned over very easily, without injury to the occupant, or damage to the boat ; and, although quite alone, he managed to right the boat ; but in descending the Mount at railway pace, he ran down a cow, the bowsprit touched the cow's buttocks, when she, fortunately, jumped clear of the road, or much mischief might have ensued.

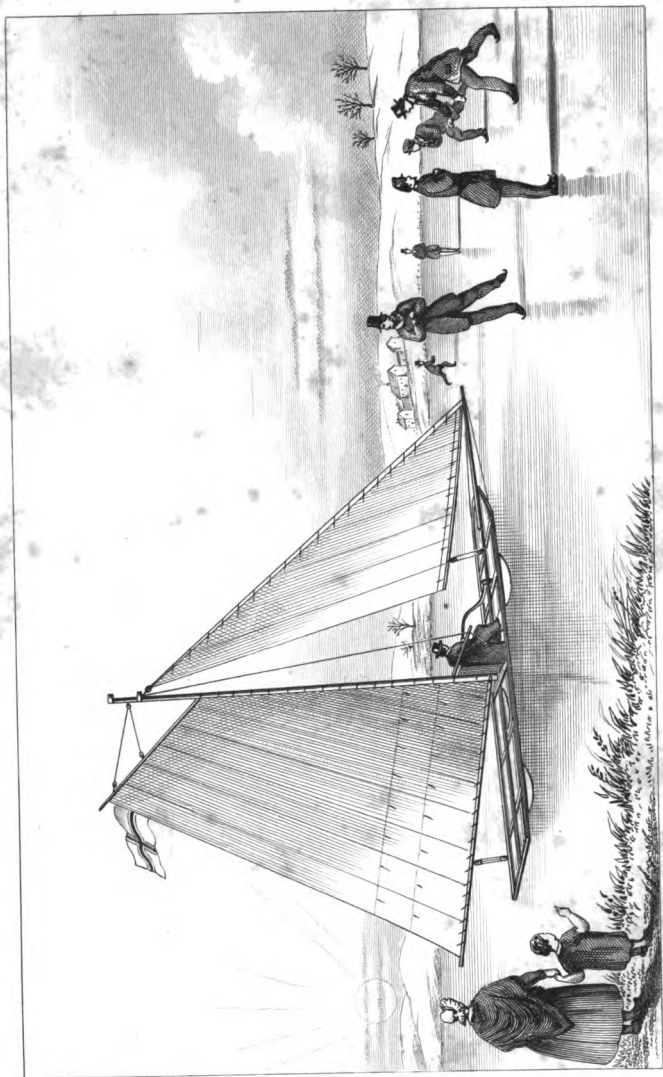
The inventor, on one occasion, carried as many as thirteen persons on his schooner-rigged land-sailing-boat.

ICE-BOATS.

Though seldom seen or employed in England, ice-boats are a good deal used in Canada, when the smaller lakes are frozen over. When ingeniously constructed, they sail at a flying pace, and the slightest wind suffices to move them along.

Canadian ice-boats consist of a long, narrow, light boat placed upon a cradle, with three large skates attached to the bottom, one on the fore part, and the other two aft ; and forming, as it were, an acute angled triangle. The boat is provided with an iron rudder, or sharp blade, which can be dug deeply into the ice. They are generally fitted with sails after the sloop rig, but other fore-and-aft boat sails are sometimes used. They can be put about with ease, sail at great speed, and may be kept very close to the wind.

Ice-boats are also common in Holland, where they are found very useful in winter, in conveying goods and passengers many miles along the dykes and lowlands of that country. These boats are placed in a transverse form upon a two-and-a-half or three inch deal board ; at the extremity of each end are fixed irons, which turn up in the form of skates. Upon this plank the boat rests, and the two ends serve as outriggers to prevent oversetting ; ropes are fastened at the ends of the plank, and leading to the head of the mast in the nature of shrouds ; others are passed through a



ICE BOAT

block across the bowsprit. The rudder somewhat resembles a hatchet with the head downwards, which, being pressed down, cuts the ice, and serves all the purposes of a rudder in the water, by enabling the helmsman to steer.¹⁷

Wilkins¹⁸ also mentions, that in Holland small ice-boats are used, having sledges instead of wheels, and being driven with a sail. But the bodies, he describes as being like little boats, so that if the ice should break, they might still hold the occupant safely upon the water.

THE SHULDHAM ICE-BOAT.

This boat is constructed in the simplest manner possible, a mere framework of wood, to which skates are affixed; and with only one seat like an arm-chair, which is made very strong, and permits a long ash spreader to be bolted to it at the back; at the extremities of the spreader, notches are made to receive two strong shrouds, which are secured to the body of the boat. In the centre line of the frame, two cutting skates are affixed, the foremost one serving for a rudder, and both being allowed to penetrate the ice just enough to prevent the boat making any lee-way. The rudder has a handle, and the helmsman steers the boat similar to a Bath invalid chair.

According to the engraving, it will be seen that the seat for the helmsman is placed just before the mast, which latter is supported by it. The boat is rigged with only two sails, (main sail and jib,) an arrangement which admits the carrying very low canvas, as the main-sail almost sweeps the surface of the ice. The sails are laced to booms, the jib-boom just shortened enough to clear the helmsman's knees. In turning to windward, the sails work themselves; which is very necessary, from the quickness with which the boat performs all her evolutions.

¹⁷ Encyclo. Brit.

¹⁸ Mathematical Magic. By Bishop Wilkins.

In consequence of the weight of the boat being so light, the inventor made her answer with common skates, which saved the trouble and expense of getting others made for the purpose: he placed them on three parallel lines; the after skate and the rudder one, were fixed on the midship line or rail, and those were the only two that were allowed to cut into the ice, in order to prevent lee-way, and were ground as sharp as hatchets. All the others (four in number) were curved so as to offer the least possible resistance; two of them were fixed to the midship rail, and two to each of the side rails, which formed the breadth of the boat; only the midship ones were a little deeper, just sufficient to raise the weather skates clear of the ice, and also to prevent the cutting skates from penetrating too deeply into it, merely enough to prevent lee-way; thus the boat was moved upon four curved skates, and two cutting ones. The inventor found this arrangement answered admirably. The chief resistance to the propelling power was in the longitudinal line, amidships, as it ought to be.

The manner in which the ice-boat was stopped under full sail, and brought to anchor, was very simple; by merely turning a screw amidships, with a conical point, the handle of which was placed within reach, so that it could be turned with the right hand whilst steering with the left. The point scraped the ice when screwed downwards, and cleared it when turned upwards, thus giving the power to stop her as gradually as possible, and eventually to anchor without the necessity of shortening sail. The arrangement of this boat would answer for others on a larger scale, by employing longer skates in proportion to the greater weight of the boat; but as the climate of England is unsuitable to ice-boats in general, they could only be used for amusement; and it would be convenient to construct them as lightly as possible, and only large enough to contain one or two persons.

FOREIGN BOATS.

CANOES.

THE term "canoe" is generally understood to imply a boat made by hollowing out the trunk of a tree; and such was a means employed not only by the ancient Britons, but also by the natives of various distant countries; and it still is, in many remote islands, the only mode of construction known.

In most books of voyages and travels, canoes are mentioned. Various nations have different methods of constructing them; even the wild savages have their notions of ingenuity, and form boats out of solid timber, with no other tools than sharp stones and shells, which they find on the coast. With some the process is, first to make a fire round the bottom of a tree suitable for the purpose, and burn it down: the trunk is then stripped of its branches, and hollowed out with the flints and shells, fires being also made in the trunk, when fallen, to burn it hollow. With others, the process of hollowing is accelerated by burning gums on the part they wish to clear away: some having iron at hand, and larger tools than others, proceed as follows:—they cut down a large, long tree, and square the uppermost side; then turn it upon the flat side, and shape the bottom, the outside, the head, and stern; three holes are then bored in the bottom,—one in the middle, and one at each end, down to the thickness they wish to leave it; for without some such gauging they would cut away the inside, and leave the bottom thinner than intended. The tree is turned after the gauging holes are bored; and being propped or fixed with the flat side uppermost, the tedious process of hollowing is commenced. It is usual, in a middle-sized canoe, to leave the bottom three inches thick, and the sides two inches at the lower part, and one and a half at the top: the ends are

afterwards sharpened and finished off to a point. With some islanders, after the tree is hollowed out, the sides are raised with additional planks, which are actually stitched and pegged to the other part of the canoe with tough thongs.

Considerable time and labour is often expended in carving figures on the stem and stern of canoes; which in some countries are ornamented with ivory.

The canoes belonging to Hudson Straits and Greenland are flat-bottomed and flat-sided, and do not represent much of a sea-going quality, still it is surprising the sea they go through when under the skilful management of the natives: the sails of these are made of skins and intestines of the walrus.

The canoes of the South Sea Islanders are larger, stronger, and better constructed than those of other islanders.

The canoes of West Africa are made from a large tree hollowed out to a great width amidships, but gradually tapering at each end, to narrow elevated points.

North American canoes are built of bark in a similar manner to those of Terra del Fuego.

The Caribbee canoe is merely the trunk of a tree hollowed out by burning.

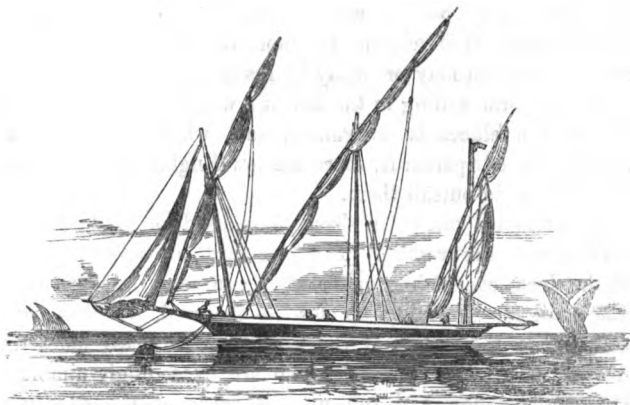
The canoes used by the natives of the Molucca Islands, and by those on the coast of Guinea, are very indifferently constructed out of the trunk of a long narrow tree, *slightly* hollowed; and are in fact little else than floating logs. From their very narrow and heavy form they are constantly liable to be upset; an event of little importance, as they will not sink; besides which, the expert swimming negroes soon right a capsized canoe.

A small-sized canoe is used on the coast of Labrador, capable of accommodating one person only, being entirely covered in fore and aft; leaving a round open space sufficient to admit the body of its occupant on a seat placed in the centre. The length of these canoes which are flat-bottomed, is from sixteen to twenty feet,—the depth about fourteen inches: the breadth at the middle part (where it forms two angles) is about two feet; from these angles it gradually inclines to a point at

each end. These canoes are composed of a very light framework covered with walrus skins ; and may be swept along, with the double paddle, very fast : and they are so light that they may be carried on the head, by one person, many miles. A similar description of canoe is also found in other parts of North America.

Models of many of these canoes are deposited in the United Service Museum, at Whitehall ; and will well repay inspection by those interested in such curiosities.

There are also many other kinds of canoes, some of which will form the subject of a more lengthened description in the subsequent pages of this work.



BOATS OF THE MEDITERRANEAN.

THE FELUCCA.

FELUCCAS have been known in the Mediterranean through centuries past. They were formerly much used by Greek pirates. Those of the present day are improvements upon the original model, and are the largest and fastest description of boat belonging to that sea. The felucca is decked and

provided with cabin accommodation, and possesses, in many respects, the comforts and conveniences of a yacht. It sits low in the water, but is graceful looking and buoyant; it has a high stem and raking stern. In addition to the sails, these boats are often assisted by long powerful oars, during the frequent calms in the Mediterranean.

The rig of the felucca consists of three masts with lateen sails, and a jib. The main-sail and fore-sail are very large and lofty: the yards used to spread the sails are each sometimes in two, three, and four parts, the stoutest spar being in the middle, and the smaller ones lashed securely at the outer ends of the larger; thus the much admired bend of the lateen yard is greatly assisted, and a lighter yard is formed than if all in one spar; besides which, when an accident occurs to the long yard, it is seldom that more than one part is broken at a time, which may promptly be restored.

For general sailing in the smooth waters of the Mediterranean, the felucca is unrivalled; and, when manned by an active crew of Spaniards, there are few English boats of equal size that could outsail them.

Feluccas are various in size; those used as coasters being smaller than the traders. The long yards are seldom lowered to the deck, except in a gale; when two sails only are used. Each sail can be brailled up close to the yard, by brail ropes attached to the throat of the sail; but they are more frequently frapped by the crew, who are exceedingly nimble in that performance, and run up the yards of the sails with peculiar agility.

VENETIAN GALLEYS.

Very similar to the felucca, not only in form and size, but historically, were the Venetian galleys.

So early as the ninth century, the galleys of Venice exceeded those of any other nation: indeed, they were then the most beautiful, as well as useful, vessels of the age. But

those were the days of ancient greatness of the Venetians—days which have long since passed away; and the beautiful galleys of Venice are now superseded by the more sombre, but graceful and interesting gondolas.¹

MALTESE GALLEYS.

The galleys belonging to the Island of Malta were originally important vessels of war used in the Mediterranean Sea. These also can be traced back to the same age as the felucca and Venetian galley; and, in the three, there has always been a strong similarity. In fact, the galley of Malta may stand as the representative of the whole class.²

The Maltese galley is a vessel furnished with a great number of oars, which are found very useful in the frequent calms of the Mediterranean; it is a long, low vessel, that draws but little water, has usually three masts, and huge, lateen sails, the middle one, or main-sail, being the largest. The interior of the galley is arranged with rowing benches, where the crew, who work the long, unwieldy oars, often undergo great hardships. The word of command is given by a person standing at the stern, who blows a whistle for that purpose; and thus a regular stroke of the oar is maintained. Some of the galleys are upwards of 150 feet in length, and thirty feet wide, and are propelled by fifty oars; each oar is fifty feet long, and balances in the row-lock, which enables it to be used with greater facility.

When about to board an enemy, the crew assemble on the "rambade," or platform, erected across the prow.

In sultry weather, a striped awning is spread above the deck, over the whole length of the galley, from stem to stern.

There is in the United Service Museum, a beautiful model of a Maltese galley of the 18th century.

¹ Described at page 165.

² "Charnock's Marine Architecture," vol. iii.

THE MISTICO,

Or Mystaco, of the Archipelago, is a decked vessel, with a long, low hull, used formerly by the Greek pirates; it was rigged with two short, stumpy masts and large lateen sails, much resembling the felucca of the Mediterranean.

PORTUGUESE CAHIQUES.

These are fast-sailing pleasure-boats, used at Lisbon, and other places, on the coast of Portugal. The cahique rig consists of two lateen sails—a jib and mizzen; the latter is set in a similar manner to a Bermudan sail, the yard serving the purpose of mizzen-mast. It is an exceedingly pretty rig, and answers well on boats having a long or narrow-shaped hull.

THE CATRIA.

The catria is a fine open sea boat, employed by the fishermen and pilots of the Douro. These boats are thirty feet long, by about six feet in breadth amidships, and three feet deep; they have, also, considerable sheer. When under oars they are pulled by twelve or more, the rowers sitting double banked. Being often exposed to heavy seas and broken water, they are provided with a large rudder, extending deeply below the bottom of the boat.

The catria is rigged with a large lateen, or rather settee, sail; which is all set inboard, the tack being hooked in the bows, and the clew sheeted astern.

These boats are managed, by the native boatmen, with skill and dexterity; particularly when crossing broken water, or going over a bar under oars.



THE GONDOLA.

“ Now rest thee here, my gondolier,
 Hush, hush ! for up I go
 To climb yon light balcony's height,
 While thou keep'st watch below :

Ah ! did we take, for Heaven above,
 But half such pains as we
 Take day and night for woman's love,
 What angels we should be.”

CHRISTOPHER GRUM.

AMONG the objects of interest at Venice, that beautiful “ city in the sea,” none are more worthy of observation than the pleasure boats.

The streets being principally canals, the gondola is the chief means of conveyance to and from all parts of the city. The gondola is usually about thirty feet long, and five broad ; of peculiarly light and elegant form ; and very elaborate workmanship is often displayed in the exquisitely carved and

high peering prow. A closed compartment is constructed in the centre, for passengers, and fitted with windows, cushions, carpet, and curtains; much resembling the interior of a gentleman's carriage. The gondola is as much in request on the Venetian canals, as cabs and omnibuses in London; and for the same purpose of moving from house to house, and from street to street. The oar of the gondola is worked in an iron rowlock, at the quarter. The gondolier impels and guides the boat, with graceful motion.

The annual procession of the brides of Venice, is thus poetically alluded to in an old legend.

“ And through the city in a stately barge
Of gold were borne, with song and symphonies
Twelve ladies, young and noble. Clad they were
In bridal white, with bridal ornaments,
Each in her glittering veil, and on the deck
As on a burnished throne, they glided by.”

There was formerly so much rivalry in the magnificent manner in which these boats were painted and gilded, that it was considered expedient to establish a law, prohibiting any other colour than black being used on the exterior of all passenger-vessels plying within the waters of Venice; except the gondolas of state, they only being allowed gay colours, the head and stern of which in some instances are gilded and richly ornamented; and the interiors are also fitted up in a very costly manner.

The only equipage of the noble Venetian is his elegant gondola; in which he, his lady, family, and friends, are conducted to all places of amusement, visits to friends, and other excursions. Coachmen and grooms are supplanted by gondoliers and boatmen; the rattling sound of carriage wheels is never heard; and nothing save the splashing oar and musical cry of the gondolier, denotes the movement of the lifeless craft.

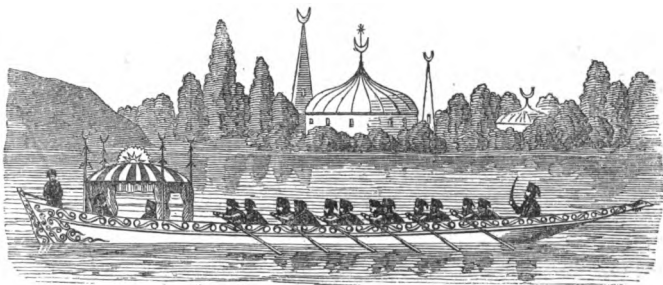
“ And gliding up her streets as in a dream,
So smoothly, silently,—by many a dome.”

There are also smaller gondolas, large enough for one or

two persons only, called gondolettes; these are light and buoyant as the little wherries of the Thames, and are moved about very swiftly and with little exertion; but are rowed differently to boats of other European nations; the rowers always sit facing the prow, and with a light pair of sculls, by a reverse motion, forcing the handle *from* the chest, instead of drawing it in, the boat is worked swiftly ahead. On approaching cross canals and corners, the gondoliers, by a musical cry, signal their approach, in order to avoid collision with other gondolas, which would otherwise frequently be inevitable.

The Grand Canal is the fashionable parade of the Venetians; and in fact, what the Champ Elysées are to the French, or Rotten Row and Hyde Park in London, to the English. Gondolas are their equipages; the sombre colour of which, shows off to advantage the gay dresses of the fair occupants, and the smart liveries of the gondoliers.

The gondolas belonging to certain Venetian corporations or societies, such as those of Chiozza, are of more elegant and attractive proportions than those which are commonly let for hire; and they are generally decorated in an antique style.



TURKISH CAIQUES.

SOME of the caiques of Constantinople and other cities on the Bosphorus, are as pretty, light, and graceful, as any boats in the world; so that it would appear there is something to

learn in boat-building, even from the Turks. In some respects they are more slender and fragile than our famous Thames wager-boats. They are very flat-floored, and swim buoyantly on the water, though often heavily laden with eight or nine of a crew, besides two or three passengers. They are lined throughout, and fitted up with exquisite taste,—elegant cushions to sit upon, and a carpet for the feet.

The prow and stern are particularly graceful, and often exhibit some elaborate carving, with the figure of a peacock, pheasant, or some other beautiful bird, as an ornament to the prow. The upper sides are also finely carved; so also is the stern, back-board, and cross rail at the bows: and these are also sometimes covered with gilt. Even the stretcher-boards for the feet of the crew, are beautiful and delicate pieces of carved work, which yield with a springy elasticity to the muscular force of the rowers. The oars are very light and slender, and of a scientific and beautiful form; they are worked in brass rowlocks, neatly covered with leather.

The caiquechees (native boatmen) are extremely cleanly and particular about their boats, in which they take great pride and delight; keeping sponges and brushes for wiping and cleaning them, and soft leathers for rubbing the brass and ornamental work: and they always take off their shoes before stepping in, and never suffer any one to get in with dirty boots.

No one is allowed to sit in the raised parts of the graceful bows and stern, for fear of disturbing the trim of the boat, besides disfiguring the elegant carving and fancy-work. Strangers are assisted into the caiques with the greatest care; and luggage is removed with nice caution and quiet management.

On landing their passengers, the caique crew always turn the boat, and back ashore stern foremost. But notwithstanding the elaborate manner in which these caiques are fitted, the boatmen are generally very poor, but obliging and grateful; though very cunning, for they never appear to have any change.

BOATS OF THE ITALIAN LAKES.

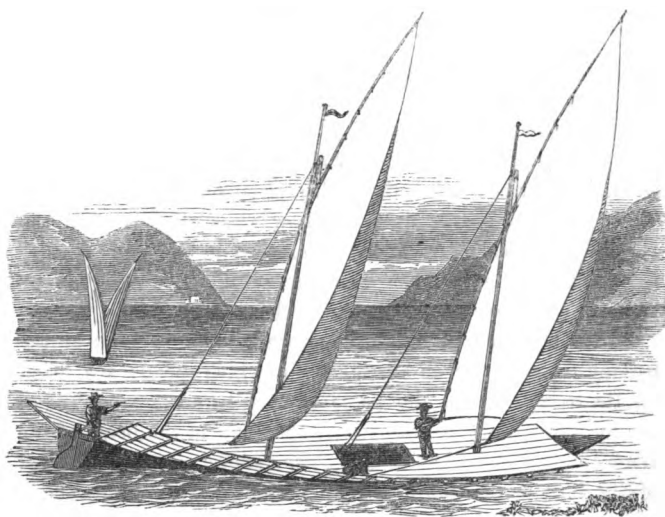
THESE boats, like those of the Swiss lakes, have considerable rake both fore and aft: thus, a boat twenty-three feet long by seven feet wide has five feet rake at the stem, and three feet at the stern; the bottom being flat, and only three feet wide, but flaring out to the width of seven feet at the top. The sides are nearly flat: the bow rises slightly, and the stern a good deal. A canopy is erected in the middle, over the seats destined for the passengers; this is supported by three broad hoops in the form of arches.

In the boats of the Lago Maggiore, the lower part of the canopy frame forms an outrigger, in which thowls are fitted for the oars; and in the boats of this lake the canopy is in the fore-part, and the frame is slight; but in the boats of the Lago Lugano and Lago di Como the canopy is in the aft part of the boat, supported by a stout framework of broad wooden hoops.

These boats all have large rudders, and are generally sailed under a square-sail, made of a thin homespun flax or hemp; the mast, in some of the boats, is without either block or sheave-hole, but has a fork at the top, over which the halliards go, and are belayed in the aft part of the boat, in the most primitive manner.

The sheet is usually hitched to one or other of the hoops forming the archway of the canopy. The sail is sometimes ornamented with coloured stripes or checks, and not unfrequently is bent to the yard with straps of its own material.

Many gentlemen, who have villas on the banks of these lakes, are provided with boats of English form and rig; these are not, however, copied by the natives.



BOAT OF THE LAKE OF GENEVA.

BOATS OF THE SWISS LAKES.

THESE boats are, for the most part, rude and clumsy contrivances, possessing few features worthy of remark ; still they are a rather numerous class, and should not be altogether passed over without notice.

Sailing boats, or rather *barges*, are very numerous on the lake of Geneva ; some of which, with their large lateen sails, are pretty-looking craft. There is also a peculiarity about them such as is seldom seen in the boats and barges of other nations. They are considerably broader on deck than elsewhere, the deck being constructed so as to flange out

considerably over the sides like a platform (see engraving), and thereby giving them an immense space of broad deck for stowage and conveyance of light goods.

These barges have high bows and sterns, and are usually rigged with two masts and lateen sails.

There are also at Geneva rowing and sailing boats in imitation of English models.

On Lake Constance are also sailing barges, rigged with tall masts and large square sails; and some are rigged with main-sail and fore-sail. The rowing-boats on these lakes are very similar to those of the other lakes referred to.

There is a great similarity in the passenger-boats employed on the Lakes of Lucerne, Brienz, and Thun; they are built of fir planking, an inch thick, fastened to great heavy timbers and knees; they are awkward and ancient-looking craft, sloping at the sides, and raking at the ends. These boats are about twenty-two feet long, and five feet wide at the top; but at the bottom, which is flat, they are about sixteen feet in length by three feet in breadth. The passengers sit in the middle, under a canvas awning, like the tilt of an old-fashioned road-waggon.

When the wind is fair, these boats are sometimes sailed with a lug-sail; but they are ticklish and unsafe under sail, except in very light winds.

Sailing-barges, a good deal like the Thames river barges, are also sometimes employed on the Lake Lucerne.

NORWEGIAN BOATS.

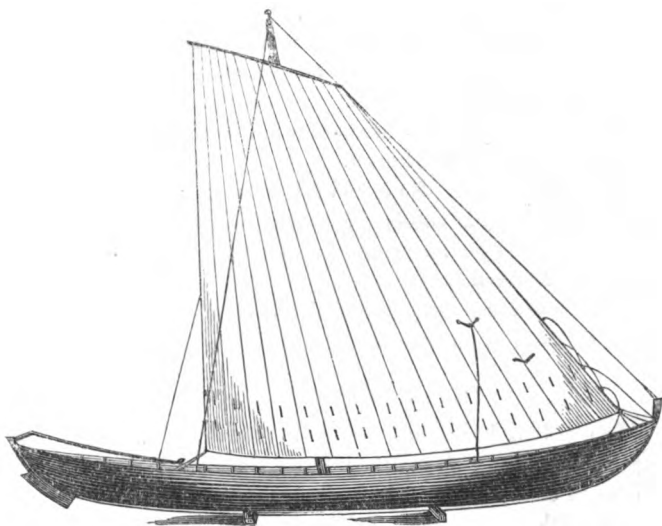
NORWAY YAWLS.

“For now in our trim boats of Norway deal,
We must dance on the waves with the porpoise and seal;
The breeze it shall pipe, so it pipe not too high,
And the gull be our songstress whene’er she flits by.”
Claud Halcro’s Norse Ditty.—“PIRATE.”

THE Norwegians are in no way behindhand with other nations in the ingenuity displayed in boat-building: the peculiarities of their yawls and praams are very remarkable, and entirely original. They are built without nails; the planking being entirely joined and fastened throughout with wooden pegs. In shape and model the two classes are totally different; the one is of wedge-like form at the bottom, whilst the other is so round at the bottom as to be nearly flat. The small Norway yawls are used chiefly for rowing, and the larger ones for sailing: the general rig under which they are sailed is a single lug-sail, the lower part of which is considerably broader than the upper.

SONDMORE BOATS.

The fishing-boats of Søndmøre, on the West coast of Norway, are very fine and beautiful open boats, alike at both ends, but long and graceful-looking, though built without a gunwale or top-rim; the boat merely consisting of ribs or timbers, and the outside planking, with high stem and stern pieces, and rather deep keel. The rudder is large and broad at the bottom, extending below the keel; and provided with a very long tiller, which fits into an elbow that is attached to one side of the rudder’s head. The tiller is long enough to reach nearly to midships; so that there is no occasion for



any one to burden the boat by sitting in either of the ends ; and the helmsman may sit just abaft the mast, and steer the boat with ease—a great advantage when under sail in a heavy sea.

The rig of these boats is an excellent one ; it consists of a very useful and ingeniously contrived lug-sail, very narrow at the top, but broad at the bottom (see engraving) ; and there are the same number of cloths in the top of the sail as in the lower part, the canvas being worked gradually narrower all the way up. The mast is rather lofty, and there are several shrouds or stays, leading from the top above the short little yard, that are held down or made fast in the boat (not to the gunwales or rim, for there are none) with little wooden fids or thumb cleats ; there is also a fore-stay, that is made fast to the stem.

The sail is set in the fore part of the boat (see engraving), and the tack is made fast at the stem. The reef-tackle forms

an important part of the rigging, and is very complete; two reefs reduce the sail considerably, it being so much wider at the bottom than at the top. The lower part of the boat is divided into several compartments by bulk-heads, that reach up to the level of the thwarts.

The ballast (which consists of a few large stones) is placed amidships, just abaft the mast.

These boats are also provided with long powerful oars, which are worked, not between thowls, but in straps, with a pin or standing-bit for leverage; the loom of the oar, where it chafes, is squared.

A very complete and beautiful model of one of these boats was exhibited in the International Exhibition of 1862.

NORDLAND FISHING-BOATS.

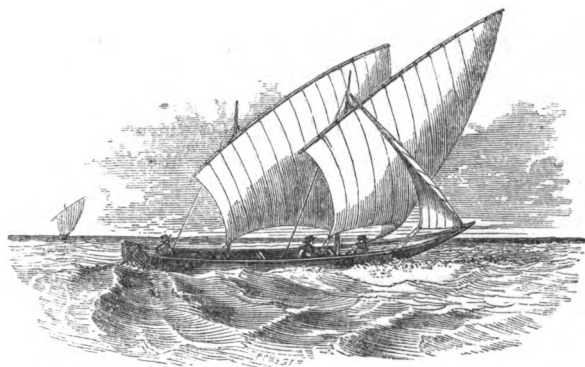
The fishing-boats of Nordland (North of Norway) are nearly similar in form, though not in rig, to those of the Western coast, but not quite so raking at the bows. They are rigged with a lofty mast and lug-sail; though not near so pretty in shape as the Sondmore boat-sail: it is in fact a mere longitudinal square-sail without a peak, but a little broader below than at the top. These boats also have bulk-heads similar to those last described.

FINMARKEN FISHING-BOATS.

The boats of Finmark are similar to those of the West coast of Norway, as regards the form of the boat, and being *minus* both rim and gunwale; the stem, however, is by no means raking, but rather inclining inwards at the top, and therefore presenting a full, or prominent, fore-gripe. They are rigged with an ugly, square-shaped lug-sail.

The fishing-boats of Drontheim are nearly identical with the above, both in form and rig.

The larger, or decked, fishing-boats of Finmark are of stronger build than the others, and have square sterns; they are rigged as cutters, but not very gracefully. They have a cabin-house amidships, that occupies the entire breadth of the boat.



THE PRAAM.

This is also a boat of Norwegian contrivance, and of peculiar form : the praams, like the yawls, are built without a single nail, being tacked together by means of wooden pegs ; notwithstanding which they are very secure, as well as strong and durable.

The shape of the praaam is very similar to a boat-scoop without a handle, the greatest breadth of beam being at the stern ; the head and bows gradually rising forward, and sometimes finishing with a carved figure-head. Small praams are used exclusively for rowing ; these, in general, have no keel.

The sailing-praaam requires a good depth of keel at the stern, on account of its very flat floor, and the buoyancy with which it sits upon the water. Some of the largest of these boats sail remarkably well, when judiciously rigged and ballasted.

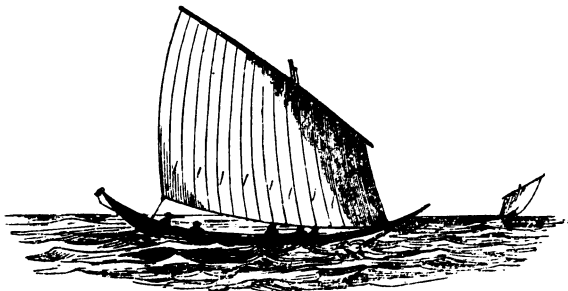
The engraving represents a large sailing-praaam, fitted with two masts and lug-sails, running before the wind, with the fore-sail hauled aweather—a common method of steadying a boat in a seaway.

The boats employed on the coast of Denmark are very similar to the Norway boats, especially the praams, some of which are longer and narrower than the Norway praams, and have a long tapering bow.

NORWEGIAN PILOT-BOATS.

The pilot-boats of the South-west coast of Norway are rigged with three sails; viz., sprit-sail, fore-sail, and jib; but the sprit-sail is rather unsightly-looking, inasmuch as it has no peak, but is purely a longitudinal square: it is fitted with four reefs; the fourth or uppermost reduces the sail to less than half its size.

These boats are decked all over, and have no bulwarks, but simply a stout rail on each side, amidships, extending nearly to the bows forward, and within a few feet of the stern. The rudder is large, and broad at the lower part.



FAROE ISLANDS FISHING-BOATS.

The fishing-boats of the North Faroe Islanders are much like those of the Western coast of Norway, having very high peering stem and stern. The prow and stern are alike; they have a deep keel, and rather deep rudder. These boats are rigged with a lug-sail of English-like shape, widest at the bottom; but the tack is not hooked so far forward in the bows as usual with English boats.

The oars are heavy-looking, with square-shaped loom, and are worked in leather straps, secured to the sides of the boat with standing bits for leverage, similar to the Norway fishing-boats.

The Faroe Islands boats sail remarkably well; the large-sized ones in particular, which are fine-looking craft, and good sea-boats.

LAPLAND BOATS.

All the boats of the Laplanders (even the large ones) are extremely light; they are made of a few very thin planks of fir, attached to a keel and ribs; the planks being sewn together with sinews of the reindeer, which resembles English catgut. This light form of boat is rendered necessary by reason of the number of cataracts that are so frequently met with on the rivers and lakes of Lapland.

In some parts of that country, where the natives are enabled to procure cord or twine, they use it in boat-building, instead of the sinews of reindeer; but before doing so, they generally rub or dress it with a kind of red glue, which they prepare from the skin or scales of fish.³ The cord or twine, on being dressed with the glue, is rendered much more durable, and is impervious to air and water.

The small fibrous roots of the fir are also sometimes used by the natives in boat-building, when neither cord nor sinews can be procured.⁴

The skiffs of the Laplanders are of peculiar form; the keel is about one fathom in length at the bottom, but extending to two fathoms upwards, being equally high and pointed at each end; the floor is of flat form; the ribs, or timbers, on each side are only three or four in number, and very small; these are covered and doubled with thin fir planks, that are joined with the sinews or other cordage before mentioned. When completed, after the strongest manner, a man might crush them between his arms; and, on stepping in and out, care must be taken to tread only on the bottom, for any hard pressure on the sides would injure them.

³ Regnard's Journey to Lapland.

⁴ Ehrenmalm's Travels in Lapland.

FINLAND BOATS.

The boats of Finland are, also, extremely light, being built after the same manner as those of Lapland.

They are made of thin strips of fir, sewn together with the sinews of the reindeer, or dressed cord, and fastened to a skeleton composed of a keel and ribs; they are, also, buoyant and flexible. But, notwithstanding the lightness and flexibility of these, it would appear, they are very strong; for, when borne by the violence of the torrent, they are exposed to rough usage, and frequently strike against stones and rocks which abound in the rivers of that country; but they appear to bear such shocks without injury.

Many a stranger would be struck with alarm on first witnessing the perils to which these boats and their occupants are exposed, when borne rapidly down a torrent of waves, foam, and stones, in the midst of a cataract; the noise of which is so deafening that it is useless for any one of the crew to attempt speaking to the other; all must be done by signs, and the boat controlled by individual skill and courage. The Finland boatmen are particularly bold and skilful in the management of their boats in cataracts, and are everywhere else ingenious in steering small vessels. One dauntless Fin stands at the stern, and steers with an oar, whilst two others row as hard as they can, in order to escape the danger of quick following waves, that threaten to overwhelm them. The perils encountered on passing through some of these cataracts are astonishing, and the boats are driven along with great rapidity, apparently diving into waves and raging torrents of foam, as if never to rise out of them; and then again appearing on the crest of a lofty wave, with keel exposed to view, and disappearing again as if going down endways; and during these perils, there are others around, in the shape of rocks, stones, and other obstructions, which, by judicious steering, the pilot has to avoid.⁵

In the travelling-boats, the Fins always insist on the

⁵ Pinkerton's Voyages and Travels, vol. i. Journey of Maupertius.

passengers landing when they come to a dangerous cataract ; and in this they consult their own interest, as well as the safety of the passengers ; for, the lighter and more buoyant their boat, the less is the danger incurred.

Most of the larger boats are furnished with a lug-sail, and a mast that can be raised and lowered at pleasure. Sails are found of great service to the boats of this country, where smooth rivers and lakes abound, though often intercepted by dangerous cataracts ; and it is in passing through these that the calmness, courage, and skill of the Fins are particularly conspicuous.⁶

HOLLAND BOATS.

FISHING SCHUYTS.

Dutch boats have a very antiquated appearance, not only in the form of the hull, but also in the rig or form of their sails. They are built with great breadth of beam in proportion to length, and are very flat-floored, have little or no keel, considerable fore-gripe and dead-wood ; with a lee-board on each side, which, when let down, hangs a foot or more below the bottom of the vessel, and answers the purpose of a keel in preventing lee-way. These boats come from their country to the north and east coast of England, with the fair winds of March, to fish in the Channel in company with the English and French vessels.

Before making a voyage, Dutchmen are said to wait not only for a fair wind to blow them to their destination, but

⁶ Trans. from Journal of a Voyage to the North, by M. Outhier.

also to wait a few days after the fair wind arrives, to see if it holds. Having once arrived at the fishing-ground, be the weather ever so rough, they seldom make for a harbour; but generally ride out the gales at anchor, or drive about wherever the wind blows them, laid-to, with foresail hauled to windward. When provisions are wanted, the schuyts are run bump ashore on the beach, where they remain until the crew have made all necessary purchases.

BOËYER RIG.

Dutch boats are rigged in a very antiquated style. The Boëyer rig is that used for the lesser craft, the main-sail of which has a very short gaff attached to the top of the sail, which, in small boats, is only a foot or two in width; the mast is placed in a perfectly upright position; a large foresail is worked on the fore-stay, but no jib; the rig is confined to two sails set upon a lofty mast; and the whole of the lower part of the canvas is inboard.

But little paint is used on the outside of Dutch boats, the preference being given to bright *varnish*. A number of polished wooden fenders are usually hung over the bows; and whilst they are useful in saving the hull of the vessel from scratches and bruises, they are also useful and ornamental to the craft. One peculiarity in Dutch vessels is, that the fore-stay is held taut by an enormous dead-eye, or block of wood attached to the stem, and having a number of holes pierced, through which lanyards are neatly rove; forming at least a dozen separate parts, and holding, very securely, the fore-stay to the stem of the vessel.

SPIEGEL RIG.

The Spiegel differs from the Boëyer in two respects only; in the first place, the sail is a little wider at the head than in the Boëyer-rigged boats, consequently a rather longer gaff is

used, and with more bend, so that the sail is not only wider, but more arched at the head than in the other, but nevertheless very narrow in proportion to English sloops and cutters; in the next place, the Spiegel-rigged boats have a short bowsprit, and carry a very small jib. A pleasure-boat of this rig, although forty feet in length and thirteen in width, drawing only four feet of water aft, by three-and-a-half forward, carries a mast fifty-six feet above the deck. An ugly rudder (necessarily large and broad at the bottom) is attached to all Dutch vessels: the boom seldom extends over the stern; but Dutchmen are careful sailors, and never overcrowd their boats with canvas, a firm deck footing being studiously adhered to under all circumstances.

Dutch boats and yachts are invariably sailed under one or other of the two rigs: much labour is often expended on the stern and aft body of Dutch pleasure vessels, the carved workmanship of which is beautifully polished, and ornamented with gildings. They are also furnished with lee-boards, as already described; these are hauled up when sailing free, and lowered on a wind, by means of a tackle attached to the outer part of the bulwarks.

CHINESE BOATS.

“I saw their boats, with many a light,
Floating the live-long yesternight.”

THERE is no other nation in the world where so great a variety, and such countless thousands of boats are met with, as on the rivers, and at the sea-port towns in China. But it is very remarkable, that although there has been considerable advancement and improvement of late years in the art of boat

building by the Chinese—some of their modern boats being as perfect in model, and as elegant in form, as those of any nation—yet the advancement of the Chinese, in naval architecture, in no way keeps pace with the sister trade of boat building. The same form of Chinese junk which furrowed the deep centuries ago, with its high towering stern and lofty mat-sails, still navigates the seas in that same antiquated style; and this too, though the Chinese shipwrights have daily before their eyes the splendid ships of England, France, America, and other nations alike famed for skill and advancement in naval architecture, whose vessels are adapted for quick voyages, capacity and convenience as trading ships, and comfort and safety at sea in hard gales. And notwithstanding that the Chinese people are fully aware of these facts, they appear totally careless and indifferent of such advantages, and adhere to their favourite old junks and lorchas, as if they possessed a superiority over every other form of vessel in the universe.

It is stated by Dr. Downing, in his “*Fan-qui*,” that “the boats on the Chinese rivers are, in the most essential respect, the same as houses of other nations on the land. They may be compared to the habitations of people living in a great city, and you soon become accustomed to look upon them in the same light.”

“From Whampoa to Canton the river is like a crowded highway, on which junks, ships, and boats, jostle one against another like ‘busses and cabs in a London street, managing, with similar dexterity, to avoid collision.’”⁷

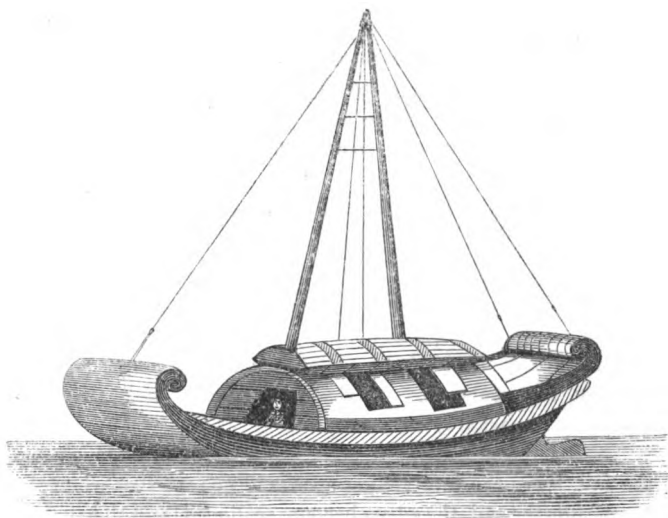
At Canton, the crowd of boats, of all sizes, shapes, and colours, passing to and fro, with the “hubbub and clamour of ten thousand different sounds, coming from every quarter, and with every variety of intonation, make an impression almost similar to that of awe upon the first visit of the stranger.”⁸

⁷ “*Recollections of a Three Years’ Residence in China*,” by W. T. Power, D.A.C.G. 1853.

⁸ “*The Fan-qui in China*,” by C. T. Downing, Esq. 1838.

When lighted up at night, with their variegated and coloured lanterns (for which the Chinese are famous), the scene is a peculiarly attractive and interesting one, particularly when viewed from a commanding position.

At Canton every boat is registered, whatever its size. In the year 1833, the whole number on the river, adjacent the City at Canton, was eighty-four thousand, a large majority of which were tankeä (*i.e.* egg-house) boats.⁹ Many thousands of these have recently been swept away, and completely destroyed by the terrible typhoon of the past year (August, 1862), when upwards of forty thousand persons were destroyed through that memorable disaster.



CHINESE RIVER JUNK.

Although tens of thousands of boats be assembled together on the Chinese rivers, yet good order and decorum is kept to a certain extent. All boats with the same kind of goods

⁹ Chinese Repository, vol. i.

are moored together in tiers, at a certain place on the river allotted to them by the local magistrates, and watchers and other officials are appointed and set both by night and day,¹⁰ to enforce regularity and conformity to the orders of the magistrates. Fairs and markets are also held on the river ; and trades of all kinds are carried on by the people, who form the aquatic population of the Chinese rivers.

Many of the boats which are brought out at regattas, and on gala days, are beautiful models of light and elegant workmanship. But most of these are copies of English wager-boats, galleys, and such like ; and the imitation is admirable. The Chinese boat-builders are said to keep the few English wager-boats they have, as models from which to make their own. The only astonishment is, that the ship-builders have not done the same as regards ships.

All the Chinese *sea-going* craft have a great goggle eye painted on each side of the bows ; the river craft, alone, are without that ornament. And, instead of iron, they use generally *wooden* anchors, which, though clumsy and inelegant in appearance, are tolerably efficient.

CHINESE SAILS.

The sails of most of the Chinese boats, although made of matting and bamboo, are decidedly neat ; and they are generally remarkably well cut.

The bamboos, which are passed across the mast to stretch the sail, leave the intervening spaces loose ; so that when the breeze fills the sail, the edges become scalloped, and the surface presents a pretty wave-like appearance. The Chinese manage these sails with great dexterity, and some of their boats sail very well.

In one of the plates to Sir George Stanton's China, is an engraving styled "Economy of Time and Labour." It

¹⁰ Nieuhoff's China.

exhibits a Chinese waterman sitting at ease in a sailing-boat, steering it with one hand, managing the sail with the other, and pulling a large oar with his foot, at the same time smoking his pipe with perfect *nonchalance*. It appears that very large boats are frequently managed by one man, who, with great dexterity, will run, full sail, through a whole fleet of vessels, in the manner represented.

Instead of pitch, the Chinese use a cement like putty, which Europeans call "Chinan."¹¹

CHINESE MODE OF PADDLING.

The paddles used by the Chinese boatmen are made of light wood, shaped after the manner of a garden spade. They use the paddle on one side only of the boat, holding it with one hand on the top, and the other about a third from the bottom. In this manner the upper hand steadies it, whilst the lower one drags it horizontally backwards. By a repetition of strokes, the boat is drawn ahead; and in order to prevent its driving too much on one side, on account of the force being all on one quarter, the paddle is inclined more or less at the close of each stroke, so as to guide the boat as if by a rudder. Persons unaccustomed to such a mode of paddling would find it difficult, as the little *san-pan* would spin round and round like a whirligig; but the Chinese manage them with a good deal of skill. Dr. Downing says he has seen a little girl ply the paddle so dexterously as to send the boat straight as an arrow across the current of the river, when an English sailor has given up the experiment in despair, and drifted fast to leeward.

FAMILY BOATS.

On entering a Chinese port, a stranger would think half the population lived in boats; and, indeed, there are thousands of families, living, throughout the summer, entirely in boats.

¹¹ Osbeck's China.

These family boats are constantly coming up and down the river in pairs. When at anchor they are arranged in regular rows forming long streets in small craft to move up and down among them, and broad channels for the larger vessels. The lesser craft are managed with uncommon dexterity by rowers and girls, who paddle and push their boats threading their way through the mass among the never-ceasing noise and patter of countless thousands of voices. Screaming and countless little children are also crawling about the decks with a few gourd or bladder tied at the back of their necks as "float-buys" when they tumble overboard, and with those they are taught, in infancy, to swim.

The family boats all have wooden or bamboo cabins or houses, of a size proportioned with the boat.

CHINESE FAST-BOATS

The hull of a Chinese fast-boat is about thirty feet in length, by eight or nine in breadth: much higher out of the water abaft than forward, and fitted with a large flat-roofed cabin in the centre, the other parts being decked over with moveable boards, and arranged into compartments as cupboards and kitchens. They are exceedingly comfortable inside. When there is little or no wind, oars are employed, and a cabin-house is erected overhead, as a protection from the intense heat of the sun. This is taken down when a breeze springs up, and the sails are hoisted.

These boats are rigged with two masts and sails, the masts being stepped, and supported by shrouds and stays in the usual way. The sails consist of squares of matting sewn together, and spread with a bamboo yard at the top (as lug-sails), also with cross pieces of lighter bamboo below, at the distance of two or three feet rounder all the way down the sail, for the purpose of stretching and strengthening it.

Toward China. The Fan-qui.

These family boats are constantly cruising up and down the river in pairs. When at anchor, they are arranged in regular tiers, forming liquid streets for small craft to move up and down among them, and broad channels for the larger vessels. The lesser craft are managed with uncommon dexterity by the men and girls, who paddle and push them about, threading their way through the maze among the never-ceasing noise and jabber of countless thousands of voices. Screaming and helpless little children are also crawling about the decks, with hollow gourds or bladders tied at the back of their necks as life-buoys¹² when they tumble overboard; and, with those they are taught, in infancy, to swim.

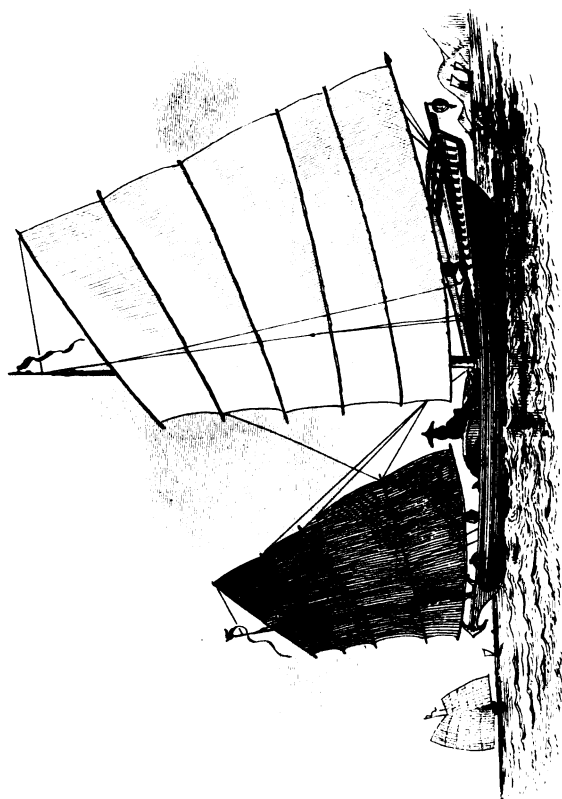
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¹² Power's China. Downing's Fan-qui.



CHINESE PLEASURE BOAT.

The halliards are rove through a block at the mast head, and the yard is slung in the usual way as for a lug-sail. As the sail is hoisted, pieces of string fasten, in a loose manner, the cross pieces to the mast; keeping the sail from flying away, but yet allowing it to traverse freely up and down.

The sail is controlled by numerous guys, or pieces of cord fastened to the yard end, or peak of the sail, and the extremities of the cross pieces; these are all collected, as a bridle, by means of clumsy wooden blocks without sheaves, and ultimately terminate in a single rope or main-sheet, which is belayed upon deck. The rudder is large, and, as Dr. Downing describes it, would be well represented by a moderate-sized grating, with a small capstan-bar attached. It is made to lift up, so as not to exceed the draught of the vessel in shallow water.

Although these fast-boats are by no means strongly built, they are so light and buoyant as to be able to live at sea in the roughest weather, when strong heavy vessels of the same size would assuredly go to the bottom.

In these boats the Hong Kong merchants make voyages to Canton and other places.

The voyage to Canton, a distance of eighty miles, is generally performed in about thirty-six hours; but it is no very uncommon thing to be attacked on the passage by pirates. To guard against such outrages, the fast-boats are generally provided with guns, pistols, and other weapons.

CHINESE PLEASURE-BOATS.

Some of the sailing pleasure-boats of the Chinese are admirable models, though original and peculiar in appearance, with their broad and high sterns, and low-pointed prows; and they sail splendidly when rigged in the most approved style, and fitted with mat sails of a superior make and texture. Many of these boats are fitted up in a very

costly manner, with elaborate carving and gilding. When caught in a breeze, their lofty bamboo sails bear them along at a tremendous pace.

The mat sails are generally preferred by Chinamen to those of any other material; because they hold a better wind, and may be kept flat and stiff. When first turned out from the maker's hands, these mat sails really look very neat and pretty, particularly those made of the superior sort of matting.

Occasionally, a sailing-boat may be seen with sails made of blue nankeen; but it is not generally liked, as it soon gets out of shape, and is not so durable as the mat material.

The sails and rigging of the pleasure-boats are fitted and controlled in a similar manner to those of the Chinese fast-boats already described.

FLOWER-BOATS.

These fairy-like vessels are objects of great attraction to strangers in the Chinese rivers. They resemble large floating bird-cages; though they are of the same shape, and in some respects similar, to the clerk's boats, only larger and more splendidly decorated. Some of them are carved and gilded in a very costly manner; and on the roof, or top, are placed pots and vases of the gayest and most fragrant flowers and shrubs, arranged with great taste; and round and about them are pretty balconies, communicating, by flights of steps, with the beautifully fitted interior. At night, when lighted up with coloured lanterns, they are even more picturesque than by day. These boats, although the greatest ornaments to be seen on the Chinese rivers, are nevertheless one of the greatest disgraces—being occupied by women of easy virtue, and frequented by the Chinese fast-men and wealthy rakes of the cities, who shape their course to them under covered sampans, by the great highway of the river; and

when once aboard, they are secure from public supervision or intrusion; and there they are indulged with the luxuries of fragrant tobacco, opium, choice wines, and *liqueurs*; music, song, dance and mirth, all lending their aid to influence the imagination of the willing victims, and render them an easy prey to the wiles of the fascinating creatures who inhabit them.

None but Chinese are admitted aboard these mysterious vessels. On several occasions when Europeans have attempted to obtain admittance, they have been robbed and maltreated by the men aboard them.¹³

MANDARIN-BOATS.

These are the prettiest of all the river craft; very light and elegantly proportioned, and fitted and ornamented with neat carving and fancy work, and painted with exquisite taste. They are said to resemble "a delicate insect upon the surface of the water."

On board each of these boats a mandarin is stationed, with about sixty or seventy soldiers under his charge, who, with the mandarin, live aboard the boat; and their office is to cruise about the river, and seize all the smugglers they can catch, as well as those who commit offences against the laws. These boats are well fitted with weapons of war, and several small swivel guns.

The manner in which they are usually painted is as

¹³ *Vide* Dr. Downing's Fan-qui. Also Power's China. Mr. Power says:—"Mr. Thom, late Consul at Ningpo, trusting to his knowledge of the language, attempted to gratify his curiosity, and introduced himself in a Chinese garb, but he was detected and captured. He was stripped, and carried on a pole through the suburbs of Canton, exposed to all the insults of the crowd. The ill-usage he met with nearly cost him his life, and quite cured him of any desire to peer too closely into the ways of his neighbours."

follows :—the upper part, outside, bright blue, and all the lower part snowy white. In the blue strake are oval port-holes for the oars ; the latter are white, and the interior of the port-holes red. Sometimes as many as thirty oars are employed on each side the boat.

The mandarin-boat is well decked with a brown hard wood, which is kept bright and polished. The crew squat on the deck, but the mandarin sits at the stern on a handsome mat, and appears to spend the greater portion of his time in smoking.

A light and elegant wooden roof, of gothic form, is supported several feet above the deck, by tall round pillars at each of the four corners. The interior of the roof is painted and ornamented with good taste ; and some of the mandarin pleasure-boats are lined with crimson silk, and the roof edged with gilded carving ; the outside is also neatly painted, the edges being adorned with vermilion and gold-leaf, and fringed with scallops of prepared leather. The seats and couches are covered with blue and scarlet velvet.

During the hottest weather in summer, and the coldest in winter, the roof is covered with mats of paddy-straw, neatly and regularly placed over the top, and thereby giving it the appearance of a newly-thatched cottage-roof.

The mandarin-boat is fitted with two masts, and long tapering top-masts ; ornamented with little flags, pennants, and golden balls.

The sails are made of a very fine sort of matting, neatly sewn together, "and are somewhat of the shape of an acute-angled triangle."

From the ensign staff at the stern hangs a beautiful white flag, marked in the centre with bright scarlet devices.

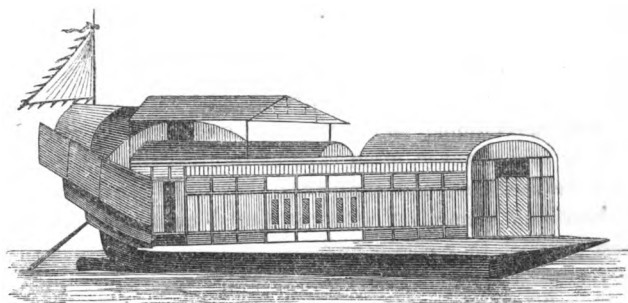
At the stern there is a rail somewhat similar to the tail-board of a hay-cart, projecting upwards in a slanting direction from the deck ; which not only lengthens the vessel, but also serves as a cool reclining place.

These gay-looking boats are exceedingly pretty objects,

and, as they pass swiftly up and down the rivers, are much admired by strangers.¹⁴

CLERK'S BOATS.

These are among the most ornamental of the Chinese boats. They are fair specimens of the floating houses of the Chinese gentry. "They may be said to be genteel residences, the counterpart of which you would expect to find in a small retired street of London." - A good deal of taste is displayed about them in lattice-work, carving, and painting; and the roof is ornamented with pots of choice flowers. These boats are moved about with a pair of large oars, and sometimes a small sail is set above the roof of the building.¹⁵



HOPPO'S BOAT.

The engraving above will more clearly illustrate the superior kind of house-boat used by the middle class of Chinese people.

¹⁴ Downing's Fan-qui. Power's Three Years Residence in China, &c.

¹⁵ For a full and minute description of these remarkable floating residences, see the Fan-qui, vol. i. p. 160.

FISHER-BOATS.

Some of these are remarkably fine and powerful boats : and in point of form and sailing qualities are infinitely superior to the junks and larger craft. It is usual for each boat, or pair of boats, to be the only house, home, and habitation of a whole family, who rely for support entirely on their exertions and success in fishing.

"Clumsy, slow, and ugly as are the junks, and larger vessels, there is nowhere to be seen finer fishing-boats, or river craft better adapted for work, or of greater variety of build." ¹⁶

Fishing is a calling that is exercised by the Chinese with great industry, and on a scale almost unexampled ; the pursuit is conducted not so much on the open sea, amid tempest and peril, but by numerous individuals in the lowest ranks, whose boats are their only abode, and who spend their lives and find their support upon the waters.¹⁷ They do not venture very far upon dangerous seas, but pursue their avocations chiefly on the lakes, rivers, and sheltered bays of the empire—forming, as it were, a nation by themselves.

They are, however, often exposed to great risk ; and after all their hardihood and enterprise, they have great difficulty in providing themselves with the actual necessities of life, most of which they obtain by barter.

DRAGON BOATS.¹⁸

These are so called from their resemblance to dragons. The prow of a dragon boat is in the form of a dragon's head and shoulders, and the stern resembles the tail ; they are very long and narrow. Some of these are used for rowing as well as sailing. For the latter purpose they are rigged with three

¹⁶ Power's China.

¹⁷ Crawford and Murray's China. Barrow's Travels, &c.

¹⁸ Nieuhoff's China. Du Halde's ditto.

masts and light sails of finely wrought bamboo ; and the masts are ornamented with tassels of hair, silk flags, and long feathers : the prow, stern, and gunwales are also hung around with gold and silver fringe : and they are among the lightest, most beautiful, and gayest-looking of any of the Chinese boats.

The festival of dragon boats is observed during the fifth moon, in honour of Kieuh-yuen, a virtuous statesman who drowned himself during the dynasty of Chau (about 2,300 years since), to avoid the displeasure of his Sovereign. The festival lasts several days. The boats used on this occasion are very long, and some pull from eighty to one hundred oars. The festival is kept on all the rivers of the empire. After the races, the boats are buried in mud or sand (to prevent their becoming warped and spoiled), where they remain till the following year. This festival is principally kept up by the different public officers, who frequently stake considerable sums on their boats.¹⁹

SERPENT BOATS, SNAKE BOATS, OR LONG-SCHONS.²⁰

These are very similar to the dragon boats, but are shaped so as to resemble a water-snake, and curiously painted in bright and gay colours. Some of them are one hundred feet in length, although only two feet wide ; they are so narrow that there is only sufficient width for the crew to sit singly : as many as fifty or sixty individuals compose the crew of each boat ; and, by means of paddles, appear to shoot over the water with great rapidity. A regular stroke of the paddles is maintained by keeping time to the beat of a gong, on which one of the crew performs.

Races with these boats take place every year at Canton, and other large towns in China.

¹⁹ Chinese Repository, vol. xx. p. 89, 1851.

²⁰ Downing's Fan-qui. Du Halde's China.

CENTIPEDES, OR SMUG-BOATS.

These are the boats of the Chinese smugglers. They are called Centipedes from the great number of oars employed in calm weather, each of which is fastened to a bamboo thowl-pin ; and so silently do they creep about at night, that they seem as it were to walk the waters with their oars, as if upon so many legs. They are also termed by the mandarins "fast crabs" and "scrambling dragons,"²¹ terms by no means inappropriate, for it is seldom the mandarins are able to overtake them.²² But these boats are not entirely dependent on their oars ; they are rigged with two masts, and mat-sails, similar in shape to those of the mandarin boats, but of a coarser material.

They are large flat-looking decked boats, with considerable breadth of beam, and very smooth bottoms ; they have also high bulwarks, with weapons of war lashed inside. They are all provided with a house or covering made of common bamboo-tiles of matting. They have high sterns, and a tail-board similar to a mandarin's boat, which, without the elegancies, they resemble ; there is no bright paint about them ; they present a shabby brown colour.

They are altogether strongly built, and very fast. The fellows who man them are a desperate set ; and as to mandarins, customs' officers, and others, they set them all at defiance. A small fleet of these boats has been known to frighten and drive off all the Government boats, and then attack and pillage Canton itself.²³

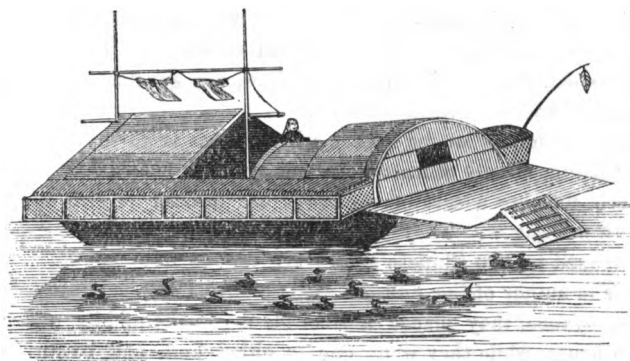
DUCK BOATS.

The duck boats must be ranked among the curiosities of Chinese craft. They are large flat-bottomed boats, with low

²¹ A new class are called "Muscle-shell boats," *vide* Chinese Repository, vol. i. p. 159.

²² Downing's Fan-qui. Powell's China, &c.

²³ Power's China.



sides, and a house erected upon them : there is a broad deck, or duck walk, covered with lattice-work, extending the whole length on each side of the covered parts.

The after, or best part of the cabin or house, is given up to the accommodation of the ducks ; whilst the proprietor, with his wife and family, content themselves with a miserable apartment at the fore-part. At sunrise in the morning the doors of the duckery are thrown open, and the feathered occupants are permitted to waddle round the premises at their pleasure.

Stepping-boards are also let down at the sides, or fore-part of the boat, towards the water on one side, and towards the land on the other. Up and down these the ducks waddle to and fro as inclination dictates, or their watchful owner directs. At roost time, they are called aboard the boat by a whistle, to the sound of which they are taught to return home ; and when they are all on board, the stepping-boards are drawn up, and the birds are then made secure for the night.

TANKEA BOATS.

The tankeâ, or egg-house boats, are probably more numerous than those of any other class. They are generally not more

than twelve or fifteen feet long, and about six broad—some of them much smaller. They are the oddest things of the kind ever seen—nearly as broad as they are long, flat-bottomed, wall-sided, and very shallow, the gunwale standing about six feet above the surface.

In the middle of the boat is a little house or covered cabin, consisting merely of matting spread over half-hoops, or lattice-work, forming a sort of archway; the interior is also lined with matting.²⁴ They are kept very clean, and are each managed by two Chinese girls.

The author of the *Fan-qui* says:—"As I stepped into this conveyance, they placed a stool for me in the middle, under the house, while they went, one to either end, and began to work stoutly at the sculls. Their dress was of blue nankeen, and one of them wore a hood; their hair was not shaved, but divided and plaited down the back, with a scarlet string interwoven near the end. They were good-natured, pretty looking young women, and smiled frequently, exhibiting beautiful teeth."

BARBER'S BOATS.

These are among the very smallest of all Chinese boats. They are about eight or ten feet long, and just wide enough at the stern to admit the body of the Chinese barber, whilst his legs are stretched forward towards the bows; and, in order to trim the little craft, a large stone is placed in the bows to keep the proper bearings.

The barber propels his little boat with a wooden paddle, shaped just like a spade, with which he moves swiftly among the shipping, digging his way, and attracting the attention of his customers by every now and then striking a flat piece of metal with a small steel instrument like a musical fork; with the notes of which, all who have visited Chinese ports are familiar.

Dr. Downing, in the *Fan-qui*, says of these curious little

²⁴ Chinese Repository, vol. i.

boats :—" In leaning over the side of the poop, on a clear fine day, protected from the burning heat of the sun by the awning spread above you, you will frequently hear a sound like that of a large grasshopper, or the striking of a musical fork, proceeding from the water. On looking about to discover the cause, you see a Chinaman, dressed in a blue frock, with one of the great umbrella hats on his head, directly under your eye. This is the barber, in his boat, who, at the same time, turns up his head to see if he has attracted any customers."

WASH BOATS.

These remarkable boats, which are in great numbers at Whampoa and other places, are each under the control and occupation of three or four Chinese girls; who scarcely ever leave them to go ashore, throughout the whole year, unless to attend the gosh-house. They are about twenty feet in length, and of proportionable breadth, with square or cropped ends, and flat bottoms. They are managed by two oars, one at the side and one at the stern; the inside is decked, or covered over with boards, many of which are made removable at pleasure, so as to form extensive cupboards between the deck and flat bottom of the boat. In addition to the deck there is also a house or roof, raised above the gunwales, and supported by wooden pillars. The roof, or covering, is composed of a rough kind of matting formed of thin pieces of bamboo, woven together and arched, and supported on ribs of stiffer portions of the bamboo. Two or three of these coverings are placed upon the framework, one portion overlapping the other, and the whole forming a very good protection from the sun and weather. During cold or wet weather the interior of this remarkable domicile is hung with a lining of rough cloth; and every night, or whenever the inmates seek privacy, the open front of the house is closed with a curtain of matting or cloth. The interior is also provided with a square of matting and a wooden pillow for each inmate. The author of the *Fan-*

qui says : " The meanest beggar in England would shrink from being confined to such a place, yet these girls seem not only content, but even cheerful and happy ; their red, good-natured faces, are to be seen peeping out of the matting, and always with a smile or a laugh at your service."

They apparently get their living by washing clothes for the sailors and petty officers of ships ; they also undertake needle-work and mending for any of the crew, always remaining with their boat fast to the ship until the articles are returned.

CHINESE CANAL BOATS.

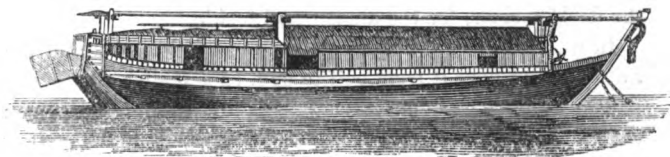
When the wind is foul these boats are propelled by a class of men called "Trackers," who, for a trifling remuneration, are compelled by poverty to perform the part which horses act in other countries—that of towing the vessel a-head by means of a rope, which is laid across the shoulders of the men. These trackers are often subject to great hardships, and severe ill-usage, the labour they have to perform being very laborious ; besides which, they are frequently obliged to wade across small rivulets and delfs of mud, whilst towing the vessels against a rapid current and foul wind. A petty officer attends them with a long lash, which is often used with brutal cruelty on the backs of the poor helpless trackers, should they tire or flag in their arduous labour.

SANPANS, OR SAMPANS.

The Sanpan is a small boat without a keel ; it is used in various occupations in all parts of the Chinese rivers and bays, and in great variety as regards size. Some of the pleasure sampans are of superior form ; the fishing sampan is the largest. Those used for general purposes, as ferry-boats, pilot-boats, egg-boats, fruit-boats, &c., are the smallest, and these are seldom sailed. The fishing sampans are rigged

with one, two, and three masts, the smallest being always placed in the bows ; the sail is usually of square or lug-like shape.

Children of both sexes are taught the managment of the sampan as soon as their strength enables them to pull an oar, hoist a sail, or trim a sheet—and it is no uncommon thing to see a female tugging at the oar with an infant tied to her back, or a fishing sampan sailed entirely by a crew of females.



JAPANESE BOATS.

THE Japanese sailing vessels are very curious, and differ in model from those of any other nation. They have high stem and stern, and both ends of the keel stand out of the water considerably. The form of the hull is tapering from the middle towards the stem ; the stern is the most peculiar part about them, being broad and flat, and provided with a wide opening, extending, in some vessels, more than half-way down to the water's edge, and laying open to view the inside of the vessel. It is supposed that the opening at the stern was originally intended for the management of the rudder ;

but is now, it is said, forced upon the people by a penal law of the state.¹

It is a practice aboard the boats and vessels of this nation to turn the aperture at the stern to advantage (for landing and access to and from the vessel), by unshipping the rudder and using it as a bridge, the large opening at the stern facilitating the proceeding. When lying in port, the rudder is triced up, in the manner represented in the engraving.

These vessels are rigged with one mast and sail, the mast being stepped abaft the middle with a slight aft rake; it depends almost entirely for security on a powerful stay, and is fitted so as to be raised and lowered with pulleys; when lowered it rests above the cabin roof, on bearers provided for the purpose, as shown in the engraving.

The deck consists simply of deal boards, laid loosely, without any fastening; but it is almost entirely covered with the cabin, which stands nearly six feet high. The fore part of the deck is clear, for the anchor and tackle.

In wet weather, and at night, the mast is lowered, and the sail is spread over it from end to end, as a shelter.

The watermen's benches for rowing are towards the stern; the stem is adorned with a knot of fringes, or black strings or hair.

In the common mode of rigging, parts of the sails go to lace and unlace, so that they are reduced, instead of being reefed, by unlacing and removing some of the lower strips of canvas.

Captain Belcher, in his *Voyages*, speaks of the smaller boats he met with, off Nangasaki, having a sail which appeared to traverse by rings on the yard, and to be divisible into four parts, by unlacing them.

In the larger vessels, the sails are divided into many parts in this manner; and, as the outer cords are securely attached to their bolt-ropes, the middle cloths must be those that are removed by reefing.

¹ "Kaempfer's History of Japan," translated by Scheuchzer, 1727. "Belcher's Voyages," &c.

Captain Belcher says the Japanese vessels are not wanting in sailing qualities, if properly fitted with light spars and canvas, but they are crippled by the enormous mast that is put into them.

The pleasure-boats of Japan are a very numerous class, exhibiting a variety almost equal to those of the Chinese. The size, shape, and finish of Japanese boats depends on the taste, wealth, and dignity of the owners. They are, however, generally contrived for rowing rather than sailing.

The class of large boats have upper and lower decks and cabins; the upper one is the principal, being of lofty dimensions, with handsome decorations, windows, and folding screens, and so divided into several partitions; the under cabin is low, and used by the crew.

The prow of the principal boats is generally ornamented with a tassel of long black strings; and when persons of distinction are on their voyages, the cabin is hung about with cloths on which their armorial bearings are embroidered.

Japanese boats are generally gaily decorated with flags of various shades and colours; even the coast-guard boats are so adorned.

The crews of the latter are, usually, all dressed alike; and the boats are propelled with sculls, which they use as oars, the rowers keeping time to a monotonous song.

BOATS OF THE INDIAN ARCHIPELAGO.

THE boats of the islanders of the Indian or Oriental Archipelago are a very numerous class, varying in size from the smallest canoe hollowed out of the trunk of a tree for the navigation of the rivers, to vessels of forty and fifty tons, which navigate the Archipelago from one extremity to the

other. Their variety in form and construction is not less remarkable. Those of each nation or tribe have a distinct character; and the vessels of the same tribe, sometimes on principle, but oftener from caprice, are very various.

The smaller class of vessels are usually safe and swift, the large ones generally dangerous—the failure of the naval architect being proportionate to the magnitude of his attempts.²

FLYING PROAS OF THE LADRONE ISLANDS.

THE flying proas of these islands are among the swiftest and most extraordinary sailing-boats in the world. Through ages past they have been the only vessels used by the natives.

The invention of the flying proa is one which would do honour to the most educated and ingenious of mankind; and yet it appears to have originated entirely with an unlettered people, the inhabitants of a few small, outlandish, and remote islands. The natives are no less dexterous in the management of the proa, than in the building of it.

The flying proa is admirably adapted to the peculiar navigation of the Ladrões, lying as they do, all of them, nearly under the same meridian, and within the limits of the trade winds, and thereby requiring vessels, for passing from one to the other, to be more particularly fitted for sailing with a side wind: and when we examine the uncommon simplicity and ingenuity of its construction and contrivance, and consider the extraordinary speed at which it sails, we shall, in each of these particulars, “find it worthy of our admiration, and meriting a place amongst the mechanical productions of civilized nations, where arts and sciences have most eminently flourished.”³

* “History of the Indian Archipelago,” by John Crawfurd, F.R.S., 1820.

³ Lord Anson’s “Voyage Round the World.”

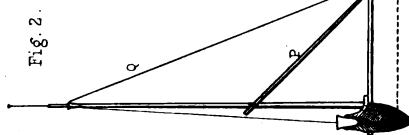


Fig. 1.

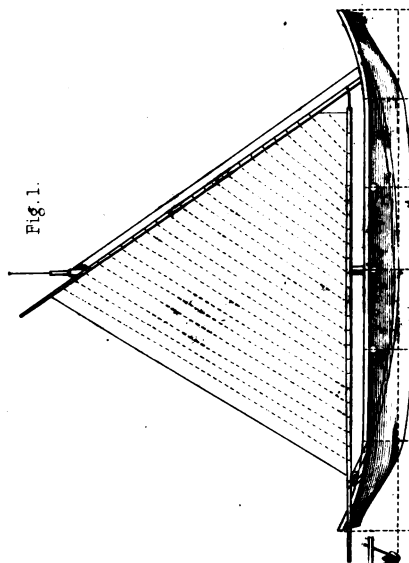
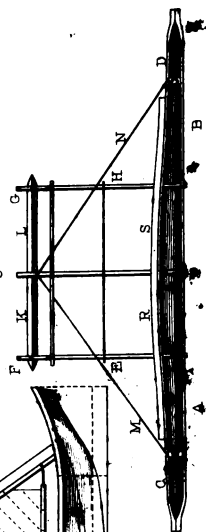


Fig. 3.



FLYING PROA.

Explanation of the Plate.

Fig. 1. Represents the proa with her sail set, as she appears when viewed from the leeward.

Fig. 2. Is a view of the proa from the head (or stem on), with the outrigger to windward.

Fig. 3. Is a plan of the whole, where A B is the lee side of the proa; C D the windward side; E F G H the outrigger, or frame, laid out to windward; K L the boat at the end of it; M N two braces from the head and stern, to steady the frame; R S a thin plank placed to windward, to prevent the proa from shipping water, and for a seat for the man who bales out the water; and sometimes goods are placed on it. T is the part of the middle outrigger on which the mast is fixed. The mast itself is supported (Fig. 2) by the shore P, and by the shroud Q, and by two stays, one of which may be seen in Figs. 1 and 2; the other is hid by the sail.

The flying proa, besides being an object of curiosity, may well furnish both the shipwright and sailing-master with wrinkles of no mean or contemptible character. The Rev. Richard Walter, M.A.³ observes, with reference to the swiftness with which these proas sail: "From some rude estimations made by our people of the velocity with which they crossed the horizon at a distance, when we lay at Tinian, I cannot help believing that, with a brisk trade wind, they will run near twenty miles an hour."

The form and construction of the flying proa is in direct opposition to the principles of boat-building, as adopted in England; and, indeed, in almost every other country in the world. For, as the rest of the world make the two sides of the vessel as exactly alike as possible, those of the flying proa are totally different, one side being round, or bilge-shaped, whilst the other is flat and perpendicular as a wall; but the head and stern of the proa are as exactly alike as possible, and the vessel may be sailed with either end fore-

³ The Chaplain of His Majesty's ship "Centurion," in Lord Anson's expedition.

Fig. 2.

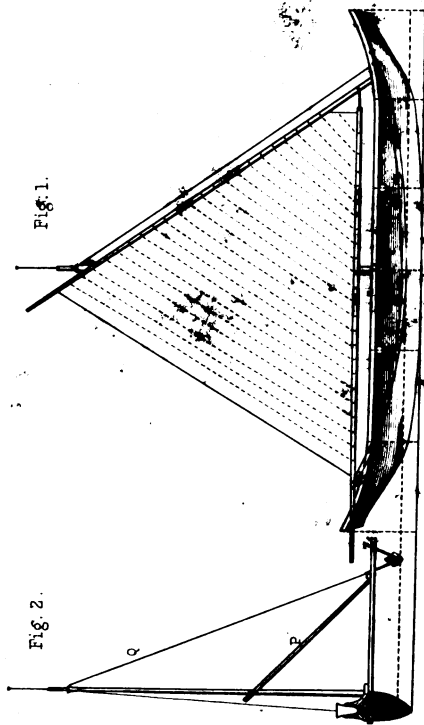


Fig. 1.

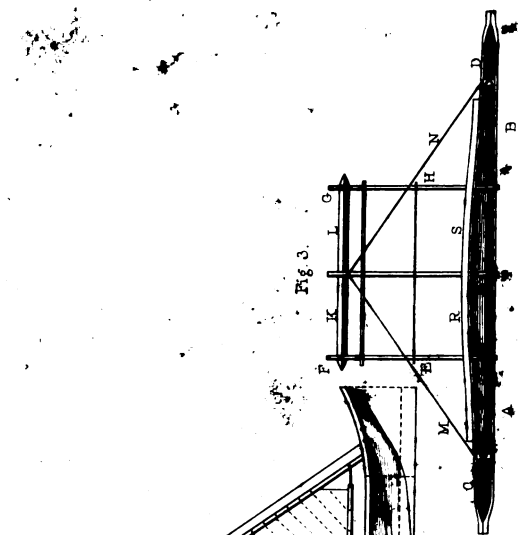
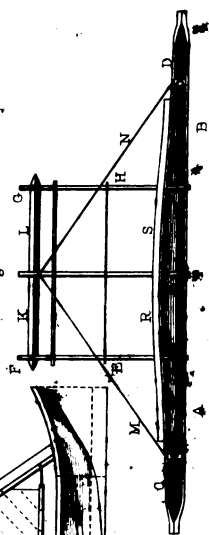


Fig. 3.



THE FLYING PROA.

Fig. 1. Represents the proa when it is in the water, and appears when viewed from the seaward.

Fig. 2. Is a view of the proa from the land, and shows with the outrigger to windward.

Fig. 3. Is a plan of the vessel, showing the position of the proa; C D the windward side, which is the outrigger or frame, laid out to windward. I J the mast, and K L M N two braces from the mast to the outrigger. P Q R S a thin plank placed to windward, and raised up from shipping water, and for a sail to be hoisted on, and out the water; and sometimes goes up as high as the top of the part of the middle outrigger on which the mast is set. The mast itself is supported Fig. 2. by the shroud Q, and by two stays one of which may be seen in Figs. 1 and 2; the other is not shown.

The flying proa, besides being an object of great interest, well furnish both the shipwright and sailing-master with wrinkles of no mean or contemptible character. Thus Mr. Richard Walter, M.A.³ observes, with reference to the business with which these proas sail: "From some true observation made by our people of the velocity with which they cross the horizon at a distance, when we lay at Tinsel, I cannot help believing that, with a brisk trade wind, they will run near twenty miles an hour."

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³ The Chaplain of His Majesty's ship "Centurion," in L. expedition.

most, but always with the rounded side to windward, and the flat one to leeward ; therefore, on the return voyage, the *boat* is not turned round, but the *sail*. Indeed, the boat never need be turned round but twice a year, and that only when the trade winds shift, which they do once a year only, blowing six months in one direction and six in the opposite.

Being, therefore, always enabled to know for a certainty the quarter from whence the wind will blow, and that wind being always a side one throughout the whole range of the Ladrões, the islanders are enabled to turn such knowledge to useful advantage ; and so they find it unnecessary that their boats should possess the power of tacking and working to windward, or of sailing before the wind, but exclusively that of reaching, or sailing with a side wind ; therefore the Ladrone inventor places a very large sail on a very narrow form of hull—indeed, such a sail as would inevitably upset the proa in a breeze, unless it were counterpoised against the force of the wind by some artificial contrivance.

It has been already stated that the lee side of the proa is flat, or, in other words, the shape of the hull is something like one-half of a boat severed longitudinally, and then the whole open side boarded up from stem to stern with straight planking ; all which materially contributes to render it the more ticklish, and easy to capsize. But in order to prevent such a catastrophe, the proa is fitted with an outrigger, which forms a most important feature in the boat, and one of its greatest peculiarities.

The outrigger consists of a frame made of bamboo poles, affixed to the proa, and standing out to windward : at the extreme end of the frame is fastened a log of wood slightly hollowed and fashioned into the shape of a small boat. The weight of the frame and log is intended to counterpoise the proa against the force of the wind acting upon the sail, and to prevent the proa from being capsized.

The hull of the proa is very neatly made and put together ; the bottom is of one piece, made like the bottom of a little canoe, very neatly dug, and left of a good substance ; this

bottom part is instead of a keel, twenty-six or twenty-eight feet long; the under part of the bottom is made round, but inclining to a wedge, and smooth; and the upper part is almost flat, having a very gentle hollow, and is about a foot broad. From this bottom, or keel, both sides of the boat are carried up to about five feet high with narrow plank, not above four or five inches broad, and each end of the proa turns up very prettily.

The breadth of the proa depends, in a measure, upon the length; but it seldom exceeds four or five feet amidships, and is much less at the ends.

The mast, it should be observed, although placed amidships, longitudinally, is not so latitudinally, but stands in the bilge of the proa close to the weather gunwale, and is fixed to the middle beam or bamboo of the outrigger.

The sail is triangular, or lateen-shaped, but much wider at the foot, and less lofty than the lateens of other nations; and, although it looks not a large sail in the proa, still, when it is considered how long and narrow the hull of the proa is, it will be found to be an enormous sail in proportion to the stability of the boat, when viewed independently of the outrigger.

The fore end of the yard is secured in a socket made purposely to receive it at each end of the proa. The sail is also fitted with a boom, so that it stands nearly as flat as a board; and the boom is also used for furling the sail, and rolling up a portion of it when the wind is heavy; and such is the mode of reefing the sail of the proa.

The sail is made of matting stuff; and the mast, yard, boom, and outrigger, are all of bamboo.

When the proa alters her tack for the return voyage, her crew bear her away a little to bring her stern up to the wind, then, by easing the halyard, raising the yard, and carrying the heel of it along the lee side of the proa, they transfix it in the opposite socket; whilst the boom at the same time, by easing off one part of the sheet, and hauling on the other, shifts into the contrary, or required position; when that

which was the stern then becomes the head, and the proa is trimmed on the other tack.

The proa generally carries a crew of six or seven, two of whom are stationed, one at the head, and another at the stern; these steer the vessel alternately, with a paddle, according to the tack on which she goes. The duties of the others are to bale out the water, when she leaks or accidentally ships any, and to trim, adjust, and attend to the sails.

From the description given of these extraordinary vessels, it will be seen how admirably they are adapted for ranging the Ladrone islands; and from the great power they possess in the very large sail, the flatness of their lee-side, and their narrow form of hull, assisted by the outrigger, they are enabled to fore-reach with immense power, and without any apparent lee-way.⁴

Vessels bearing some resemblance to the flying proa are often met with in various parts of the East Indies, but none of them appear to have any pretensions equal to those of the Ladrone islanders, either in the ingenuity of their form and construction, or in the speed at which they sail. Voyagers have, therefore, asserted their belief that the flying proa was the original invention of some genius of the Ladrone, and was afterwards imperfectly copied by neighbouring nations.

This peculiar mode of counterpoising the canoe, when the pressure of the wind acts forcibly on the sail, is a very primitive notion. It is mentioned by Pliny, eighteen hundred years ago, that vessels navigating the seas to the west of Taprobane (Ceylon), had prows at either end, to avoid the necessity of tacking.⁵ And the same fact is also mentioned

⁴ The engraving at page 208, and explanation above given, have been taken from a description of one of these remarkable vessels which fell into the hands of the crew of His Majesty's ship "Centurion," when on an expedition to the South Seas, under the command of Lord Anson, in the year 1744.

⁵ "Ob id navibus utrinque prorœ ne per angustias alvei circumagi sit necesse." Pliny, lib. vi. c. 24.

by Strabo, who says they were built with prows at each end, but without holds or keels.⁶ Still, this mention by Pliny and Strabo does not appear to clash with the general impression of the voyagers, that the original notion was derived from the Ladrone islanders.

Even at the present day, with all our boasted improvements in marine architecture, it may be fairly questioned whether the flying proa is not incomparable as a vessel for swift sailing, close to the wind, without making lee-way; and the flatness of the sail, and the perfection of its form and rig, with reference to the peculiarly long narrow hull, with a flat lee side, would seem to be the most perfect combination of simple but surpassing excellence for the purpose.⁷

The only attempt we ever heard of constructing a flying proa to sail in English waters, was made in the year 1860, by a member of the Royal Mersey Yacht Club, who, assisted by the ingenuity of an able mechanic and boat-builder on the Mersey, appears to have been highly successful in turning out a veritable flying proa; which was, of course, one of the greatest attractions and extraordinary novelties ever seen on that river. This Anglo-Ladrone proa, which is still to be seen in that locality, was described as being "highly scien-

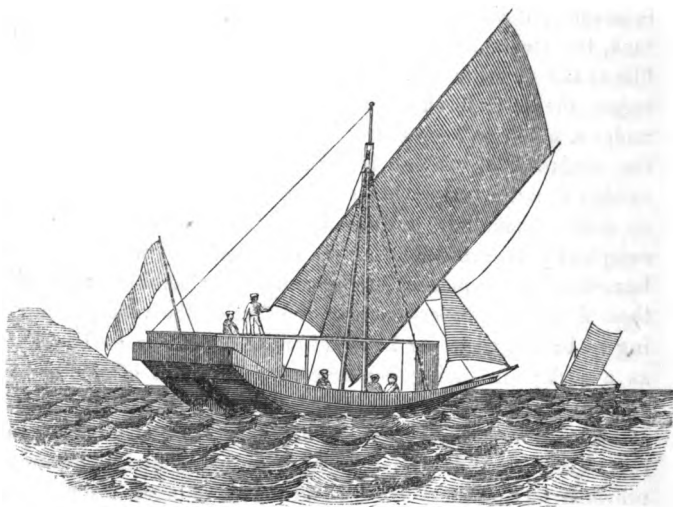
⁶ Strabo, lib. xv. c. 15.

⁷ In Pigafetta's "Voyage round the World," A.D. 1519 to 1522, the flying proa is thus described:—"The chief amusement of the Ladrone Islanders consists in sailing about with their wives, in canoes similar to the gondolas of Fusine near Venice, but they are still more narrow; all of them are painted either black, white, or red. The sail is made of the leaves of the palm tree sewed together, and has the shape of a latine sail. It is always placed on one side; and on the opposite side, to form an equipoise to the sail, they fasten a large wooden log, pointed at one end, with poles laid across and fixed in it, which keeps the boat steady, and admits of their sailing without apprehension; their rudder resembles a baker's shovel—that is to say, it consists of a pole fastened into a plank. They make no difference between head and stern, as they have a rudder at each end. They are excellent swimmers, and have as little fear of the sea as dolphins."

tific, and solving many problems in the science of marine architecture. She is diametrically the opposite, in every particular, to all European vessels. She has much greater length, in proportion to her beam, than any other vessel whatever. Her two ends are exactly alike, turning up prettily in the prow-like manner of the ancient galley, ending in a handsomely carved scroll. The most surprising feature is, that one of her sides is flat, acting as a lee-board, and it is on this side that she always sails when by the wind. The other side, which is the windward one, is moulded, rounding in the manner of other vessels. Her bottom, or midship section, is of a rounded, wedge-like form, and has great length of floor. On this side there is a strong framework, or outrigger, firmly built to the vessel; and at the end, parallel with her keel, a log of wood, shaped into the form of a small boat, is strongly attached, made hollow and decked over. By this arrangement she is kept from over-setting, it acting also as her ballast, and giving the required stability. This quality can be increased *ad libitum*. It also acts antagonistically to the pressure of the wind on her sail when by the wind: as the pressure increases, so does she gain stability by gradually feeling the whole weight of the log boat, as it is lifted out of the water. She only draws about fifteen inches of water; yet, the flatness of her lee side, and its clear run, prevents deflection from her course when on the wind. Her mast is placed exactly in the middle of her length, perpendicular, but stepped on one side, and can be lowered on deck at pleasure. She has one sail, which, in its principle and working, is as surprising and as novel as the boat herself. It will turn round to any point of the compass, and, working on a centre, it can be so adjusted as to act as a steering sail, as well as a most powerful means of propulsion. She can be run upon any rhumb line of the compass, in a direct course, merely by varying the angle of the sail. It can be reefed to any degree required to meet the power of the wind, and yet retain its original shape; this is effected by the simple and perfect process of rolling it up

at the foot on the lower yard. The general simplicity of the construction of the proa, and ingenuity of contrivance in her sail and tackle, is beyond that of any other vessel; one mast, one sail, one shroud, three blocks, one halyard, two sheets and braces, comprise all her tackle. The seamanship, or manner of sailing her, too, is not the least part of the novelty. Two hands, in moderate weather—one sitting at each end—have perfect command over the vessel; and she is steered by them alternately, with a broad paddle, according to the tack she is upon. She sails either end foremost, that which was her bow becoming her stern, and *vice versâ*. She is never ‘put about;’ but when required to go on the opposite tack, the steersman merely ‘keeps her away,’ until her sail fills at the opposite end; that which was the fore-leech becomes the after-leech, and the end of the yard is passed under a notch to keep down the tack of the sail. Thus all the confusion of tacking is done away with. In case of a sudden squall, a single halyard let go brings down the sail on deck immediately, and the danger of the gaff-sail and its complexity is avoided; nor is there any gybing, and its hazardous consequences, to encounter. Combined with all these desirable properties, from the fact of the vessel requiring no ballast whatever beyond her stores, she is as buoyant as a cork; and being constructed throughout of wood, she can never founder, and is a lifeboat in principle. Her prow at each end rising boldly up, with curved lines, enables her to rise to the sea, and her length of floor and general buoyancy prevents any of those sudden and heavy plunges which take place with vessels loaded with lead or iron ballast and heavy spars. She will rise with the utmost promptitude on the top of the wave, or, in scudding before the sea, go over it, from her superior lightness. Swimming on an even keel, and drawing so little water, she can run over banks and get into shallows, or make harbour, when other vessels are compelled to keep at sea or wait a tide. We have said nothing of her velocity, or capability of passing through the water; yet this is her most surprising quality. From the general curvature of her

bottom, and bold flanging stems, she can be propelled through the water with a speed which would send any other sailing vessel bows under. From the above description, it will be very evident, to the scientific reader, that her capability of speed is something very considerable. We shall conclude with the statement that she carries *three times* more canvas, according to the resistance of the vertical midship section immersed, than the fleetest of our yachts."⁸



BORNEO BOATS.

SOME of the boats of Borneo and Celebes are exceedingly well made. The sailing-boats have high and very broad sterns, and long raking bows. They have a double or shear-mast, and long-shaped square-sail; they have also a stage

⁸ Vide "Bell's Life" for June, 1860.

or gallery, and other peculiarities, which will be better understood by reference to the engraving opposite.

Some of the canoes which Captain Mundy⁹ met with in the Gulf of Boni he describes as of very long shape, propelled by fifteen paddles, and ornamented both at prow and stern with carved wood.

The small sailing-boats he describes as fitted with wooden outriggers, which, weighted with men, enabled them to carry a sail of enormous size. On the inland rivers, some of the canoes are forty or fifty feet long, by only two-and-a-half wide, and covered with a small kajang or mat.

Some of the canoes are so small as scarcely to float a child of five years of age—in fact, but a hollowed log. But mothers do not fear to trust their children upon them, as they swim like fish.¹⁰

BOATS OF BRUNI.

One of the greatest novelties at Bruni is the floating-bazaar. There being no shops in the city, the market is held every day in covered canoes. These are moored in tiers, forming lanes in the river, up which the purchasers ply in paddling canoes, look on, and make their purchases.

The trading-boats come in at sunrise every morning from all parts of the river, laden with fresh fruit, and every other commodity produced in the vicinity.¹¹

The floating market is thus alluded to by Mr. Spencer St. John, in his interesting work on Borneo:—

“Several hundred canoes, each containing one or two women, covered over with mat hats a yard in diameter, floated up and down about the town, pulling through the water lanes, and resting for a while in the slack at the back of the houses.”

⁹ “Borneo and Celebes.” By Captain Mundy, R.N., 1848.

¹⁰ “Life in the Forests of the Far East.” By Spencer St. John, F.R.G.S., F.E.S., &c., 1862.

¹¹ “Borneo and the Indian Archipelago.” By F. S. Marryat, 1848.

PRAHUS OF BORNEO.

These are swift sailing vessels, though made of the trunk of a tree hollowed, and sharpened at both ends; they have neither keel nor rudder, but are steered over the quarter with a very long paddle, thirty feet in length, and about twelve or fourteen inches wide in the blade. The thwarts or seats are placed across the boat, above the gunwale. To prevent their capsizing under sail, they are fitted with outlagers; two long poles being run out on each side, one across the forepart, and one across the aft part of the prahu; the outer ends of the poles are run into, or lashed to, a large bamboo. When the wind is heavy, part of the crew run out on the windward outlager to keep the boat upright. These prahus are fitted and sailed with a large lateen-shaped sail, made of matting.¹²

BAJU AND BALIGNINI PIRATE PRAHUS.

The Balignini, which are of the Bajow or sea-gipsy tribe, have large prahus, with crews of seventy or eighty men, who sometimes row double banked; and to each prahu, a long and fleet small boat is attached, which will hold from ten to fifteen men.

They seldom carry large guns like the Illanans, but lelahs (small brass guns); also swords, spears, stones, and other instruments of attack.

The Balignini and Baju pirates used to be the terror of the Indian isles. Mengkabong is the head-quarters of these lawless people.

The Baju prahus are rigged with tripod masts, which consist of three tall bamboos, the two foremost being fitted on a cross beam, the other loose; so that when a heavy squall threatens, the masts can be immediately struck.¹³

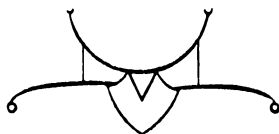
¹² "Voyage To and From the Island of Borneo." By Captain Daniel Beeckman, A.D. 1718.

¹³ Spencer St. John's "Borneo, &c."

The arrangement of the tripod mast fitted to the Tartar galley (and described at page 214) is somewhat different to this of the Baju boats, the *fore-leg* of the mast being the one by which the tripod of the Tartar galley can be struck.

SULU CANOES.

The canoes of Sooloo, or Sulu, differ from those of other islands, not only in shape, but also in the outrigger. A Sulu canoe is made of a single log or tree; and it is seldom large enough to carry more than two persons at a time. The outrigger is a double one, *i.e.*, it extends on both sides of the canoe,



so as to give it extra stability. (See the Section.) A railing is also placed above the sides of the canoe, with supports, which rest upon the bearers of the outrigger. This, however, and the canoe itself, will be better explained on reference to the engraved section.

These canoes are propelled by means of a double-bladed paddle or sweep, with which they may be managed entirely by one person. Small canoes of this kind are never sailed. They are sometimes built upon and enlarged; this is done with boards or upper sides, secured to the trunk.



THE SULU OR ILLANAN PRAHUS.

These are the largest description of prahus used in the Archipelago; they belong to the pirates of the Sulu Isles, who go under the name of Lanans. These prahus are from

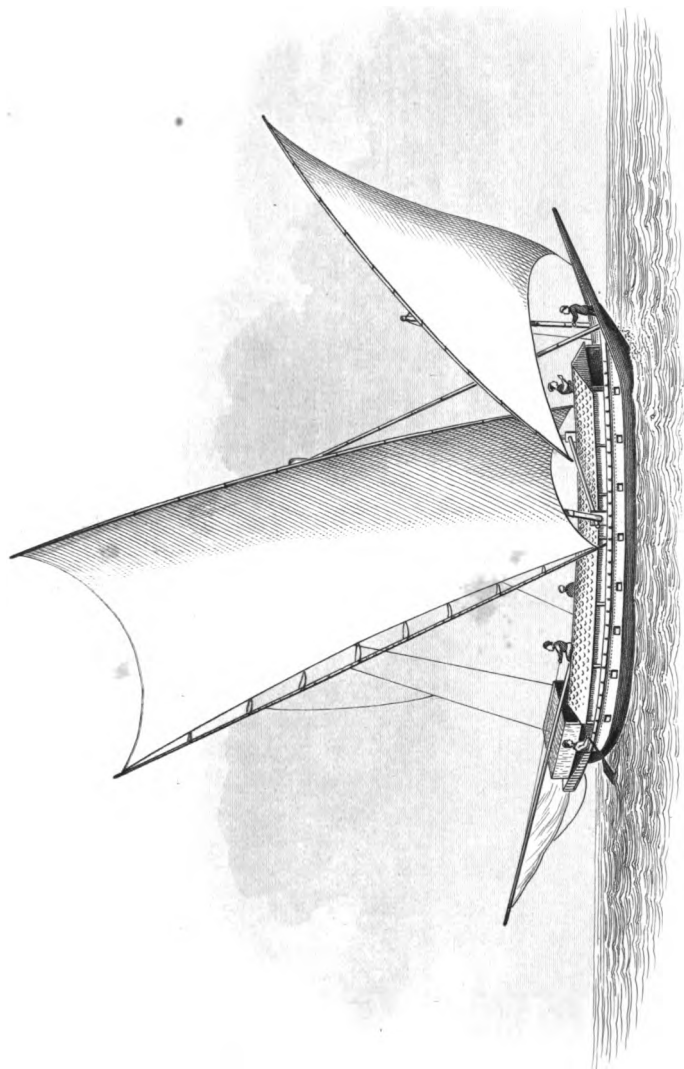
twenty to thirty tons burthen, nearly a hundred feet in length, and of considerable breadth of beam, with a sharp hollow bow; the lower part of the hull is strongly built of timber; but the bulwarks, decks, and internal fittings, are chiefly of bamboo, ingeniously fastened together. The crew is generally a very numerous one, sometimes from forty to fifty. A raised platform is constructed on both sides, for the convenience of the fighting pirates. These boats are propelled both by sails and oars; sometimes fifty oars are used at once; they draw little water, are fast under sail, and well adapted for navigating those dangerous seas. They are rigged with two separate shear-masts, each consisting of two spars lashed together at the top; the heels of the foremost are set in a base, which partly revolves, and can be also raised and lowered at pleasure; the object is, that when attacking a vessel, the sail can be lowered, and the shear-mast directed so as to fall on the vessel's side; it then forms a ladder for the pirates to climb from the prahus to the deck of the vessel. The shear-mast can also be dropped on the bank of a river, so as to form a bridge; and it may be used for scaling walls, and other marauding practices. The sails are made of matting and bamboo canes, and are of very large size; they generally carry a square red flag at their fore-mast head, and assemble in a numerous fleet when on their piratical adventures.

The Sulu are said to be the boldest, and most cold-blooded pirates in the Archipelago. They infest the straits of Macassar, the sea of Celebes, and the Sulu sea.

TARTAR GALLEY.

(See Engraving.)

The vessel used by Capt. Forrest, in his voyage along the coast of New Guinea, was called a Tartar galley, but was, in fact, a Sulu boat, about 10 tons burthen. This boat had a kind of gallery built on each side, from stem to stern, pro-



TARTAR GALLEY.



jecting about thirty inches over each gunwale, upon which the rowers sat, sometimes twenty in number.

Although the Tartar galley was but twenty-five feet long on the keel, she overhung so much forward and abaft, that her length, over all, was forty feet; draught of water, about three and a-half feet.

This boat had a tripod mast, made of three stout bamboos: the feet of the two which stood abreast were bored at the lower end, across, with holes about three inches in diameter; and these two holes received the two ends of a piece of timber, which, like a main shaft, was placed across the boat from side to side; on these the two spars turned, as on a hinge.

The fore spar of the tripod mast was fixed forward to a knee amidships, with a forelock; by unlocking which, the mast could be struck with ease by three men.¹⁴

The mainsail was a large four-cornered one, called by the Malays *lyre tanjong* (pointed sail). The boat also carried a foremast, on which a lateen sail was set; also a mizzenmast, on which another lateen could be set. When the wind was heavy, the lyre tanjong¹⁵ was lowered, and a smaller sail (a lateen) was set; and the vessel then resembled the rig of the Mediterranean galleys.

The advantages of the lyre tanjong appear to be, that it is a very powerful sail, and may be very quickly reduced or reefed by luffing into the wind, easing off the sheet, and then rolling or winding up the sail, by turning the winch or crossbar that is fixed to the inner end of the boom.¹⁶ By this means the sail may be entirely rolled up, so that the boom and yard meet, and lay side by side together, as in the engraving of the paduakan on the next page, and so

¹⁴ Capt. Forrest's Voyages, 1792.

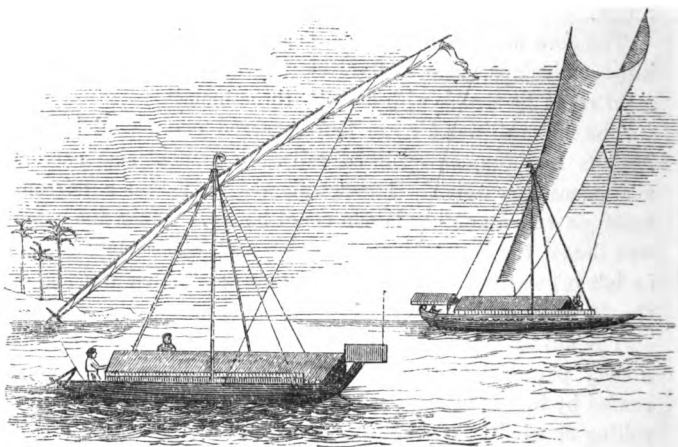
¹⁵ Capt. Forrest suggests, that if two London wherries were lashed together, and the double vessel, thus formed, were fitted with the lyre tanjong, it would beat the fast sailing boats at least three to two.

¹⁶ The Kolay, Malay Jellore, and other boats, are also reefed in the same manner.

furled. In the same manner, by turning the winch the other way, it may be unfurled, and as quickly set, or half set, according to the weather.

The cabin, or covered part of the Tartar galley, was thatched with the leaves of a palm tree called *nipa*, being the same material as the natives cover houses with on the south-west coast of Sumatra, and in most of the Malay countries. The small apartment abaft was covered with boards, and was called by the Malays *koran*.

The Molucca proas and vessels of burthen are all fitted with the tripod mast and lyre tanjong.



PADUAKANS OF CELEBES.

These singular boats of the Island of Celebes are called "Bugis paduakans:"¹⁷ they are built at Bera, at which place were formerly the chief building-yards of the Macassars. The paduakans are remarkably well built boats; they are made very tight, by the system of dowling the planks to-

¹⁷ "Forrest's Voyage to Mergui," &c.

gether, as coopers do the parts which form the head of a cask; and they put the bark of a certain tree between the planks, in the place of oakum, which swells and fills them tight.

Their system of boat-building is the very reverse of the English, inasmuch as they put the planks together first, and then secure them to the timbers (or ribs); whereas we set up the timbers first, and then secure the planks to them.

The natives are much bigoted to old models, and seldom venture on new designs. The bow of these vessels is lower than the stern, and in that respect they are rather unsightly; and the bow is often driven under water in rough winds. In order to keep off the spray at the bows, a bulk head is raised, a few feet abaft the stem; in other respects they are not decked, but have a roof or covering extending all over the vessel from the bows.

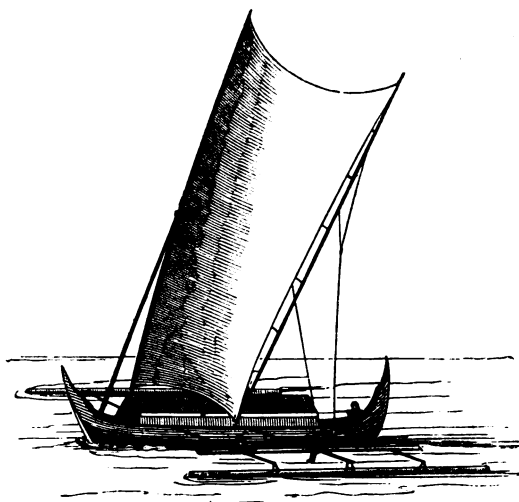
The paduakan is rigged with a tripod mast, and high pointed sail. The tripod mast is formed of three stout bamboos, and fitted in a similar manner to that of the Tartar galley, described at page 214.

The woodcut represents two of these curious vessels; one with its sail furled, the other under full sail.

THE COROCORA.

A corocora is a curious-looking boat or vessel, used chiefly by the inhabitants of the Molucca Islands; and the Dutch have fleets of them at Amboyna, which they employ as *guarda costas*.

The corocora has a very high stem and stern, and is generally fitted with outriggers, in the manner explained by the engraving. They vary in size from small boats to above ten tons burthen. On the cross pieces which support the outriggers, are often put, fore and aft, planks, on which part of the crew sit and paddle when there is no wind; besides which, others, who sit in the vessel, use long oars. In smooth water, they are propelled in this manner with considerable velocity, by a good number of hands, or banks



THE COROCORA.

of rowers. They are steered with two commoodies (broad paddles).

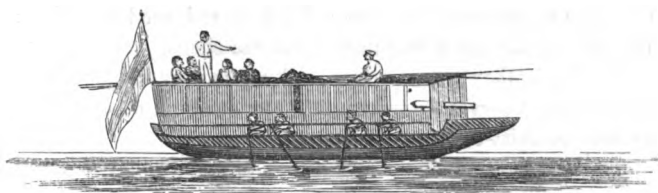
Captain Forrest, in his "Voyage to New Guinea," mentions a corocora he met with, which had six banks of paddles—three on each side, ranged on planks laid across the outriggers, and was manned by Papua men. This vessel belonged to one of the Chiefs of those islands.

MALAY AND DYAK CANOES.

These canoes, which are made out of a hollowed tree, are long and narrow, and capable of being propelled with considerable swiftness. It is not unusual to find them so heavily laden, that any inclination on either side would let the water in over the gunwale. One of the crew is generally employed in baling out the water, with a scoop made of a banana leaf.

The Dyaks' canoes are similar to those of the Malays. The Dyaks may often be seen standing upright in their canoes, and propelling them with their lanoes.¹⁸

¹⁸ "Borneo," by F. S. Marryat, 1848.



MALAY PRAHU.

The Malay prahus, or war-boats, are curious. Being constructed for the purposes of war, they have a strong flat roof, or fighting platform, extending the entire length of the boat, and raised upon pedestals standing about five feet from the deck. Their principal weapons are the kris and spear; to use which with effect, requires elbow-room.

These vessels are built of timber at the lower part of the hull; but the upper part is of bamboo, rattan, and kedjang (the dried leaf of the Nepa palm). They are also provided with a strong gallery, which runs outside the bends, about one foot from the water-line: upon this the rowers sit, with their legs crossed. At the aft part of the prahu is the cabin for the chief who commands.¹⁹

DYAK WAR BOATS.

These are long-shaped canoes, of more substantial construction than the Malay prahus; and are, besides, sufficiently capacious to hold from seventy to eighty men.

They are made with a flat keel, having a curve or sheer of hard wood. A long keel does not exceed six fathoms, and upon such an one they build a boat of eleven fathoms, over all. The extra length is brought up with a sheer. The seams are caulked with a bark that is plentiful in the jungle. No other fastenings but rattans are used.

These boats are painted red and white; when they have

¹⁹ Marryat's "Borneo." &c.

no ochre for the red, they use a kind of red seed pounded; the white is simply a lime made from sea-shells.

The bark they employ for caulking is very tough; when beaten out, it serves to make useful and comfortable coverlets, as well as waist-cloths and head-dresses.²⁰

The Dyak war boat has also a roof to fight from; and the stern is ornamented with feathers.

Notwithstanding the heavy top-weight of gallery and fighting men, these boats, as well as the Malay prahus, are remarkably swift under oars.²¹

MALAY PIRATE PRAHUS.

These are of much smaller size than those of the Sooloo pirates, being only from ten to twelve tons burthen; but, in proportion, better manned, and ply with more efficiency their oars or paddles.

These infest the Straits of Malacca, Cape Romania, the Carimon Islands, and neighbouring straits; and they sometimes visit the Straits of Rhio.

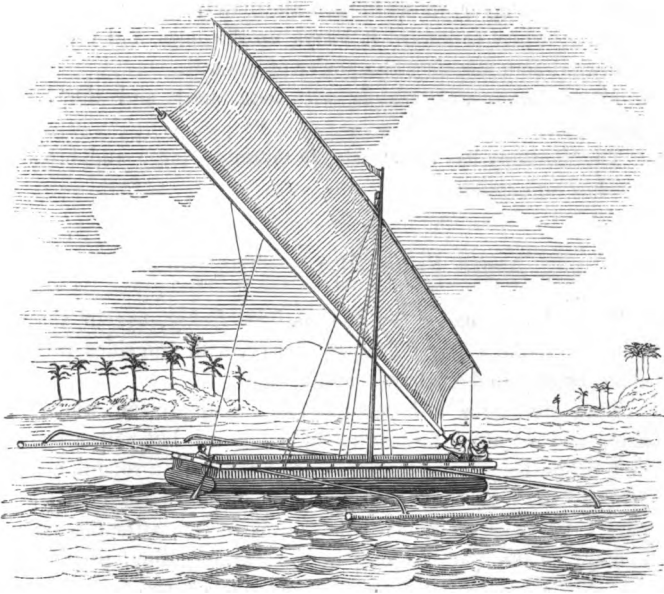
They have a long, low hull, and are provided with several guns, though not very large ones. They seldom attack unless the sea is calm, whereby they can be more certain of success. They are generally found in small flotillas of from six to twenty.

The rig of these prahus consists of two masts, with sails of matting. The crew are a cold-blooded race, and sometimes so numerous, and well armed with steel weapons, as not easily to be beaten off by the crews of ships, when once they get so near as to be out of danger of the ship's guns. A platform is erected over the bows, and extending some distance beyond; this platform is as wide, or wider, than any other part of the boat, and is the rendezvous of the pirates when attacking a vessel.

²⁰ "Life in the Forests of the Far East," by Spencer St. John, F.R.G.S., F.R.S., &c. 1862.

²¹ Marryat's "Borneo," &c.

BOATS OF SUMATRA.



JELLORES AND BALLELLANGS.

THESE curious boats are of a long and narrow form, and are fitted with double outriggers, which stand out at a considerable distance from the sides. They are sometimes rigged with two masts, and lyre tanjong sails; though sometimes with lug-sails. The ballellang is rather broader than the jellore, but still comparatively narrow; yet they mount two small swivel guns, and carry twenty or thirty men. They sail remarkably fast in light winds, and in strong winds too, if the water is smooth. The engraving above represents a jellore with the sail partly rolled up, after the manner of reefing adopted in those parts. The wide-spreading outrigger

enables these vessels to carry a very large sail in smooth water. The form of the jellore will be better understood by reference to the preceding engraving. Jellores have sometimes only one outrigger, which is then alternately to windward and leeward; when to leeward it buoys up the boat, and when to windward counterpoises the power of the sail.

Panchallangs are vessels with one mast and the lyre tanjong.

The *Bantang* has two masts, is tolerably large and broad, and has no outrigger. ¹

ACHEN FISHER BOATS.—THE KOLAY.

These boats, called by the natives *Kolay*, are in shape something like a large Thames wherry, with the sides raised about twenty inches above the gunwale.

They are rigged with one mast and a square sail, which is slung and set after the manner of a lug-sail, but with a boom and bridle below: to the fore-end of the yard a rope is made fast, for the purpose of hauling it down and peaking the aft part. When it blows hard the sail is reefed, after the same manner as the lyre tanjong in the Tartar galley, page 214, *i. e.* with a cross-stick or pin, which passes through the inner end of the boom; and the sail is thereby easily rolled up, sheet and all, passing the lower end of the trunnel forward; and it is unrolled as the wind slacks. This seems to be a very convenient rig, and mode of reefing a small sail. Captain Forrest says he never saw anything so convenient in any European boat. In putting about, the sail is dipped in the same way as a lug-sail.

The *kolay* is steered with an oar, which is passed through a lashing on the aft quarter of the boat.

¹ See Forrest's "Voyage from Calcutta."

BOATS OF THE PHILIPPINE ISLANDS.

THE Indians at *Manilla* are very skilful in the arts of building and managing boats.

On the river, and along the shore, may be seen a number of prettily built boats, with sharp bows, and furnished with bamboo outlagers,² like the yard of a square-rigged vessel, but laid across the boat. When it blows hard, one, two, or more of the crew, are put out on the windward end of the outlager, to counterpoise the effect of the wind upon the sail. This contrivance, however, does not always ensure safety, for at times the bamboo yards, which form the outlager, break; in which case the boat is seldom saved from capsizing, and the whole crew are lost.³

PANGUES.

These are small ferry-boats, used chiefly for crossing the rivers; they are made of the hollow trunk of a tree. Those generally used are large enough for two or three persons only, though a few of larger size are to be met with in the island. They are generally propelled by oars, assisted sometimes with a small sail.

MINDANAO PLEASURE PROA.

Many of the boats of the Indian Archipelago are extremely curious. Dampier describes a pleasure proa he met with at Mindanao, belonging to the Sultan of that island. It was large enough to carry fifty or sixty persons, or more. The hull was neatly built, with a round head and stern; and over the hull

² These must not be confounded with the *outriggers* of the canoes of some other islanders; there is no outrigger to these canoes, but merely an *outlager*.

³ Dampier's "Voyages."

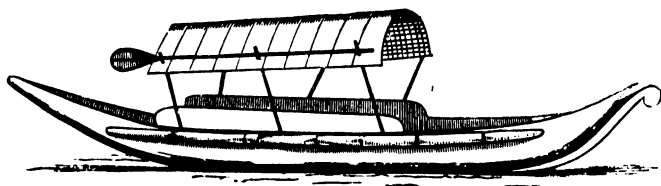
there was a small, slight house, built with bamboos, about four feet high, with neat little windows of the same, to open and shut at pleasure. The roof was almost flat, but neatly thatched with palmetto leaves. The house was divided into two or three small partitions, or chambers, one in particular for the Sultan himself, which was neatly matted underneath and round the sides, and furnished with a carpet and pillows. The second was the ladies' room, or that for the Sultan's wives, and was much like the former. The third for servants, who attend the ladies with tobacco and betel-nut. The fore and aft part of the vessel was for the crew; but the outlager is the place for them, when navigating the vessel.

The proa was fitted with outlagers on each side, but very different to the flying proas of the Ladrone Islands. The Mindanao proa has no flat side; both sides are rounded or bilge-shaped, like an English skiff, and there is no outrigger boat; beams are placed across the protruding bamboos, and they do not touch the water on either side, like the Ladrone proa, but are two, three, or four feet above the surface, and serve for the boatmen, or paddlers, to sit and row, or paddle from. If the proa reels or lists on one side, the beams dip in the water, and so the paddlers are often wetted and dipped up to their waists.⁴

MANILLA BANCAS.

THE passage-boats of Manilla are termed Bancas; they are much used on the canals, and although made of the single trunk of a tree, are prettily formed, and are a very useful kind of boat. The trunk is carefully hollowed, so that the boat is very light, and rows easily. The bottom is narrow and round-shaped; therefore the banca is easily upset, under careless or awkward management.

⁴ Captain Dampier's "Voyages," vol. i.



MANILLA BANCA.

The thwarts for the passengers are placed very low, so as to keep the weight of the occupants as close to the bottom as possible, and preserve the stability of the vessel. A stout bamboo is attached to the top rim of the banca, outside, which serves the purpose of a fender on going alongside, and also renders the vessel more buoyant when heavily laden, or when lurching on one side. The banca is also fitted with a light top or awning (as shewn in the engraving), to protect the passengers from the heat of the sun.

These vessels are steered with a large shovel-like paddle (see engraving).

THE MANILLA SARABOA.

The Saraboa is a fishing raft employed by the Manillas ; it is composed of two or more layers of bamboos, or light wood, laid transversely one above the other, with a little gothic hut at one end

The native fishing is chiefly performed in rafts of this kind. They take fish at the mouth of the river, in nets suspended by the four corners, from hoops attached to a crane, by which they are lowered into the water.

The Paroa is a larger kind of trading, or passage-boat, used at Manilla.

FEEJEE CANOES.

THE sailing canoes of the Feejees, or Fiji islanders, are of a very superior class. The largest size are as much as a hundred feet in length, and are of the double form, consisting of two canoes of different sizes united by beams, on which a platform is laid (see the *Frontispiece*). But although these are called "double canoes" by Europeans, the second, or attached canoe, is merely an outrigger composed of a tree hollowed out for the sake of buoyancy, like the canoe itself. The mast and sail are fitted to the larger of the two canoes; the smaller one, as above stated, serving as an outrigger to the other.

The bottom of each of the canoes is formed of the trunk of a single tree, hollowed out and built upon with considerable ingenuity. The sides and coamings are fitted and secured to the canoe by dovetailing, and drawn closely together with lashings of cocoa-nut plait, passed through flanges left for the purpose on each of the planks. The joinings and crevices are filled and closed with gum taken from the bread-fruit tree, which is also used as an outer dressing, in the place of tar or paint. The planks are secured inside to small ribs, or timbers, which are placed at regular intervals, according to the usual mode of boat-building. These vessels are generally built of the Vas wood.

The fore and aft ends of the two vessels, forming the double canoe, are decked over; the larger one to the extent of about twenty feet at each end, and the smaller one somewhat less. This is done to prevent the shipping of seas in rough water.

The platform is firmly secured over the top of the open part of the canoes, resting on the raised sides (or coamings), and extending, on the outer sides, two or three feet beyond them. The depth of hold under the platform is about seven feet.

On the platform, nearly amid-ships, between the two parts of the canoe, there is a stage about eight feet square by four or five high, with a railing on each side; on the stage there is space for several persons to stand or sit, more particularly for those who give directions as to the steering and management of the canoe. Beneath the stage is a small thatched house or cuddy, for the crew when seeking protection from the weather.

The covered fore and aft part of the canoes belonging to the chiefs of the Feejee Islands are prettily ornamented with shells, and the sails with white flags and streamers. The sail is of triangular shape, and so large as to appear quite out of proportion to the vessel; it is made of tough, yet pliable, matting.

The mast is about half the length of the longer of the two vessels composing the canoe; it is stepped in a chock on the platform. The yard is nearly twice as long as the mast, and the boom somewhat less than the yard.

The halliards are passed over a crescent at the top of the mast; they are bent on to the yard, nearly in the middle, or so that when the sail is hoisted with the fore-end of the yard secured on deck, the seizing on the yard comes upon a level with the top of the mast.

The mast is supported to windward by two ropes or shrouds, fastened to a bit or rail fixed to two posts and fitted to the platform, so that the heels of the posts are secured to the outer side of the coamings of the smaller of the two vessels.

The Feejee islanders are very expert in the management of these vessels: both ends being alike, they are sailed either end foremost. When under sail it is necessary that the smaller canoe or outrigger should always be on the weather side; the canoe itself is never tacked, but the sail. The process of tacking will be better understood if given in the words of one who has seen it performed:—

The operation of tacking was effected by luffing into the wind, “when the rake of the mast, which is stepped on a

kind of hinge, and always inclines forward, was reversed; and at the same time a number of men, clapping on the tack of the sail, or the point where the yard and boom meet, hauled it aft; the yard, being nicely poised in the slings, and hoisted over a fork at the mast-head, then swung round, and the 'unwilling tack' was dragged to the loop, or becket, into which it was inserted at the other end of the vessel. The business was conducted apparently with very little order, and the process occupied a much longer time than that of tacking ship would have done with us."⁵

Under the skilful handling of the natives these canoes are enabled to carry sail in heavy winds, and to travel fast, preserving an almost upright position: this is maintained by several of the crew squatting on the windward side of the platform, or in the hold of the outrigger; and thus, by the extra weight, counterbalancing the pressure upon the sail.

It is customary for the chief to hold the end of the sheet; and it is, therefore, his task to prevent the upsetting of the canoe, or the carrying away of the mast. The canoe is steered with a large-bladed oar of stout proportions.

In smooth water, these vessels sail very swiftly; but when any extra pressure of sail is put upon them, it is found that their hulls are scarcely equal to the strain, and they become leaky, and require one or two hands to be constantly baling out the water.

They, nevertheless, make long voyages of hundreds of miles—to Tonga, Kotuma, and the Samoan Islands.

The building of these canoes frequently occupies several years; one is never built under three or four months. The tools employed are of the simplest kind, consisting chiefly of an axe, gimlet, chisel, and knife; and a few nails have been used of late years.

The carving is performed with tools made of the teeth of small animals, set in hard wood, and yet the workmanship is excellent.

⁵ Erskine's "Western Pacific Islands," p. 139.

Dimensions of a Feejee double canoe of the most common size.⁶

Length of larger canoe	...	...	...	70 feet.
„ smaller canoe	...	...	...	55 „
Distance of the canoes apart...	...	...	...	7 „
Length of platform	...	...	...	30 „
Breadth of do.	...	...	...	15 „
Length of cuddy	...	...	...	15 „
Breadth of do.	...	...	...	6 „
Height of platform above water-line	...	...	...	10 „
Draught of water	...	...	...	2 to 3 „
Length of mast	...	...	...	35 „
Length of yard	...	...	...	60 „
Booms	...	...	...	15 and 35 „

A canoe of this description will carry, conveniently, forty or fifty persons.

CANOES OF THE FRIENDLY ISLES.

THERE is a great similarity between the canoes of these islanders and those of the Feejees; and there can be no better evidence as to the ingenuity of the inhabitants than the art displayed in the construction of these vessels; which, in point of neatness and workmanship, are stated, by Captain Cook,⁷ to exceed anything of the kind he met with in the Western Pacific. The planks, or pieces of which they are composed, are sewn together with bandages in so neat a manner, that on the outside it is difficult to see how they are joined; all the fastenings being on the inside, and passing through cants or ridges, which are wrought on the edges and ends of the several boards which compose the vessel. At each end is a kind of deck, one third part of the whole length of the vessel, and open in the middle. In some of these

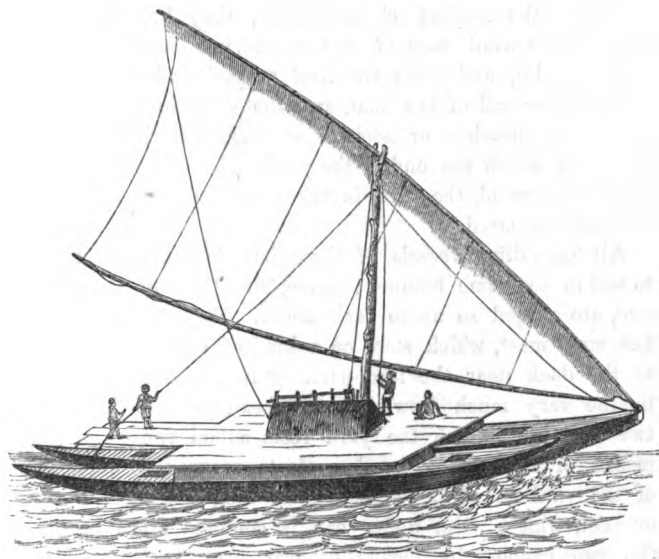
⁶ Wilkes' United States Exploring Expedition.

⁷ See Cook's Voyage towards the South Pole.

canoes the middle of the deck is ornamented with white shells, stuck in a row on little pegs, wrought out of the same piece which composes it.

Those which are called single canoes all have outriggers, and are sometimes navigated with sails ; but more generally with paddles, the blades of which are short, and broadest in the middle. The single canoes are from twenty to thirty feet in length, and about twenty or twenty-two inches broad in the middle ; the stern terminates in a point, and the head in a wedge-like form.

The double canoes of these islands are composed of two vessels, each about sixty or seventy feet long, and four or five broad amidships, and each end terminates nearly in a point similar to the single canoe ; but those which compose the double canoe have sides or coamings round the middle or open part, in the form of a long trough, composed of boards closely fitted together, and well secured to the body of the vessel. The double canoe is formed of two of the last described vessels fastened together, parallel one with the other, about six or seven feet apart. The joining together is effected by means of strong cross-beams, supported by stanchions fixed to the canoes, and secured by bandages and lashings of sennit made of cocoa-nut bass. Upon these beams a boarded platform is laid and fixed, extending from side to side, and a little beyond the outer side of each canoe, the width from outside to outside being thirteen feet nine inches ; and thus the double canoe is made very strong and burthensome, but light and buoyant as the nature of the work will admit, and so it becomes a vessel of burthen fit for distant navigation. (*See the illustration on the opposite page*). The double canoe is rigged with one mast, the heel of which is stepped and secured through the platform, into the fore part of one of the canoes, and can be raised or lowered at pleasure ; the sail used is made of matting, and is of lateen-like shape, fitted to a long yard and boom. On the platform of the double canoe a cabin or hut is erected, which generally contains a moveable fire-place, or



FRIENDLY ISLES CANOE.

trough of wood filled with stones. There are hatchways leading through the platform into each hold of the canoes.⁸

Captain Cook observes that the only tools used by the natives of the Friendly Isles in constructing their boats, are hatchets, or rather thick adzes (made out of a smooth black stone which abounds at Tofoa), augers made of sharks' teeth fixed on small handles, and rasps of the rough skin of a fish, fastened on flat pieces of wood. With tools such as these they contrive to build their boats not only neatly, but strong and durable.

SAILING AND MANAGEMENT OF THE DOUBLE CANOES OF THE FRIENDLY ISLES.

These canoes are sailed under a sort of lateen rig, as before mentioned. The slings, by which the sail is hoisted, are attached to the yard nearly in the middle.

⁸ Captain Cook's Second Voyage.

When the natives change tacks, they luff the canoe up into the wind, ease off the sheet (just as the Ladrone islanders do), and bring the heel or tack-end of the yard to the other end of the boat, and the sheet in like manner. There are notches or sockets at each extremity of the vessel, in which the end of the yard fixes. When they sail before the wind, the yard is taken out of the socket, and the sail is squared.

All the sailing vessels of these islanders are *not* rigged to sail in the same manner; some, and those of the largest size, are rigged so as to tack about. These have a short but stout mast, which steps on a kind of roller that is fixed to the deck near the fore part. "It is made to lean or incline very much forward; the head is forked, on the two points of which the yard rests as on two pivots, by means of two strong wooden cleats secured to each side of the yard at about one-third its length from the tack or heel, which, when under sail, is confined down between the two canoes, by means of two strong ropes passing through a hole at the head of each canoe; for it must be observed that all the sailing vessels of this sort are double. The tack being thus fixed, it is plain that in changing tacks the vessels must be put about; the sail and boom on the one tack will be clear of the mast, and on the other, will lie just as a whole mizzen.

"The outriggers and ropes used for shrouds, &c., are all stout and strong. Indeed sail, yard, and boom are altogether of such an enormous weight, that strength is required."⁹

In order to form some idea as to the rate of sailing of these canoes, Captain Cook tried experiments on board one of them with the log-line, and found the rate to be about seven knots an hour, the canoe being close-hauled and the wind very strong.

⁹ Cook's Voyage towards the South Pole, vol. ii. p. 17.

TONGESE CANOES.

The canoes of the Island of Tonga (or Tongataboo) are precisely similar to those of the Feejees; indeed, the large Tongese canoes are generally built in the Feejee Islands, as Tonga produces no timber fit for the purpose, and the Feejeeans are said to have acquired the art of building their large double canoes from the Tongans.¹⁰ In all these canoes, whether double or single, there are small hatchways at both ends, with high coamings; and it would appear as if they constantly leaked, for whenever they are seen under way, there is always some one in each of the end-hatchways baling out the water.

The double canoes, or those with a heavy outrigger, sail much faster on a wind than before one; and it is obvious that these vessels require a peculiar kind of seamanship, but in which the Tongese are particularly expert. When there is no wind they propel the canoe by a mode of sculling that is peculiar to the Tongese and Feejees; the sculler, instead of using the oar in the regular way, stands behind it, holding it perpendicularly, in a manner so as to throw the whole weight of his body upon the oar, and so assist his strength in using it: the oar is thrust through a hole in the platform, and so confined whilst being employed. There is generally one oar at each end; these oars are ten feet long, and have broad blades. When several oars are used, the Tongese scullers are kept strictly in time by a tune, in which all join: this custom does not appear to be practised among the Feejeeans.

These apparently fragile vessels are navigated, in the face of trade winds, voyages of hundreds of miles to and fro between the Fiji, Samoan, and Tonga islands.

Wilkes (in his *U.S. Exploring Expedition*, vol. iii.) speaks of a canoe he saw off the island of Ovalau, belonging to Tanoa, the King of Anibau, which was under the management of a crew of forty Tongese. He says:—"It had a magnificent appearance, with its immense sail of white mats; the pennants

¹⁰ Erskine's *Western Pacific Islands*, p. 439.

streaming from its yard denoted it at once as belonging to some great chief. It was a fit accompaniment to the magnificent scenery around, and advanced rapidly and gracefully along. It was a single canoe, one hundred feet in length, with an outrigger of large size, ornamented with a great number (2,500) of the *Cypræa ovula* shells. Its velocity was almost inconceivable, and every one was struck with the adroitness with which it was managed, and landed on the beach."

The same author also adds:—"I was told that Tanoa frequently amuses himself when sailing, by running down canoes, leaving those who belong to them to recover their canoe and property the best way they can."

The platform is the general resort of those aboard the canoe: the hull, even of the main canoe, is seldom occupied by any one except those employed in baling out the water at sea.

These canoes, as also those of the Fiji islanders, are carefully protected when not in use, and placed under cover of the lofty canoe sheds, which are erected on the beach.¹¹

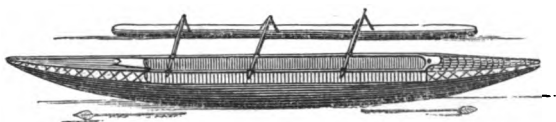
The smaller canoes of the Tongese are similar to those of the Samoan islanders.

FULANGESE CANOES.

The island of Fulanga is one which produces fine timber; and it is, therefore, much resorted to by the Vavao and Friendly Islanders as the depôt for canoe building.

Commander Wilkes, who visited this island about the year 1840, makes mention of three large canoes he saw in the process of construction under a long shed. One of these vessels, on measurement, was found to be 102 feet long, seven feet wide, and five feet deep; and he speaks of it as "a beautiful model;" the other two were somewhat smaller. The builders informed him they were intended for a Vavao chief, and the work was being performed under a contract, the price agreed on to be paid in whales' teeth, axes, guns, &c.

¹¹ Erskine's Western Pacific, Wilkes's Voyages, &c.



SAVAGE ISLAND CANOE.

The main part of these canoes are made from a single tree, wash streaks being afterwards fastened to the low sides; the fore and aft parts are covered over, and handsomely carved. They are from twenty to twenty-four feet long, and furnished with an outrigger composed of a log of wood, which floats in the water parallel with the canoe, and is supported by three spars placed transversely, and forming a sort of platform, on which they lay their spears and other implements.

The contrivance of the outrigger is absolutely necessary to enable these narrow canoes to maintain their equilibrium in the smoothest water; but the outrigger must not be confounded with that of a spar projecting to windward when under sail, to enable the canoe to be ballasted or counterpoised by some of the crew going out and sitting upon it,¹² as described under the head Samoan canoes, and others. Outriggers are common to most of the canoes of the Pacific.

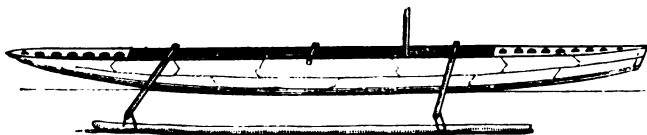
The paddles employed in these canoes are very prettily shaped, like a plantain leaf; they are short and concave in the blade, and handsomely carved.

NEW CALEDONIA AND LOYALTY ISLAND SAILING CANOES.

Captain Erskine¹³ alludes to some large canoes he met with in these islands, rigged with two sharp-headed sails of matting, laced to long flexible yards, and having somewhat the appearance, when set, of the sprit-sails in common use among the Thames barges. These appeared to be fitted for long voyages, and had earthen fire-places constructed on deck. They were each apparently capable of accommodating thirty persons.

¹² Erskine's Western Pacific Islands.

¹³ Ibid.



SAMOAN CANOE.

The natives of Samoa, or Navigator Islands, display very considerable skill in the construction of their canoes.

The largest of the Samoan canoes are from thirty to sixty feet in length, and capable of carrying from ten to fourteen paddlers, besides a sitter. They are formed of planks fastened together with sennit. There is no regularity in the length or breadth of the planks. On one of the edges of each plank a ledge or projection is formed, which serves to attach the sennit, and to connect and bind it closely to the adjoining one. The labour bestowed in joining so many small pieces of plank is surprising. Before the pieces are joined, gum from the bark of the bread-fruit tree is used to stick them close, and prevent leakage. On the outside the pieces are so neatly joined as to require a close examination before the seams can be detected; and this perfection of workmanship is the more astonishing when it is considered, that the only tools they use are a gimlet or piercer, and a piece of iron tied to a staff, thus forming a sort of adze.

These canoes are long and narrow, and their shape approaches even to elegance. They are decked fore and aft, and provided with an outrigger, as shewn in the engraving.

When propelled with paddles the natives sit two abreast, and the canoe is guided by a steersman. The seat of honour is on the forward deck, in the centre of which is a row of pegs, which are covered with the large white ovula shell, by way of ornament. The striking peculiarity

of these canoes (and in which they differ from those of other islanders) is, that they have both prow and stern; and therefore the sail cannot be shifted without tacking the boat; consequently, the outrigger that constitutes their safety under sail is alternately to windward and leeward; when to leeward it is not half the protection to the canoe, in preserving its stability, that it is when to windward. These canoes, however, carry less sail than those of the other natives of Polynesia; and in order to guard against the danger of upsetting when the outrigger is to leeward, the precaution is taken of rigging a *suati* (i. e. a sprit or boom), which projects from the windward gunwale; and, in some respects, answers the purpose of an outrigger. When the wind is heavy, one or two of the crew go out upon the *suati*, and so counterpoise the canoe against the force of the wind.

The sail is made of matting, in the manufacture of which considerable labour is sometimes bestowed. It is of triangular shape, with its apex below, and standing about ten feet high. The mast is stepped at about one third from the bows. The matting used in the sail-making is made entirely by hand, by interlacing the fibres. Some of the finest textures are as soft and pliable as cotton canvas. These canoes are not calculated for long voyages.

The Samoans have no large double canoes such as those of the Tongans and Feejeeans, except those which they procure by barter with those islanders.¹⁴

The usual Samoan fishing-canoe is made of a single tree, with a small outrigger to balance it.

MANUAN CANOES.

THE canoes of the Island of Manua are described by Wilkes, in his "United States Exploring Expedition," as the best among the islanders of the Samoa group.

¹⁴ Erskine's Western Pacific, Wilkes's Exploring Expedition, &c.

They are from thirty to forty feet in length, and sufficiently capacious to accommodate twenty or twenty-five men.

The bottom of these canoes consists of a log, or trunk of a tree, hollowed and shaped as a foundation, and then raised upon with boards or strips of wood. Both ends of the canoe are partly covered or decked. The canoe is also provided with an outrigger, but which does not stand out so far from the canoe as the outriggers usually attached to the canoes of other islanders.

The Manuan canoe is propelled with great swiftness by hand paddles, in the usual way.

The seat of honour in these canoes, or place where the chief sits with his legs crossed, is on the forward platform or deck.¹⁵



KINGSMILL CANOE.

The canoes of the Kingsmill and Ellice's group, Western Pacific Islands, are somewhat different to those of the other islanders.

They are commonly from twelve to fifteen feet in length, from two to three feet in depth, varying from fifteen inches to two feet in width; those in the northern islands being much larger, some of them sixty feet in length.

¹⁵ Wilkes, vol. ii.

These canoes are very well modelled, and, in some respects, better built than those of any other islanders in the Pacific; they have considerable sheer, and are built in frames, each canoe having six or eight ribs or timbers. The sides are in irregular pieces of cocoa-nut plank, varying in length, from a few inches to six or eight feet, and from five to seven inches in width. They are joined very neatly, and sewn with sennit; and, in order to make them water-tight, they insert slips of the pandanus leaf under and between the planking, in the same way as brown paper is used in boat-building.

The manner in which they attach the upright to the flat timbers displays considerable ingenuity; they are so secured as to possess all the virtue of a double joint, making them easy in a sea-way, and capable of withstanding the force of the waves. One side of the canoe is nearly flat, and in this respect they resemble the proa of the Ladrone Islands. They are also provided with an outrigger, but of smaller proportions than those of other islanders, and the stage or platform covers less space.

These canoes are tacked and sailed in the same manner as those of the Feejee islanders, the outrigger being always kept to windward, so that they sail with either end foremost.

Wood being an exceedingly scarce article with the natives, the mast and yards are of several pieces neatly joined together.

The shape of the sail is triangular, and very similar to that of the Feejee islanders; the mast is stepped in a fore-raking position, and the boom is considerably elevated.

The natives are very expert in the management of these canoes, and always avoid using paddles as much as possible.

Their paddles are made of a piece of cocoa-nut board, or tortoiseshell, about six inches square, and secured to a round stick for a handle. Some of their canoes are built entirely of wreckage wood, which is always a great prize to them.

Notwithstanding the odds and ends of which these canoes are composed, they are strong and durable, and also elegant



in shape. One of them, of a suitable size for ten persons, occupies five or six months in building.¹⁶

BOATS OF TAHITI.¹⁷

The canoes or boats of Tahiti (or Otaheite), and the neighbouring islands, may be divided into two general classes—Ivahahs and Pahies.

The ivahah, in shape, is wall-sided and flat-bottomed; it is used for short excursions to sea. The pahie is bow-sided and sharp-bottomed, and is used for longer voyages.

The ivahahs are all after the same shape or pattern, but vary in length from ten feet up to seventy or more. The breadth is in no way proportionate; those of ten feet long are about one foot wide, whilst those of seventy feet long are scarcely two feet wide.

Ivahahs may also be classified thus:—There are war ivahahs, fishing ivahahs, and travelling ivahahs: the war boat being by far the largest, with the head and stern peering in a crescent-like form to the height of seventeen or eighteen feet in some of the boats, though the body be only three feet deep. These ivahahs never go to sea singly, but two are lashed together side by side, and kept at a distance of about three feet apart, by strong poles laid athwart the gunwales; and upon these, in the fore-part, the fighting stage or platform is raised on pillars about the height of six feet, the stage being about ten or twelve feet long, and somewhat wider than the boats.

The fishing ivahahs vary in length from ten feet to about forty; all that are of the length of twenty-five feet and upwards occasionally carry sail. These boats are sometimes joined together, and have a house on board; but this is not common.

The travelling ivahahs are always double, and fitted with

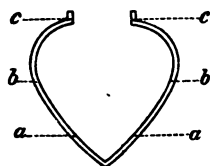
¹⁶ Wilkes, vol. v.

¹⁷ See Dr. Hawkesworth's *Voyages*, vol. ii. cap. 18.

a small neat house about five or six feet broad, and six or seven feet long; it is erected on the aft part of the platform, for the convenience of the principal occupants of the boat, who sit in it by day and sleep there at night.

All these ivahahs have high peering sterns; in those twenty-five feet long the stern rises about four feet, and so in proportion. The bows and stem are covered with a board, which projects forward about four feet, and forms a sort of fore-deck or standing-place, the only advantages of which appear to be, the convenience it offers for stepping in and out from the beach.

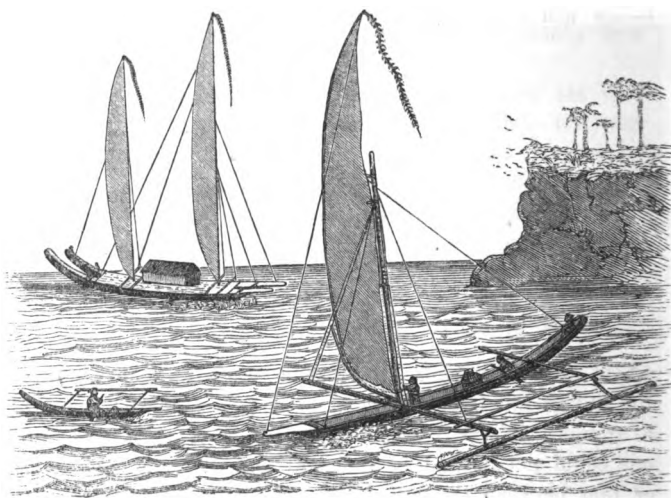
The pahie also varies in length from thirty up to sixty feet; but, like the ivahah, is very narrow. The sectional form of the pahie is rather full, as the transverse section below will explain, and will also illustrate the manner in



SECTION OF PAHIE.

which they are built. The first stage or kelson, below the dotted line *a a*, is formed of a tree hollowed out like a trough; this part sometimes consists of three trees, as one could not be had of sufficient length. The next stage, or that between the dotted lines *a a* and *b b*, is formed of straight planks about four feet long, fifteen inches broad, and two inches thick. The third stage, or that between the dotted lines *b b* and *c c*, is, like the bottom, formed out of the trunks of long trees hollowed into its bilging form. The upper part, or *coamings*, as they are termed in English boats, is formed of straight planking.

To form all these parts separately, without saw, plane, chisel, or any other iron tool, may well be thought no easy task; still less so to join them together; but they do so with considerable ingenuity, by sewing, clamping, and lacing them with strong thongs of plaiting: and the nicety with which this is done may be inferred from the fact of their being sufficiently water-tight for use without caulking. As the plaiting soon rots in the water, it is renewed at least once a year; in order to do which, the vessel is taken entirely to pieces. The head and stern are rude with respect to design, but very neatly finished and highly polished.



PAHIES.

The pahies, like the large ivahahs, are used for fighting, but principally for long voyages. Two of these are fastened together, side by side, in the same manner as the ivahahs.

They are also fitted with a stage or platform, proportionably larger than those of the ivahah, as the form of their bottoms enables them to sustain a much greater weight.

Those that are used for sailing are generally double, and the middle-sized are said to be the best sea boats. The natives are sometimes a fortnight or more at sea in these boats, and could go longer voyages if they had more stowage for provisions and water.

Those that are sailed singly are fitted with a log of wood, or outrigger, which is fastened to the end of two poles that are placed across the vessel, and project over to windward from six to ten feet beyond the sides, according to the size of the pahie. The outrigger is essentially necessary to preserve the stability of these vessels, when the wind is heavy.

Some of them are rigged with one mast, and some with two. When the length of the pahie is 30 feet, that of the mast is somewhat less than 25; it is stepped through the frame that is placed across the canoe.

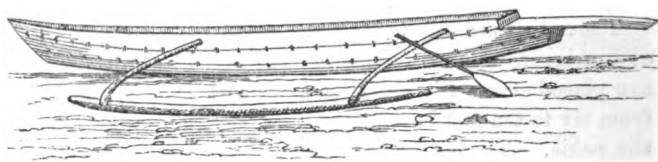
The sail, which is made of matting, stands about one-third higher than the mast, and very much peaked at the top, square at the bottom, and curved at the side—somewhat resembling what we term a shoulder of mutton sail. The sail is completely framed with wood, and there is no contrivance either for reefing or furling it. At the top of the mast, and peak of the sail, are placed a bunch and streamer of feathers, as shown in the engraving. They have no other contrivance for steering than the paddles, which two or more of the crew use at the aft part of the pahie.

The only thing in which these remarkable vessels excel, appears to be in the great advantages they possess, with their high peering sterns, for beaching or landing, and putting off from the shore in a heavy sea, without shipping water.

The pahies are kept with great care in boat-houses, built specially for their reception. These are formed of poles set in rows in the ground, the tops being drawn together and fastened with cord, forming a sort of Gothic arch, which is

then thatched quite to the ground, the ends only being left open. Some of these canoe-sheds are 50 or 60 yards long.

When on their long voyages the natives steer by the sun during the day, and by the stars at night, many of which they distinguish by names.¹⁸



TAHITIAN CANOE.

The common canoes of the Island of Tahiti are of better form and construction than those of the Disappointment Islanders.

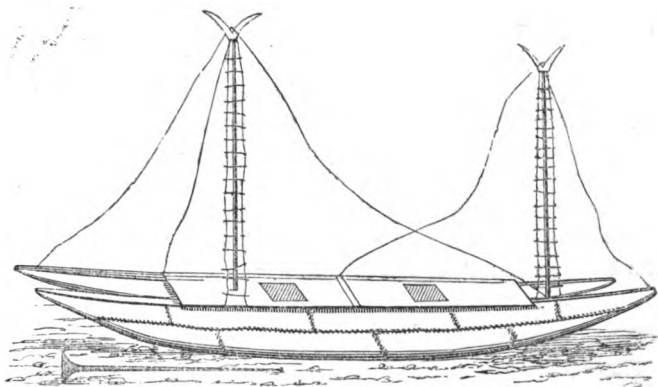
The outrigger in some of these canoes is neatly secured on one side of the canoe, by thrusting the pieces which support it *through* the upper strake of the canoe, instead of over the top. The common Tahitian canoe is also provided with a similar landing stage to that of the ivahahs and pahies, except that it is placed at the stern ¹⁹ instead of the bows.

The trading canoes of Tahiti are somewhat different to the last described, inasmuch as the outrigger of the trading canoe is attached in the more usual way, the bearers being lashed across the top gunwales.

These canoes are propelled with small paddles, which are employed on the opposite side to that on which the outrigger is placed.

¹⁸ Hawkesworth's Voyages, vol. ii.

¹⁹ Wilkes, vol. ii.



DOUBLE CANOE OF THE PAUMOTA GROUP.

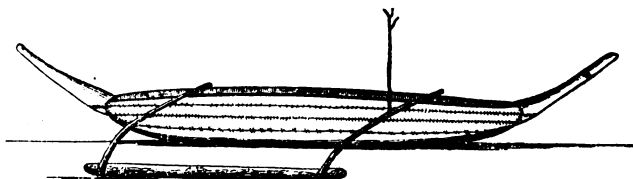
The natives of the Paumotu Islands, in the Low or Dangerous Archipelago, disdain to use nails in the construction of their canoes, or hempen ropes for their sails and rigging. Their double canoes are remarkably curious. They are composed of two canoes, each 35 feet in length, by $4\frac{1}{2}$ in width. The two vessels are connected by a strong framework of wood, and a platform which extends nearly over the whole surface of the two canoes. When long voyages are undertaken in the double canoe, a temporary hut is erected on the platform.

Every part of the canoe is neatly put together, and secured with twine and sennit made of cocoa-nut fibres; no iron or metal of any kind is used in their construction.

The sails and rigging of these canoes are also very remarkable. They are provided with two masts, the shrouds and rigging of which are merely tough branches of vine, and these supply the place of ropes. Each mast is fitted with a forked top-piece, in shape resembling the extended wings of a bird at the moment of alighting. Over the forked top-piece other vine ropes are placed, and by these the sails are hoisted and lowered.

The sails of the Paumotu canoe are made of matting and cocoa-nut fibres, and the canoe is steered with a large oar, the shape of which resembles the flat section of a trumpet. In these canoes the islanders undertake voyages of many miles to neighbouring lands, steering by the sun in the daytime, and by the stars at night, when out of sight of the coast.

The natives carry on a small trade in their double canoes ; but these are principally the Chain Islanders, who supply themselves at Tahiti with various small articles, in exchange for their cocoa-nut oil and dried fish.²⁰



WYTOOHEE CANOE.

The canoes of Wytoohee, one of the Islands of the Paumotu group, are very small, being only from twelve to fifteen feet in length. They are built of strips of cocoa-nut wood sewn together, and are so light that two persons may carry the canoe on their shoulders.

These canoes have projecting stem and stern pieces, as shown in the engraving ; and it is by aid of these projections that the natives are enabled to get into them from the water without upsetting them.

They are also provided with an outrigger of a very simple kind, but which gives additional stability to the vessel. The paddles by which they are propelled are curved backwards.²¹

²⁰ Wilkes, vol. i.

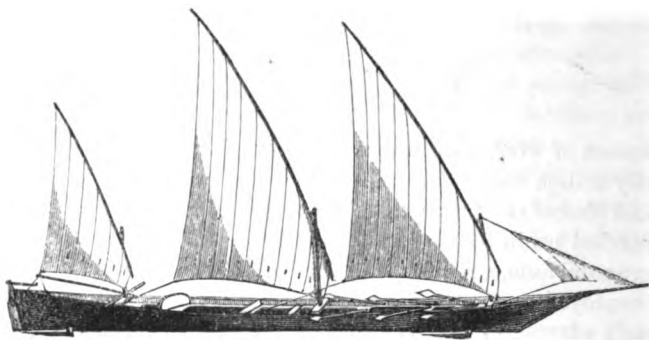
²¹ Ibid.

BOATS OF HINDUSTAN.

BOATS of every size, and as numerous in variety as those of any nation under the sun, are met with on the noble rivers and shores of Hindustan. The chief traffic of the country is carried on in boats, some of which are of symmetrical and graceful form, and able to sail very fast in smooth water. Those employed on the Ganges have high stems and sterns, and are well adapted to the peculiar and intricate navigation of that locality. These boats have necessarily a large rudder, and even that is often rendered powerless, by reason of the rapidity of the current. Great presence of mind is sometimes necessary in the navigation of boats on the Ganges, particularly by the man at the prow; who, with surprising agility, has to dart his long bamboo pole to the opposite bank, and turn the boat, or fend off, to prevent accidents. The boats employed between the Gulf of Cutch and the Sinde, or Indus, are chiefly Bugalos, Naodees, Muchoos, Coteyahs, and Gungos.

Almost all Indian boats have considerable rake both at stem and stern; the fastest have long, sharp, and generally hollow bows, and the stern is often as sharp and tapering as the bow; but those used for the purposes of trade have less graceful proportions.

The Indian method of construction differs from the European in many respects. Instead of nailing and clinching the planks, they are carefully rabbetted together in a neat and durable manner; and although such a method occupies a vast amount of time and labour, the vessels so constructed seldom require caulking, all the seams and interstices being filled with cotton stuff, and the whole exterior served over with pitch, or other resinous substance. The wood used by the Indians in boat and ship-building is chiefly teak; which is well known as hard, tough, and durable.



INDIAN PLEASURE-BOAT.

Indian pleasure-boats sail very fast with a free wind, but, generally speaking, are too long and tapering for turning to windward with effect; they are usually flat and full amidships, but drawn out to great extremes at each end.

The common mode in which Indian boats are rigged is the lateen and lug.

The style of rigging the lateens of India differs from that of the Mediterranean. The Indian lateens have seldom more than two masts, of which the hinder one is much the smaller; they have also a bowsprit; and the sails, instead of being right-angled triangles, like true lateens, have the foremost angle cut off, so that they are nearer the principle of the old settec sail.

BOATS OF THE PUNJAUB.

The pleasure-boats of Punjaub are of a long and graceful form, but broad amidships, with a cabin or house; but they are very shallow, and drawn out to great length at the bows: there is a railing all along each side of the vessel, and it is decked all over, except the part occupied by the cabin. These boats are gilded, painted, and ornamented with good taste.

BOMBAY FISHING-BOATS.

The Bombay fishing clippers rank among the fastest of their class in India, and are worthy of imitation in many respects. Some of them are splendid sea boats, but not so quick in staying, nor so convenient for turning to windward in narrow channels, as the boats of some other nations. Their greatest draught of water is forward—which is the reverse of the English and American system; they have also a very sharp, hollow bow; the breadth of beam is carried well aft, and finishes with an overhanging stern. The rig consists of two lateen sails; but no bowsprit, nor sail of any kind is set beyond the stem; both masts rake forward; the mizzen-mast is generally a trifle more than half the length of the main-mast, and the yards in the same proportion. Some of these boats measure from twenty to thirty tons, the main-yard for which is nearly a hundred feet in length; the head, or fore-leach of the sail is of the same dimensions as the aft-leach. They have neither shrouds nor rigging, the main-halliards being the only stay to the mast; the tack of the mainsail is managed with a luff-tackle purchase at the stem, in the same manner as the clew by the main-sheet; the double block being seized into the tack, and the single block strapped to the mast. In wearing, both sheet and tack are eased off; but in tacking, the sail and yard are shifted to either side of the mast without being lowered; the mast, in such case, must be rather above the yard at the slings, to allow a little play for the halliards, and must also have considerable fore-rake;¹ the manœuvre could not be performed with advantage on an upright mast.

In the International Exhibition of 1851, was a model of the Bombay yacht "Wave;" the model was taken from a fishing-boat of Bombay.

¹ This method is adopted in some of the lateen boats of the Mediterranean.

One remarkable peculiarity about this vessel is, that the keel is curved in a crescent-like form, so that the fore-end is two feet lower than the level of the keel amidships, and so acts as a gripe, or lee-board, the intention being to make the boat weatherly. This vessel has a comparatively flat floor, hollow entrance, and a sharp flat run; its length over all is forty-six feet, entrance breadth twelve feet, and depth amidships three feet eight inches.

Mainmast	36 feet long.
Main-yard.....	65 ,,
Mizzen-mast	22 ,,
Mizzen-yard	40 ,,

The sails are of lateen shape, and made of drill sewn in narrow cloths.

The "Wave," it is stated, was built as a pleasure yacht, but more particularly for the regattas, for which Bombay is famous; and when ballasted, has won many prizes. No boat of European form and construction had been found to compete with her in point of sailing in moderate weather.²

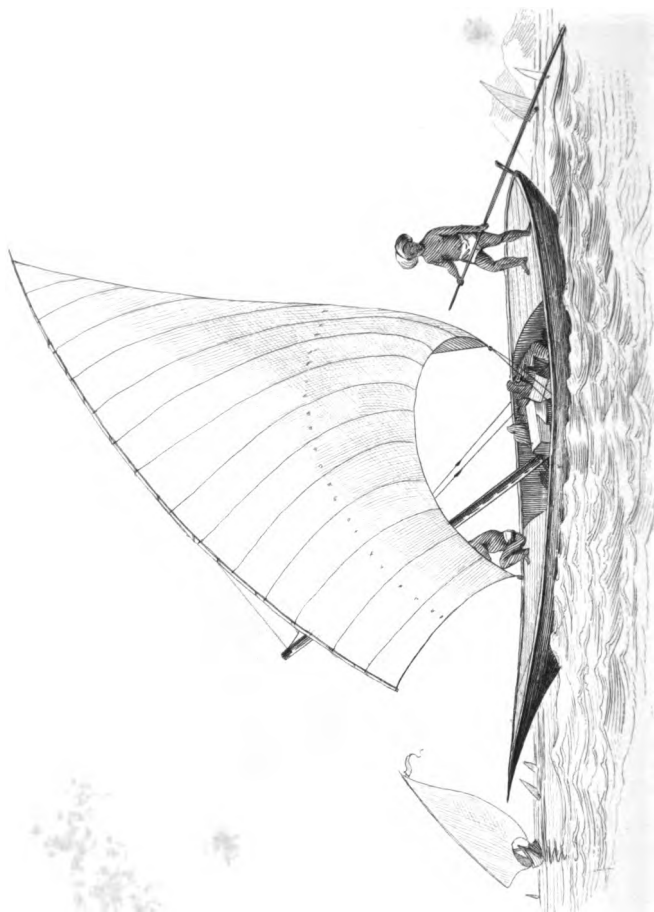
BOMBAY COTTON-BOATS.

This description of boat belongs entirely to the port of Bombay; they are so called on account of their being invariably employed in conveying cotton from the shore to ships bound for China and Great Britain, loading with that article.

These are the only boats made use of in loading and unloading the numerous kinds of outward and inward cargoes of ships visiting the port.

They are from twenty-five to thirty-five feet in length, ten to thirteen feet in breadth, and three and a half to four feet in

* *Vide* Official Catalogue of the International Exhibition, 1851.



BOMBAY DINKY.

depth: very rudely but strongly built, and the largest of them will carry fifteen tons of dead weight. They are also employed in bringing the produce of the Island of Salsette, such as grain, grass, vegetables, &c., to Bombay; also for the conveyance of troops, with their baggage, to and from Panwell.

The inside of the cotton-boat is lined with bamboo matting, to protect the cargo from bilge water. They are generally navigated by a crew of six men and a *tindal*—principally Mahomedans—who live in the boat.

On one side of the mast is a fire-place, and on the opposite a cask or tank, containing fresh water. The bottom is annually, or oftener, paid over with a mixture of chunam, or lime, and vegetable oil, which hardens, and is a good protection against worms. They have one mast, which rakes forward, and a yard of the same length as the boat. The cost of one of the best of these boats, complete, is about seven hundred rupees. They are mostly hired by the day, at a rate varying from two to five rupees, according to their size and the season of the year.

BOMBAY DINGHY.

(See Engraving.)

The Dinghy, Dingee, or bum-boat of Bombay, is a small boat from twelve to twenty feet in length, five to seven feet in breadth, and eighteen inches to two feet in depth. It is very similar in form and size to the cotton-boat before described, but smaller. They are rigged with a raking mast, and a yard the same length as the boat; and navigated by three or four men who, very frequently, are joint owners of the boat.

Dingees sail very well, and are employed in carrying persons to and from vessels in the harbour; they also carry persons desirous of visiting the Islands of Elephanta, Caran-

jah, and others in the harbour of Bombay. It is generally the practice of captains and commanders of ships to hire one by the month, at the rate of forty or fifty rupee.

BATELLES OF BOMBAY AND SURAT.³

These belong principally to the merchants, and are decidedly the best constructed, and better found in fittings and stores, than any other description of boat of Western India. They are built entirely of teak-wood, well planked and fastened with iron nails and bolts; they have a great rise of sheer forward, and a regular stem with madoes abaft. Some are fitted with a cabin under the poop, but the majority of them carry bamboo decks, over beams fitted for the purpose.

They are from thirty-five to fifty feet in length, fifteen to twenty feet in breadth, and five to seven feet in depth, and from twenty-five to one hundred tons burthen.

They are lateen rigged, having main and mizzen masts, both raking forward; also a boom or bowsprit, on which a jib is set; the main-yard is a little longer than the extreme length of the boat. They invariably have a break in the top-sides, from the fore part of the poop to the luff of the bow, nearly level with the beams, for the facility of taking in and out heavy cargos. At sea this break is stopped up with bamboo mats inside, and outside with soft mud or puddle. This excludes water, and is as water-tight as any other part of the hull. It is a remarkable fact, that no one ever hears of damage being done to the cargo from any defect in this part of the vessel, although, when the boat is fully laden, the break is about one foot or eighteen inches above water. These boats import cotton from Surat, Broach, Cambay, and other cotton growing districts, to Bombay; and teak timber from the northern forests, which is extensively used in ship-building at Bombay.

There is a great similarity between these vessels and the Arab Batelles, described and illustrated at page 273.

³ *Vide* Official Catalogue of the International Exhibition, 1851.

THE BUDJEROW.

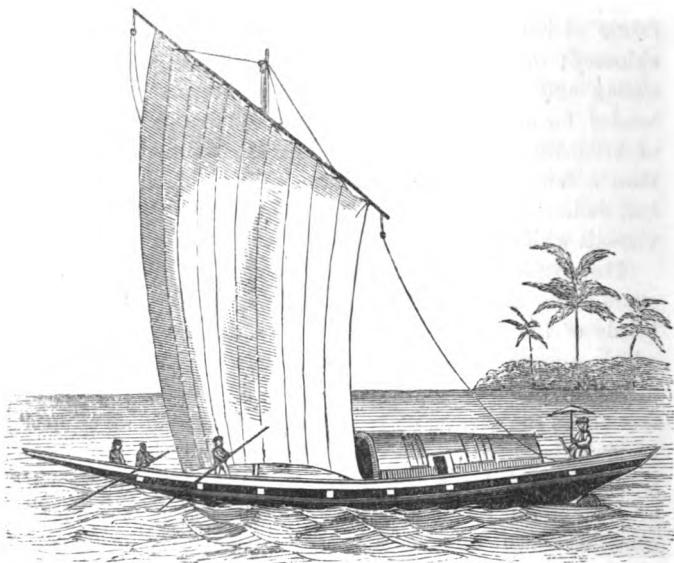
These boats, sometimes called Bengalee boats, are a good deal used by Europeans for travelling on the Ganges, and in various other parts of India. They have very high sterns, are full amidships, and round-bottomed. They are of various sizes, from twenty-five up to sixty feet in length. Some of the longest are rowed by as many as twenty oars, and are steered with a large oar or sweep, extending nearly ten feet abaft the stern. They are sailed with a square sail, and square topsail in fine weather.

The budjerow is decked all over, throughout its whole length, with bamboo. On the deck is erected a low, light fabric of bamboo and straw, like a small cottage without a chimney; this is the cabin and baggage room, and also the sitting and sleeping apartment for passengers; and if intended for a cooking-boat, there are one or two small ranges of brick-work, like English hot hearths, but not rising more than a few inches above the deck, with small, round, sugar-loaf holes, like those in a lime-kiln, adapted for dressing victuals with charcoal.

The roof of the cabin being by far too fragile for men to stand upon, and as the apartment itself takes up nearly two-thirds of the vessel, upright bamboos are fixed by its side, which support a platform of the same material, immediately above the roof; on which, at the height of about six or eight feet above the surface of the water, the boatmen sit or stand to work the vessel. They have for oars long bamboos, with disk-shaped blades, a longer one of the same sort being used to steer with; a long rough bamboo for a mast, and one, or sometimes two sails of a square form (or rather broader above than below), of very coarse and flimsy canvas, which is set on a bamboo yard.⁴ Some of the fastest of these boats are of a somewhat superior form and construction, and have three square sails one above another.

⁴ *Vide* Bishop Heber's "Narrative," vol. i. p. 84, A.D. 1828; Forrest's "Tour of the Ganges," &c.

With a fair wind these vessels sail merrily over the water, though they make but slow progress with a foul wind. The English at Bengal and elsewhere, have made great improvements of late years in the building or form of budjerows, by introducing a broad flat floor, square stern, and full bow; and in that form these vessels are safer and faster, and enabled to carry more sail; and, being of less draught of water, they do not so often run aground. Some of them have cabins six or seven feet high, and are very commodious.⁵



GANGES SAILING-BOAT.

⁵ "Voyage to the East Indies," by J. S. Stavorinus, Esq.; translated from the Dutch by S. H. Wilcocke, 1798; Forrest's "Tour of the Ganges," &c.

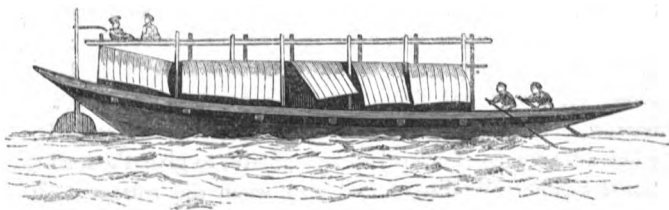
A gentleman in his budjerow is usually attended by a *pulwah*, or large open boat, for the accommodation of the kitchen, and a smaller boat called a *panchway*, for taking him ashore.

THE PANCHWAY.

The larger sized *Panchway*, or passage-boat, used on the Ganges, is a very characteristic and interesting kind of vessel; it is large and broad, and shaped like a snuffer-tray; it is also decked fore and aft, and the middle part is covered with a roof of palm branches, over which is lashed a coarse cloth, the whole forming an excellent shade from the sun.

The *serang*, or master, stands on the little aft deck, and steers with a long oar, the loom of which is lashed to a stanchion standing above the deck on the port side of the stern; another of the crew has a similar oar on the starboard quarter: there are also six rowers, who sit cross-legged on the deck upon the tilt, and ply their short paddles with much dexterity; but in the manner of oars, resting them on bamboos which are fixed to the sides, instead of rowlocks or thowls.

When the wind is fair, they use a large long sail of thin transparent sack-cloth in three pieces, very loosely tacked together and secured to a bamboo yard, and also hoisted on a mast of bamboo. These singular boats are sometimes taken on very long voyages, plying from various places on the Ganges, to and from Calcutta. For ten or fifteen rupees, the boatmen will convey passengers by them a hundred miles or more.



GANGES ROWING-BOAT.

THE BHAULEA.

The Bhaulea, or Ganges boat, is very much like the Budjerow. A peculiarity among many of the different forms of boats of the Ganges is the gallery, upon which light goods are carried ; and the manjee or steersman also sits there to guide the boat, as shown in the engraving at the foot of the preceding page.

CUTCH COTIYAHs.

These boats belong to the ports of Cutch, Mandivee, Poar Bunder, and some to Kurachee in the newly acquired territory of Sindé ; and trade between Bombay and those ports. They are very well built with a square tuck, and many of them have a regular built stern with ports, and are handsomely carved. Some have a deck fore and aft, but more commonly they have framework between the beams to ship and unship, for the facility of stowage, and a bamboo deck. Their general length is from thirty to fifty feet, by twelve to twenty-three in breadth, and seven to ten feet deep.

They are navigated by a crew of fifteen to twenty men, and a tindal. They are lateen rigged, with a main and mizzen sail, both masts raking forward, to keep the ponderous yards clear of the mast in lowering and hoisting.

THE PATTAMAR.

The Pattamar or Pattamach is a trading boat of Hindustan, employed by the natives of the Deccan and at Bombay, for conveyance of rice and other articles of merchandise ; it is remarkable for its peculiar shaped keel, which forms part of a circle, the hollow being directly under the midship section, where there is much less keel than at the extremities ; the extent of curvature varies according to the form of the boat ; those with very flat floors have the greatest curve, but in narrow and sharp bottomed boats the curve is less. The chief advantages of a keel so formed are, to give them additional weatherly qualities ; and also, when the pattamar gets aground (as it is very liable to do in the Deccan) it may be



THE PATTAMAR.

got off again without difficulty. The pattamar has a hollow bow and very raking stem; it is rigged with two masts, which rake forwards, the fore-mast being the larger, as customary with Indian boats; and the main-yard is considerably longer than the boat. The pattamar is sailed under large lateen sails, and is a stiff and fast sailing boat off wind, but not very well adapted for tacking or turning to windward.

THE MUCHOO,

or Muchwa, is a fishing-boat of the Deccan, bearing strong resemblance to the pattamar, but is much smaller. The Muchoo sails remarkably fast, even when close hauled, but from its small size is better suited to smooth water.

BUNDER BOAT.

This is a stout and strongly built boat, used in and about the Indian harbours; hence the name from *bundur*, a harbour. They are generally rigged with two masts, and sailed under two lateen sails.

THE BHOLCO.

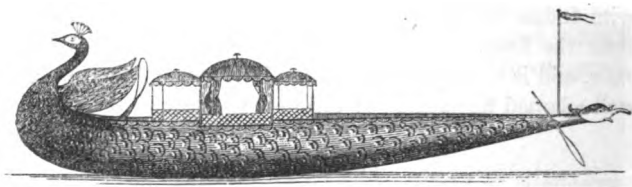
A small light boat, employed in the Upper Provinces of India, for the conveyance of passengers to short distances.

THE BUGALO.

These are a useful class of boats, trading between Bombay and the Persian Gulf; they are of various sizes, have high sterns, and are usually sailed with one large triangular sail. A smaller class of bugalos are employed on the Indus.

CHEEP BOATS AND PARINDA BOATS.

These are Bengal river boats—long, narrow, and shallow—with a hood or cabin abaft, and lattice deck or raised grating made of bamboo.



MOHR PUNKEE.

The Mohr Punkee, or Peacock-boat, is so called from its being built to resemble a peacock, having at its prow the figure of that bird, with its tail extending the whole length of the boat; the plumage on each side is beautifully painted and varnished, in imitation of the gaudy feathers of the peacock, and the stern finishes with the head of some ferocious animal. The extreme length of some of these boats is eighty feet and upwards; and the extreme breadth, which is toward the front, is about nine feet, from which they gradually

diminish to the stern, which terminates in a point. Over the broadest part is erected a pavilion, the canopy of which is six feet high, and covered with crimson velvet richly embroidered with gold, as are also the curtains which hang on each side; the whole being supported by several varnished pillars, the lower part of which are surrounded by a light rail. A narrow balcony, extending beyond the sides of the pavilion, serves as a receptacle for confectionery, fruit, sherbet, and other refreshments.

The floor of the pavilion is carpeted and cushioned; all persons sitting upon it with their feet doubled under their bodies.

The throne, or seat of eminence, is in front of the pavilion, where the Nabob, or person of the highest distinction sits.

A canopy of velvet and gold is spread above the throne, and supported by pillars and the pavilion.

The boat is propelled by a crew of thirty or forty, who sit behind the pavilion, with faces fronting the direction in which the boat is to go. The paddles are each furnished with two brass rings on their handles, which, clashing and rattling together, serve to keep the rowers in time.

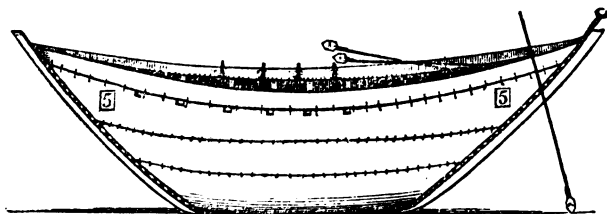
The boat is steered by a long oar fastened on the larboard side, near the stern. It glides with great velocity, and draws very little water. At the head and stern a flag-staff is set up, on which streamers of crimson silk are displayed.

The crew are directed by a pilot, who stands at the prow, and generally makes use of the branch of some plant to regulate their rowing, using much gesticulation, and telling his story to excite alternately laughter and exertion. These boats belong alone to the princes and nabobs of the country. They are very expensive, owing to their great length, and the carving and costly decorations with which they are fitted.

MASSOOLAH BOATS.

The Massoolah boats (sometimes spelt Mussulah, also Masuli,) are chiefly remarkable for their peculiar construction

and capabilities for beaching and putting off in the surf at Madras, to ships in the roads there. Neither nails, pegs, nor bolts, are used in any part of the structure of a Massoolah boat; but the planks are corded and laced throughout with the strong fibres of the cocoa-nut tree, layers of cotton being placed between the planks; and over the seams inside a flat narrow strip of tough fibrous wood is placed; the whole being then drawn tightly together with cordage. The object in building them in this manner is, that boats are so severely knocked about and bumped on the shore by the heavy surfs at Madras; and if put together with nails, in the usual way, they would soon be dashed to pieces. The Massoolah boat, however, remains as uninjured by rough usage, in broken water and on the beach, as if made of leather. The wood of which the Massoolah boat is composed is also tough and flexible.



In shape, the Massoolah boat is by no means comely; it resembles no other; varies in size, but is very deep and broad at the top, in proportion to its length and the small size of the bottom. It is flat-bottomed, has very high flaring sides, and abruptly pointed ends.

An elevated rowlock is fixed at the stern, so as to give a leverage for twisting the boat. The steersman stands on a sort of deck at the stern, and guides the Massoolah boat with a paddle or pole, which has a circular blade at the lower end, about a foot or foot and a half in diameter. A good deal of courage and nerve are requisite in the steersman of a Massoolah boat. It is rowed by six oarsmen, who sit facing the

prow, and row double-banked, *i.e.*, two on a thwart. The passengers sit aft on a bench, not so elevated as the seats of the rowers.

There is sometimes a good deal of risk in landing; but these boats are considered the safest form that can be used for the purpose on that dangerous beach. There is also the same risk when putting off from the beach; indeed, the launching of the Massoolah boat is an art only to be acquired by constant practice. The naked and hardy mariners who effect it cannot always, with all their expertness, prevent the water from breaking over the boat and drenching the passengers to the skin.⁶

These boats all carry their numbers distinctly painted on each bow.

MADRAS CATAMARANS.

Strangers, on visiting the coast of Coromandel, are generally much amused with these remarkable and fragile-looking rafts, and the manner in which they breast the heavy breakers at Madras; which, though nothing very formidable to look at, their force is irresistible.

The catamarans of Madras are formed of three logs of timber—not rounded logs, but shaped and roughly squared on the upper side. The length of the raft is from 20 to 25 feet, and breadth $2\frac{1}{2}$ to $3\frac{1}{2}$ feet; the logs are secured together by three spreaders and cross lashings; the centre log being much the largest, and fitted with a stem-piece having a curved surface at the fore end, which turns and finishes upwards to a point. The side logs are similar in form, but smaller, having their sides straight, and fitted to the centre log.

These well-known floats are generally navigated by two men, though sometimes by one only, with the greatest skill and dexterity; as they think nothing of passing through the

⁶ Col. Napier's "Wild Sports of Asia," 1844; Hall's "Fragments of Voyages and Travel," &c., &c.

BOATS.

MOONEY.

The Mooneys of Ceylon are very remarkable. The largest is one of the largest; it is employed to transport timber to various parts of the Indian coast. The pegs are used in constructing the hull, which is neatly and mechanically joined together; a caulking of soft matting stuff is used between the seams; the hull is low, but wide at the top, and of wedge-shaped form, somewhat resembling a Norway yawl, but without the rise at stem and stern: it has a considerable fore and aft rake; the top of the hull is formed of bamboo canes, but the deck is supported by cross beams or spars are placed under the deck fore and aft; to the projecting ends of the hull, the girds, sheets, and tacks of the sails are made fast. They are fast sailing vessels, and are usually fitted with long outriggers, suspended and fitted in the same manner as the outriggers of other Ceylon boats.

The Mooneys of Jafnapatam are differently constructed to those of Ceylon, being fastened with nails and iron bolts; they have no outrigger, and therefore carry smaller sails; they are awkward-looking boats.¹

BOATS OF THE CAVERY IN MYSORE.

The boats on the Cavery river are called "Donies" or "Mooneys." They are of a circular form, eight or ten feet in diameter, and are covered with leather.

Mr. Emerson, in his description of Ceylon, says:—"The Singalese Mooney is a large vessel, the hull of the Galle, with its gunwale is covered with clay, to prevent the water from getting in."

surf at Madras and at other parts of the coast, while boats of the country could not live on the waves.⁷

Catamarans are used chiefly for conveying letters, messages, &c., to and from ships in the roads; and when first seen at a distance, the men appear as if they were treading or kneeling upon the water, performing evolutions with a racket, and bobbing about in the surf.⁸ They carry letters in their conical-shaped skull-caps, which are tied and waxed to their heads; and these also ease the force of the heavy breakers, under which they are often obliged to dive their heads: sometimes they appear completely buried in the surf, which is constantly breaking over them: notwithstanding which, they are seldom known to be lost or swept from the raft; although they frequently venture out to sea many miles from the shore. But if they chance to be washed off by a heavy sea, woe betide them! for sharks abound on that coast. If the catamaran-man escapes the voracious jaws of the shark, he quickly regains his position on the raft; and, generally, without losing his paddle, an instrument the catamaran-men use with singular dexterity.

When they come ashore the raft is untied, and the logs left on the beach to dry.⁹

FISHING CATAMARAN.

The fishing catamarans used off the same coast are larger than those last described, being composed of four logs of flatter form, and the outside ones are curved or raised a little,

⁷ Exhibition Catalogue, 1851, vol. ii. p. 909.

⁸ In the log-book of one of the early Indian voyagers is the following entry, made whilst lying off in the roads at Madras:—"This morning, six a.m., saw distinctly two black devils playing at single stick. We watched those infernal imps above an hour, when they were lost in the distance. Surely this doth portend some great tempest."

⁹ "Letters from Madras," by a Lady, 1846; "Voyages and Travels," &c.

so as to form sides. They are, nevertheless, as shallow as a dripping-pan; the stem or fore part inclines upwards, and the stern also turns up a little.

There is a rope or wire for holding on by, which leads up the middle of the raft, being made fast to the lashings which hold the logs together fore and aft. The logs are shaped so as to be thin and light at the fore part, and heavier and thicker at the aft part. A wooden anchor, with a stone for a shank, completes the tackle of these remarkable rafts.

CEYLON CATAMARAN.

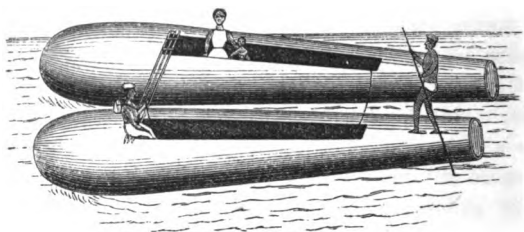
These are nearly identical with the catamarans of Madras. The Ceylon catamaran, however, has no separate stern-piece, but the middle log itself inclines upwards at the fore end, and forms a sort of stern. The side logs too, are rather wider and more substantial than those of the Madras raft. The Ceylon catamaran is employed in the same service, and answers every purpose of a Madras catamaran.

MADRAS SAILING CATAMARAN.

These are a very superior kind of raft, composed of three carefully-shaped solid logs of wood, turning up at each end with considerable sheer; each log is flat on the upper side, but rounded at the bottom; the fore ends are also rounded; the stern ends are flat. The middle log, which is much the largest, is placed lower than those at the sides, and hangs down like a round-shaped keelson; the two other logs, therefore, form sides and seats for the crew of the raft, and the middle one being lower than the side ones, a sort of interior is formed. The three logs are held together by three cross pieces, firmly secured upon the upper surface of the logs; besides which, the logs are bolted together, and form a very pretty and graceful-looking raft, of narrow form, apparently much too narrow to carry sail without some artificial

contrivance; and therefore an outrigger is fitted to the raft on one side, with a heavy solid log neatly shaped and smoothed in the form of a long narrow boat.

These beautiful and ingeniously contrived rafts sail remarkably fast, and will hold a course to windward equal to that of a vessel with a deep keel. The natives fearlessly venture out to sea upon them many miles, and seem to regard strong winds as ordinary weather; their catamaran being a life-boat in principle; and there is no fear of its foundering.



THE LANGADY.

The Langady, or native ferry-boat of Nursapore, is somewhat curious, as will be seen by the engraving. These twin boats are made from the trunks of Palmyra trees. They are quite hollow inside, and have a slit or opening at the top for passengers and goods. The stern ends are fitted with circular pieces, like the head of a cask. Two or three poles are lashed across the head to hold them at a proper distance apart; at the stern they are merely held together by a rope. The ferryman stands on the aft part of one of the pontoons, and conducts and propels the langady with a setting pole, in the manner represented by the engraving.

CEYLON BOATS.

THE DHONEY.

SOME of the trading boats of Ceylon are very remarkable. The Dhoney or Yatrave is one of the largest; it is employed for conveyance of merchandise to various parts of the Indian Archipelago. No nails or pegs are used in constructing the dhoney, but the planking is neatly and mechanically joined and secured by lacings; a caulking of soft matting stuff is ingeniously laid over and interwoven between the seams; the hull is short and shallow, but wide at the top, and of wedge-like form below; somewhat resembling a Norway yawl, but with less curve amidships, and less rise at stem and stern: they have considerable fore and aft rake; the top of the cabin or hold is formed of bamboo canes, but the deck is wood. Several cross beams or spars are placed under the gunwale and deck fore and aft; to the projecting ends of which the shrouds, sheets, and tacks of the sails are made fast. These are fast sailing vessels, and are usually fitted with shifting outriggers, suspended and fitted in the same manner as the outriggers of other Ceylon boats.

The dhoneys of Jafnapatam are differently constructed to those of Ceylon, being fastened with nails and iron bolts; these have no outrigger, and therefore carry smaller sails; they are awkward-looking boats.¹

DHONEY OF THE CAVERY IN MYSORE.

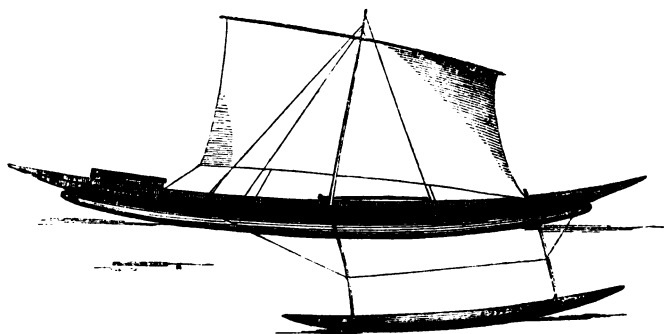
The ferry-boats on this large river are called "Donies" or Dhoneys. They are merely baskets of a circular form, eight or ten feet in diameter, and covered with leather.

¹ Sir Emerson Tennent, in his work on Ceylon, says:—"The Singhalese dhoney, south of Colombo, is but an enlargement of the Galle canoe, with its outrigger so clumsily constructed that the gunwale is frequently topped by a line of wicker-work smeared with clay, to protect the deck from the wash of the sea."

In these fragile vessels men and women, goods and merchandize, are transported with tolerable safety.

THE BALHAM.

The Balham, or Ballam, is found in the estuaries and shallow lakes around the northern coast of Ceylon: it is made by carving and hollowing out the trunk of a single tree of enormous size, and is the largest description of boat so constructed. The balham has a deck or covered compartment fore and aft, and is chiefly employed for carrying rice to different parts of Ceylon; it is rigged with a single lug-sail.



SAILING CANOES OF CEYLON.

The Ceylonese (also called Singhalese) or Point de Galle canoes, are very remarkable, both for the ingenuity displayed in construction, as well as the manner in which they are rigged. They are sometimes called double canoes, though one portion is merely a balance log, carried at the extremity of two flexible outriggers, each 18 feet long. They have always been made without nails; and although such

was the practice of the remotest period, it is retained to the present day.²

These curious canoes are life-boats in principle ; they carry no ballast ; but their peculiarly buoyant nature, and ingenious form of construction, enables them, under the skilful handling of the natives, to be as safe a kind of boat as any in which to venture out to sea. Their dimensions are from 18 to 30 feet in length by only $2\frac{1}{2}$ feet in breadth, and from two to three feet deep, exclusive of the wash-boards, which are from 10 to 18 inches deep, and sewn to the gunwale with coir yarns, loose coir padding being bound over the joinings.

The hull is formed from the trunk of the doop-wood, or pine-varnish tree, hollowed and smoothed in the usual way, but not left so open at the top as in other canoes ; the hollowed trunk being more like a cylinder, with a narrow slit about 8 or 10 inches wide, cut out of the top. The sides are slightly flattened, and both the ends turned up. The leeward side, in some of these canoes, is bow-sided and inclining to flatness, the weather side being fuller and rounder ; and in this respect they seem inclined to the principle of the flying proa of the Ladrone islands. Upon the cylindrical part of the canoe they build a sort of trough extending from one end to the other ; the flat sides of which, standing up above the cylinder, considerably add to the ticklish nature of the whole fabric ; but in order to prevent the canoe from oversetting, and to preserve its stability on the water, an outrigger is fitted on one side of the canoe in the following manner :—two curved poles or spars are laid across the gunwale or trough of the canoe, at right angles to its length, and extending on one side some 12, 15, or 20 feet beyond it, where the ends are secured to a log of buoyant wood, about half the length of the canoe, and placed parallel with it ; the log being shaped as a boat, and sharpened and turned up at each end. The outrigger thus fitted acts as a powerful lever to the canoe, and preserves its stability under great pressure of sail.

² Sir Emerson Tennent's "Ceylon," vol. ii.

The mast is stepped exactly amidships, and secured by shrouds extending from the mast head to the gunwales: the canoe is sailed under a large square-sail, the yard of which is slung precisely in the middle: the tack of the sail is made fast in the bows of the canoe, and the clew is held by a main-sheet, which hauls from the stern.

The sail of the canoe is generally so large in proportion, that the tall, slender mast has to be further supported with backstays: a shroud or stay is also attached to the outrigger, and secured to the top of the mast. The mast is thus well supported, but not more so than is necessary for so unequal a spread of canvas.

On tacking or putting about, they do not turn the boat, but merely change the tack of the sail into the clew, and *vice versá*. The stern then becomes the prow, and thus the outrigger remains on the windward side of the canoe.

In a strong breeze, one or two of the crew walk out to the extremity of the outrigger, keeping their footing on the spars, and holding on by a man-rope, which is purposely fixed about breast-high to the mast; and in this manner the extra pressure of the wind is counterpoised.

Flanging seats are formed in the aft part of the top-sides of the canoe, so that the crew sit out beyond the sides or wash-boards, and facing each other, their legs hanging down in the trough or interior of the canoe.

These narrow and fragile-looking boats are managed by half-clad natives, who venture out to sea with them distances of twenty miles or more from land—for a Cingalese canoe will live at sea in any weather; and they skim along among the shipping at Point de Galle with fearless velocity, bringing up now and then to offer for sale beautiful, but unfamiliar fruits, and fishes of extraordinary colours and fantastic forms.³

Vessels passing the southern coast of Ceylon are generally boarded by these boats, even at the distance of twenty-five miles from shore.

³ Sir Emerson Tennent's "Ceylon," vol. ii.

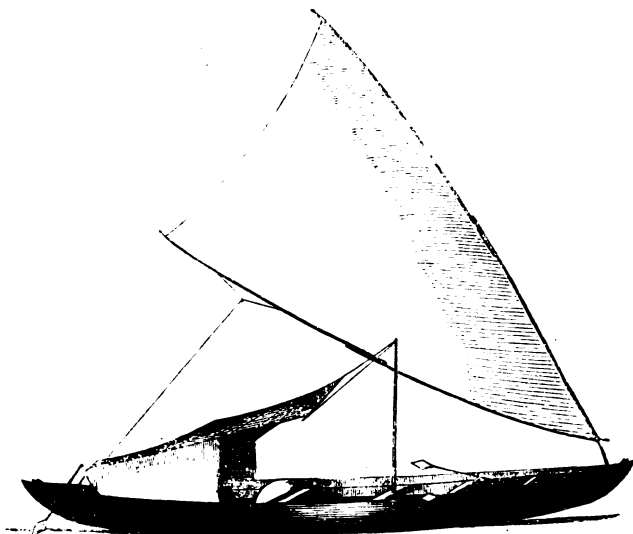
They will sail at the rate of ten miles an hour in strong winds, which are generally prevalent there; and, with a crew of five men, carry a cargo of vegetables, which are great luxuries to the crew and passengers of a ship, after a long voyage from England to Bombay or Bengal.

Major Forbes, in his "Eleven Years in Ceylon,"⁴ describes the outrigger in these canoes as being always placed to *leeward*, which would seem to be a direct contradiction of all other authorities as to the outrigger being to *windward*. The gallant Major thus describes the native canoe:—"From Chilaw I procured a canoe which conveyed me with great speed to Colombo, bounding over the waves without any of that jumbling motion which has caused me to feel, in every other kind of vessel, so sickening and uncomfortable. These canoes are of a very peculiar construction, the principal part consisting of a long tree hollowed out; on this a high mast, and still higher sail, appear quite disproportioned to the vessel, which is prevented from being upset by a log of wood, called the outrigger; this is a sort of miniature of the canoe, only it is solid, and is attached and kept parallel to the canoe by means of two curved elastic sticks. The outrigger is always to leeward; and, as both ends of the canoe are shaped alike, the change of direction is accomplished with little delay, by simply shifting the sail, and proceeding with the former stern as its head. The motion of these canoes somewhat resembles a horse at full gallop; and, if the outrigger is well secured, they are able to keep the sea in ordinary weather, which could hardly be expected, judging from their simple form and skeleton-like appearance."

⁴ Vol. i. p. 269.

JAVA, &c.

THE TAMBANGAN.



THE Tambangan, or Sourabaya passage-boat, is a curiously rigged vessel, very broad and shallow, but uncommonly roomy, convenient, and useful for its purpose.

The tambangan is partly flat-bottomed, *i. e.* it has a small flat floor without a keel, but is flanged and round at the lower sides like an English doble.

The sail, which is a very light one of triangular shape, is set independently of the mast, the yard (which is in two parts) being apparently self-supporting.

A light canvas awning is set up in the aft part of the boat, in the manner represented by the engraving.

MALDIVE AND LACCADIVE BOATS.

The sea-boats used by these islanders are of a very grotesque and antiquated appearance, and have many peculiarities; in the first place, cocoa-nut is the wood used chiefly in their construction, there being no other in the island suitable for the purpose. The planking is pegged together with hard wooden pegs: the larger boats are thus made very strong. The form of the Maldive boats is not much adapted for fast sailing, but they are wide and safe-looking vessels, rigged with two sails—lug main-sail, and large triangular fore-sail; the main-sail has a boom, the fore-end of which works in a rest or cross-piece, a few inches before the mast; and that sail generally stands very flat: the fore-sail is large, and set out on a bowsprit. The stem of the Maldive boat is aquiline and curious.

BIRMAN WAR-BOATS.

The Birmese war-boats are constructed out of the solid trunk of the teak tree, which is hollowed partly by fire and partly by cutting: the largest of these boats are from eighty to one hundred feet in length; but the breadth seldom exceeds eight feet; and even that width is only produced by extending the sides after the trunk has been hollowed. These boats are manned by fifty or sixty rowers; the oars used are short ones, and they are worked on a spindle. The steersman of a Birman boat is called the "leedegee."

The prow is a solid portion of the tree, and has a flat surface, on which, when they go to war, they mount a piece of ordnance—a 6, 8, or 12-pounder. Each rower is provided with a sword and lance, which are placed by his side in the boat. In addition to the crew, there are usually twenty or thirty soldiers aboard, armed with muskets and other weapons. When they advance upon their foes they sing a war-song, to encourage the men and regulate the strokes of the oars. They display considerable dexterity in the management of

these boats ; and when they retreat they row backwards, so as to keep the large gun bearing upon the enemy.

The largest of these boats draw only three feet of water. When a person of rank is aboard, there is usually a tilt or canopy placed over the part intended for his special accommodation. In Birman boats the prow is always the place of dignity.

Some of these boats are gilded and ornamented down to the water's edge ; but gilded boats are only permitted to princes of the blood and persons holding high stations, such as a Maywoon of a province, and a minister of state.⁵

SANDWICH ISLANDS CANOES.

These canoes are well made, considering that they are constructed by hollowing out the trunk of a tree. They are, however, very ticklish and easily upset: those which are used singly are fitted with a log of wood for an outrigger, which stands out several feet beyond the canoe.

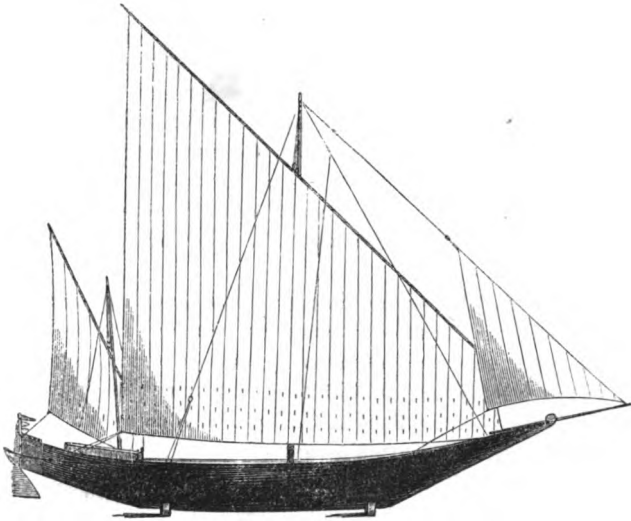
The double canoes of these islands consist merely of two ordinary canoes held together at a distance apart, of from four to six feet, by very strong beams, which are arched and stand up above the canoes. These are sometimes assisted with a small sail, the mast for which is stepped in one of the canoes, and the sail is sheeted in the other.

ARABIA.

ARAB BATELLE.

ARAB batelles were the boats principally used by the Joaseme pirates of the Persian Gulf, who, in years gone by, were a terror to the native mariners, until exterminated by the united efforts of the King's ships and the Honourable

⁵ Syme's "Embassy to Ava," A.D. 1795; "Voyages and Travels," &c.



ARAB BATELLE.

East India Company's vessels of war. They were very destructive to trade, because no vessel could escape them; and their weatherly qualities prevented square-rigged ships from capturing them, except in strong breezes.

But their fame has now passed away by the introduction of steamers, previous to which it is said "there was no vessel ever built that could sail so close to the wind."⁶

It is also stated that when the pirates were supposed to be on the point of surrendering, they used to make off in gallant style within gun-shot of the ship, and were chased and pursued in vain.

Batelles have very raking bows and stern, the latter inclining upwards in a most peculiar manner from the keel (*see engraving*); and the rudder consequently hangs lower than the under part of the stern.

⁶ Official Catalogue, International Exhibition 1851.

These vessels are exceedingly well built, upon scientific lines; they have good beam, a very sharp and hollow floor, very clean run, and a perfectly wedge-like entrance, which offers little or no resistance to the water.

To the after part of the rudder is fixed the tiller, which has a curve pointing upwards; the ropes are led inboard by means of an outrigger at the side, by which the helmsman steers the batelle. The rudder requires very little head, as, indeed, it is confined to a certain point by spreaders nailed on the stern-posts.

These batelles are lateen rigged, and have three suits of sails made of very fine cotton canvas, woven by hand at Bahrein. In calms they are propelled by sweeps or long oars. The largest size batelle is 150 tons, and now used only by the Arab chiefs of the Persian Gulf on state occasions and visits of ceremony. The main-yard of the batelle is longer than the vessel itself, and the sails are very large and pointed.

A beautiful model of one of these remarkable vessels was exhibited in the International Exhibition of 1851. This model, from which the above engraving was made, is now in the Indian Museum, and is considered perfect in every respect, as a whole; and as to the detail, the making of it is said to have been superintended by an Arab from the Persian Gulf. (See also Batelles of Bombay, page 252.)

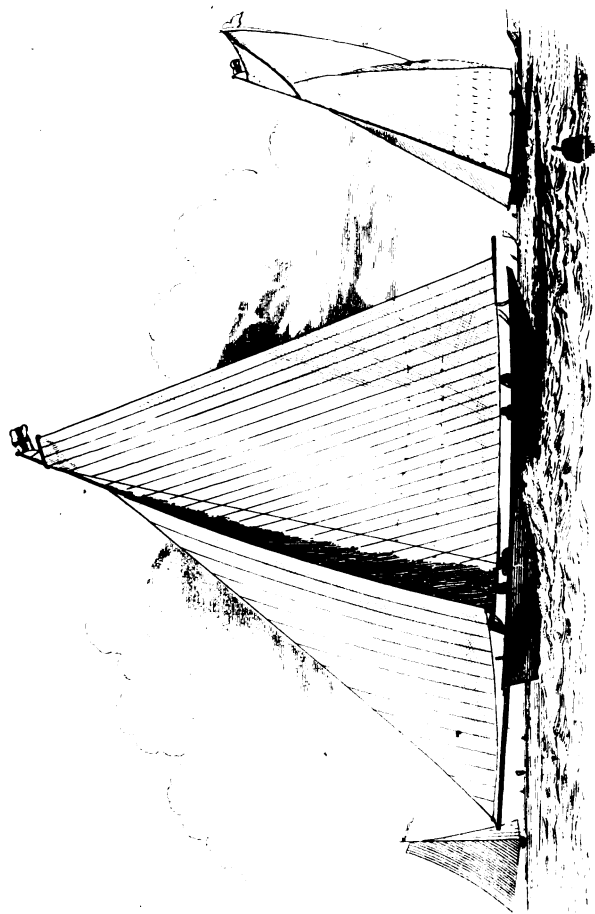
BERMUDAN BOATS.

“Where the remote Bermudas ride,
In the ocean’s bosom unespied.”

A. MARVELL.

THE Bermudas or Summer Islands, although mere specks in the wide Atlantic Ocean, have many interesting circumstances connected with their history:

“Those leafy islets on the ocean thrown,
Like studs of emerald on a silver zone.”



PERUVIAN BOATS.

It was at Bermuda Tom Moore wrote many of his most beautiful poems, during the time that he held a Government appointment there. There, also, the lofty cedar thrives in luxuriant moisture, whilst the clear blue waters of the ocean dash their hissing foam against the rocky shores and coral reefs of the coast.

“The morn was lovely, every wave was still,
When the first perfume of a cedar hill
Sweetly awak'd us, and with smiling charms
The fairy harbour woo'd us to its arms.”

The navigation of Bermuda, with its hundred and fifty islets, is very intricate and dangerous; but the natives have acquired a world-wide fame for their sailing boats, which are of a superior form and beautiful rig, and possessing great powers of sailing to windward. It has often been questioned whether the Bermudan rig is not the most effective in the world for sailing to windward. The schooner rig was formerly the favourite with those islanders, and up to the present day some of the Bermudan boats are rigged with two masts; but the favourite rig consists of one mast only, as shown in the engraving.

The Bermudan boats are built chiefly of the native cedar, as are also many large vessels belonging to the islands. The “Driver” sloop-of-war, in which the English poet was conveyed to Bermuda, is described as a “vessel built of cedar.”

Most of the boats are protected by a deck or water-way, about eighteen inches wide, which extends from the galley or forecastle to the stern. The larger ones are decked all over; they have a very deep keel and deep form of hull; short and wide on deck, with immense fore-gripe and aft dead-wood.

The Bermudans are particularly skilful in the management of their boats, and also in cutting and fitting the triangular sail. The present rig was introduced among them many years ago, by an experienced boat sailor, the Honourable H. G. Hunt; who, having lost a schooner race for a large sum, had an impression that the one mast would have a

decided advantage over the schooner rig. He therefore challenged his antagonist to another race for a similar wager ; meanwhile he secretly proved the superiority of the single main-sail by a private race with a schooner at midnight ; and on the following day was eminently victorious in a public sailing-match. From that period, the chosen rig of the Bermudan boatmen has been that with one mast only. Boat racing became a favourite amusement with the islanders ; and a first-rate boat sailor was a *sine qua non*. A yacht club was established, and yacht-racing afterwards became general, and was looked forward to from time to time with as much interest as horse-racing is in England. Boats were constructed exclusively with regard to speed ; the cedar bottoms polished, so as to present the smoothest possible surface to the water : they had neither gunwale nor cabin, nor even bulkheads. The deck was of convex form, and made of the lightest wood, just strong enough to bear the weight of one or two of the crew. Such consummate skill was displayed both in construction of the hull, and the cut and fit of the sail, added to which, the exquisite seamanship of a well disciplined racing crew, that a boat of thirteen feet in length, upon the improved plan, would beat one of the old fashioned form of twenty-five, in fair ordinary weather. This was the case in the year 1840, when a large boat, twenty-five feet in length (and which a short time before had won a challenge cup), was beaten by a small one, the "Alarm," of thirteen feet ; which boat, in a very light breeze, worked three miles to windward in thirty-six minutes.

The mast of the Bermudan boat is spiral, a beautiful tapering spar of white spruce, placed very far forward ; so much so, that the heel is stepped into the lower part of the stem-piece, with considerable aft rake. A single shroud on each side bears the extra pressure of the canvas, and is secured within a few feet of the mast head ; from the same berth the jib-halliards are rove through a tackle ; the main-sail is triangular, or nearly so, and there being no gaff to sway the sail to leeward, and only one other sail before the

mast, the great spread of canvas abaft enables the boat, when properly handled, to go within three points of the wind in ordinary trim; and when well equipped, and judiciously ballasted, will work within the seven points. A top-sail is occasionally used when going free in very fine weather, which is of different shape, and set in a different manner to anything of the kind used for other boats (*see engraving*); a large square sail is also used when running directly before the wind in a race.

A model of the Bermudan yacht "Undine" was exhibited by Lieut. Taylor, of the 39th Regt., in the International Exhibition of 1862.

There is also in the United Service Museum, London, a model of a celebrated Bermudan boat, the "Lady Ussher," of fifteen tons, presented by Mr. Triscott. This boat was wrecked in the hurricane of September, 1839; her dimensions were as follows :—

Length of keel ...	...	...	...	24 feet
„ over all ...	...	...	...	32 „
Breadth of beam ...	...	...	...	12 „
Draught of water ...	...	...	...	8 „
Length of mast ...	...	...	...	64 „
Hoist of main-sail ...	...	...	...	56 „

It will thus be seen that these boats have considerable breadth of beam in proportion to length, which is necessary, to enable them to carry the lofty mast required to spread the peculiar and graceful-looking triangular main-sail.

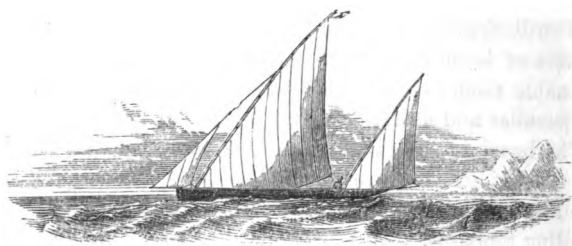
The boom, in some of the boats, extends considerably abaft the taffrail. They have generally a jib-boom as well as a standing bowsprit, and the foresail extends to the end of the standing bowsprit.

Much of the success of a Bermudan boat depends on the experience and skill of the crew; and there are no boats so useless in the hands of persons who are not accustomed to their peculiarities; hence, there is great difficulty in affording

them a chance in the English matches, added to the circumstance of our races being generally *round a course*, and not, as in Bermuda, dead to windward and leeward, which are the two most advantageous courses for these boats.

When a boat is entered for a race at Bermuda, there is no restriction as to sails; the owner being at full liberty to put any canvas upon her, or prepare her in any way he may think proper; but only two sails are used to windward, and a square sail to leeward; in ordinary trim they have main-sail, jib, flying-jib, gaff-top-sail, and square-sail. The stake boats are placed dead to windward and leeward, and the start takes place from the latter, so that the racing boats are sure to start on even terms.

The Bermudan rig ought not to be extended to vessels above eighteen tons, because any mast fit to carry a proportionate area of canvas would be too ponderous for its position. It is unquestionably a superior mode of rig, and if not one of the fastest in the world, is considered as unequalled for working to windward in smooth water; but in a sea way the lofty peak is not so effective. These boats are said to sail nearer the wind than cutters; this, and the quickness with which they tack, would seem to make up any deficiency of speed on a bowline.



AMERICA.

NEW YORK PILOT-BOATS.

THESE are among the very fastest boats of America; and their sea-going qualities are truly astonishing. They are employed on those parts of the coast of the United States where the seasons are alternately very fine and very rough. They vary in size from 50 up to 100 tons, are long and low in form, with round sterns and sharp flaring, long, and hollow bows; their greatest breadth of beam is generally some feet before the main-mast.

They are schooner rigged; the general complement of canvas being main-sail, fore-sail, and stay-sail; but in light winds they use a large flying stay-sail which is set from the head of a short main-topmast to the head of the fore-mast, and sheeted home half way to the deck.

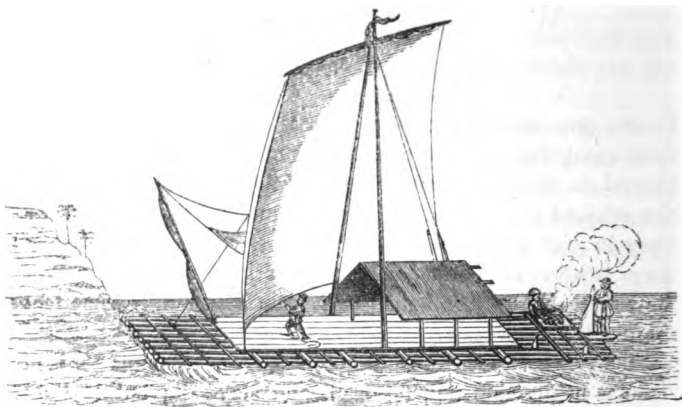
In very strong winds and heavy seas they sail under nothing but a close reefed foresail; and yet they shoot ahead and glide past large and swift-sailing vessels with apparent ease. They are constantly at sea, looking out for vessels requiring the aid of a pilot.

In fine weather, smooth water, and light winds, these boats skim over the blue waters of the Atlantic with all the grace and velocity of light fairy-like pleasure-boats.

BALZAS.

The Balza or Balsa signifies a raft; balzas are termed by the natives *jungadas*, or *janjades*: they are composed of five, seven, or nine beams, or trunks of trees called balza. The natives of Darien call the tree Puero. The balza is a soft, whitish wood, and very light.

Over part of the balza is a strong tilt formed of reeds. Instead of a mast, the sail is hoisted on two poles or shears of mangrove wood, (see engraving on next page); and those which carry a foresail have two other poles erected on the fore-part of the raft in the same manner.



SAILING BALZA.

Balzas are not only used on rivers, but short sea voyages are made in them.

They are of various sizes; some are fishing balzas, some are used for carrying all kinds of goods from the custom-house to Guayaquil, and from thence to Puna, the Saltode Tumbez, and Paita: and others of a more curious and elegant construction are employed in removing families to their estates and country houses.

Some of these are formed of very large beams of 2 feet or $2\frac{1}{2}$ in diameter, and from 60 to 70 feet in length. These beams are fastened or lashed together with bejucos or withes; and so securely that with the cross pieces, or smaller logs, which are also lashed with all possible strength, they resist the rapidity of the currents and heavy seas in their voyages to the coast of Tumbez and Paita. The width of the raft varies from 15 to 20 and 30 feet, according to the size of the logs.

The thickest beam of those which compose the balza is the longest and middle one, so that it projects beyond the others; and to this are firmly lashed the first beams on each side, and thus successively till the whole are secured; that in the middle being the principal piece, and thence the number of beams is

always odd. The larger balzas carry between 400 and 500 quintals, or 25 tons, of merchandise without damage by proximity with the water; for the waves of the sea never sweep over the balza, neither does the water splash up between the beams, the balza always following the motion of the water.

But the greatest singularity of this floating vehicle is, that it is sailed, tacked, and worked in contrary winds with the facility of a vessel with a keel, and it makes but little lee-way. This advantage it derives from another method of steering than by a rudder, viz., by guaras, or thin boards three or four yards in length, and half a yard in breadth; these are placed vertically, both in the head and stern, between the main beams; and the crew, by thrusting some of them deep in the water and raising others, bear away, luff, tack, lie to, and perform all the other motions of a regular ship.

A guara being shoved down in the forepart of the balza causes it to luff, and by taking it out and dropping one astern, the balza bears away or falls off.

On a guara being shoved down at the stern, and the fore one drawn up, the balza will bear away, and by taking it out of the water the balza will luff, or keep nearer the wind. Such is the method used by the South American Indians in steering the balza, and sometimes they use five or six guaras to prevent the balza from making leeway: it is also evident that the deeper they are under water the greater resistance the side of the vessel meets with; the guaras performing the office of leeboards, such as are used in Dutch vessels, billy-boys, and barges.

The method of steering by these guaras is so easy and simple, that when once the balza is put in her proper course one only is made use of, raising or lowering it as occasion requires; and thus the balza is always kept in her intended direction. Balzas are tacked and wore with a degree of precision truly wonderful.

Near both ends of the central spar of the balza there is cut a perpendicular slit about two inches wide by one or two feet in length; through these the guaras are put, and they may

be thrust down to a depth of 10 or 12 feet, or drawn up entirely.

When it blows hard, and the balza is sailing with a side wind, several guaras are kept down to make the raft hold a better wind and sail faster.¹

PERUVIAN BALSA.

The Balsas employed on the south coast of Peru are famed for their qualities as surf boats. They are used for crossing the surf off the coast of Mollendo, and are of tubular construction. They are described by Capt. Hall² as made of two entire seal-skins inflated, placed side by side, and connected by cross pieces of wood and strong lashings of thongs; over all a platform of cane-mats forms a sort of deck about four feet wide and six or eight feet long. At the fore-part the person who manages the balsa kneels down, and by means of a double-



bladed paddle, which he holds by the middle and strikes alternately on each side, moves it swiftly along through the heavy breakers and surf on the coast. The passengers or goods are placed on the platform behind him.

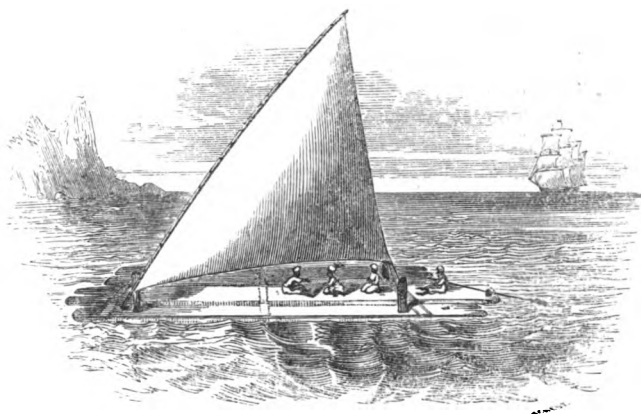
The buoyancy of these balsas enables them to cross the surf in safety, and without wetting the passengers, at times when an ordinary boat would inevitably be swamped. All

¹ Vide "Relation Historica del Viage a la America Meridional necho de Orden, de S. Mag., &c.," Impresa de Orden de Roy en Madrid, 1748; Ulloa's "Voyage to South America;" Hall's "Fragments of Voyages and Travels," &c.

² "Journal written on the Coasts of Chili, Peru, and Mexico." By Capt. Basil Hall. 1824.

sea-borne goods destined inland at this part of the coast, are landed in this manner. The great bars of silver and the bags of dollars also, which are shipped in return for the merchandise landed, pass through the surf on these tender, though secure conveyances.

They can keep the sea in any swell or surf in the hands of the natives, but are slow in their progress through the water on account of their spare length.³



CATAMARAN OF PERNAMBUCO.

The Brazilian fishermen of Pernambuco, South America, use a raft or catamaran consisting of three straight logs of wood, lashed together, but not curved upwards at the fore part like the Ceylon catamaran. The sail very much resembles the Bermudan one, being triangular, and set on a very light mast, raking considerably aft; the heel of which can be shifted at pleasure, several inches on either side the raft. The crew generally consists of two or three persons,

³ "Travels in Peru and Mexico." By S. S. Hill, 1860.

who sit in an elevated posture on a four-legged stool, permanently fixed to the outer logs, or they squat themselves down on the raft, and so go to sea in any weather, sometimes five or six miles from land; the sail is always kept well wetted when going to windward: and in a fresh breeze, when close hauled, they sail at the rate of three or four knots an hour, and much faster when going free; on which occasion the sail is boomed out with a bamboo, and otherwise trimmed by stepping the heel of the yard (which is also the mast) in such of the holes under the seat as may be best adapted for that purpose, and according to the position of the raft with regard to wind.

The catamaran anchor consists of a straight piece of wood with a sharp wooden barb lashed to one end for a fluke, and a heavy stone attached to the other for stem and sinking weight. The same description of anchor is used by the Chinese, and by various islanders in the Oriental Archipelago.

CANADA.

BIRCH BARK CANOES.

“He hollow'd a boat of the birchen bark,
Which carried him off from shore;
The wind was high and the clouds were dark,
And the boat returned no more.”

T. MOORE.

THESE canoes are chiefly remarkable for the material of which they are composed, and for their extreme lightness and buoyancy. The whole of the outside or planking consists of the bark of the American birch tree: a material tough enough in its nature, if carefully employed, but easily ripped or injured when coming in contact with hard substances.



CANADIAN BIRCH BARK CANOE.

The Iroquois Indians are the most expert makers of the birch bark canoes: the largest are called "North Canoes," some of which are thirty-six feet in length by about six feet in breadth. These are propelled by a crew of sixteen men (besides a bowman and steersman), who, with paddles, by the short quick stroke, drive the canoe along at the rate of eight miles an hour when going with the stream.

The bowman is the chief navigator; he "fends off" at the bow with his pole when running rapids, and directs the crew as to the paddling of the canoe. These boats are sometimes sailed when the wind is free, but not otherwise. Small sails only are used; for they are tender under sail, and require very careful management.

These canoes are used for travelling through the wild and unsettled parts of North America; and voyages of nearly one hundred miles per day are sometimes performed in them. It is in a canoe of this kind that the Governor of the Hudson's Bay Company performs his annual trip to the Red River settlement. They were also formerly used by the old "North West Company," for the purpose of conveying the peltrie (i.e. furs) from inland parts, and as far north as Hudson's Bay to Montreal.

These canoes are very ingeniously constructed, and so light that they are often carried on the shoulders several miles by four men.

The smallest of the birch bark canoes are about twelve feet long; these may be carried by one man, and are what are termed "light hunting canoes." They are also used as postage canoes.

In building the birch bark canoe, a skeleton is first made of light wood ; the casing or planking of bark is then put on—not lengthways, like the streaks of an ordinary boat, but transversely, passing under the boat and ending at the gunwales—the broadest strips being placed amidships. The bark is sewn together with the fibrous roots of the fir-tree, and the seams are well dressed with the gum of the balsam-tree. When they become leaky the place is stopped by warming the gum, and plastering over it a thin piece of birch bark.

The bark of which these canoes are made is stripped from the birch-tree in winter, the largest trees affording the most useful bark. The canoes are also sometimes trimmed and strengthened with wicker wreaths, and ribs or timbers of cedar, which are almost as light as cork ; the wicker wreaths are about as thick in substance as a crown-piece, but the birch bark is as thick as two crown-pieces, and the cedar ribs as thick as three or four. On the inside of the boat a rim or gunwale of tough wood is fitted, and the top ends of the cedar ribs are enclosed in it.

It should be observed, that these canoes have neither stem nor stern-piece, but run to a point at both ends. Neither have they keels ; and neither nails nor pegs are used in the whole structure.

The lightness and easy transport of these boats overland is a great advantage in Canada, where the navigation of the rivers is continually interrupted by cataracts, waterfalls, and shallows.

A full-sized Montreal bark canoe, from which the preceding engraving was made, was exhibited in the International Exhibition of 1851, and described as made from the bark of the white birch, and one of the largest class of canoes used in the north-west country. Previously to its being forwarded to England it made a voyage in the spring of the previous year of upwards of 3,000 miles, with a crew of twenty men and their stock of necessaries and provisions.

Being exceedingly light, the crews are enabled to carry these canoes when it is essential to avoid the falls and rapids ; and,

for months together, they form the homes of the hardy and daring voyagers during their transit to and from the far west.⁴

Birch bark canoes are also used by the Sioux (who purchase them of the Chippeways) in gathering wild rice, which grows about the rivers and lakes of the north country. The manner of gathering it is curious: one woman paddles the canoe, whilst another, with a stick in each hand bends the rice over the canoe with one and strikes it with the other, and so shells it into the canoe, which is constantly moving along until it is filled.⁵

They are also used for shooting wild-fowl, which are abundant in those parts, and feed and fatten on the rice.

BIRCH BARK CANOES OF SOUTH AMERICA.

The bark canoes of South America are formed of the whole unbroken bark of a tree. The tree used for the purpose is called by Stade *yga-ywera*. They take off the bark in one piece; then, keeping the middle straight and stretched by means of thwarts, they curve and contract the two ends by fire, and the boat is made. The bark is about an inch in thickness; the canoe is commonly about four feet wide, by forty in length: some of them are capable of carrying forty persons, but they seldom proceed farther than half a league from the coast in these canoes. In bad weather they land, and carry the canoe on their shoulders.⁶

BIRCH-BARK CANOES OF PRINCE EDWARD'S ISLAND.

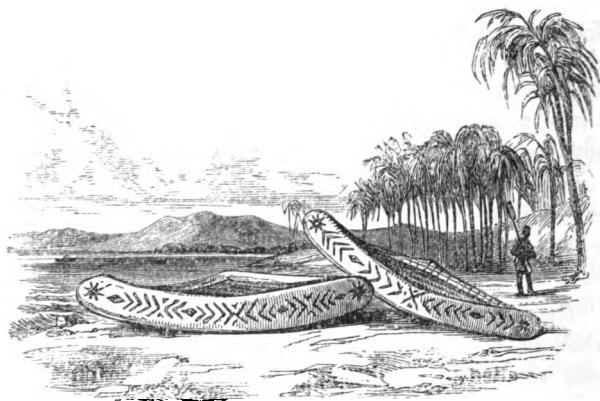
These are exceedingly well made; the interior or skeleton is formed of flat ribs a quarter of an inch in thickness and an inch or more in breadth; these are placed nearly close together throughout the whole structure, and the bark is then put on outside and laced or sewn together, the seams being

⁴ Exhibition Catalogue, 1851, vol. ii.

⁵ "Letters and Notes of the Manners, Customs, and Condition of the North American Indians." By George Catlin. 1841.

⁶ Southey's "History of Brazil."

well dressed with gum. There are no thwarts or benches, but five or six cross-pieces or stretchers at the top, as if to keep the canoe in shape. The top-rim or gunwale is neatly worked with wicker. The paddles are small and short.



BIRCH BARK CANOES OF NOVA SCOTIA.

The birch bark canoes of Nova Scotia are very remarkable, though similar in many respects to those above described ; and they are curiously wrought on each side with fancy work woven in a variety of colours. The material used in construction very much resembles wicker, and appears to be worked in a similar manner ; and they are no heavier than large wicker baskets. Light weight is an important consideration, as they have frequently to be taken out of the water and carried on the shoulders past rapids and other obstructions.

Those of New Brunswick are precisely similar to these.

CEDAR BOATS OF THE WILLAMETTE.

These boats are built at Okonagan, and are somewhat after the model of a whale-boat, but much larger. They are remarkable for their lightness and capacity, although capable of

carrying three tons of merchandise, with a crew of eight men and a padroon. They may be carried easily on the shoulders by three or four of the crew, which is often necessary on passing over the portages.

They are clinker-built, and very strong and buoyant. The usual length is thirty feet, by five and a half feet in breadth, with a sharp bow and stern; the planks and gunwale are of cedar, and extend the whole length of the boat. They have no knees, but flat oak timbers are bolted to a flat keel, at distances each about one foot apart. The rowlocks are made of birch.

One very remarkable peculiarity in the construction of these boats is, that they are merely rivetted at each end, and the seams being well dressed with gum of the pine tree, they require no nailing or other kind of fastening; in case of accident they are easily repaired, a supply of gum being always carried in the boat.

The crews who man these boats are chiefly Canadians, with the exception of about one-fourth, who are Iroquois Indians. When the wind is fair they set a small square sail, but they rely chiefly on their oars.

AMERICAN BATTOES.

Battoes⁷ are a kind of flat-bottomed boat much employed in Albany. They are used chiefly for carrying goods up and down the rivers, where the birch-bark canoes would be unfit, by reason of their slender and delicate construction.

Battoes are constructed of boards of white pine; the bottom is flat, in order to enable them to go with facility into shallow water. In form they are sharp at both ends, and somewhat lower amidships than fore and aft. They are of various sizes, from three to four fathoms in length, and about three feet six inches in breadth; the sides are almost perpendicular, and in depth from one foot eight inches to two feet.⁸

⁷ From the French *batteaux*.

⁸ Kalm's "Travels in North America."



CANOE OF OREGON INDIANS.

These canoes, although made from the single trunk of a tree, are really of elegant form, with gracefully peering bows and stem, full midships section, and rather tapering stern, with a spring both fore and aft.

The interior is so carefully hollowed that the sides are only three-fourths of an inch in thickness, and the thwarts are ingeniously fitted to the interior so as to prevent the sides from warping or getting out of shape.

These canoes are preserved with great care, and when not in use are never allowed to lie exposed to the sun, for fear of injury. But where cracks and rents occur, they are repaired with much ingenuity. After boring holes on each side of the crack, small withes are passed, crossed, and pegged in such a manner as to draw the crack close: when the tying is completed the whole is well dressed with gum of the pine tree, and the damage is thus neatly and effectually repaired.

VANCOUVER'S ISLAND CANOES.

The canoes which Captain Cook met with at Nootka, in his "Voyage to the Pacific," he describes as "well calculated for every useful purpose." Many of them are forty feet long, seven feet broad, and about three feet deep; capable of carrying twenty persons or more, and formed of the trunk of a single tree.

From the middle, towards each end, they become gradually narrower; the after part or stern, ending abruptly or perpendicularly, with a small knob on the top; but the fore part

is a protuberance stretching upwards, and ending in a notched point or prow, considerably higher than the sides of the canoe, which run nearly in a straight line. (*See preceding engraving of Oregon Indian Canoe.*) For the most part they are without any ornament; but some have a little carving, and are decorated with seals' teeth, set on the surface like studs, as is the practice on their masks and weapons. A few have likewise a kind of additional head-piece or frame, like a large cut-water, which is painted with the figure of some animal. They have no seats, nor any other supports to the insides than several round sticks, little thicker than a cane, placed across at mid-depth. They are very light, and their breadth and flatness enables them to float firmly without an outrigger, which none of them have—a remarkable distinction between the boats of all the American nations and those of the southern parts of the East Indies, and the islands of the Pacific. Their paddles are small and light, the shape in some measure resembling that of a large leaf, pointed at the bottom, broadest in the middle, and gradually losing itself in the shaft, the whole being about five feet long. The natives have acquired great dexterity in the management of these boats and in the use of the paddles, sails forming no part of their art of navigation.



FUEGIAN CANOE.

The canoes of the natives of Terra del Fuego are very curious.

The engraving above will convey an idea of their shape, and the manner in which they are made.

They are constructed of bark, sewn together with shreds of whalebone, seal-skin, and twigs.

The top rim, or gunwale, is formed by binding canes or small poles to the sides. The canoe is kept in shape by stretchers lashed across from one gunwale to the other; the two longest and strongest being placed amidships across the top of the vessel.

They have no seats or thwarts; the occupants squat on some dried grass strewn about the bottom.

Frail as this canoe is, a fire is generally carried in the bottom (on a few stones and ashes), in the deepest part of the canoe, which always contains water, and surrounds the fire. One person is generally employed in baling out the water, and attending to the fire.

The Fuegians "seldom venture outside the kelp, by the aid of which they pull themselves along; and their paddles are so small as to be of little use in propelling their canoes, unless it is calm." ⁹

AUSTRALIAN BARK CANOES.

THE canoes which are employed by the natives on various parts of the coast of Australia, are made of the bark of the large gum tree, which has a thick and tough coating.

The manner in which these canoes are constructed is as follows:—The tree is first girdled; the bark is then slit to the size required for the canoe, and then stripped from the tree with great care. The usual size is fourteen feet long by seven feet wide.

The piece of bark thus peeled from the tree is then charred on the inside; each end is then folded, joined, and fastened with cords and wooden rivets; a wooden stretcher is next placed across amidships, and the canoe is complete; and,

⁹ Wilkes's "American Exploring Expedition," vol. i.

although only three feet wide in the broadest part, will convey six persons.

The paddles employed are of different sizes and lengths—from two to five feet; when the short ones are used, one is taken in each hand.

These canoes are seldom seen without a fire, which is made on a layer of gravel, in the middle of the boat. In this custom the natives resemble the Fuegians. The canoes of the latter, however, far excel those of the Australians.

GREENLAND BOATS.

THE OOMIAK.

THE boats chiefly used by the Greenlanders are of two sorts—the oomiak (or umiak), which is specially for females, and is called the woman's boat; and the other, the kaiak, or kajak (also sometimes spelt kayak), is exclusively the men's boat; the lightest and smallest kajaks are the boats they use in their seal-hunting, fowling, and fishing exploits.

The umiak is considerably the larger boat of the two, and is commonly from twelve to twenty feet in length, and of a proportionate breadth and depth, varying from four to six feet in width, and from two and a half to four feet in depth. Umiaks are all flat-bottomed; the prows and sterns of some of them are sharp and pointing upwards; others are bluff, or even square, and have two projecting arms at each end, like a bier, only pointing upwards.

The form and construction of the umiak is very singular. It is made of slender laths about two inches broad, fastened together cross-wise with whalebone, and covered over with tanned¹ seal-skins. The bottom consists of three main

¹ The natives understand the art of tanning, and are able to dress their skins very fairly with a mixture of oil and colouring.

beams ; the principal one runs up the middle of the boat as a keelson, and the others, one on each side, are bent so as to meet at each end.

Upon these three main beams some thin cross-pieces are laid and morticed. They then fix stout timbers or ribs to the outer beams, and secure them at the top to the gunwale, and also to a second or lower beam, to which the rowing benches are attached ; and in this manner a good strong frame-work is got up, though all are secured with wooden rivets and whale-bone bands. Indeed, they decline to use iron nails, because they would rust, and fret the holes in the skins. It is not too much to say of the ingenuity of these people, that they perform their work with considerable mechanical skill and neatness. They use neither rule nor square, but, with the eye alone, mete out the due proportions. The only tools they use are a little lock-saw, a chisel (which, when fastened to a wooden haft, serves also for a hatchet), a small gimlet, and a pocket-knife. When the skeleton of the boat is completed the men's work is done, and the women commence theirs, by covering the whole with newly-dressed seal-skins, which are as soft and pliable as chamois leather ; and they caulk the seams with old grease, and finish off their work in a neat and proper manner, making the boat perfectly water-tight. In truth these boats, with careful handling, are not so liable to leak as wooden ones.

They are always taken out of the water when not in use, and when in the water the stitches swell as soon as wetted and fill up any slight leakage which, when dry, might exist. If they accidentally touch upon a stone on the shore, and so rip a hole in the boat, they directly do it up by patching a piece of soft seal-skin over the place ; but they handle their boats with so much care, that they are seldom injured through carelessness or neglect. Almost every year they renew the outer covering of the boat.

Lieutenant Hooper, in his "Arctic Boat Expedition in search of Sir John Franklin," often made use of these oomiaks ; and with reference to one which he purchased, he

observes:—"We purchased an oomiak (woman's or family boat), made of two walrus skins sewed together, and stretched tightly over a light wooden frame, with paddles complete. Its price was a large butcher's knife, a looking-glass, and a quantity of tobacco and beads.

"She differed much in shape from those of the Tuski, which are a little wider at top than bottom. This one measured thirty-seven inches at the greatest breadth of floor, and forty-three where widest between the gunwales; a species of keel was formed by carrying the wood-work to an edge below.

"Workmanship was displayed in the manufacture of her frame, which would not have disgraced a skilful carpenter; it was particularly neat and well fitted, which made it much stronger than its lightness would lead one to suppose."

These boats (umiaks) are rowed and managed entirely by women, generally five to each boat, four of whom ply the oars or paddles, and one steers. The men never interfere in the management of the oomiak, except under very trying and peculiar circumstances.

The oars they use are short, but broad in the blade, and of a shovel-like shape; they are secured to their place on the gunwale by straps of skin. In fair winds they use a small square-sail, which they hoist from a mast they set up in the fore part of the boat; the sail is generally made of skins from the intestines of fish and animals. Some of the more wealthy, or trading Greenlanders, have sails made of white linen striped with red.

In boats of this kind, the Greenland women sometimes perform voyages along the coast of many miles, carrying with them their tents, goods, family, and whole substance; but on these long voyages the men keep with them near at hand in their kajaks; and in heavy seas they attend the umiak very cautiously, sheltering it from the heaviest waves, and holding the gunwales to keep it upon a balance. They frequently travel as many as twelve leagues a day in these boats; and every evening when they halt, they go ashore, unload their

boats, and draw them to land, turn them upside down, and load the beams fore and aft with stones, that the wind may not blow them away. When they come to any difficult or impassable part of the coast, they carry the boat on their heads over land to more navigable water.

THE KAJAK, OR KAIK.

In this light little craft the Greenlander passes the greater part of his life, performs most of his ordinary and extraordinary sporting adventures, and thereby obtains a sustenance throughout the year for himself and family.

In appearance the kajak bears some resemblance to the wild-fowlers' gunning-punts of the eastern coast of England,² only the ends of the kajak turn up, which is not so in the gunning-punts.

The kajak is about eighteen feet in length, eight inches deep, and only sixteen or seventeen inches wide. It is built with a slender wooden keel, long side laths, with cross hoops not quite circular, bound together with whalebone, and covered over with fresh-dressed seal-skins, as the umiak; only the leather encloses the frame like a bag, on all sides, over the top as well as the bottom. Both the sharp ends at the head and stern are fortified with an edge of bone, and finished with a knob at the top, in order to protect it from damage when it touches against ice-bergs and rocks. In the centre of the kajak, on the top of the covering, is a circular hole formed with a rim or hoop of wood or bone, similar to the Esquimaux and Labrador canoes described at page 299. Into this hole the Greenlander slips, when about to proceed on his aquatic excursions. Having stretched his legs out under a covering, he finds a seat on a board covered with soft skin; and in that position the rim reaches just above his hips. He then tucks and arranges the lower part of his water pelt, or seal-skin coat, so tightly round the rim of the

² *Vide* p. 69.

hole, that no spray or water can penetrate, though the waves should fly over him and well souse both the boat and outer garment of its occupant.

The coat is, at the same time, very closely folded and buttoned about his face, shoulders, and arms, with bone buttons.

On each side of the top of the kajak he places his lances, and secures them under straps of leather, affixed for the purpose. In front of him he places the running line for seal hunting, rolled up on a small circular seat or platform purposely placed for it, and behind him he places the bladder.

His *pautik*, or double-bladed oar, is made of solid red deal, bound at each end of the blades with a thin plate two or three inches in width; and it is further protected from the ice by the edges of the blades being inlaid with bone. The Greenlander uses the *pautik* after the same manner as the sweep, or double-bladed oar is used in the Welsh coracle, but with great skill and precision of time. So equipped, he is enabled to propel his tiny boat with great velocity. He fears neither storms nor heavy seas; his kajak is light and buoyant as a cork. When a wave strikes it a broad-side, he balances himself and his boat with the *pautik*. And if he happens to be capsized, (which is not an uncommon occurrence,) by a peculiar and difficult muscular exertion, to which the Greenland lads are trained from their youth, he swings his body by aid of the oar, and rights himself and his kajak in a moment, though he be turned over with boat bottom upwards. He keeps fast hold of the *pautik*, for if he loses or breaks it in a strong wind or heavy sea, his chance is almost hopeless, if far from shore or assistance.³

Lieutenant Hooper, in his "Ten Months among the Tents of the Tuski," gives a very amusing picture of these remarkable little boats, and the skill and dexterity of the Greenland seal-hunters.

³ *Vide* Scheffer, *De Militia Navali Veterum*; M. Hassæus, *D. Leviathan Jobi*; Crantz' *Greenland*; *Modern Voyages and Travels*, &c.

"Five men, who launched their tiny and beautiful kiacs, accompanied us for a while; the light vessels, propelled by a double-bladed paddle, dancing over the rippling waves, light as a gossamer, and keeping pace with us with perfect ease, although we were favoured by a light breeze, and used both oars and sail. Two of these men continued with us for a considerable time, and amused us greatly by their proceedings. They frequently darted their fish-spears, or harpoons, at imaginary prey, to demonstrate the mode of using them. These spears were about a yard long, of thin drift-wood, with double or triple barbs of ivory: they were thrown from a hand-board, into which they fitted in a socket, and had a bladder attached to them, to keep them afloat in the water, the hunter picking them up with great ease as he passed rapidly onwards in his canoe."

Kajaks were also much used by Sir Leopold M'Clintock and his party, for the purposes of shooting and fishing when in the Arctic regions; and it appears they are used by pilots as well as fishermen and seal-hunters. It is stated, in M'Clintock's narrative, that when passing Godhaven, the pilot was launched off the deck of the "Fox" in his little kyak, without stopping the ship, and without the addition of a capsize, a feat which it would be difficult for an Englishman to perform in any boat of English construction.

M'Clintock also observes, with reference to these extraordinary little boats, and the manner in which they are managed, that when the "Fox" was at anchor in the neighbourhood of Whalefish Island, watching an opportunity to enter Godhaven, notwithstanding the hard-blowing weather, some natives came off in kajaks, about five miles from land, to the "Fox;" the water washed over their little craft, and kept the kayakers' seal-skin coats streaming with wet. And he adds—"this part of their dress seems rather part of the kayak, as it is attached to it round the hole in which the kayaker sits, so that no water can enter."



ESQUIMAUX CANOE.

These very fragile boats are made of seal and walrus skins stretched over a light framework of wood. They are entirely covered over at the top, with the exception of a circular hole, higher at the back than in front, to admit the body of the occupant. A reference to the above engraving and section will more clearly explain these curious little boats.

It would appear to be almost impossible for any kind of boat to be of lighter form; and they are propelled with great rapidity by the native Esquimaux.

LABRADOR CANOES.

There is a great similarity between these canoes and those of the Esquimaux, with the exception that they have not such high stem and stern-posts as the latter. In other respects they are nearly identical, being built in the same way, and with the same sort of materials. Some of them are large enough to carry two or three persons, a separate hole being formed in the top for each person.

BOATS OF THE TUSKI,⁴ OR TSCHUKTSCHI.

These very curious boats serve alike for the transport of goods and the purposes of the chase.

They are light but very tough, being composed of walrus skins, which are beautifully prepared, sewn over a light frame of wood. They are flat-bottomed, and nearly wall-sided;

⁴ "Ten Months among the Tents of the Tuski, with Incidents of an Arctic Boat Expedition in search of Sir J. Franklin." By Lieut. W. H. Hooper, R.N.

about three feet and a half in breadth at the widest part of the gunwales, and about two feet and a half below, decreasing in width at the ends, which are just wide enough to admit a man's body. The top rim is neatly worked over with strips of hide or whalebone. These boats are propelled with great speed by paddles, which are also used in bow and stern to direct their course. The smooth surface of the walrus skin offers little resistance to their passage through the water ; and they are carefully preserved from any incrustation, being carried on shore when not in use, turned over and beaten with stout sticks, to drive off the moisture and keep the skin in shape, as it would otherwise be liable to bag.

Similar boats to these are used by some of the North American Indians, who cover the frames with seal skins, and stain them with a cherry colour.

BOATS OF KAMTSCHATKA.

The Kamtschatkas have two sorts of boats—one is called *koaihtahta*, and the other *tahta*. The former is very similar to an English peter boat, except that the prow and stern are higher and the sides lower.⁵

The *tahta* has the prow and stern of an equal height : the sides are not rounded outwards, as in most boats, but rather incline inwards at the midships ; consequently they are very unsafe, and soon fill and swamp in rough water.

The Kamtschatka boats are built chiefly of poplar wood.

BAIDARS.

The boats of the Kurilski Islands and Lopatka are called *Baidars* : these are built with a keel similar to an English boat, with the exception that the planks are sewn together, and the seams are caulked with moss. The Kuriles have no proper wood for boat-building, except that which is washed

⁵ "History of Kamtschatka." Translated by Dr. Grieve, A.D. 1764.

ashore by the sea. In transporting their goods to and from these islands, they frequently lash two baidars together, and form a sort of bridge across them, in which way they carry them safer and drier. The northern inhabitants of Kamtschatka—the settled Koreki and Tchuktcha—for want of proper timber and plank, make their baidars of the skins of sea animals stretched on poles.

AFRICAN CANOES.

THE canoes of the Inland Africans appear to be seldom sailed: they are chiefly propelled by paddles. Dr. Livingstone describes those of Bakoba as very primitive craft, hollowed out of the trunks of single trees by means of iron adzes; and if the tree has a bend so has the canoe. The natives make fires in them; and prefer sleeping in them while on a journey, to spending the night on shore. "On land you have lions," say they, "serpents, hyenas, and your enemies; but in your canoe, behind a bank of reeds, nothing can harm you."¹

The fishing canoes of the Bakurutse he describes as—simply large bundles of reeds tied together.

Mr. Andersson also describes the reed raft as having sidings and uprights of the same material. There is also an engraving of it in his "Lake Ngami." It appears to be an ingenious contrivance; and, in localities where wood is scarce, answers the purpose admirably.

The chase of the hippopotami is also sometimes conducted on a large reed raft, capable of conveying six or eight men, and two small canoes. There is also an engraving of this remarkable contrivance in the same work.

One of the canoes in which Dr. Livingstone ascended the Leeambye he describes as thirty-four feet long, by twenty inches wide. This was propelled by six paddles; and it was

¹ Dr. Livingstone's "Travels in South Africa."

not the largest size : some of them had a crew of ten. They stand upright, and keep stroke with great precision, though they change from side to side as the course demands, the men at the head and stern being selected from the strongest and most expert. These canoes are flat-bottomed, and thereby enabled to go into very shallow water ; and whenever the bottom can be felt, the paddles are used as setting poles, to set the canoe along instead of paddling. Dr. Livingstone's fleet consisted of thirty-three canoes and about 160 men.

AFRICAN BARK CANOES.

On the Chikapa, Dr. Livingstone met with some bark canoes which he describes as made out of a single piece of bark, sewed together at the ends, and having sticks placed in it at different parts to act as ribs. The word "Chikapa" means "bark," or "skin;" and the Doctor says that was the only river in which he saw this kind of canoe used.

AFRICAN HUNTING CANOES.

These are very small, and can carry only two persons. They are made quite thin and light, and as sharp as English racing skiffs, being used chiefly for hunting the hippopotami and other animals in the water.²

CANOES OF LAKE ANEGUE.

The canoes used on the Anegue Lake are of a singular construction ; quite flat-bottomed, very light draught, about fifty feet long, and not more than two feet wide ; consequently they are ticklish craft. The oarsmen stand up, and use paddles seven feet long, with which they can propel one of these boats at a very good rate. But these canoes are very easily capsized, the gunwale being but a few inches above the water ; they are not, however, very often tipped over, though the negro paddlers stand up at their work.³

² Dr. Livingstone's Travels.

³ "African Travels." By M. Chaillu. 1861.

GELVES.

THESE remarkably ancient vessels are used chiefly on the Red Sea. They are built entirely out of the cocoa tree, which, on being sawn into planks, are sewn together with a kind of twine, which is spun out of the bark of the same tree, and also twisted into ropes and cables. The gelve thus built is tough and flexible; and if it chances to strike against a rock or run aground, it receives comparatively little or no injury.

The sails of the gelve are made of the broad leaves of the cocoa tree stitched together.

It has often been remarked, that out of the cocoa tree alone a vessel of the Red Sea may be built, rigged, and fitted with mast, sails, cordage, and ropes; and victualled with bread, water, wine, sugar, vinegar, and oil.



THE BEACON-LIGHT.

THERE'S a signal well known to the mariner brave,
His guide o'er the waters from perils to save ;
'Tis hailed with a welcome wherever in sight,
And the mariner calls it "the Beacon-light,"

A true friend in distress, ever brilliant and gay,
Whether fixed or revolving will ne'er lead astray ;
Tho' many brave hearts on a thick foggy night,
Have quaked at th' eclipse of "the Beacon-light."

When distant at sea and a beacon is seen,
A warning he takes of near danger I ween ;
Then he warily steers whilst he still keeps in sight
With his monitor dumb, "the Beacon-light."

When darkness fast creeps o'er the wide-ranging sea,
And the grim shark is lurking close under his lee,
Tho' gloomy the prospect and dismal the night,
His fears are dispelled by "the Beacon light."

Should a gale overtake him and danger appal,
The mariner skilful may weather the squall ;
But how gladly he looks thro' the darkness of night
On his guide and director "the Beacon-light."

The gale may increase, spreading terror around,
And 'neath the blue waves gaping sands may abound ;
Still his faith rests in heaven, his hope is in sight,
'Midst the bright gleams that fall from "the Beacon-light."

When far out at sea on the perilous deep,
And the high-swelling waves appear lull'd into sleep ;
There's the compass to steer by, and stars shining bright,
But he's sad at the loss of "the Beacon-light."

When parted from friends and dear ones at home,
(For long he's been toss'd on the white surgy foam),
How cheer'd is his heart when welcom'd at night
With a happy return, by "the Beacon-light."

SAILING RULES AND REGULATIONS.

1.—*Cutters* to be allowed the use of four sails only, viz., main-sail, fore-sail, jib, and gaff-top-sail. *Yawls* : the same sails as cutters, with the addition of a mizzen. *Sloops* : main-sail, fore-sail, and gaff-top-sail. *Schooners* : main-sail, fore-sail, fore-stay-sail, jib, main-gaff-top-sail, fore-top-sail, and fore-top-gallant-sail. *Lateens* : main-sail, fore-sail, and mizzen. *Luggers* : three lugs, jib, and top-sails. *Skiffs and open boats* to use only two sails.

2.—Three yachts or boats to start, or no race.

3.—All entries to be made on or before six o'clock P.M. on the day previous to the race, and each yacht or boat to carry, during the race, a distinguishing colour at the mast-head.

4.—Trimming the ballast, or shifting it, is not allowed ; and no ballast to be thrown overboard during the race.

5.—That lots be drawn for choice of stations.

6.—No jibs or fore-sails to be boomed out ; and no sweeps, boat-hooks, or setting booms, to be used during the race ; nothing to be used for sounding the depth but a lead line.

7.—Any boat wilfully fouling another in the same match, to forfeit all claim to the prize.

8.—That boats on the port tack give way to those on the starboard tack. Boats sailing free to give way to those on a wind.

9.—Any boat bearing away or altering her course to leeward, thereby compelling another to go out of her course, shall forfeit all claim to the prize.

10.—If two or more boats (by the wind) are approaching the shore, or any other obstruction, and are so close to each other that the leewardmost cannot tack clear of the weathermost, and by standing farther on would risk her running foul or ashore, the weathermost boat, on being requested to put about, is immediately to do so ; but the leewardmost must also tack at the same time as the one she hails.

11.—No anchoring allowed during the race, except to prevent accident ; in which case the anchor shall be weighed and taken aboard again.

12.—Each boat to go fairly round the mark buoys without touching the same, the boat nearest thereto to be considered headmost ; and should any other boat in the race compel the headmost boat to touch any mark buoy, the boat so compelling her to forfeit all claim to the prize, and the headmost boat to be held harmless for such contact.

13.—All protests to be delivered, in writing, to the secretary, judge, or other person appointed to receive the same, within two hours after the race.

14.—No person to be allowed to go aboard, or leave a boat in the race, for any purpose whatever, except in case of getting aground, when all who jump overboard must again return to the boat.

15.—Any boat or yacht failing to comply with any or either of the above rules and regulations, to forfeit all claim to the prize.

NAUTICAL VOCABULARY.

AFT, ABAFT, or ASTERN, towards the stern of a vessel.

A-LEE.—The side opposite to the wind; the situation of the helm when in an opposite direction to that from which the wind blows.

AMIDSHIPS.—Any part of the middle of a vessel with regard to her length and breadth.

ATHWART-HAWSE.—Across the direction of a vessel's head, under the bowsprit.

ATHWART-SHIPS.—Across a vessel from side to side.

A-WEATHER.—On the weather side, or towards the wind, the situation of the helm when in the same direction as that from which the wind blows, opposed to A-lee.

BACK-STAYS.—Ropes or stays extending from the mast-head to the sides of the vessel, abaft the mast.

BEAM.—The width of a vessel at the widest part.

BEAR.—To *bear-up* is to turn a vessel from the wind, by putting the helm up, or towards the quarter from which the wind blows. To *bear-down* is to pursue a vessel from a windward quarter.

BEARINGS.—The lower part of a vessel, or that which is below the water line when in proper trim.

BELAY.—To make a rope fast to a cleat by two or more turns without hitching it.

BEND.—To make fast; to *bend a sail* is to attach it to the mast or yard by lashings or otherwise.

BERTH.—A sleeping apartment aboard a vessel. A safe anchorage is termed a good berth.

BIGHT.—A noose or folded part of a rope; any part may be called the bight excepting the ends.

BILGE.—The protuberant part of a cask; the breadth of a vessel's bottom, or that part on which a vessel rests when lying aground on her side.

BINNACLE.—A box containing a mariner's compass.

BITTS.—Perpendicular pieces passing through the deck on each side the heel of the bowsprit; the cable is generally made fast to the bitts if there is no windlass.

BOARD.—The distance a vessel goes upon any one tack when beating to windward. The mast of a vessel is said to go by *the board* when it snaps off and falls over the bulwarks. To *make a good board* is to sail a long distance in a straight line when close hauled. To *make short boards* is to tack frequently.

BOBSTAY.—A small chain or rope used to keep the end of the bowsprit down.

BOLSTERS.—Pieces of soft wood tacked on each side the mast above the cheeks on which the eyes of the shrouds rest.

BOOM.—A spar used to spread the foot of a main-sail.

BOOM-REST.—A shifting iron of the same shape as an iron rowlock, in which the outer end of the boom is laid when the sails are furled.

BOWLINE.—A rope used in square rigged vessels for holding out the leach when sailing close hauled. Sailing on a bowline, is when the wind is on the quarter and the sheet is not close hauled.

BOWSE.—To pull or haul a rope on a tackle.

BOWSPRIT-SHROUDS.—Ropes on each side for protecting the bowsprit.

BRAILS.—Ropes connected with the throat of a sail, by which the canvas is drawn up close to the mast or yard.

BREAKERS.—Waves of white spray denoting shallow water; also a small cask containing water.

BRIDLE.—A rope is called a bridle when the two ends are made fast to the leaches or yard ends of a sail.

BROADSIDE.—The side of a vessel lengthways.

BULK-HEAD.—A temporary partition separating different parts of a vessel.

BULL'S-EYE.—A piece of wood the shape of a block, but with a hole in the centre for a rope to reeve through, and without a sheeve; also a thick piece of glass for a cabin light.

BULWARKS.—The boarding round the sides of a vessel above the deck.

BUMKIN, OR BOOMKIN.—An iron for setting out the fore-sail or mizzen beyond the boat's stem or stern.

BUM-BOATS.—Are those which attend alongside vessels with provisions, fruit, &c.

BUNTING.—Woollen stuff for making flags, burgees, &c.

CABLE.—A large rope for holding a vessel at anchor. A ship's cable is usually 120 fathoms, or 720 feet, in length.

CARRY AWAY.—To break a spar or rope.

CARVEL-BUILT.—A vessel is carvel-built when the planks are laid in smoothly and not lapped over as in clench, or clincher-built vessels.

CAT-HEADS.—Strong timbers projecting from the bows of vessels, to which the anchor is secured when heaved up.

- CAT'S-PAW.**—A very light current of air occasionally seen during a calm.
- CAULK.**—To fill the seams of a vessel with oakum.
- CHANNELS.**—Strong pieces of wood attached to a vessel's side, to which the shrouds or rigging are secured.
- CHEEKS.**—Small carved projections on each side the mast, upon which the trestle-trees rest.
- CHINSE.**—To fill the deck seams with oakum and serve them over with melted resin.
- CLENCH, OR CLINCHER-BUILT.**—When the planking of a vessel is lapped over, similar to weather-boarding.
- CLAMP.**—A mast-clamp is an iron for securing a boat's mast to the thwart.
- CLEAT.**—A small iron or wooden piece for belaying ropes to.
- CLEW.**—The lower corner of a square sail, and the after or outer corner of a fore-and-aft sail. To *clew-up* is to haul up the clew.
- CLEW-GARNET.**—A rope used in square-rigged vessels for hauling up the clew of a sail.
- CLOSE-HAULED.**—When the sheets are hauled in close, and the sail set as flat as possible.
- CLOVE-HITCH.**—(See *Knots*, &c., p. 106.)
- COMPANION-WAY.**—The staircase leading to and from the cabin.
- CRANK OR CRANKY.**—Not stiff. A vessel is said to be crank when inclined to roll on her side and is unable to carry but little canvas.
- CROSS-TREES.**—An iron or wooden piece secured to the mast above the cheeks for steadying and spreading the top-mast shrouds.
- CROWN.**—To crown a knot is to pass the strands over and under, so as to form a crown above the knot.
- CUDDY.**—A small cabin in the fore part of a boat.
- CUTWATER.**—A sharp projection forward of the vessel's bows.
- DAVIT.**—A projection of wood or iron over a vessel's side with a sheave in the end, used for hoisting up boats. A *fish-davit* is a short spar with a sheave in the end, used for fishing the anchor.
- DEAD-EYE.**—A small solid circular block with three holes through it, for the lanyards of rigging to reeve through; they are used instead of blocks.
- DEAD-WATER.**—The eddy under a vessel's counter.
- DOUBLE-BANKED.**—When two rowers sit upon the same thwart.
- DOUSE.**—To lower a sail suddenly.
- DOWN-HAUL OR DOWN-HAULER.**—A rope attached to the peak-end or other top part of the sails, for hauling down.
- DRAUGHT.**—The depth of water required for a vessel to float in.
- DRIVE.**—To drift down a tideway or scud before the wind.
- DRUM-HEAD.**—The top of a capstan.

EYE.—A loop in the end of a rope or stay.

EARING.—A rope attached to the aft-leech-thimble of a sail for bending the sail to the boom end, or for reefing purposes.

FENDER.—A soft substance of rope or tow covered with canvas, used for hanging over the side of a vessel, to protect it from striking or chafing when alongside.

FISH-DAVIT.—(See *Davit*.)

FLUKES.—That part of an anchor which holds in the ground, forming the arms and flat, triangular tips.

FORE-AND-AFT.—From stem to stern. A fore-and-aft rigged vessel has no square sails.

FORE-CASTLE.—(Pronounced foaks'l). A small cabin before the mast in the bows of a vessel; called in yachts, the galley.

FORE-REACH.—To pass a vessel when close-hauled and on the same tack.

FORE-GRIPE.—An additional piece secured to the lower part of a vessel's stem.

FOUNDER.—To fill with water and sink.

FRAP.—To pass a lashing round a sail to prevent it from blowing about.

FULL-AND-BY.—Sailing close to the wind without shaking any part of the sails.

FURL.—To roll up a sail close to the yard and secure it with lashings.

GAFF.—The top spar of a cutter's main-sail.

GALLEY.—A long-shaped boat; also the fore-castle aboard a yacht.

GAMMON.—An iron hoop or ring by the side of a vessel's stem, through which the bowsprit is run out.

GANGWAY.—An opening in a vessel's bulwarks, for convenience of getting from the vessel to a boat.

GARBOARD-STRAKE.—The strake or planking of a vessel nearest the keel on each side.

GASKETS.—Pieces of plaited yarn, used for lashings when the sail is furled.

GOOSE-NECK.—An iron joint connecting the boom with the mast.

GOOSE-WINGED.—A term applied to schooners or vessels with two masts, when running before the wind with sails boomed out on each side.

GORES.—Angles at one or both ends of such cloths as increase the width of the sail; goring-cloths are sometimes added when a sail is required with a narrow head and wide foot.

GRAPNEL.—A kind of anchor with four or more claws or barbs, used for securing the earings of fishing nets when spread in the water.

GROMMET.—A rope ring, much used in boats and ships; it is formed by laying round a single strand of rope (see *engraving*, p. 106).

GROUND-TACKLE.—Consists of anchors, cables, &c.; anything used for anchoring and securing a vessel at anchor.

GUNTACKLE-PURCHASE.—A rope working through two single blocks.

GUNWALE, or GUNNEL.—The inside piece leading from stem to stern on each side the boat, and to which the top strake is nailed.

GUY.—A rope for steadying a spar or boom, and for keeping it in its place.

GYBE.—To shift a boom sail from one side of a vessel to the other.

HALLIARDS, HALYARDS, or HAULYARDS.—Ropes or tackles used for hoisting and lowering sails.

HALF-HITCH.—(See *Knots*, p. 106.)

HATCHWAY.—The opening in a vessel's deck, leading to the cabin or hold.

HATCHES.—Shifting boards fitting over the hold of a vessel.

HAWSER.—A large rope for securing a vessel either at anchor or otherwise.

HEAD-SAILS.—Sails used before the mast, as fore-sail and jib.

HEAVE-SHORT.—To heave the vessel close up to, or over the anchor, by drawing in the cable.

HEAVE-TO.—(See *Lay-to*.) To haul the fore-sail to windward, so that the vessel makes little or no headway.

HELM.—The tiller or wheel by which a vessel is steered.

HITCH.—(See *Knots*, &c., p. 106.)

HOG.—A kind of broom used for scrubbing vessels' bottoms.

HOLD.—The interior of vessels where the cargo is placed.

HORNS.—That part of the gaff to which the trucks are attached, and which fits close to the mast; also called *Jaws*.

HOUNDS.—The shoulders at the mast-head, where the trestle-trees rest.

HOUSED.—The top-mast is housed when lowered down and secured at the heel by a lashing.

HOUSING-LINE.—A small cord used for seizings.

HULL.—The body of a vessel.

JAWS.—Gaff-ends hollowed out to fit the masts. (See also *Horns*).

JIB.—A triangular sail set out on the bowsprit.

JIB-BOOM.—A spar rigged out beyond the bowsprit.

JIGGER.—A small tackle for hoisting or hauling.

JUMPER-STAY.—An iron used in vessels with two masts, connecting one with the other at top.

JUNK.—Old rope cut up for oakum, &c.

JURY-RIG.—A temporary rig when a mast is carried away. A temporary mast is called a jury-mast.

KEDGE-ANCHOR.—An anchor with a shifting stock.

KELSON, or KEELSON.—A strong piece running the whole length of the vessel, and to which the keel is attached.

KEVEL-HEAD, or KEVEL.—A strong piece bolted to some part of the bulwarks for belaying ropes to.

KNEES.—Bent pieces used in boat-building for holding the planks together.

KNIGHT-HEADS.—Strong timbers near the stem on each side the bowsprit.

KNOT.—A mile as marked on the log line.

LANYARD.—A small rope, one end of which is made fast, whilst the other is used for securing anything to its place.

LARBOARD.—(Now obsolete), *Port* being the term applied. The left side of a vessel looking forward.

LAY-TO.—(See *Heave-to*.)

LAY-ALOFT.—An order for some part of the crew to get up the rigging.

LEACHES.—The outer edges of sails fore and aft.

LEADING-WIND.—A free wind ; when the wind is abeam, or nearly right aft.

LEE.—The opposite side to that from which the wind blows.

LEE-BOARD.—A board attached to the side of flat-bottomed vessels, which turns on a pivot, and is lowered when the vessel is going to windward ; it is intended to answer the purpose of a keel.

LEE-WAY.—The distance lost by a vessel drifting from the wind.

LEEWARD.—From the wind, the opposite to windward.

LIST.—To lay on one side by the pressure of the wind upon the sails.

LOG-BOOK.—A journal kept aboard ship, as to working the vessel, winds, weather, distances, &c.

LOG, LOG-LINE, and LOG-SHIP.—Used for regulating and ascertaining the rate at which the ship sails.

LOGGERHEAD.—A block with a small neck and larger head, for making ropes fast to.

LUFF.—To bring the ship near to the wind by putting the helm down ; the fore-leech of sails.

LUFF-TACKLE.—A purchase comprising a double and a single block.

MARLINE.—A kind of spunyarn, small soft stuff used for lashings.

MARLINSPIKE.—A wooden or iron pin gradually tapering to a sharp point ; used for splicing ropes, and various other purposes.

SHIPS.—The middle or broadest part of a vessel.

MISS-STAYS.—When the helm is put down, and the vessel fails to come about to the other tack.

MIZZEN.—A mizzen-mast, or mizzen-sail, is a small mast or sail abaft the main-mast.

MOORING.—Securing by two anchors placed in different directions.

MOUSE.—To tie a small spunyarn round the hollow of a hook, to prevent it from slipping off.

NEAP-TIDES.—Small or low tides, occurring at the middle of the moon's second and fourth quarters.

NOCK.—The upper end forward of a boom-sail.

OAKUM.—A sort of tow made by picking old rope to pieces.

OFFING.—Distance from shore.

OUTHaulER.—A rope used for hauling out the tack of a jib, or the clew of a boom-sail.

OUT-RIGGER.—A spar for spreading an extra sail beyond the hull of the vessel. A boat with projecting rowlocks. Also a weight suspended over a vessel's side.

OVERHAUL.—To examine; to slack a rope through a tackle, by letting go the fall and pulling on the leading parts.

PAINTER.—A rope secured in the bows of a boat, and used for making fast to a ship or otherwise.

PALM.—A sailmaker's thimble.

PARCELLING.—Winding tarred canvas round a rope, to prevent its chafing.

PARRAL.—A rope for securing a yard to a vessel's mast at its centre.

PAZAREE.—A rope secured to the clew of a fore-sail, and leading through a block on the boom; used for guying out when scudding.

PAWL.—An iron bar for securing the windlass or capstan from turning back.

PAY-OFF.—To haul the jib or fore-sail a-weather for the purpose of turning the vessel's head from the wind. To *Pay-over*, to cover with pitch or tar. To *Pay-out*, to slack out a cable from the vessel.

PEAK.—The upper aft corner of a main-sail.

PENDANT OR PENNANT.—A long narrow streamer carried at the mast-head. A *Swallow-tailed pennant* is called a *Broad pennant*, and is only hoisted by the Commodore of a club or fleet.

PINTLE.—An iron bolt belonging to a rudder.

PORT.—To port-the-helm is to put it to the left or contrary way to that of starboard.

PORT-HOLE.—A small opening in the bulwarks for pointing cannon.

PURCHASE.—A mechanical power applied to ropes rove through blocks where extra power is required.

QUARTER.—A term applied to that part of a vessel's sides which extends from the main channels to the stern.

QUARTER-DECK.—Applied to large vessels; that part of a vessel's upper deck abaft the main-mast.

RATLINE.—A kind of rope ladder formed across the shrouds by means of a small line called ratline.

REACHING.—Sailing on a side wind.

REEF.—To reduce a sail by tying up the reef points.

REEF-EARINGS.—Ropes attached to the reef thimbles of the aft leech of a sail.

REEVE.—To pass a rope through a block or sheave hole.

RIGGING.—A general term applied to shrouds, stays, and other ropes of a vessel. *Running-rigging* applies to such ropes as lead through blocks, and can be altered at pleasure. *Standing-rigging* applies to shrouds and stays which seldom require hauling down or slacking.

RING-TAIL.—A light studding sail set abaft the main-sail, by a yard slung from the gaff end.

ROACH.—The fore-leach of a sail which appears to draw by the pressure of the wind.

ROLLING-TACKLE.—An extra tackle for steadying sails in a heavy sea.

ROWLOCKS.—A rest for the oars of a boat to work in.

RUN.—The hollow or narrow part of a vessel's stern.

SCUD.—To run before the wind in a gale with little or no sail.

SCULL.—To propel a boat by means of a single oar at the stern. A *Scull*, a small oar.

SCUPPERS.—Openings in the lower part of a vessel's bulwarks for the water to run off the deck.

SCUTTLE.—A small hatchway. To *scuttle*, to bore holes in a vessel's bottom for the purpose of sinking her.

SCUTTLE-BUTT.—A cask kept on deck from which the water is taken for daily use.

SEAMS.—Spaces between the planks of a vessel.

SEIZE.—To secure by means of small stuff called seizings.

SERVAGEE.—A strong neat strap made by marling several rope yarns together.

SERVE.—To wind small marline or spunyarn round ropes to prevent chafing.

SERVING-BOARD.—A small board or mallet for winding taut, and putting on the service stuff.

SHACKLE.—An iron link with a shifting pin for connecting chains together.

SHEAVE.—The roller or wheel in a block on which the rope runs.

SHEAVE-HOLE.—A hole in a spar or block for a rope to reeve through.

SHEER.—The line of plank under a vessel's gunwale.

SHEET.—A rope attached to the aft clew of a sail by which it is worked from one side to the other.

SHEET-ANCHOR.—The largest anchor a vessel carries.

SHROUDS.—Ropes for protecting the mast, leading from the mast-head to the sides of the vessel.

SISTER-BLOCK.—A sort of double block with two sheaves one above another.

SKY-SCRAPER.—A light triangular sail used in large ships, properly called a sky-sail; it is set above the royal.

SKEET.—A scoop with a long handle used for wetting sails.

SNATCH-BLOCK.—A single block with a hole in its side for the bight of a rope to reeve through.

SNORTER.—A small rope strap into which the heel of a spreet is set.

SPENCER.—A sail used in schooners, set on the fore-mast, with gaff, but no boom.

SPLICE.—To join the ends of two ropes by interweaving the strands.

SPREET.—A light spar for setting a sprit-sail.

SPRING-A-LEAK.—To commence leaking.

SPRING-A-LUFF.—To force a vessel closer to the wind.

STANDING-RIGGING.—Shrouds and other ropes made fast, and not very often requiring alteration.

STARBOARD.—The right hand side of a vessel looking forward.

STAY.—A rope for protecting the mast. *To stay*, to tack a vessel, or put about. *In stays*, the situation of a vessel at the instant of tacking, when all sail is shaking in the wind's eye.

STAY-SAIL.—Implies a sail set upon a stay.

STEERAGE.—That part of a vessel (between decks) forward of the cabin.

STEM.—The fore piece of a vessel, and to which the two sides or bows are united.

STERN-BOARD.—The back motion of a vessel moving stern foremost.

STERN-POST.—That to which the rudder is fixed; the extreme end aft of a vessel.

STERN-SHEETS.—The after part of a boat, abaft the rowers, intended for passengers.

STEVE.—The angle of a bowsprit when raised more or less from the horizontal.

STIFF.—An essential quality in the hull of a sailing vessel, which enables large sails to be used in safety, or without causing the vessel to list on one side.

STRAND.—A yarn, or number of yarns twisted together.

STRIKE.—To strike the mast is to lower it down into the boat.

SWEEP.—A long oar of extra size used in small sailing vessels when becalmed, or requiring to be forced ahead.

TACKING.—Turning, or beating to windward. To *Tack* is to put a vessel about by putting the helm a-lee, thereby bringing the vessel round on another tack. The tack of a main-sail is the corner nearest the goose-neck.

TACKLE.—A purchase formed by a rope rove through a block or blocks.

TAFFRAIL.—The rail, or top-piece, leading round a vessel's stern, level with the side rail above the bulwarks.

TAIL-BLOCK.—A block with a rope spliced into, and hanging from, the end.

TARPAULIN.—Canvas rendered waterproof by being dressed with tar.

TAUT.—To haul tight.

TELL-TALE.—A compass suspended from the beams of a cabin.

THIMBLE.—A metal ring, with concave surface, used in sail-making, and for splicing into rope-ends when required.

THROAT.—That part of a sail where the brails catch and draw it close to the mast; the inner end of a gaff near the *Jaws*.

THWAETS.—All seats across a boat are so called. (See also *Athwart*).

TILLER.—The lever by which the rudder is moved.

TOPPING-LIFT.—A rope leading from the upper part of the mast to the outer end of the boom, and by which the boom is topped up or raised.

TRAVELLER.—An iron, or metal ring, with a hook below and an eye at top. A *Traveller* leads up and down the mast, or along a bowsprit, for keeping the sail close to the spar.

TRESTLE-TREES.—Strong pieces placed horizontally, and fore and aft of the mast-head, to support the cross-trees and top.

TRICE-ROPE.—A rope used for hauling up the main-tack.

TRICE-UP.—To haul up the main-tack by means of the trice-line.

TRIM.—To arrange the ballast, or cargo, in the most advantageous manner. To *Trim* a sail, to set it in the best and most effective position with regard to wind.

- TRUCK.**—The flat circular piece, on the very top of the mast-head, through which signal halliards are rove.
- TRYSAIL.**—A fore-and-aft sail set with boom and gaff at the main-mast of a brig; also a small extra stout gaff-sail used for cutters in a gale.
- TYE.**—A rope belonging to a boat's yard, with a hoisting tackle at one end.
- UNBEND.**—To untie, to cast off.
- UNDER WAY.**—A vessel is under way when moving through the water by the action of the wind upon the sails.
- UNSHIP.**—To take out of a vessel, to remove out of its place.
- VEER.**—To turn a vessel from the wind and bring her round on another tack.
- WAIST.**—The middle part of a vessel's deck, between the quarter-deck and galley.
- WAKE.**—The track of a vessel in the water.
- WALES.**—Strong planks in the upper part of a vessel's sides, running the length fore and aft.
- WARP.**—A strong rope for securing a vessel.
- WASH-BOARDS.**—Shifting side boards used in boats for preventing the water coming in over the gunwales when sailing in a rough sea.
- WEATHER-BOARD.**—That side of a ship which is to windward.
- WEATHER-HELM.**—A vessel carries a weather-helm when the tiller requires to be slightly inclined towards the windward to keep the vessel on her course and prevent her from flying into the wind.
- WEIGH-ANCHOR.**—To draw the anchor up from the bottom.
- WHIP-PURCHASE.**—A purchase formed by a rope rove through a single block. To *Whip*, to secure the end of the rope from unravelling by winding twine round it.
- WORMING.**—To wind cord or marline, spirally between the strands of a rope to give it a neat appearance.
- YARD.**—A spar tapering at each end for spreading a lug or square sail.
- YOKE.**—A top-piece fitting on the boat's rudder, with lines attached for steering: it is chiefly used for small boats instead of a tiller.

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